Lectures on TQFT

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Contents

Preface

This text is based on the notes for a course taught by the author at Stony Brook University in the Spring of 2026.

This is a work in progress, to be updated frequently.

The goal of this course is to give a review of mathematics related to Topological Quantum Field Theory and more generally, extended TQFT — from its origin in physics through the initial definition of Atiyah to notions of extended TQFT following works of Lurie and possibly to recent developments related to factorization homology. We also include constructions of TQFT via state sums (also known as lattice TQFT), at least in examples.

Our goal is to make it a readable overview; therefore, in many places we only give a sketch of the proof, or even of a definition (for example, we do not give all details of construction of (∞, n) categories), referring the reader to original papers for details. Instead, we concentrate on examples of TQFTs, such as

- Gauge theory with finite group G (also known as Dijkgraaf–Witten theory)
- 2d TQFT using symmetric Frobenius algebras, which we present both using generators for the category of 2d bordisms (Schommer-Pries) and using state sums (Fukuma–Hosono–Kawai theory)
- Turaev–Viro 3d TQFT
- Crane–Yetter 4d TQFT and the closely related Chern–Simons 3d TQFT

Of all the examples mentioned above, Chern–Simons theory is by far the best known, and there are numerous expositions of this theory. Partly for this reason, we do not plan to spend too much time on it; in particular, we will not be discussing quantum invariants of knots and links.

Part 1

Basic definitions

Part 1 overview

In this part, we introduce the first definitions of TQFT.

In Chapter 1, we briefly review the main ideas of quantum field theory in physics, most importantly path integral formulation of QFT due to Feynman. This section is extremely brief and is used as motivation only; no precise definitions or statements are given there. However, we hope that it helps the reader understand the physical roots of TQFT.

In Chapter 2, we introduce the key mathematical tool which will be extensively used throughout this book: the notions of a monoidal category, monoidal functor, and related topics.

Finally, in Chapter 3 we define the category of cobordisms and use it to give a definition of TQFT as a functor $\mathbf{Cob}_n \to \mathbf{Vec}$. This definition is essentially due to Atiyah. We also study some basic properties of TQFTs, in particular showing that a TQFT Z defines, for any closed $(n-1)$-manifold N, an action of the mapping class group $\Gamma(N) = \pi_0(\mathrm{Diff}_+(N))$ on the vector space $Z(N)$. Finally, we give a classification of 1-dimensional TQFTs. This is, of course, trivial, but serves as a foundation for all classification results which appear later in the book, such as the cobordism hypothesis.

CHAPTER 1

Physical motivation

Overview of classical and quantum field theory. N is the configuration space of the theory, $\dim N = n-1$; $M = N \times [t_0, t_1]$ is the spacetime.

1.1. Classical/quantum mechanics

The table below summarizes the main structures in classical and quantum mechanics:

	Classical mechanics	Quantum mechanics
Position	$x \in N$ (configuration space)	$\psi \in H_N = L^2(N)$
Trajectory	$\gamma\colon [t_0, t_1] \to N$	$U_{t_0,t_1}\colon H_N \to H_N$
Action	$S[\gamma] = \int L(\gamma, \dot\gamma) dt$	
Laws of motion	Least action principle	$(U\psi)(x_1) = \int_N (K(x_0, x_1)\psi(x_0) dx_0$
	$\delta S[\gamma] = 0$ (fixed endpoints)	$K(x_0, x_1) = \int_P e^{\frac{i}{\hbar}S[\gamma]} D\gamma$
		$P = \{\gamma : \gamma(t_0) = x_0, \gamma(t_1) = x_1\}$

Table 1.1.1. FIXME

Note that in the limit $\hbar \to 0$, the integral in the definition of $K(x_0, x_1)$ is dominated by the contribution from the neighborhood of critical points of the action. For finite-dimensional integrals of this form, method of stationary phase allows one to write asymptotic expansion in powers of $\hbar$.

1.2. Classical/quantum field theory

In field theory, a point $x \in N$ is replaced by the field φ, which is typically either a function on N (with values in some target space V) or a section of a vector bundle or a connection. We will denote by $\mathcal{F}_N$ the space of all fields on N.

	Classical field theory	QFT
	$\varphi \in \mathcal{F}_N$	$\psi \in \mathcal{H}_N = L^2(\mathcal{F}_N)$
Trajectory	$\varphi \in \mathcal{F}_M$, $M = N \times [t_0, t_1]$	$U_{t_0,t_1}\colon \mathcal{H}_N \to \mathcal{H}_N$
Action	$S[\varphi] = \int_M L(\varphi, \partial_\mu\varphi) d^n x$	
Laws of motion	Least action principle	$(U\psi)(\varphi_1) = \int_N (K(\varphi_0, \varphi_1)\psi(\varphi_0) D\varphi_0$
	$\delta S[\varphi] = 0$ (fixed restriction to	$K(\varphi_0, \varphi_1) = \int_P e^{\frac{i}{\hbar}S[\varphi]} D\varphi$
	$t = t_0$,$t = t_1$)	$P = \{\varphi \in \mathcal{F}_M : \varphi(t_0) = \varphi_0, \varphi(t_1) = \varphi_1\}$

Table 1.2.1. FIXME

The above formulas freely use integrals over infinite-dimensional spaces of paths/fields, which are very hard to define rigorously; as of now, QFT cannot be defined rigorously as a mathematical theory except very few special cases. Usually the best one can do is to write the asymptotics formulas, expanding the operator U or the kernel K as a series in powers of $\hbar$, and even that requires a lot of work.

However, if we ignore that and assume that all the above makes sense, we get the following structure (again, at the moment we choose to ignore details)

(1) For every $n-1$-dimensional manifold N, we have a Hilbert space $\mathcal{H}_N$. Moreover, since $\mathcal{F}_{N_1\sqcup N_2} = \mathcal{F}_{N_1} \times \mathcal{F}_{N_2}$, we get

$$\mathcal{H}_{N_1\sqcup N_2} = \mathcal{H}_{N_1} \otimes \mathcal{H}_{N_2} \tag{1.2.1}$$

(2) The definition of operator U_{t_0,t_1} immediately generalizes, using the same formulas, to arbitrary n-dimensional manifold M and not only the cylinder $N \times [t_0, t_1]$: if $\partial M = N_0 \sqcup N_1$ (with appropriate orientations), then U_M is an operator

$$U_M : \mathcal{H}_{N_0} \to \mathcal{H}_{N_1} \tag{1.2.2}$$

In particular, for a closed spacetime (i.e. $\partial M = \varnothing$), we get $U_M \in \mathbb{C}$ is a number; in physics it is usually called the *partition function* and denoted by $Z(M)$.

(3) So defined operators satisfy the gluing axiom: (FIXME)

So far we haven't specified what we mean by "manifold". Usually in physics the spacetime is an oriented pseudo-Riemannian manifold, with metric of signature $(1, n-1)$. However, we will be interested in a very special case, namely when all the structures above do not use the metric on the space; thus, we only require M to be an oriented smooth manifold. Such theories are called *topological*. The most famous example of such a theory is Chern–Simons theory (see [**Wit1989**]), in which the spacetime M must be 3-dimensional, fields are $\mathrm{SU}(n)$ connections on M (which, after choosing a trivialization of the bundle, can be described by one-forms with values in the Lie algebra $\mathfrak{su}(n)$) and the action is given by

$$S[A] = g \int_M \mathrm{tr}(A \wedge dA + \tfrac{2}{3} A \wedge A \wedge A) \tag{1.2.3}$$

We can take properties above as axioms.

Preliminary Definition 1.2.1. An n-dimensional topological field theory (TQFT) is the following collection of data

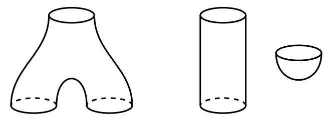
- For every $(n-1)$-dimensional oriented smooth manifold N, a vector space $Z(N)$
- For every n-dimensional oriented manifold M such that $\partial M = N_0 \sqcup N_1$, a linear operator $Z_M : Z_{N_0} \to Z_{N_1}$,

This data has to satisfy the properties above (gluing axiom, disjoint union).

(Clearly, there is a lot more detail to be filled in, which we will do later).

This definition is due to Atiyah ([**Ati1988**]).

Having this definition, instead of defining a TQFT by a path integral or other global construction (which can be very hard to do), we can try to construct TQFTs by defining $Z(M)$ for "elementary blocks" and gluing every manifold from them. For $n = 2$, these blocks are disk, cylinder, and pair of pants.



Of course, we need to have some compatibility conditions to ensure that different ways of gluing the manifold from pieces give the same result; essentially, we are trying to define manifolds by "generators and relations". For $n = 2$ it is not too hard (we will do so next week); for larger n, it is harder — e.g. for $n = 3$ we can consider so-called Heegard decomposition (which already requires infinitely many blocks, all handlebodies of arbitrary genus) or use surgery along knots (again, to be discussed later).

However, there is another idea: why stop at manifolds with boundary? If we allow manifold with corners of any codimension, the set of "elementary blocks" is greatly simplified: any manifold can be glued together from just one block, an n-dimensional simplex. Such theories are called fully extended (or sometimes local) TQFTs; see [**Lur2009a**].

But the price you pay is algebraic complexity: for closed n-manifold, the theory gives you a number; for a codimension 1 manifold, a vector space. What is the next level?

One way to answer is that codimension 2 should give you a category, and so on; thus, doing it properly requires a notion of n-categories (or a variation of that, called (∞, n) categories). This gets confusing quite fast...

CHAPTER 2

Algebraic prerequisites: monoidal categories

In this chapter we introduce some of the algebraic language necessary for describing TQFTs, most importantly the notion of a monoidal category. We only give a very brief introduction, referring the reader to books [**ML1998**], [**EGNO2015**] for details.

2.1. Category theory basics

We assume that the reader is familiar with basic notions of categories and functors. We will completely ignore all set-theoretic difficulties related to the definition of a category; if necessary, the reader can assume that all sets of objects, morphisms, etc in all the categories discussed are subsets of a large enough universal set U.

For a category $\mathcal{C}$, we will use the notation $X \in \mathcal{C}$ to show that X is an object of $\mathcal{C}$. For two objects $X, Y \in \mathcal{C}$, we denote by $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ the set of morphisms from X to Y (we will drop the subscript $\mathcal{C}$ when there is no ambiguity).

Given categories $\mathcal{C}, \mathcal{D}$, we denote by $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ the set of all functors $\mathcal{C} \to \mathcal{D}$.

Examples:

- **Set**: category of all sets.
- $\mathbf{Man}_n$: category of smooth manifolds of dimension n. There are different versions: oriented manifolds, framed manifolds, etc.
- $\mathbf{PLMan}_n$: category of PL manifolds of dimension n. Note that in dimensions $n \leq 6$, every PL manifold has a canonical smooth structure and vice versa. In general, it is not so: every smooth manifold has a unique PL structure (Whitehead's theorem), but a PL manifold might not have a compatible smooth structure, or if it has one, it might not be unique (Milnor's exotic spheres). We will talk about PL manifolds in more detail later (see Chapter 11).
- $\mathbf{Vec}_{\mathbf{k}}$: category of vector spaces over a field $\mathbf{k}$.
- $R-$**mod**: category of left modules over an associative ring R.

One special case of a category, which we will use frequently, is a groupoid.

Definition 2.1.1. A groupoid is a category in which all morphisms are invertible.

Example 2.1.1. Let X be a topological space. Define its *fundamental groupoid* $\pi(X)$ as the category whose objects are points in X and morphisms are homotopy equivalence classes of paths between points, with the obvious composition.

One common theme in all our constructions is equality vs isomorphism. When talking about points in a set (e.g., elements of a ring), we can require them to be equal; for example, we can require $a + b = b + a$ for elements of a ring. However, while it is technically possible to require two objects in a category to be equal, it rarely makes sense to do so. For example, in the category of finite sets, sets $A \times B$ and $B \times A$ are not equal. On the other hand, requiring two objects to be isomorphic is typically too weak: e.g., any two sets with the same number of elements are isomorphic.

The common way to resolve this is the following convention.

Convention 2.1.1. A collection of objects X_α of category $\mathcal{C}$, together with a collection of isomorphisms $f_{\beta\alpha}\colon X_\alpha \xrightarrow{\sim} X_\beta$, such that $f_{\alpha\beta}f_{\beta\gamma} = f_{\alpha\gamma}$, $f_{\alpha\alpha} = \mathrm{id}_{X_\alpha}$, is treated as a single object in $\mathcal{C}$.

This convention, though rarely stated explicitly, is actually widespread. Virtually every math textbook treats sets $A \times B$ and $B \times A$ as the same set, even though they are technically not the same object of category **Set**; from the point of view of the convention above, it is justified by noting that we have a canonical isomorphism $A \times B \xrightarrow{\sim} B \times A\colon (a, b) \mapsto (b, a)$.

Formally, this convention can be justified by constructing a category $\widehat{\mathcal{C}}$, whose objects are such collections $(X_\alpha, f_{\alpha\beta})$, and observing that categories $\mathcal{C}$, $\widehat{\mathcal{C}}$ are equivalent; see, e.g., [**BK2001**, Lemma 1.1.12].

2.2. Abelian categories

Definition 2.2.1. An abelian category is a category in which for every pair of objects X, Y, the set of morphisms $\mathrm{Hom}(X, Y)$ is an abelian group, composition is additive, we have a zero object, the direct sum of two objects, and for every morphism, we have the notion of image and kernel satisfying the usual properties (see [**EGNO2015**, Section 1.3] for details).

In such a category, we can define the notion of a simple object (which has no subobjects other than 0 and itself), semisimple one (isomorphic to a direct sum of simple objects), etc. We can also define the notions of short exact sequences, Jordan–Hölder series (which in turn allows one to define the notion of length of an object), injective and projective objects, and all other usual constructions of homological algebra.

An abelian category is $\mathbf{k}$-*linear* if all sets $\mathrm{Hom}(X, Y)$ are vector spaces over a field $\mathbf{k}$, and composition of morphisms is linear over $\mathbf{k}$.

Example 2.2.1. Below are some examples of $\mathbf{k}$-linear abelian categories.

(1) $\mathbf{Vec}_{\mathbf{k}}$;
(2) category $\mathrm{Rep}_{\mathbf{k}}(G)$ of representations of a group G;
(3) category $\mathrm{Rep}_{\mathbf{k}}(\mathfrak{g})$ of representations of a Lie algebra $\mathfrak{g}$;
(4) category $A-\mathbf{mod}$ of modules over an associative algebra A over $\mathbf{k}$.

Typically, one imposes some finiteness conditions for $\mathbf{k}$-linear categories. We will usually assume that all linear categories, unless explicitly stated otherwise, are *locally finite*, i.e.,

- All spaces $\mathrm{Hom}(X, Y)$ are finite-dimensional
- Every object has finite length

(see [**EGNO2015**, Section 1.8]).

One can also impose stricter requirements, for example semisimplicity (every short exact sequence splits), finiteness (there are finitely many isomorphism classes of simple objects), and more. We will discuss it in more detail when it becomes necessary.

One useful fact about $\mathbf{k}$-linear categories is the following version of the famous Schur's lemma in representation theory.

Lemma 2.2.1. *Let $\mathcal{A}$ be a locally finite $\mathbf{k}$-linear category.*

1. *If $X, Y \in \mathcal{A}$ are simple objects, then any morphism $f\colon X \to Y$ is either zero or invertible.*
2. *If $\mathbf{k}$ is algebraically closed, then for any non-zero simple $X \in \mathcal{A}$, $\mathrm{Hom}_{\mathcal{A}}(X, X) = \mathbf{k}$.*

As with the usual Schur's lemma, the proof is immediate from the fact that the only finite-dimensional division algebra over an algebraically closed field $\mathbf{k}$ is $\mathbf{k}$ itself (see, e.g., [**Pie1982**, Lemma 3.5]).

2.3. Monoidal categories

One of the most important notions is the notion of a monoidal category, which informally is described as a category with an associative operation; the model example is the category of vector spaces with the operation of tensor product. In this section, we review this notion; as before, we refer the reader to [**EGNO2015**, Chapter 2]; [**Lei2004**, Section 1.2] for a more in-depth discussion.

Before defining monoidal categories, let us look at simpler notion: a monoid.

A *monoid* is a set together with an associative multiplication operation and a unit:

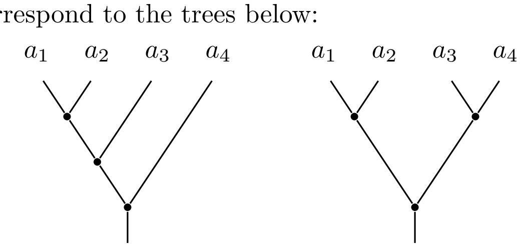
$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$
$$1 \cdot a = a \cdot 1 = a$$

(if we also add the requirement that every element has an inverse, we get the definition of a group).

Theorem 2.3.1. *Let M be a monoid and $a_1, \dots, a_n \in M$. Consider all possible expressions obtained by writing $a_1, \dots, a_n$ (in this order), then inserting some copies of 1 and then putting parentheses so that multiplications are just binary products, e.g., $(a_1 \cdot 1) \cdot (1 \cdot a_2)$. Then all of them are equal: if X_1, X_2 are two such expressions (with the same $a_1, \dots, a_n$), then $X_1 = X_2$.*

It may seem obvious, but in fact it is not quite trivial; essentially it is the statement that if we construct a graph whose vertices correspond to different expressions written above, and edges to application of associativity and unit relations, then this graph is connected.

If we only include $a_1, \dots, a_n$ but no 1's, then such expressions are naturally described by planar binary trees. For example, the bracketings $((a_1 a_2) a_3) a_4$ and $((a_1 a_2)(a_3 a_4))$ correspond to the trees below:



Adding ones requires a minor modification, adding appropriately colored leaves to the tree.

For categories, the analogous structure is that of a monoidal category: a category with a functor $\otimes\colon \mathcal{C}\times\mathcal{C}\to\mathcal{C}$ and a unit object $\mathbf{1}\in\mathcal{C}$.

One can try to mimic the definition above.

Definition 2.3.1. A *strict monoidal category* is a category $\mathcal{C}$ together with a functor $\otimes\colon \mathcal{C}\times\mathcal{C}\to\mathcal{C}$ and a unit object $\mathbf{1}\in\mathcal{C}$ such that

$$\begin{aligned}&(A\otimes B)\otimes C = A\otimes(B\otimes C),\\ &\mathbf{1}\otimes A = A\otimes\mathbf{1} = A.\end{aligned}$$

However, this definition is too strict: objects in a category are almost never equal, just isomorphic. For example, if V is a vector space over $\mathbf{k}$, then $\mathbf{k}\otimes V\neq V$ (but is canonically isomorphic to V).

We can try to replace equality by isomorphism, but then it is too weak: e.g., any two vector spaces of the same dimension are isomorphic.

The proper way is to require the existence of a distinguished (canonical) isomorphism which should be made part of the structure. Moreover, these "canonical" isomorphisms should themselves satisfy some compatibility conditions, as suggested by Convention 2.1.1.

Definition 2.3.2. A monoidal category is a category $\mathcal{C}$ together with the following data:

- An object $\mathbf{1}\in\mathcal{C}$
- a functor $\otimes\colon \mathcal{C}\times\mathcal{C}\to\mathcal{C}$
- functorial isomorphisms

$$\begin{aligned}&l_A\colon \mathbf{1}\otimes A\xrightarrow{\sim} A\\ &r_A\colon A\otimes\mathbf{1}\xrightarrow{\sim} A\\ &\alpha_{A,B,C}\colon (A\otimes B)\otimes C\xrightarrow{\sim} A\otimes(B\otimes C)\end{aligned}$$

satisfying compatibility conditions below:

(1) Triangle axiom: for any $A,B\in\mathcal{C}$, the diagram below commutes

$$\begin{tikzcd}(A\otimes\mathbf{1})\otimes B \arrow[rr, "\alpha_{A,\mathbf{1},B}"] \arrow[dr, "r_A\otimes\mathrm{id}_B"'] & & A\otimes(\mathbf{1}\otimes B) \arrow[dl, "\mathrm{id}_A\otimes l_B"] \\ & A\otimes B & \end{tikzcd} \tag{2.3.1}$$

(2) Pentagon axiom: for any $A,B,C,D\in\mathcal{C}$, the diagram below commutes

$$\begin{tikzcd}[column sep=small] & (A\otimes(B\otimes C))\otimes D \arrow[r] & A\otimes((B\otimes C)\otimes D) \arrow[dr] & \\ ((A\otimes B)\otimes C)\otimes D \arrow[ur] \arrow[drr] & & & A\otimes(B\otimes(C\otimes D)) \\ & & (A\otimes B)\otimes(C\otimes D) \arrow[ur] & \end{tikzcd} \tag{2.3.2}$$

(all arrows are given by the associativity isomorphism α.)

EXAMPLE 2.3.1. The following are examples of monoidal categories.

(1) Category **Set**, with $\otimes$ given by disjoint union and $\mathbf{1}$ being the empty set
(2) Category **Set**, with $\otimes$ given by the Cartesian product and $\mathbf{1}$ being a one-point set
(3) Categories **Man**, **PLMan** with respect to disjoint union
(4) Category **Vec** with the usual tensor product, $\mathbf{1} = \mathbf{k}$
(5) Category of bimodules over a fixed ring R, with respect to $\otimes_R$, and $\mathbf{1} = R$

EXAMPLE 2.3.2. Let G be a group. Define the category $\mathbf{Vec}_G$ of *G-graded vector spaces* as the category whose objects are finite-dimensional vector spaces V together with decomposition

$$V = \bigoplus_{g \in G} V_g.$$

This is a monoidal category with tensor product given by

$$(V \otimes W)_g = \bigoplus_{h \in G} V_h \otimes W_{h^{-1}g}$$

and unit object given by $\mathbf{k}$ considered as a G-graded vector space with grading equal to $1 \in G$. The associativity isomorphism is trivial.

THEOREM 2.3.2 (Mac Lane's coherence theorem). *Let $\mathcal{C}$ be a monoidal category. Let $F_1, F_2 \colon \mathcal{C}^n \to \mathcal{C}$ be two functors obtained by compositions of tensor product functor $\otimes$ and functors $X \mapsto X \otimes \mathbf{1}$, $X \mapsto \mathbf{1} \otimes X$, e.g.,*

$$F_1(A, B, C) = (A \otimes B) \otimes (\mathbf{1} \otimes C),$$
$$F_2(A, B, C) = A \otimes ((B \otimes C) \otimes \mathbf{1}).$$

Then

(1) *There exists a functorial isomorphism $F_1 \simeq F_2$, obtained by composition of associativity isomorphism α, left and right unit isomorphisms l, r, and their inverses.*
(2) *Any two such isomorphisms $F_1 \to F_2$ are equal.*

In other words, if we construct a cell complex X_n whose vertices are functors $\mathcal{C}^n \to \mathcal{C}$ as in the statement of the theorem, edges correspond to applications of the associativity and unit isomorphisms, and 2-cells correspond to commutative diagrams given by the pentagon axiom, unit axioms, and functoriality of α, l, r, then this complex is simply-connected.

This is a natural generalization of coherence theorem for monoids (Theorem 2.3.1); in this language, Theorem 2.3.1 is exactly claim (1) of the theorem, i.e., the statement that X_n is connected.

REMARK 2.3.1. If we only consider functors obtained by composing the tensor product functor (thus, no units), the corresponding complex is a 2-skeleton of a convex polytope K_n, called the *associahedron*, or the *Stasheff polytope*. The vertices of K_n are indexed by binary trees with n leaves; thus, the number of vertices is the Catalan number $C_{n-1} = \frac{1}{n}\binom{2(n-1)}{n-1}$ (so K_5 has $C_4 = 14$ vertices). This polytope was introduced by Jim Stasheff in the 1960s. It can be realized as a convex polytope in Euclidean space; see, e.g., [**Lod2004**].

EXERCISE 2.3.1. Draw the associahedron K_5 (it has 14 vertices) and verify that it is simply-connected (more precisely, its 2-skeleton is topologically a sphere S^2). We will return to it later, when talking about monoidal 2-categories.

DEFINITION 2.3.3. Let $\mathcal{C}, \mathcal{D}$ be monoidal categories. A monoidal functor $\mathcal{C} \to \mathcal{D}$ is a functor $F\colon \mathcal{C} \to \mathcal{D}$ together with functorial isomorphisms

$$J\colon F(X \otimes Y) \to F(X) \otimes F(Y)$$
$$\varphi\colon \mathbf{1}_{\mathcal{D}} \to F(\mathbf{1}_{\mathcal{C}})$$

satisfying suitable constraints (see [**EGNO2015**, Section 2.4]).

We say that categories $\mathcal{C}, \mathcal{D}$ are monoidally equivalent if there exist monoidal functors $F\colon \mathcal{C} \to \mathcal{D}$, $G\colon \mathcal{D} \to \mathcal{C}$ such that $F \circ G \simeq \mathrm{id}_{\mathcal{C}}$, $G \circ F \simeq \mathrm{id}_{\mathcal{D}}$.

EXAMPLE 2.3.3. Consider the category of representations of a group G. Then the forgetful functor from this category to the category of vector spaces is a monoidal functor. More generally, given a subgroup $H \subset G$, the restriction functor $\mathrm{Res}^G_H\colon \operatorname{Rep} G \to \operatorname{Rep} H$ has an obvious monoidal structure.

EXAMPLE 2.3.4. If G is a connected, simply-connected Lie group, then the category of finite-dimensional representations of G is equivalent, as a monoidal category, to the category of finite-dimensional representations of its Lie algebra $\mathfrak{g}$.

EXAMPLE 2.3.5. Recall the category of G-graded vector spaces $\mathbf{Vec}_G$ defined in Example 2.3.2. This category has a generalization. Namely, let $\omega\colon G \times G \times G \to \mathbf{k}^\times$ be a 3-cocycle, i.e., a function satisfying the condition

$$\omega(g_1g_2, g_3, g_4)\omega(g_1, g_2, g_3g_4) = \omega(g_1, g_2, g_3)\omega(g_1, g_2g_3, g_4)\omega(g_2, g_3, g_4).$$

Consider the category $\mathbf{Vec}_G^\omega$ which coincides with $\mathbf{Vec}_G$ as an abelian category; moreover, it has the same tensor product functor $\otimes$ and unit object $\mathbf{1} = \mathbf{k}_1$. However, the associativity isomorphism is different: namely, $\alpha\colon (V \otimes W) \otimes U \to V \otimes (W \otimes U)$ is given by

$$\alpha((v \otimes w) \otimes u) = \omega_{g,h,m} v \otimes (w \otimes u), \qquad v \in V_g, w \in W_h, u \in U_m.$$

The cocycle condition immediately implies that so defined α satisfies the pentagon axiom.

EXERCISE 2.3.2. Let $\omega\colon G \times G \times G \to \mathbf{k}^\times$ be a 3-cocycle. Assume that ω is a coboundary, i.e., there exists a function $\beta\colon G \times G \to \mathbf{k}^\times$ such that $\omega = d\beta$, where the coboundary operator d is given by

$$(d\beta)(g_1, g_2, g_3) = \frac{\beta(g_2, g_3)\,\beta(g_1, g_2g_3)}{\beta(g_1g_2, g_3)\,\beta(g_1, g_2)}.$$

Show that then category $\mathbf{Vec}_G^\omega$ is monoidally equivalent to $\mathbf{Vec}_G$.

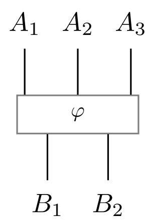
More generally, categories $\mathbf{Vec}_G^\omega$ and $\mathbf{Vec}_G^{\omega'}$ are equivalent, with the equivalence preserving the G-graded dimension, if and only if $\omega' = \omega \cdot d\beta$ for some $\beta\colon G \times G \to \mathbf{k}^\times$ (see [**EGNO2015**, Section 2.6]).

We can restate Mac Lane's coherence theorem as follows.

THEOREM 2.3.3. *Any monoidal category is monoidally equivalent to a strict monoidal category.*

If one knows the concept of a skeleton of a category, then Mac Lane's coherence theorem can be understood as the construction of the skeleton by looking at the isomorphism classes of objects in a monoidal category.

Convention 2.3.1. We will frequently use graphical presentation of morphisms in a monoidal category, representing a morphism $\varphi\colon A_1 \otimes \cdots \otimes A_k \to B_1 \otimes \cdots \otimes B_m$ by a box with k inputs at the top and m outputs at the bottom:



$$\varphi\colon A_1 \otimes A_2 \otimes A_3 \to B_1 \otimes B_2$$

Figure 2.3.1. Graphical notation for a morphism in a monoidal category.

2.4. Symmetric monoidal categories

For the category of vector spaces, the tensor product is not only associative, but also "commutative": one has natural isomorphisms $V \otimes W \simeq W \otimes V$. The definition below gives a proper generalization of this to arbitrary monoidal categories.

Definition 2.4.1. A symmetric monoidal category is a monoidal category $\mathcal{C}$ together with a functorial isomorphism

$$c_{X,Y}\colon X \otimes Y \xrightarrow{\sim} Y \otimes X$$

such that

$$c_{X,Y} c_{Y,X} = \mathrm{id}$$

and which satisfies the hexagon axiom:

$$\begin{tikzcd}
(A \otimes B) \otimes C \arrow[r, "\alpha_{A,B,C}"] \arrow[d, "c_{A,B} \otimes \mathrm{id}_C"'] & A \otimes (B \otimes C) \arrow[r, "c_{A,B\otimes C}"] & (B \otimes C) \otimes A \arrow[d, "\alpha^{-1}_{B,C,A}"] \\
(B \otimes A) \otimes C \arrow[r, "\alpha_{B,A,C}"'] & B \otimes (A \otimes C) \arrow[r, "\mathrm{id}_B \otimes c_{A,C}"'] & B \otimes (C \otimes A)
\end{tikzcd} \tag{2.4.1}$$

The hexagon axiom looks complicated; to make it easier to understand, one can drop all associativity isomorphisms, replacing the hexagon axiom with the one below

$$\begin{tikzcd}
A \otimes B \otimes C \arrow[rr, "c_{A,B\otimes C}"] \arrow[dr, "c_{A,B}"'] & & B \otimes C \otimes A \\
& B \otimes A \otimes C \arrow[ur, "c_{A,C}"'] &
\end{tikzcd} \tag{2.4.2}$$

Example 2.4.1.

(1) Category **Set** with respect to disjoint union is a symmetric monoidal category; same is true for categories **Man**, **PLMan**.

(2) The category of vector spaces over $\mathbf{k}$ is a symmetric monoidal category. Same is true for categories of representations of a group G or a Lie algebra $\mathfrak{g}$.
(3) (Super vector spaces). Consider the category **SVec** whose objects are $\mathbb{Z}_2$-graded vector spaces $V = V_0 \oplus V_1$. This has an obvious monoidal structure. Define the commutativity isomorphism $c_{V,W} : V \otimes W \to W \otimes V$ by
$$c(v \otimes w) = (-1)^{pq} w \otimes v, \quad v \in V_p, w \in W_q.$$
Then this defines on the category **SVec** a structure of a symmetric monoidal category.

Note that one can also define the symmetric structure on **SVec** without the $(-1)^{pq}$ factor; this illustrates the fact that a monoidal category can have several different symmetric structures.

Remark 2.4.1. There are monoidal categories that do not have a natural symmetric structure; for example, the category $\mathbf{Vec}_G$ of G-graded vector spaces, or a category of bimodules over a ring. There are also "braided" categories where we do have isomorphisms $c_{X,Y} : X \otimes Y \to Y \otimes X$, but these isomorphisms fail to satisfy condition $c^2 = \mathrm{id}$. The most famous example of a braided category is the category of representations of a quantum group. We will discuss braided categories later (see Chapter 20).

As before, the real importance of the hexagon axiom is the following coherence theorem, which is an analog of Mac Lane's coherence theorem, but for **symmetric** monoidal categories.

Theorem 2.4.1 (Mac Lane's coherence theorem for symmetric monoidal categories). *Let $\mathcal{C}$ be a symmetric monoidal category. Let $F_1, F_2 : \mathcal{C}^n \to \mathcal{C}$ be two functors obtained by compositions of*

- *tensor product functor $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$*
- *functors $X \mapsto X \otimes \mathbf{1}$, $X \mapsto \mathbf{1} \otimes X$*
- *permutation functor $\mathcal{C} \times \mathcal{C} \to \mathcal{C} \times \mathcal{C} : (A, B) \mapsto (B, A)$*

e.g.,
$$F_1(A, B, C) = (B \otimes A) \otimes (\mathbf{1} \otimes C)$$
$$F_2(A, B, C) = C \otimes ((B \otimes A) \otimes \mathbf{1}).$$
Then

(1) *There exists a functorial isomorphism $F_1 \simeq F_2$, obtained by composition of associativity isomorphism α, left and right unit isomorphisms l, r, commutativity isomorphism c, and their inverses.*
(2) *Any two such isomorphisms $F_1 \to F_2$ are equal.*

As before, this theorem can be understood as a statement that a certain cell complex is simply connected; if we only use permutations and tensor product (i.e., no units), then this complex is called *permutoassociahedron*. (It took me 3 tries to spell that correctly; in the end, I had to use copy and paste from another source.)

CHAPTER 3

Category of cobordisms and definition of TQFT

In this chapter, we give the key definition of this book, namely the notion of a TQFT, following work of Atiyah (see [**Ati1988**]). To do that, we first define the notion of a cobordism.

Unless specified otherwise, "manifold" means a compact oriented smooth manifold, possibly with boundary. For a manifold M, we denote by ∂M its boundary, with the orientation determined by the orientation of M. If $\partial M = \varnothing$, we say that M is *closed*. Morphisms between manifolds are smooth maps; unless stated otherwise, "diffeomorphism" means an orientation-preserving invertible smooth map.

We denote by $\overline{M}$ the manifold M considered with the opposite orientation.

The main references for this chapter are [**Koc2004**] and [**Mil1965**].

3.1. Category $\mathbf{Cob}_n$ and definition of TQFT

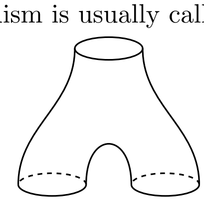
Definition 3.1.1. Let N_0, N_1 be closed $(n-1)$-dimensional manifolds. A *cobordism* from N_0 to N_1 is an n-dimensional manifold M together with an isomorphism

$$\varphi\colon \overline{N_0} \sqcup N_1 \xrightarrow{\sim} \partial M.$$

We will refer to N_0 as the incoming boundary (or simply the in-boundary), and to N_1 as the outgoing boundary (out-boundary).

Following [**Koc2004**], we will write $N_0 \overset{M}{\Longrightarrow} N_1$ to indicate that M is a cobordism from N_0 to N_1.

The picture below shows a 2-dimensional cobordism between a circle and a union of two circles; this cobordism is usually called *pair of pants*.



Convention 3.1.1. In our pictures, cobordisms go "top to bottom": the in-boundary is at the top, and the out-boundary is at the bottom.

Definition 3.1.2. An isomorphism of two cobordisms $M_1, M_2 \in \mathbf{Cob}_n(N_0, N_1)$ is a diffeomorphism $f\colon M_1 \to M_2$ which makes this diagram commutative:

(3.1.1)
$$\begin{tikzcd} & M_1 \arrow[dd, "f"] \\ \overline{N_0} \sqcup N_1 \arrow[ur] \arrow[dr] & \\ & M_2 \end{tikzcd}$$

Given two cobordisms $N_0 \xRightarrow{M_1} N_1$, $N_1 \xRightarrow{M_2} N_2$, it is natural to ask if they can be glued together in a single cobordism $M = M_2 \circ M_1$

$$N_0 \xRightarrow{M_2 \circ M_1} N_2.$$

Clearly, as a manifold, M should be obtained by gluing M_1 and M_2 along the common boundary N_1: $M = M_1 \cup_{N_1} M_2$. However, there is a problem: it is easy to define M as a topological space, but the smooth structure is not unique. There are several ways to deal with this problem.

- One can switch to the category of PL manifolds and cobordisms, where the problem does not arise: the PL structure on $M_1 \cup_{N_1} M_2$ is uniquely determined by the PL structures on M_1, M_2.
- One can consider *collared* manifolds: instead of isomorphism $\overline{N_0} \sqcup N_1 \to \partial M$, we can require in the definition of cobordism an isomorphism of $(\overline{N_0} \sqcup N_1) \times [0, \epsilon)$ with a neighborhood of ∂M in M.

However, the simplest way is to note that while the smooth structure is not unique, different smooth structures give isomorphic cobordisms.

THEOREM 3.1.1. *The smooth structure on $M = M_1 \cup_{N_1} M_2$ which agrees with the given smooth structures on M_1, M_2, is unique up to isomorphism: if α, α' are two different smooth structures on M, then (M, α), (M, α') are isomorphic as cobordisms.*

You can find a proof of this in [**Koc2004**, Theorem 1.3.12].

DEFINITION 3.1.3. The cobordism category $\mathbf{Cob}_n$ is the category whose objects are closed $(n-1)$-dimensional manifolds, and $\mathrm{Hom}(N_0, N_1)$ is the set of isomorphism classes of cobordisms $N_0 \xRightarrow{M} N_1$. The composition of cobordisms $N_0 \xRightarrow{M_1} N_1$, $N_1 \xRightarrow{M_2} N_2$ is given by

$$M_2 \circ M_1 = M_1 \cup_{N_1} M_2$$

(by the previous theorem it is independent of the choice of smooth structure). The identity morphism id_N is the cylinder $N \times I$.

Similarly one defines the PL version of cobordism category.

This category has a natural structure of a symmetric monoidal category with respect to disjoint union.

DEFINITION 3.1.4. An n-dimensional TQFT is a symmetric monoidal functor $Z\colon \mathbf{Cob}_n \to \mathbf{Vec}$.

This definition was first given in [**Ati1988**] in slightly different language.

One can also generalize this definition, allowing the TQFT to take values in other categories.

DEFINITION 3.1.5. Let $\mathcal{C}$ be a symmetric monoidal category. Then an n-dimensional TQFT with values in $\mathcal{C}$ is a symmetric monoidal functor $Z\colon \mathbf{Cob}_n \to \mathcal{C}$.

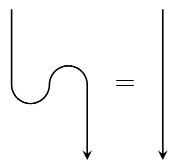
3.2. 1D TQFT

Now that we have a definition of a TQFT, let us try to construct examples. We begin with the simplest possible case, that of a 1-dimensional TQFT.

THEOREM 3.2.1. *A 1-dimensional TQFT is the following collection of data:*

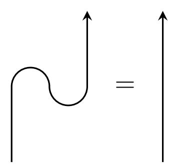
- *A vector space* $V_+ = Z(\bullet^+)$
- *A vector space* $V_- = Z(\bullet^-)$
- *A linear map* $\mathrm{ev} = Z(\cup)\colon V_+ \otimes V_- \to \mathbf{k}$
- *A linear map* $\mathrm{coev} = Z(\cap)\colon \mathbf{k} \to V_- \otimes V_+$

satisfying relations below.



$$(V_+ \to V_+ \otimes V_- \otimes V_+ \to V_+) = \mathrm{id}_{V_+}$$

(3.2.1)



$$(V_- \to V_- \otimes V_+ \otimes V_- \to V_-) = \mathrm{id}_{V_-}$$

PROOF. Since Z is symmetric monoidal, $Z(N_1 \sqcup N_2) \simeq Z(N_1) \otimes Z(N_2)$, so Z on objects is determined by its values on connected 0-manifolds. The only connected 0-manifolds are the positively and negatively oriented points $\bullet^+, \bullet^-$, hence Z on objects is determined by $V_\pm := Z(\bullet^\pm)$.

Similarly, Z on morphisms is determined by its values on connected cobordisms. Up to isomorphism, every connected oriented 1-cobordism with non-empty boundary is one of the following:

- the intervals $\bullet^+ \to \bullet^+$ and $\bullet^- \to \bullet^-$, on which Z is the identity of V_+ resp. V_-;
- the *cup* $\bullet^+ \sqcup \bullet^- \to \varnothing$, on which Z is a linear map $\mathrm{ev}\colon V_+ \otimes V_- \to \mathbf{k}$;
- the *cap* $\varnothing \to \bullet^- \sqcup \bullet^+$, on which Z is a linear map $\mathrm{coev}\colon \mathbf{k} \to V_- \otimes V_+$.

The only connected closed 1-cobordism is the circle S^1; its value $Z(S^1) \in \mathbf{k}$ will turn out to be determined by the data above.

It remains to identify the relations these data must satisfy. The snake cobordism on the left-hand side of the first equation of (3.2.1) is diffeomorphic (rel boundary) to the straight interval on the right-hand side; applying Z to this diffeomorphism gives the first snake identity. The second is obtained analogously by reversing orientations.

Conversely, given data $(V_+, V_-, \mathrm{ev}, \mathrm{coev})$ satisfying (3.2.1), one defines Z on objects and connected cobordisms by the formulas above and extends to general objects and morphisms by tensor product and composition. The relations (3.2.1) ensure that the result is independent of the decomposition; with this, Z is well-defined as a symmetric monoidal functor $\mathbf{Cob}_1 \to \mathbf{Vec}$. □

LEMMA 3.2.2. *Let* V_+, V_-*,* $\mathrm{ev}, \mathrm{coev}$ *be as in Theorem 3.2.1. Then* V_+, V_- *are finite-dimensional, and one can identify* $V_- = V_+^*$ *so that* $\mathrm{ev}, \mathrm{coev}$ *become the usual evaluation and coevaluation maps* $V \otimes V^* \to \mathbf{k}$*,* $\mathbf{k} \to V \otimes V^*\colon 1 \mapsto \sum e_i \otimes e^i$.

PROOF. Consider the map $V_- \to (V_+)^*$, given by $y \mapsto \mathrm{ev}(-, y)$; and a map $V_+^* \to V_-$ given by $f \mapsto \mathrm{id} \otimes f(\mathrm{coev}(1))$ (thus, if $\mathrm{coev}(1) = \sum y_i \otimes x_i$, then $f \mapsto$

$\sum f(x_i)y_i)$. Then the rigidity conditions say that these two maps are inverse to each other; thus, we have $V_- \simeq V_+^*$. Finite-dimensionality follows from the fact that the elements y_i generate V_-. $\square$

Thus, a one-dimensional TQFT is completely determined by a *finite-dimensional* vector space $V = Z(\bullet^+)$; in such a TQFT, $Z(S^1) = \dim V$.

3.3. Rigid monoidal categories

In the previous section, we had shown that a 1d TQFT with values in the category of vector spaces is uniquely determined by the vector space $V = Z(\bullet^+)$, which must be finite-dimensional. What happens if we now consider TQFTs with values in an arbitrary monoidal category $\mathcal{C}$?

It turns out that there is a very similar answer, but with finite-dimensionality replaced by the following condition.

Definition 3.3.1. Let $\mathcal{C}$ be a monoidal category. A *dual pair* is a pair of objects $A, B \in \mathcal{C}$ together with morphisms

$$\mathrm{ev}\colon A \otimes B \to \mathbf{1} \qquad \text{(evaluation morphism)},$$
$$\mathrm{coev}\colon \mathbf{1} \to B \otimes A \qquad \text{(coevaluation morphism)},$$

satisfying the *rigidity* condition below.

(3.3.1)
$$\left(B \to \mathbf{1} \otimes B \xrightarrow{\mathrm{coev} \otimes 1_B} (B \otimes A) \otimes B \xrightarrow{\alpha_{B,A,B}} B \otimes (A \otimes B) \xrightarrow{1_B \otimes \mathrm{ev}} B \otimes \mathbf{1} \to B\right) = 1_B$$

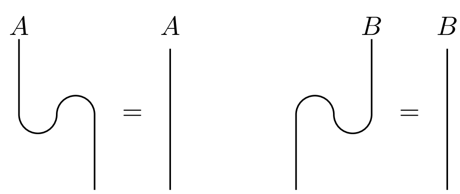
$$\left(A \to A \otimes \mathbf{1} \xrightarrow{1_A \otimes \mathrm{coev}} A \otimes (B \otimes A) \xrightarrow{\alpha^{-1}_{A,B,A}} (A \otimes B) \otimes A \xrightarrow{\mathrm{ev} \otimes 1_A} \mathbf{1} \otimes A \to A\right) = 1_A$$

In this situation we say that A is a *left dual* of B, and B is a *right dual* of A.

It is common to represent the evaluation and coevaluation graphically:

(3.3.2)


Then the rigidity equations (3.3.1) are illustrated by

(3.3.3)


It turns out that right dual, if exists, is unique up to unique isomorphism.

Theorem 3.3.1. *Let* $(A, B, \mathrm{ev}, \mathrm{coev})$ *and* $(A, B', \mathrm{ev}', \mathrm{coev}')$ *be two dual pairs with the same* A. *Then there exists a unique morphism* $\varphi\colon B \to B'$ *such that* $\mathrm{ev} = \mathrm{ev}' \circ (1_A \otimes \varphi)$, $\mathrm{coev}' = (\varphi \otimes 1_A)\,\mathrm{coev}$.

The proof is a standard snake-diagram argument: the required φ is given by the composition

$$B \xrightarrow{1_B \otimes \mathrm{coev}'} B \otimes A \otimes B' \xrightarrow{\mathrm{ev} \otimes 1_{B'}} B',$$

and uniqueness follows by chasing the rigidity equations. A similar statement holds for left dual.

This allows us to talk about the right dual of A as a well-defined object in $\mathcal{C}$. We will denote the right dual of A by ${}^*A$; similarly we will use notation B^* for left dual of B. In this notation, evaluation and coevaluation morphisms become

$$\begin{aligned} \mathrm{ev}\colon X^* \otimes X &\to \mathbf{1} \\ \mathrm{coev}\colon \mathbf{1} &\to X \otimes X^*. \end{aligned} \tag{3.3.4}$$

Clearly, in a symmetric monoidal category, the left dual is also a right dual.

DEFINITION 3.3.2. An object A in a monoidal category $\mathcal{C}$ is *rigid* if it has both left and right duals.

A monoidal category is *rigid* if every object is rigid.

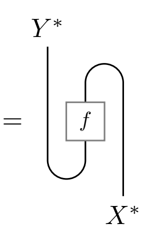
EXAMPLE 3.3.1. A vector space V is rigid as an object of **Vec** if and only if it is finite-dimensional.

EXERCISE 3.3.1. Show that in a dual pair, the coevaluation is uniquely determined by the remaining data: if $(A, B, \mathrm{ev}, \mathrm{coev})$ and $(A, B, \mathrm{ev}, \mathrm{coev}')$ are dual pairs, then $\mathrm{coev} = \mathrm{coev}'$.

Similarly, ev is uniquely determined by (A, B, coev).

EXERCISE 3.3.2. Let X^*, Y^* be right duals of objects X, Y. Show that then $X \otimes Y$ also has a right dual, and there is a canonical isomorphism $(X \otimes Y)^* \simeq Y^* \otimes X^*$.

EXERCISE 3.3.3. Let $\mathcal{C}$ be a rigid monoidal category. For a morphism $f\colon X \to Y$, define a morphism $f^*\colon Y^* \to X^*$ by the picture below.

(3.3.5) $f^* =$ 

Prove that then $(fg)^* = g^* f^*$; thus, $*$ is a contravariant functor $\mathcal{C} \to \mathcal{C}$.

After these preliminaries, we can now easily classify 1-dimensional TQFTs with values in any symmetric monoidal category.

THEOREM 3.3.2. *If $Z\colon \mathbf{Cob}_1 \to \mathcal{C}$ is a 1-dimensional TQFT with values in a symmetric monoidal category $\mathcal{C}$, then $A = Z(\bullet^+)$ is a rigid object; its dual is $Z(\bullet^-)$. Conversely, any rigid object $A \in \mathcal{C}$ defines a one-dimensional TQFT such that $Z(\bullet^+) = A$.*

3.4. First properties of TQFTs

Let us now try to establish some properties of general n-dimensional TQFTs.

THEOREM 3.4.1. *Let $Z\colon \mathbf{Cob}_n \to \mathcal{C}$ be an n-dimensional TQFT with values in a symmetric monoidal category $\mathcal{C}$. Then for any closed $(n-1)$-dimensional manifold N, $Z(N) \in \mathcal{C}$ is rigid, and $Z(\overline{N})$ is a dual of $Z(N)$.*

The proof is the same as in the one-dimensional case.

In particular, for TQFTs with values in **Vec**, for any closed $(n-1)$-dimensional manifold N, the vector space $Z(N)$ is finite-dimensional.

EXERCISE 3.4.1. Let Z be an n-dimensional TQFT with values in **Vec**. Prove that then $Z(N \times S^1) = \dim Z(N)$ for any closed $(n-1)$-dimensional manifold N.

Note that unlike the one-dimensional case, in general we cannot claim that any rigid object of $\mathcal{C}$ defines a TQFT. The proper generalization of that statement is the cobordism hypothesis, which we will discuss later.

Another property of TQFTs is functoriality. As mentioned before, it is natural to expect that a diffeomorphism $\varphi\colon N_0 \to N_1$ between two $(n-1)$-dimensional closed manifolds gives rise to an isomorphism $Z(\varphi)\colon Z(N_0) \to Z(N_1)$; however, we didn't include that as part of the definition. It turns out it is not necessary.

Let $\varphi\colon N_0 \to N_1$ be a diffeomorphism. Define the cobordism

$$N_0 \overset{C_\varphi}{\Longrightarrow} N_1 \tag{3.4.1}$$

to be the cylinder $N_1 \times I$, with the identification $\overline{N_0} \sqcup N_1 \to \partial C_\varphi$ given by $\varphi \sqcup \mathrm{id}$.

LEMMA 3.4.2. *Let C_φ be defined above. Then:*

(1) *The isomorphism class of C_φ (as a cobordism) only depends on the isotopy class of φ: if φ, φ' are isotopic to each other, then $C_\varphi \simeq C_{\varphi'}$.*
(2) *Given diffeomorphisms $\varphi_1\colon N_0 \to N_1$, $\varphi_2\colon N_1 \to N_2$, we have an isomorphism of cobordisms*

$$C_{\varphi_2} \circ C_{\varphi_1} \simeq C_{\varphi_2\varphi_1}.$$

Recall that φ, φ' are called isotopic if there exists a one-parameter family φ_t of smooth maps $N_0 \to N_1$ such that φ_t is a diffeomorphism for all t, and $\varphi_0 = \varphi$, $\varphi_1 = \varphi'$.

We leave the proof of this lemma as an easy exercise to the reader.

REMARK 3.4.1. In fact, there is a stronger statement: $C_\varphi \simeq C_{\varphi'}$ if and only if φ, φ' are *pseudo-isotopic*, see [**Mil1965**, Theorem 1.9].

As an immediate corollary, we get the following theorem.

THEOREM 3.4.3. *Let $Z\colon \mathbf{Cob}_n \to \mathcal{C}$ be an n-dimensional TQFT.*

(1) *For any diffeomorphism $\varphi\colon N_0 \to N_1$, define $Z(\varphi)\colon Z(N_0) \to Z(N_1)$ by $Z(\varphi) = Z(C_\varphi)$. Then $Z(\varphi_2\varphi_1) = Z(\varphi_2) \circ Z(\varphi_1)$.*
(2) *If φ_0, φ_1 are isotopic, then $Z(\varphi_0) = Z(\varphi_1)$.*

Part 2

TQFTs in 2 dimensions

Part 2 overview

This part is the heart of the book. Here we start the systematic study of TQFTs, classifying and studying TQFTs in dimension 2. We also introduce the main tools that will be used in all dimensions.

In Chapter 4, we discuss classification of 2-dimensional TQFT, showing that such theories are classified by commutative Frobenius algebras. Our approach is based on the use of Morse functions and families of Morse functions (Cerf theory).

In Chapter 5, we study Frobenius algebras, in particular classifying all semisimple commutative Frobenius algebras and describing the corresponding 2d TQFTs.

In Chapter 6, we describe the fundamental example of a TQFT: gauge theory with finite gauge group, also known as Dijkgraaf–Witten theory. This theory can be described in terms of G-covers of surfaces; we can also relate it to the path integral description of TQFT.

After that, we move to describing extended TQFTs, i.e., theories where we consider not just cobordisms but manifolds with corners. To give an accurate definition of such an extended TQFT, one needs to introduce the language of (weak) 2-categories, which we do in Chapter 7. We also introduce the notion of a symmetric monoidal 2-category. Our key examples of 2-categories are the 2-category of small categories (or a variation of it, the 2-category of all locally finite $\mathbf{k}$-linear abelian categories) and the category $\mathbf{Alg}$, whose objects are algebras over $\mathbf{k}$ and 1-morphisms are bimodules. Both 2-categories play an important role in future constructions.

Next, in Chapter 8 we discuss dualities in 2-categories, discussing both the notion of a dual object (which only makes sense in a monoidal 2-category) and of a dual 1-morphism; in particular, we show that for the 2-category of small categories, where 1-morphisms are functors between categories, so defined duality coincides with the classical notion of adjointness for functors. In Chapter 9, we study the duality in the 2-category $\mathbf{Alg}$ and show that fully dualizable objects in $\mathbf{Alg}$ are the separable algebras; moreover, fixing a duality between ev and coev 1-morphisms is equivalent to choosing a structure of a Frobenius algebra, which provides the link to the construction of (non-extended) TQFTs done earlier.

After completing all these preliminaries, we finally define the notion of an extended 2d TQFT in Chapter 10. Following work of Schommer–Pries, we construct the set of generators and relations for the 2-category $\mathbf{Bord}_2$ of 2-dimensional bordisms and use it to show that extended 2d TQFTs are classified by separable symmetric Frobenius algebras.

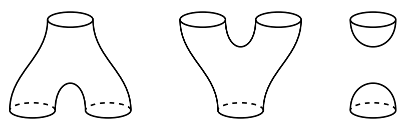
Finally, in Chapter 11–Chapter 12 we present an alternative approach to construction of extended 2d TQFTs. Instead of using Morse functions to create a presentation of a cobordism as a composition of elementary cobordisms, we use triangulations (or more generally, PL cell decompositions); instead of Cerf moves, we use Pachner moves. We show how in dimension 2, it gives a very explicit construction of an extended 2d TQFT from a separable symmetric Frobenius algebra A, providing a different way of proving the same classification result.

CHAPTER 4

Classification of 2d TQFT

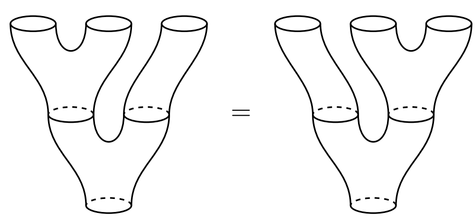
In this chapter, we present a full classification of 2-dimensional TQFTs; the final answer, given in Theorem 4.3.3, states that such theories are classified by commutative Frobenius algebras. This is a relatively simple result; it goes back to at least 1989, see Dijkgraaf's thesis [**Dij1989**], but was probably known to experts much earlier. For us, it will serve as a model example for all future constructions.

For a two-dimensional TQFT, there is only one 1-dimensional oriented closed manifold: the circle S^1, and any orientation-preserving diffeomorphism $S^1 \to S^1$ is isotopic to the identity. It also seems obvious (and in fact can be justified using Morse theory, as we discuss below) that any cobordism can be obtained by gluing pieces diffeomorphic to one of the cobordisms below together with the cylinder $S^1 \times I$:



It is also not hard to write some relations, such as the associativity relation

(4.0.1)



However, how can one show that this is the full list of relations?

There are two approaches. One, used in [**Koc2004**], uses the classification of surfaces; using that, we show that the relations we have are enough to bring any composition of generators to some canonical form. This approach does not require much beyond the classification of surfaces, but it does not generalize to higher dimensions.

There is, however, an alternative approach, based on Morse and Cerf theory. This goes back at least to [**Abr1996**], who used a handle-decomposition viewpoint to give a classification of 2d TQFTs; the Cerf-theoretic version is presented in [**MS**] and [**Fre2012**, Lecture 23], the latter of which our exposition follows. One can also find more details in the unpublished paper [**GWW2012**].

4.1. Morse theory basics

We start with an overview of Morse theory; details can be found, e.g., in [**Mat2002**].

Recall that given a smooth real-valued function f on an n-dimensional manifold M, a point $p \in M$ is a *non-degenerate critical point* if $\partial_i f(p) = 0$ and the Hessian matrix $a_{ij} = \frac{\partial^2 f}{\partial x^i \partial x^j}(p)$ is non-degenerate. It is easy to see that this is independent of the choice of coordinate system. It is also known that in such a case, one can choose local coordinates in a neighborhood of p so that

$$f(x^1, \dots, x^n) = c - (x^1)^2 - \dots - (x^r)^2 + (x^{r+1})^2 + \dots + (x^n)^2 \tag{4.1.1}$$

(this statement is usually called the *Morse lemma*). The number r of minuses in (4.1.1) is called the *index* of the critical point.

It is immediate from the definition that a non-degenerate critical point is isolated; thus, if M is compact, there can be only finitely many non-degenerate critical points.

Definition 4.1.1. Let M be a closed n-manifold. A Morse function is a smooth function $f : M \to \mathbb{R}$ such that all critical points of f are non-degenerate.

It will be more useful to impose a stronger condition.

Definition 4.1.2. Let M be a closed n-manifold. A Morse function $f : M \to \mathbb{R}$ is called *excellent* if all critical values $c_i = f(p_i)$ are distinct.

It was proved by Morse that such functions exist; moreover, the set $\mathcal{F}^0$ of excellent functions is an open and dense subset in $C^\infty(M)$ (in an appropriate topology).

We can generalize the notion of excellent functions to manifolds with boundary, or to cobordisms $N_0 \overset{M}{\Longrightarrow} N_1$.

Definition 4.1.3. Let $N_0 \overset{M}{\Longrightarrow} N_1$ be an n-dimensional cobordism. A Morse function on M is a smooth function $f : M \to \mathbb{R}$ such that

- f is constant on both in-boundary and out-boundary: $f|_{N_0} = a_0 = \min_M f$, $f|_{N_1} = a_1 = \max_M f$
- All critical points of f are in the interior $M \setminus (N_0 \cup N_1)$, and all critical points are non-degnerate.

If, in addition, all critical values of f are distinct, then f is called *excellent*.

Again, one can prove that such functions exist and form an open dense subset of all functions satisfying $f|_{N_0} = a_0$, $f|_{N_1} = a_1$.

The main result of Morse theory is that given an excellent function f, the topology of the manifold $M_t = f^{-1}((-\infty, t])$ does not change between critical values: if $c_i < c_{i+1}$ are critical values, with no other critical values between them, then for all $t \in I = [c_i + \varepsilon, c_{i+1} - \varepsilon]$, the manifolds M_t are diffeomorphic, and $M_I = f^{-1}(I) \xrightarrow{\sim} N_{t_0} \times I$, where $N_{t_0} = f^{-1}(t_0)$ is a slice for any $t_0 \in I$. Moreover, one can give an explicit description of how topology of M_t changes when one goes through a critical point (this is usually referred to as “attaching a handle”). In particular, this implies that if f has no critical points, then M is a cylinder: $M \xrightarrow{\sim} N \times I$.

Therefore, one can use Morse functions to decompose a cobordism as a composition of “elementary” cobordisms.

Definition 4.1.4. An elementary cobordism is a cobordism that admits a Morse function with a single critical point.

Thus, given an excellent function with critical values $c_1 < \cdots < c_k$, and values on the boundary $f|_{N_0} = a_0 < c_1$, $f|_{N_1} = a_k > c_k$, we can choose intermediate values $a_i \in (c_i, c_{i+1})$; then we have

$$M = M_k \circ \cdots \circ M_1, \tag{4.1.2}$$

where $M_i = f^{-1}([a_{i-1}, a_i])$ is the elementary cobordism containing critical point p_i. It follows from the result above that up to isomorphism, M_i does not depend on the choice of a_i. We will refer to (4.1.2) as a *Morse decomposition* of a cobordism.

In particular, for $n = 2$, each connected elementary cobordism is isomorphic to one of the four cobordisms shown in Table 4.1.1. Note that here we follow our convention that cobordisms go "top to bottom"; thus, the y-axis is directed downward, so that the minimal value of f is at the top, and the maximal, at the bottom. A (not necessarily connected) elementary cobordism is obtained as a

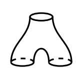
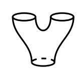
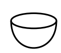
Index	Elementary cobordism
0	
1	 
2	

TABLE 4.1.1. Connected elementary 2-cobordisms

disjoint union of one of the connected elementary cobordisms and several copies of the identity cobordism, i.e., $S^1 \times I$.

Thus, we see that a choice of an excellent function gives a presentation of a 2-cobordism as a composition of the generators shown in Table 4.1.1 (and identity cobordisms).

4.2. Cerf theory

We have seen that excellent Morse functions allow one to present a cobordism as a composition of elementary cobordisms; for $n = 2$, this gives us a set of generators for the category of cobordisms.

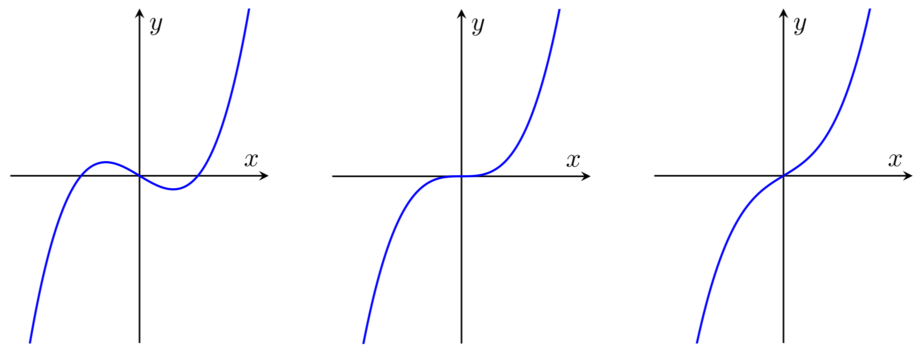
To describe relations between generators, we need to see how one can go from one such presentation to another, or equivalently, how the presentation changes if we replace one excellent Morse function by another.

It is easy to see that the space of excellent functions is not connected; thus, if f_0, f_1 are two excellent functions, and we want to connect them by a path f_s, $s \in [0, 1]$ in the space of functions, in general this path must go through some points which are not excellent. This was studied by Cerf in [**Cer1970**] and is usually referred to as *Cerf theory*.

EXAMPLE 4.2.1. Consider the one-parameter family of functions $f_s = x^3 + sx$ on a neighborhood of the origin in $\mathbb{R}$, with local coordinate x, shown in Figure 4.2.1.

Then for $s < 0$, the function is Morse with two critical points of indices 0 and 1 respectively; for $s > 0$, it is also Morse, with no critical points; and for $s = 0$,

$f_0(x) = x^3$ is not a Morse function: zero is a degenerate critical point. Thus, in this case, f_s, considered as a path in the function space, connects two Morse functions but goes through a non-Morse function $f_0 = x^3$. This path describes annihilation of a pair of critical points (or, going in the opposite direction, birth of a pair of critical points).

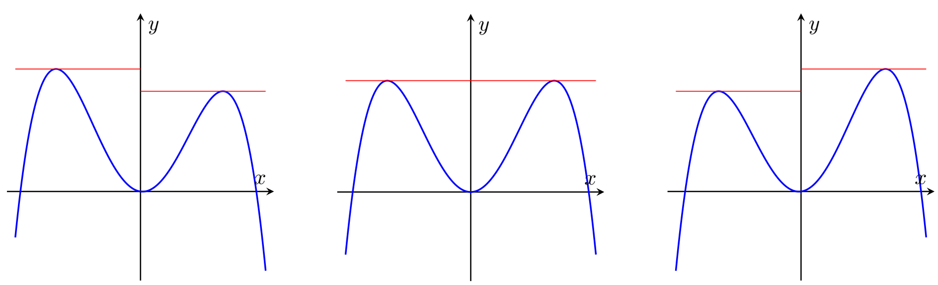


$s < 0$ $s = 0$ $s > 0$

FIGURE 4.2.1. Birth-death of critical points in family $f_s = x^3 + sx$

EXAMPLE 4.2.2. Consider the family of functions $f_s(x) = x^2 - x^4 + sx$. Then for values of s close to 0, it has three critical points, all of them non-degenerate: local minimum at p_0 close to 0, and local maxima at points $p_1 < 0$, $p_2 > 0$.

For $s < 0$, we have $c_1 > c_2$; for $s > 0$, $c_1 < c_2$, and for $s = 0$, $c_1 = c_2$ as shown below, so for $s = 0$ this function is Morse but not excellent. Thus, we again get a path connecting two excellent functions going through a non-excellent function with $c_1 = c_2$.



$s < 0$ $s = 0$ $s > 0$

FIGURE 4.2.2. Crossing of critical values in family $f_s = x^2 - x^4 + sx$

It turns out that these two model examples are enough: any two excellent functions can be connected by a one-parameter family where all non-excellent functions are at worst as in these examples. More precisely, we have the following.

DEFINITION 4.2.1. A *good* function on an n-dimensional closed manifold M is a smooth function of one of the following types:

- **Type** α: all critical values are distinct, and all critical points are non-degenerate, except for one. Near this point, there is a local coordinate system where f is given by

$$f(x^1, \dots, x^n) = c + (x^1)^3 - (x^2)^2 - \dots - (x^r)^2 + (x^{r+1})^2 + \dots + (x^n)^2.$$

- **Type** β: all critical points are non-degenerate, and all critical values are distinct except for one pair: $c_i = c_j$ for some $i \neq j$.

In a similar way one defines good function on a cobordism M.

(The name "good" is not standard.)

It was shown by Cerf that this is the beginning of a stratification of the space $\mathcal{F} = C^\infty(M)$: we can write $\mathcal{F} = \mathcal{F}^0 \sqcup \mathcal{F}^1 \sqcup \dots$, where $\mathcal{F}^0$ is the set of excellent functions, which is open and dense in $\mathcal{F}$; the set $\mathcal{F}^1$ consists of good functions and is a codimension 1 subset of $\mathcal{F}$, and so on. (One can also define higher codimension strata, but we will not need them.) A review of this theory can be found, e.g., in [**HW1973**, Section I.2].

THEOREM 4.2.1. *Let f_0, f_1 be two excellent functions on a cobordism M. Then one can find a one-parameter family f_s of smooth functions on M such that:*

(1) *f_s is excellent for all s except for finitely many values $s_1, \dots, s_p$*
(2) *At each of these special values, f_s is good.*

In other words, any two points in $\mathcal{F}^0$ can be connected by a path contained in $\mathcal{F}^0 \cup \mathcal{F}^1$.

Moreover, this path can be chosen so that it is transversal to $\mathcal{F}^1$ (we skip the precise definition, referring the reader to [**HW1973**]). This allows one to describe how the Morse decomposition changes when we cross one of the components of $\mathcal{F}^1$ ("walls"). Crossing a component corresponding to a type-α function produces a *birth-death move*: two adjacent critical points (of indices $r-1, r$) appear or disappear. Crossing a component corresponding to a type-β function produces a *crossing move*: two critical values c_i, c_{i+1} exchange their order.

THEOREM 4.2.2. *Let M be a cobordism, and let f_0, f_1 be two excellent functions on M, with corresponding Morse decompositions $\mathcal{D}_0$, $\mathcal{D}_1$ as in (4.1.2). Then $\mathcal{D}_0$ and $\mathcal{D}_1$ can be obtained from each other by a finite sequence of birth-death moves and crossing moves and their inverses.*

4.3. Generators and relations of Cob$_2$

Now let us apply the general theory above to the question of classification of 2d TQFTs.

First, choosing an excellent Morse function on a 2d cobordism M, we get a decomposition of M into a composition of elementary cobordisms. Each elementary cobordism is a disjoint union of copies of the connected elementary cobordisms; and each connected elementary cobordism is isomorphic to one of the standard cobordisms shown in Table 4.1.1: pair of pants, copants, cap, and cup.

Let us denote by $A = Z(S^1)$ the vector space associated to the circle, and by $m, i, \Delta, \varepsilon$ the linear operators corresponding to the standard cobordisms as shown below.

(4.3.1)

$m = Z($$)\colon A \otimes A \to A$

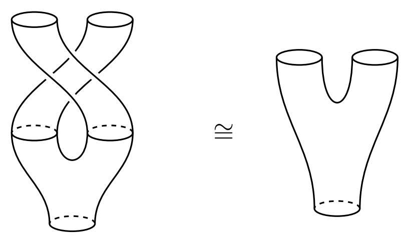
$i = Z($$)\colon \mathbf{k} \to A$

$\Delta = Z($$)\colon A \to A \otimes A$

$\varepsilon = Z($$)\colon A \to \mathbf{k}.$

Then a choice of an excellent Morse function f on M, together with a choice of isomorphism φ_i of each of the connected elementary cobordisms defined by f with one of the "standard" ones, gives us a presentation of $Z(M)$ as a composition of tensor products of operators $m, i, \Delta, \varepsilon$ (and the identity operator id_A). Therefore, the structure of TQFT is completely determined by these 4 operators. Note also that m and Δ are symmetric: $m \circ P = m$, where $P\colon A \otimes A \to A \otimes A$ is the permutation $P(a \otimes b) = b \otimes a$. Indeed, this follows from the isomorphism of cobordisms shown below: the standard copants surface admits a self-diffeomorphism (a 180° rotation about its central vertical axis) that swaps the two input boundary circles and fixes the output boundary circle.

(4.3.2)



The analogous argument applied to the pants surface (turn the picture upside down) gives $P \circ \Delta = \Delta$.

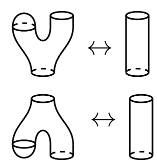
Next, observe that the choice of isomorphisms φ_i is in fact irrelevant. Indeed, since any orientation-preserving diffeomorphism φ of S^1 is isotopic to the identity, by Lemma 3.4.2 its action on A is trivial: $\varphi_* = \mathrm{id}_A$. From this it immediately follows that if M is topologically a disk, considered as a cobordism $\varnothing \Rightarrow S^1$ (i.e., a cap), then the operator $Z(M)\colon \mathbf{k} \to A$ does not depend on the choice of the isomorphism of M with the standard cap; a similar result holds for the cup.

If M is a cobordism $S^1 \sqcup S^1 \to S^1$ which is isomorphic to the standard pair of pants, there are just two isomorphisms up to isotopy; one is obtained from the other by composing with the diffeomorphism that interchanges the two circles that form the in-boundary. But since m is symmetric as discussed above, both choices of isomorphism give us the same operator $Z(M) = m\colon A \otimes A \to A$. A similar statement holds for the cobordism $S^1 \Rightarrow S^1 \sqcup S^1$.

Summarizing, we see that for any cobordism M, the presentation of $Z(M)$ as a composition of tensor products of $m, i, \Delta, \varepsilon$ only depends on the choice of an excellent Morse function and does not depend on any additional data.

Thus, to check that $Z(M)$ does not depend on any choices, we need to check that $Z(M)$ is invariant under the Cerf moves described in Theorem 4.2.2. Let us analyze what these moves look like for $n = 2$.

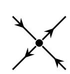
The birth-death moves (type α) are easily seen to be of one of the following two types:



Now let us consider the crossing move. Assume that we have a family of Morse functions f_s and two critical points p_1, p_2 such that for $s < 0$, $c_1 < c_2$, for $s > 0$, $c_1 > c_2$, and for $s = 0$, $c_1 = c_2 = c$. Choose some $a_0 < c$, $a_1 > c$ and consider the (non-elementary) cobordism $M = f_0^{-1}[a_0, a_1]$.

Since operators of the form $X \otimes 1$ and $1 \otimes Y$ commute anyway, if the points p_1, p_2 are in different connected components of the cobordism M, then the crossing move does not give any new conditions on $m, i, \Delta, \varepsilon$. Therefore, we only need to consider the case when the two critical points p_1, p_2 are in the same connected component of the critical level set $S = f_0^{-1}(c)$. This immediately implies that each of the critical points is of index 1.

The set S is a smooth curve except at critical points, where S looks like a coordinate cross; thus, S is a graph with two 4-valent vertices p_1, p_2. Moreover, since M is oriented, we can define an orientation on S (for example, defining the tangent vector v to S by the condition that $(v, \operatorname{grad} f)$ is a positively oriented frame). Orientation also gives a cyclic order of edges at each vertex p_i. Using Morse lemma, it is easy to see that near each critical points S looks as shown below:

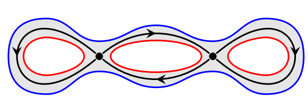
(4.3.3) 

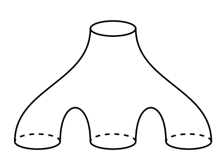
LEMMA 4.3.1. *Let S be a connected directed graph with two vertices p_1, p_2 of degree 4, together with a cyclic order of edges at each vertex, such that near each vertex it is locally isomorphic to the graph shown in (4.3.3).*

Then S is isomorphic to one of the four graphs shown in Figure 4.3.1, and their orientation reversals.

The proof of this lemma is an elementary combinatorics argument; we leave details to the reader.

Each of these graphs uniquely determines the corresponding connected component of the cobordism $M = f_0^{-1}(c-\varepsilon, c+\varepsilon)$; we can recover it by replacing each arc of the graph by a thin ribbon. For example, for graph A, the corresponding cobordism is shown below (for reader's convenience, we also show it in a more familiar form):





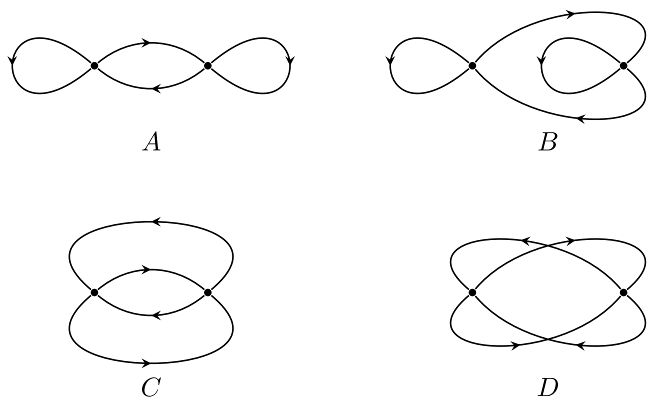
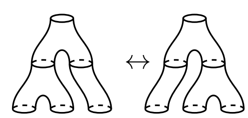
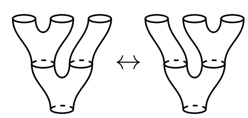


FIGURE 4.3.1. Possible shapes of critical level set S. Note that for the last graph, intersection points are not vertices.

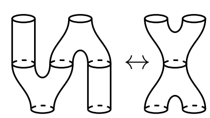
In this case, one easily sees that the corresponding Cerf move is given by



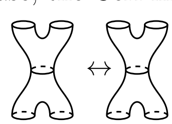
Reversing the orientation of the graph A is equivalent to flipping the picture in the vertical direction, which gives the Cerf move below:



Similarly, graph B gives the following Cerf move:

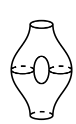


Graphs C and D are less obvious. For C, the cobordism M is of type 2-2: it is a surface of genus zero, with two circles as in-boundary, and two circles as out-boundary. However, in this case, the Cerf move looks like:



In other words, both for $s < 0$ and for $s > 0$ in the family of Morse functions f_s, the corresponding Morse decomposition of the cobordism is a composition of copants and pants. Thus, it gives no new conditions on m, Δ.

Similarly, graph D gives a cobordism of genus 1 from S^1 to S^1; again, in this case both for $s < 0$ and for $s > 0$ the Morse decomposition looks like



and thus the corresponding Cerf move gives no new conditions on m, Δ.

Summarising, we see that the non-trivial Cerf moves (both birth-death and crossings) for $n = 2$ are shown in Figure 4.3.2. In this table, we also include the algebraic identities on the operators $m, i, \Delta, \varepsilon$ (defined by (4.3.1)) corresponding to each of the moves.

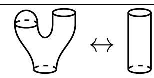
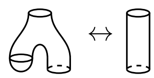
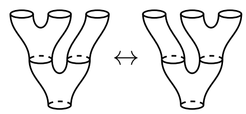
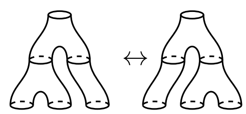
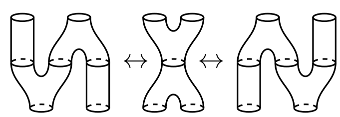
Name	**Cerf move**	**Algebraic identity**
Unit		$m \circ (i \otimes \mathrm{id}_A) = \mathrm{id}_A$
Counit		$(\varepsilon \otimes \mathrm{id}_A) \circ \Delta = \mathrm{id}_A$
Associativity		$m \circ (m \otimes \mathrm{id}_A) = m \circ (\mathrm{id}_A \otimes m)$
Coassociativity		$(\Delta \otimes \mathrm{id}_A) \circ \Delta = (\mathrm{id}_A \otimes \Delta) \circ \Delta$
Frobenius		$\Delta \circ m = (m \otimes \mathrm{id}_A) \circ (\mathrm{id}_A \otimes \Delta)$ $= (\mathrm{id}_A \otimes m) \circ (\Delta \otimes \mathrm{id}_A)$

FIGURE 4.3.2. Cerf moves in dimension 2 and the corresponding algebraic identities

It is easy to see that the identities given by the unit and associativity moves are equivalent to the statement that m, i define on $A = Z(S^1)$ a structure of an associative algebra. Similarly, coassociativity and counit say that Δ, ε define on A a structure of coassocaitive coalgebra.

Combining these identities with the symmetry relations $m \circ P = m$, $P \circ \Delta = \Delta$ established in (4.3.2), we obtain the following theorem.

THEOREM 4.3.2. *A 2d TQFT is equivalent to the following collection of data.*

- *A vector space $A = Z(S^1)$.*
- *Linear maps $m\colon A \otimes A \to A$, $i\colon \mathbf{k} \to A$.*
- *Linear maps $\Delta\colon A \to A \otimes A$, $\varepsilon\colon A \to \mathbf{k}$.*

This data has to satisfy the following conditions:

- *(A, m, i) is a commutative algebra with unit; as usual, we will use the notation ab as a shorthand for $m(a \otimes b)$.*
- *(A, Δ, ε) is a cocommutative coalgebra with counit.*
- *The Frobenius condition*

$$\Delta(a_1 a_2) = (a_1 \otimes 1)\Delta(a_2) = \Delta(a_1)(1 \otimes a_2). \tag{4.3.4}$$

REMARK 4.3.1. Since Δ, m are symmetric, any one of the two Frobenius relations implies the other; we include both for use in the more general case.

REMARK 4.3.2. Note that commutativity of m and cocommutativity of Δ do not come from any of the Cerf moves in Figure 4.3.2; they follow from the symmetric-pants argument (4.3.2) (and its mirror image), which in turn reflects the symmetric monoidal structure of $\mathbf{Cob}_2$.

A standard way to restate the above theorem is using the notion of Frobenius algebra.

DEFINITION 4.3.1. A Frobenius algebra is a vector space over a field $\mathbf{k}$ with two structures:

- A structure of an associative algebra, with multiplication m and unit $i\colon \mathbf{k} \to A$.
- A structure of a coassociative coalgebra, with comultiplication Δ and counit $\varepsilon\colon A \to \mathbf{k}$.

These two structures must satisfy the Frobenius condition (4.3.4).

Note that in this definition, we do not assume that A is commutative or cocommutative.

Then Theorem 4.3.2 can be rewritten as follows.

THEOREM 4.3.3. *A 2d TQFT is equivalent to a commutative, cocommutative Frobenius algebra.*

In the next section we will show that cocommutativity follows from commutativity, so it is enough to require A to be commutative; see Corollary 5.1.2.

CHAPTER 5

Frobenius algebras

In the previous chapter, we have shown that 2-dimensional TQFTs are classified by commutative, cocommutative Frobenius algebras. In this chapter, we will study Frobenius algebras in detail.

Recall that a Frobenius algebra is a vector space over a field $\mathbf{k}$ with two structures (see Definition 4.3.1):

- A structure of an associative algebra, with multiplication m and unit $i\colon \mathbf{k} \to A$.
- A structure of a coassociative coalgebra, with comultiplication Δ and counit $\varepsilon\colon A \to \mathbf{k}$.

These two structures must satisfy the Frobenius condition (4.3.4).

Note that the Frobenius condition can be restated as follows. Define on $A \otimes A$ an A-bimodule structure by $x(a_1 \otimes a_2) = (xa_1) \otimes a_2$, $(a_1 \otimes a_2)x = a_1 \otimes (a_2 x)$. Then the Frobenius condition is equivalent to the following:

$$\Delta\colon A \to A \otimes A \text{ is a morphism of } A\text{-bimodules.} \tag{5.0.1}$$

Note that in this definition, we do not assume that A is commutative or cocommutative.

Remark 5.0.1. Given a vector space with structures of an associative algebra and coassociative coalgebra, there are several natural choices for compatibility between these two structures. By far the most commonly used ones are

- *Bialgebra condition*: the comultiplication is a morphism of algebras. (Frequently one also requires existence of a certain involution, called the *antipode*; this gives the definition of a Hopf algebra.) Examples of bialgebras include group algebras, universal enveloping algebras of Lie algebras, and more.
- *Frobenius condition*: the comultiplication is a morphism of A-bimodules.

These are different choices: a Frobenius algebra is not the same as a Hopf algebra. They both have their uses; for 2d TQFT, we need the structure of a Frobenius algebra.

We will use a graphical presentation of various algebraic operations, representing linear operators $A^{\otimes k} \to A^{\otimes m}$ by graphs with k legs at the top and m legs at the bottom (thus, our operators act "from top to bottom").

Figure 5.0.1 shows graphs used to represent the common algebraic operations.

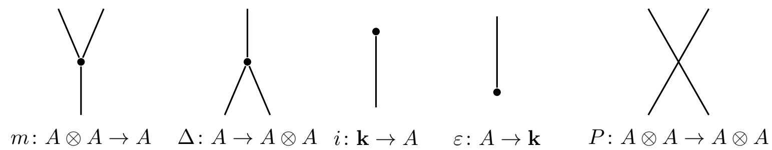
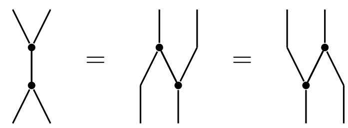


FIGURE 5.0.1. Graphical presentation of operations in a Frobenius algebra. P stands for permutation: $P(a\otimes b)=b\otimes a$.

Thus, the Frobenius relation (4.3.4) is graphically represented by

(5.0.2)


5.1. Frobenius pairing

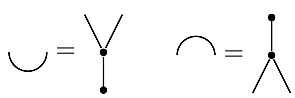
It turns out that there is an alternative definition of Frobenius algebras.

THEOREM 5.1.1. *Let A be a Frobenius algebra. Then A is finite-dimensional and the bilinear pairing*

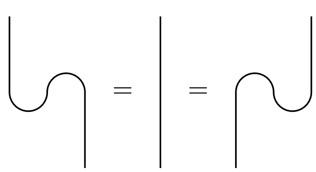
$$(a,b)=\varepsilon(ab) \tag{5.1.1}$$

is non-degenerate; the corresponding coevaluation map $\mathbf{k}\to A\otimes A$ is given by $\Delta\circ i$.

We will use "cup" and "cap" graphics to denote the pairing $\varepsilon(ab)$ and the coevaluation $\Delta\circ i$:

(5.1.2)


PROOF. Define maps $\mathrm{ev}\colon A\otimes A\to\mathbf{k}$ and $\mathrm{coev}\colon \mathbf{k}\to A\otimes A$ by (5.1.2). Then it immediately follows from the Frobenius relation that these maps satisfy the rigidity condition:

(5.1.3)


$$\sum(x,x_i)x^i=x=\sum(x^i,x)x_i$$

where x_i, x^i are defined by $\Delta(1)=\sum x_i\otimes x^i$. Thus, by Lemma 3.2.2, A is finite-dimensional, and $\varepsilon(ab)$ is a non-degenerate pairing.

□

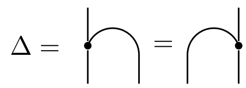
As an immediate corollary, we see that the comultiplication is uniquely determined by the unit, the multiplication, and the counit. Indeed, by Theorem 5.1.1, $e=\Delta(1)$ is given by

$$e=\Delta(1)=\sum x_i\otimes x^i \tag{5.1.4}$$

where x_i, x^i are dual bases in A with respect to the Frobenius pairing: $(x^i, x_j) = \delta_{ij}$. On the other hand, it follows from the Frobenius relation that

$$\Delta(x) = (x \otimes 1)e = e(1 \otimes x)$$

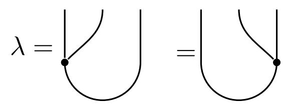
(5.1.5)



EXERCISE 5.1.1. Let A be a Frobenius algebra, and let x_i, x^i be as in (5.1.4).

(1) Define the map $\lambda\colon A \otimes A \otimes A \to \mathbf{k}$ by $\lambda(a_1 \otimes a_2 \otimes a_3) = \varepsilon(a_1 a_2 a_3)$:

(5.1.6)



Let $c_{ijk} = \lambda(x_i, x_j, x_k) = \varepsilon(x_i x_j x_k)$.

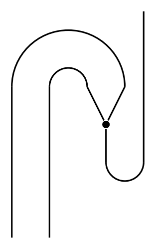
Prove that then the structure constants of the multiplication and the comultiplication are obtained by "raising indices" of c_{ijk}:

$$x_i x_j = \sum c_{ijk} g^{kl} x_l = \sum g^{lk} c_{kij} x_l,$$
$$\Delta(x_i) = \sum c_{ijk} g^{jj'} g^{kk'} x_{k'} \otimes x_{j'} = \sum c_{jki} g^{j'j} g^{k'k} x_{k'} \otimes x_{j'}$$

where g^{ji} is the inverse matrix of $g_{ij} = (x_i, x_j)$ (thus, $x^i = \sum g^{ij} x_j$). Note that we do not assume that g_{ij} is symmetric.

(2) Prove that the comultiplication is also given by the picture below:

(5.1.7)



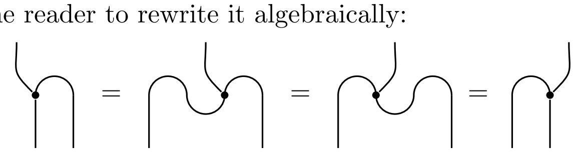
COROLLARY 5.1.2. *If a Frobenius algebra A is commutative, it is also cocommutative.*

Conversely, one can recover the structure of a Frobenius lagebra from the pairing $(\ ,\ )$.

THEOREM 5.1.3. *Let A be a finite-dimensional associative algebra and let $\varepsilon\colon A \to \mathbf{k}$ be a linear map such that the bilinear pairing $(a, b) = \varepsilon(ab)$ is non-degenerate. Let $e = \sum x_i \otimes x^i \in A \otimes A$, where x_i, x^i are dual bases with respect to $(\ ,\ )$: $(x^i, x_j) = \delta_{ij}$. Then e is central: $(a \otimes 1)e = e(1 \otimes a)$, and A has a canonical Frobenius algebra structure with the comultiplication given by* (5.1.5).

PROOF. The sequence of pictures below gives a graphical proof that e is central; we leave it to the reader to rewrite it algebraically:

(5.1.8)



Coassociativity, the counit axiom, and the Frobenius property are proved by very similar computations, which we leave to the reader. $\square$

Thus, we could give an alternative definition of a Frobenius algebra.

DEFINITION 5.1.1. A Frobenius algebra is a finite-dimensional associative algebra A together with a map $\varepsilon\colon A \to \mathbf{k}$ such that the bilinear pairing

$$(a, b) = \varepsilon(ab) \tag{5.1.9}$$

is non-degenerate.

By Theorems 5.1.1 and 5.1.3, this is equivalent to the previously given definition (Definition 4.3.1).

There are also many other equivalent definitions of Frobenius algebras; see, e.g., [**Koc2004**, Section 2.2].

Note that while in general, the pairing $(a, b) = \varepsilon(ab)$ need not be symmetric, in most examples of interest to us it is.

DEFINITION 5.1.2. A Frobenius algebra A is *symmetric* if ε is trace-like, i.e.,

$$\varepsilon(ab) = \varepsilon(ba).$$

Of course, every commutative Frobenius algebra is symmetric; however, the examples below show that an algebra can be symmetric without being commutative.

EXAMPLE 5.1.1. Let $A = \mathbb{C}[x]/(x^{n+1})$, with $\varepsilon(x^k) = 1$ if $k = n$ and $\varepsilon(x^k) = 0$ otherwise. This is a commutative Frobenius algebra which has a natural topological interpretation: it is the cohomology of $\mathbb{C}P^n$. In particular, for $n = 1$, this Frobenius algebra and the corresponding TQFT play an important role in the theory of Khovanov homology.

EXAMPLE 5.1.2. Let A be the algebra of $n \times n$ matrices over $\mathbf{k}$, with $\varepsilon(a) = \lambda \cdot \mathrm{tr}(a)$, for some $\lambda \neq 0$. Then A is a symmetric Frobenius algebra.

The following lemma is immediate from the definition.

LEMMA 5.1.4. *Let A be a symmetric Frobenius algebra, and let $e = \sum x_i \otimes x^i \in A \otimes A$ be as in Theorem 5.1.3. Then*

(1) *e is symmetric: $P(e) = e$, where $P\colon A \otimes A\colon a \otimes b \mapsto b \otimes a$.*
(2) *e is* bicentral*:*

$$\sum ax_i \otimes x^i = \sum x_i \otimes x^i a,$$
$$\sum x_i a \otimes x^i = \sum x_i \otimes ax^i.$$

5.2. Semisimple Frobenius algebras

Recall that an associative algebra is semisimple if it is a direct sum of simple algebras. As we will see later, semisimple algebras play a special role in the construction of TQFTs, so in this section, we study **semisimple** Frobenius algebras. We have already seen one example: the Frobenius algebra $A = \mathrm{Mat}_n(\mathbf{k})$ given in Example 5.1.2. In fact, over an algebraically closed field, it is the most general example of a semisimple symmetric Frobenius algebra. Namely, we have the following lemma.

LEMMA 5.2.1. *Let $\mathbf{k}$ be an algebraically closed field. Then any semisimple symmetric Frobenius algebra A has the form*

$$A = \bigoplus_i \mathrm{Mat}_{n_i}(\mathbf{k})$$

where $\mathrm{Mat}_n(\mathbf{k})$ *is the algebra of* $n \times n$ *matrices, and*

$$\varepsilon(a) = \sum_i \lambda_i \operatorname{tr}(a_i)$$

for some constants $\lambda_i \neq 0$*, where* $a_i \in \mathrm{Mat}_{n_i}(\mathbf{k})$ *is the* i*-th component of* a*.*

Indeed, by the Wedderburn theorem (see e.g., [**Pie1982**, Section 3.5]), any finite-dimensional semisimple algebra over $\mathbf{k}$ is a direct sum of matrix algebras, and it is well known that for a matrix algebra, $A/[A, A]$ is one-dimensional, which easily implies the statement of the lemma.

In particular, this gives us the following classification of semisimple commutative Frobenius algebras.

Theorem 5.2.2. *Let* $\mathbf{k}$ *be an algebraically closed field. Then any semisimple commutative Frobenius algebra over* $\mathbf{k}$ *has the following form*

$$A = \bigoplus_{i=1}^{n} \mathbf{k} e_i, \quad e_i e_j = \delta_{ij} e_i$$

with unit $1 = \sum_i e_i$ *and counit*

$$\varepsilon(e_i) = \lambda_i, \quad \lambda_i \neq 0.$$

It is an easy exercise to show that for an algebra described in Theorem 5.2.2, the canonical element $e = \Delta(1)$ is given by

$$e = \sum_i \lambda_i^{-1} e_i \otimes e_i$$

and the comultiplication is given by

$$\Delta(e_i) = \lambda_i^{-1} e_i \otimes e_i.$$

In particular, it implies that for the corresponding TQFT, we have

$$Z(S^2) = \varepsilon(1) = \sum \lambda_i$$
$$Z(T) = \dim A$$

where $T = \Sigma_1$ is the two-dimensional torus.

Lemma 5.2.3. *Let* A *be as in Theorem 5.2.2, and let* Z_A *be the corresponding 2d TQFT. Then for a closed surface* Σ_g *of genus* g*, we have*

$$Z_A(\Sigma_g) = \sum_i \lambda_i^{1-g}.$$

Proof. The cases $g = 0, 1$ have already been discussed. For general $g \geq 1$, it is easy to see that $Z_A(\Sigma_g) = \operatorname{tr}(W^{g-1})$, where $W \colon A \to A$ is given by $W = Z_A(\Sigma_{1,2})$, with $\Sigma_{1,2}$ a torus with one in-boundary circle and one out-boundary circle:

$W = Z_A\Big($  $\Big) = m \circ \Delta.$

Using the formula for comultiplication, we see that $W(e_i) = \lambda_i^{-1} e_i$, which immediately gives the statement of the lemma. □

REMARK 5.2.1. It is easy to see that the operator W defined above coincides with the operator of multiplication by the element $w = m \circ \Delta(1)$. Explicitly, this element can be described by

$$w = \sum x_i x^i \in A, \tag{5.2.1}$$

where x_i, x^i are dual bases in A with respect to the pairing. This element is sometimes called the *Euler element* and will play an important role later, see Theorem 9.3.4. (In [**LP2007**], w is called the "window element".)

EXERCISE 5.2.1. Recall that for an A-B-bimodule M, the dual vector space M^* is naturally a B-A-bimodule, with action given by

$$(b\lambda)(m) = \lambda(mb),$$
$$(\lambda a)(m) = \lambda(am).$$

In particular, the dual of an A-bimodule is again an A-bimodule.

Let A be a symmetric Frobenius algebra.

(1) Show that the map $A \to A^* \colon a \mapsto (a, -)$ is a morphism of A-bimodules.
(2) More generally, let M be a finite-dimensional left A-module. Let $M^\vee = \mathrm{Hom}_A(M, A)$; it has a natural structure of a right A-module. Show that then the map below is an isomorphism of right A-modules:

$$M^\vee \to M^*$$
$$f \mapsto \varepsilon(f(-)).$$

and the inverse is given by

$$M^* \to M^\vee$$
$$\lambda \mapsto \sum \lambda(x_i m) x^i,$$

where x_i, x^i are as in (5.1.4).

CHAPTER 6

Dijkgraaf–Witten theory

In this chapter we discuss the fundamental example of a TQFT: the gauge theory with finite gauge group, also known as the Dijkgraaf–Witten model. This theory was introduced by Dijkgraaf in his thesis [**Dij1989**, Section 8.5] (see also [**DW1990**, Section 6]). In these works, the focus was mostly on the theory in dimension 3, where it can be considered as a baby version of Chern–Simons theory, but it works the same way in any dimension. We will present the Dijkgraaf–Witten model following [**FQ1993**], [**Tel2016**].

Throughout this chapter, we fix a finite group G; for simplicity, we will use the ground field $\mathbf{k} = \mathbb{C}$.

Informally, for a closed n-manifold M, $Z_G(M)$ should count G-covers (or G-bundles) on M:

$$Z_G(M) = \text{“number of } G\text{-bundles } P \to M\text{”}. \tag{6.0.1}$$

(The reason why this is in quotes will be discussed shortly.) For people not familiar with the notion of G-bundles, we give a review of the main facts.

6.1. Principal G-bundles

Definition 6.1.1. A G-bundle over a manifold M is a finite covering $\pi\colon P \to M$ together with a right action of G on P which preserves π and such that the action of G on each fiber $P_x = \pi^{-1}(x)$ is free and transitive.

The reason for choosing a right action instead of a left one is purely technical; one could also use left action and get an equivalent definition.

We define an isomorphism of G-bundles (for the same G and M) in the obvious way. We denote by $\mathrm{Bun}_G(M)$ the category of G-bundles on M, with morphisms being isomorphisms of G-bundles. This category is a *groupoid*, i.e., a category in which all morphisms are invertible. In particular, for a G-bundle P we denote by $\mathrm{Aut}(P)$ the group of automorphisms of P.

Since G is discrete, any G-bundle comes with a flat connection: given a path γ in M connecting points $x_0, x_1 \in M$, and a choice of lift $\tilde{x}_0 \in P_{x_0}$, there is a unique way to lift γ to a path in P. This gives a holonomy map $\rho_\gamma\colon P_{x_0} \to P_{x_1}$ which commutes with the right action of G. In particular, this gives us the monodromy map

$$\rho\colon \pi_1(M, x) \to \mathrm{Aut}(P_x) \tag{6.1.1}$$

where $\mathrm{Aut}(P_x)$ is the group of all bijections $P_x \to P_x$ which commute with the right action of G.

If we choose a point $\tilde{x} \in P_x$, we can identify P_x with G; this gives an identification $\mathrm{Aut}(P_x) \xrightarrow{\sim} G$ (since any bijection $G \to G$ which commutes with the right action of G is given by left multiplication by some element of G). Thus, a pair $(P, \tilde{x})$ gives rise to a monodromy map $\rho\colon \pi_1(M, x) \to G$. It is easy to see that conversely, for connected M, any group homomorphism $\pi_1(M, x) \to G$ defines a G-bundle P together with a choice of a base point $\tilde{x} \in P_x$. In other words, we have the following lemma.

LEMMA 6.1.1. *Let M be a connected manifold; fix a choice of base point $x \in M$. Then the set of isomorphism classes of pairs $(P, \tilde{x})$, $\tilde{x} \in P_x$, is in bijection with*

$$\mathrm{Hom}(\pi_1(M, x), G).$$

Moreover, such pairs have no non-trivial automorphisms: if $(P, \tilde{x})$ and $(P', \tilde{x}')$ are two such pairs, then the isomorphism between them, if it exists, is unique.

We will refer to a pair $(P, \tilde{x})$ as in the lemma as a *based G-bundle*. One can restate this lemma as follows: the groupoid $\widetilde{\mathrm{Bun}}_G$ of based G-bundles is equivalent as a groupoid to the set $\mathrm{Hom}(\pi_1(M, x), G)$ (considered as a trivial groupoid: the only morphisms are identity morphisms).

If we change the choice of $\tilde{x}$, replacing it by $\tilde{x}g$ (for the same P), then it is easy to see that the monodromy map (6.1.1) is conjugated by g^{-1}. Thus, we get the following result.

LEMMA 6.1.2. *Let M be connected. Then the set of isomorphism classes of objects in $\mathrm{Bun}_G(M)$ is in bijection with*

$$\mathrm{Hom}(\pi_1(M, x), G)/G$$

where G acts by conjugation. Moreover, for $\rho \in \mathrm{Hom}(\pi_1(M, x), G)$, the automorphism group of the corresponding bundle P is

$$\mathrm{Aut}(P) = \{g \in G \mid g\rho g^{-1} = \rho\}. \tag{6.1.2}$$

One can restate it as an equivalence of groupoids.

DEFINITION 6.1.2. Let X be a set with an action of a group G. We define the groupoid $X//G$ as the category whose objects are points $x \in X$ and $\mathrm{Hom}(x_0, x_1) = \{g \in G \mid g(x_0) = x_1\}$.

Then Lemma 6.1.2 says that for a connected M, the groupoid $\mathrm{Bun}_G(M)$ is equivalent to $\mathrm{Hom}(\pi_1(M, x), G)//G$.

EXAMPLE 6.1.1. Let $M = S^1$. Then $\widetilde{\mathrm{Bun}}_G(S^1) \simeq G$, and $\mathrm{Bun}_G(S^1) \simeq G//G$ (in the second identity, G acts on G by conjugation). In particular, there is a bijection between the set of isomorphism classes of $\mathrm{Bun}_G(S^1)$ and the set of conjugacy classes in G; if we denote by P_g the bundle corresponding to $g \in G$, then $\mathrm{Aut}(P_g)$ is the centralizer of g: $\mathrm{Aut}(P_g) \simeq Z_g = \{h \in G \mid hgh^{-1} = g\}$.

6.2. Definition of DW theory

In this section, we define the Dijkgraaf–Witten (DW) TQFT; this definition works in any dimension n.

Naively, we want to define, for a closed $(n-1)$-manifold N, the vector space

$$Z_G(N) = \text{vector space generated by isomorphism classes of } G\text{-bundles on } N$$

and for a cobordism $N_0 \overset{M}{\Longrightarrow} N_1$, we define

$$Z_G(M)([P_0]) = \sum_P [P|_{N_1}], \qquad P_0 \in \mathrm{Bun}_G(N_0)$$

where the sum is over all isomorphism classes of bundles P on M such that $P|_{N_0} = P_0$. The gluing axiom follows immediately.

This, however, needs further work. First, the requirement $P|_{N_0} = P_0$ is meaningless: two objects in a category (in this case, two bundles) are never equal; instead, we should talk about them being isomorphic. But even if M is closed, so we do not have this problem, just counting isomorphism classes gives the wrong answer, as we completely ignore the fact that some of these bundles have non-trivial automorphisms.

To begin with, let us explain how one counts G-bundles on a connected manifold M. In this case, we can choose a base point $x \in M$; then for any bundle P on M, we have $|G|$ possible choices of lifts $\tilde{x} \in P_x$. Thus, we make the following definition

$$|\operatorname{Bun}_G(M)| = \frac{1}{|G|}|\widetilde{\operatorname{Bun}}_G(M)| \tag{6.2.1}$$

where we denoted by $|\widetilde{\operatorname{Bun}}_G(M)|$ the number of isomorphism classes of $(P, \tilde{x}) \in \widetilde{\operatorname{Bun}}_G(M)$. Note that there are no non-trivial automorphisms in $\widetilde{\operatorname{Bun}}_G$: this groupoid is equivalent to a set.

Lemma 6.2.1. *Let M be connected, and let $|\operatorname{Bun}_G(M)|$ be defined by (6.2.1). Then*

$$|\operatorname{Bun}_G(M)| = \sum_{[P]} \frac{1}{|\operatorname{Aut}(P)|}$$

where the sum is taken over isomorphism classes in $\operatorname{Bun}_G(M)$.

In other words, in this count every isomorphism class is taken with the weight equal to $\frac{1}{|\operatorname{Aut}(P)|}$.

This lemma immediately follows from the following elementary observation, whose proof we leave to the reader: if X is a set with an action of a finite group G, then

$$\frac{|X|}{|G|} = \sum_{[x]} \frac{1}{|\operatorname{Stab}(x)|}$$

where $\operatorname{Stab}(x) = \{g \in G \mid g(x) = x\} = \operatorname{Aut}_{X//G}(x)$.

Example 6.2.1. Let $M = S^1$. Then $\widetilde{\operatorname{Bun}}_G(M) \simeq G$, so $|\operatorname{Bun}_G(M)| = \frac{1}{|G|}|\widetilde{\operatorname{Bun}}_G(M)| = 1$. For an element $g \in G$, the automorphism group of the corresponding bundle is the centralizer of g: $\operatorname{Aut}(P_g) = Z_g$; thus,

$$\frac{1}{|\operatorname{Aut}(P_g)|} = \frac{|C_g|}{|G|}$$

where $C_g \subset G$ is the conjugacy class containing g.

More generally, we see that for a connected manifold M, we have

$$Z_G(M) = |\operatorname{Bun}_G(M)| = \frac{|\operatorname{Hom}(\pi_1(M), G)|}{|G|}. \tag{6.2.2}$$

Lemma 6.2.1 suggests the following definition, which now can be used for any groupoid (in particular, for $\operatorname{Bun}_G(M)$ for general M).

Definition 6.2.1. Let $\mathcal{C}$ be a groupoid such that the set of isomorphism classes is finite, and for any $x \in \mathcal{C}$, the group $\operatorname{Aut}(x)$ is finite. Then we define

$$|\mathcal{C}| = \sum_{[x]} \frac{1}{|\operatorname{Aut}(x)|}$$

where the sum is taken over representatives of isomorphism classes in $\mathcal{C}$.

Example 6.2.2. Let X be a finite set with an action of a finite group G. Then $|X//G| = |X|/|G|$. Note that we are not assuming that the action of G is free.

The above formulas for counting elements in "sets with automorphisms" (aka groupoids) have a long history; in particular, they have been extensively used in physics, when discussing Feynman diagrams. For a nice discussion of this see [**BD2001**]. In particular, we can count how many finite sets there are — try doing this yourself!

See also the blog post [**Bae0**] by John Baez.

Similar ideas can also be used, e.g., to define the Euler characteristic of an orbifold (see [**Car2019**, Section 2.4]). Note: there are several different definitions of the orbifold Euler characteristic.

Now that we have discussed the proper way of counting, we can return to the definition of DW theory.

Theorem 6.2.2. *Let G be a finite group. Then the formulas below define an n-dimensional TQFT, called the Dijkgraaf–Witten TQFT.*

For a closed $(n-1)$-manifold N, we define

$$Z_G(N) = \textit{vector space generated by isomorphism classes of } G\textit{-bundles on } N$$

For a cobordism $N_0 \overset{M}{\Longrightarrow} N_1$, we define the linear map $Z_G(M)\colon Z_G(N_0) \to Z_G(N_1)$ by

$$Z_G(M)([P_0]) = \sum_{(P,\varphi)} \frac{1}{|\operatorname{Aut}(P,\varphi)|}[P|_{N_1}]$$

where the sum is over all isomorphism classes of pairs (P,φ), where P is a G-bundle on M, φ is an isomorphism of bundles $P_0 \to P|_{N_0}$, and $\operatorname{Aut}(P,\varphi)$ is the group of automorphisms of P which commute with φ.

Note that if M is connected and N_0 is non-empty, there are no non-trivial automorphisms of (P,φ).

Equivalently, one can define

$$Z_G(M)([P_0]) = \sum_{[P_1]} \Big(\sum_{[P]} \frac{1}{|\operatorname{Aut}_0(P)|} \Big) [P_1]$$

where the outer sum is over all isomorphism classes of bundles on N_1, the inner sum is over all isomorphism classes of bundles on M such that $P|_{N_0} \simeq P_0$, $P|_{N_1} \simeq P_1$, and $\operatorname{Aut}_0(P)$ denotes the group of automorphisms of P which are the identity on N_0. Some authors instead use automorphisms that are identity on N_1; these two definitions are equivalent after conjugation by the operator $Z_G(N) \to Z_G(N)\colon [P] \mapsto \frac{[P]}{|\operatorname{Aut}(P)|}$.

Example 6.2.3.

(1) Let $M = S^n$ (we assume $n \geq 2$). Then any G-bundle on M is trivial, and $\mathrm{Aut}(P) = G$, so $Z_G(S^n) = \frac{1}{|G|}$.
(2) Let $M = N \times I$ be the cylinder over N. Then for a given bundle P_0 on N, there is a unique bundle P on M extending it (more precisely, the pair (P, φ) is unique up to unique isomorphism); thus, $Z_G(M) = \mathrm{id}$ as expected.
(3) Let M be the cylinder over N considered as cobordism $N \sqcup \overline{N} \to \varnothing$. For a bundle P on N, denote by $\bar{P}$ the same bundle considered as a bundle on $\overline{N}$. Then

$$Z_G(P_1, \overline{P_2}) = \begin{cases} |\mathrm{Aut}(P_1)| & P_1 \simeq P_2 \\ 0 & \text{otherwise} \end{cases}$$

Thus, the isomorphism classes of bundles on N form an orthogonal but not orthonormal basis with respect to the pairing given by this cobordism.

Let us now consider the special case $n = 2$. In this case, every element $g \in G$ gives a bundle P_g on S^1 with monodromy g, and $P_g \simeq P_{g'}$ iff g, g' are conjugate in G (we will denote it $g \sim g'$); in particular, P_1 is the trivial bundle. We will denote the corresponding element of $A = Z_G(S^1)$ by $[P_g]$. By general theory (Theorem 4.3.3), A must have the structure of a commutative Frobenius algebra. The following lemma gives an explicit description of this algebra structure.

Lemma 6.2.3. *The unit, the counit, and the multiplication in A are given by*

$$\begin{aligned} 1 &= \frac{1}{|G|}[P_1] \\ \varepsilon([P_g]) &= \delta_{g,1} \\ [P_{g_1}][P_{g_2}] &= \sum_{h \in G} [P_{g_1 h g_2 h^{-1}}]. \end{aligned} \tag{6.2.3}$$

The proof of this lemma is given by an explicit computation. For example, the unit is given by $Z_G(\ )$; since every bundle on a disk is trivial, and the automorphism group of the trivial bundle is G, the definition of $Z_G(M)$ given in Theorem 6.2.2 becomes $\frac{1}{|G|}[P_1]$. The other cases are given by similar explicit computations.

Corollary 6.2.4. *Let $\mathbb{C}[G]$ be the group algebra of G. Define the linear map $A \to \mathbb{C}[G]$ by*

$$[P_g] \mapsto \sum_{h \in G} hgh^{-1}.$$

Then this map is an algebra isomorphism $A \xrightarrow{\sim} \mathbb{C}[G]^G = z(\mathbb{C}[G])$ (here, z stands for the center of the algebra), which identifies ε with the map

$$\varepsilon(g) = \frac{1}{|G|}\delta_{g,1} = \frac{1}{|G|^2}\,\mathrm{tr}_R(g), \tag{6.2.4}$$

where $R = \mathbb{C}[G]$ is the regular representation.

We can describe the algebra $A = \mathbb{C}[G]^G$ in the form of Theorem 5.2.2. Namely, let $\{V_i\}$ be the set of representatives of isomorphism classes of irreducible representations of G; recall that by Maschke's theorem, any complex finite-dimensional representation of G is isomorphic to a direct sum of copies of V_i. By a well-known result of representation theory of finite groups, $\mathbb{C}[G] \simeq \bigoplus_i \operatorname{End}(V_i)$, which implies

$$A = \mathbb{C}[G]^G = \bigoplus_i \mathbb{C} e_i, \tag{6.2.5}$$

where e_i are the projectors onto irreducible representations: $e_i|_{V_j} = \delta_{ij}\,\mathrm{id}_{V_i}$; thus, $e_i e_j = \delta_{ij} e_i$.

Remark 6.2.1. It follows from orthogonality of characters that

$$e_i = \frac{\dim V_i}{|G|} \sum_{g \in G} \chi_i(g^{-1}) g,$$

where $\chi_i(g) = \operatorname{tr}_{V_i}(g)$ is the character of V_i. Thus, characters of irreducible representations, considered as elements in $\mathbb{C}[G]^G$, are (up to scalars) exactly the orthogonal idempotents. In fact, this is how characters were first introduced in [**Fro1896**]: in modern terminology, they were defined as elements of the center of the group algebra such that $\chi_i \chi_j = \delta_{ij} c_i$. The statement that they are traces in irreducible representations (and indeed the notion of irreducible representation itself) appeared later.

Since every V_i appears in the regular representation with multiplicity $d_i = \dim V_i$, (6.2.4) gives

$$\varepsilon(e_i) = \lambda_i = \frac{\dim^2 V_i}{|G|^2}.$$

We can use this to compute $Z_G(\Sigma_g)$, where Σ_g is a closed surface of genus g. Indeed, by Lemma 5.2.3, we have

$$Z_G(\Sigma_g) = \sum_i \lambda_i^{1-g} = |G|^{2g-2} \sum_i (\dim V_i)^{2-2g}$$

where the sum is over all isomorphism classes of irreducible representations. On the other hand, by (6.2.2),

$$Z_G(\Sigma_g) = \frac{|\operatorname{Hom}(\pi_1(\Sigma_g), G)|}{|G|}.$$

The structure of the fundamental group $\pi_1(\Sigma_g)$ is well-known. For $g = 0$ it is trivial, and for $g \geq 1$ it is generated by elements a_i, b_i with a single relation $\prod_i [a_i, b_i] = 1$, where $[a_i, b_i] = a_i b_i a_i^{-1} b_i^{-1}$ is the group commutator. Thus, for $g \geq 1$, we we get the following identity

$$|G|^{2g-1} \sum_i (\dim V_i)^{2-2g} = \left|\{a_1, \dots, a_g, b_1, \dots, b_g \in G \mid \prod_i [a_i, b_i] = 1\}\right|. \tag{6.2.6}$$

In particular, for $g = 1$ (6.2.6) gives

$$|\{g, h \in G \mid gh = hg\}| = |G| \cdot (\text{number of irreps of } G)$$

which can also be proved directly using representation theory methods without any reference to TQFT. This formula was first obtained by Frobenius in 1896, see [**Fro1896**].

6.3. Path integral interpretation of DW theory

One can rewrite our definition of DW theory so that it looks similar to the original description of QFT in terms of path integrals. Namely, let BG be the classifying space of the group G; this is a topological space which is obtained by taking a contractible space EG on which G acts freely and then defining $BG = EG/G$. (Explicit constructions of EG and BG are well-known; note however that typically BG is infinite-dimensional, even for a finite group G).

We have the universal G-bundle $EG \to BG$; thus, any continuous map $\gamma\colon M \to BG$ gives rise to the pullback bundle $P_\gamma = \gamma^*(EG)$ over M. It is well known that any G-bundle can be obtained in this way; more precisely, we have a bijection

$$\text{isomorphism classes in } \operatorname{Bun}_G(M) = [M, BG]$$

where $[M, BG]$ is the set of homotopy classes of maps $M \to BG$.

Thus, the definition of DW invariant of a closed n-manifold can be rewritten as follows

$$Z_G(M) = \int_{[M,BG]} D\gamma$$

where the integral is over the set of homotopy classes of maps (which is a finite set) and the measure $D\gamma$ is defined by the condition that each homotopy class $[\gamma]$, considered as a point in the finite set $[M, BG]$, has measure $\frac{1}{|\operatorname{Aut}(P_\gamma)|}$.

This also suggests that we can define a "twisted" version of DW theory. Assume from now on that M is a closed oriented n-manifold, with fundamental class $[M] \in H_n(M, \mathbb{Z})$. Let $\omega \in H^n(BG, U(1))$ be a cohomology class; then γ allows us to define the pullback class $\gamma^*(\omega) \in H^n(M, U(1))$, and pairing it with $[M]$ gives a number. Since ω is a cohomology class, the pullback $\gamma^*(\omega)$ depends only on the homotopy class $[\gamma]$, so the pairing $\langle \gamma^*\omega, [M]\rangle$ is a well-defined function on $[M, BG]$. Thus, we define

$$Z_G^\omega(M) = \int_{[M,BG]} \langle \gamma^*\omega, [M]\rangle D\gamma.$$

One should think of $\langle \gamma^*\omega, [M]\rangle$ as an analogue of the term $e^{iS(\phi)}$ in the path integral.

Note that the short exact sequence of abelian groups $0 \to \mathbb{Z} \to \mathbb{R} \to \mathbb{R}/\mathbb{Z} \to 0$ gives rise to a long exact sequence of cohomology of BG; since it is well known that for a finite group and $n \geq 1$, $H^n(BG, \mathbb{R}) = H^n(G, \mathbb{R}) = 0$ (here $H^n(G)$ is the group cohomology), we get an isomorphism

$$H^n(BG, U(1)) = H^{n+1}(BG, \mathbb{Z}).$$

Thus, the "twist" ω can also be thought of as an element in $H^{n+1}(BG, \mathbb{Z})$.

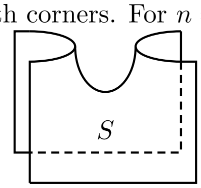
CHAPTER 7

2-categories

In the previous chapters, we classified all 2d TQFTs; the key step of this classification was the fact that any cobordism can be obtained by composing a small set of generating cobordisms (cup, cap, pants, copants). However, for higher dimensions, this approach becomes hard; even for $n = 3$, it is impossible to produce a finite set of generating cobordisms.

A possible way out is rethinking the very idea of a cobordism. After all, the initial physical motivation for TQFT was coming from path integral formulation, which shows that the invariant of a closed n-manifold can be computed by cutting the manifold into small local pieces, computing the path integral over each of them (with appropriate boundary conditions) and summing up. Thus, we just need to formalize the idea of "gluing the manifold from local pieces". For this, the notion of a cobordism is not enough.

One possible way of defining "local pieces" is using the notion of a manifold with corners. For $n = 2$, such a manifold could look like the one shown below:



This is a manifold with corners; you can think of it as a cobordism from the union of two arcs at the top to the union of two intervals at the bottom. Each of them is not a closed manifold but a 1d cobordism. Thus, for 2-dimensional TQFT, we have a structure where we have

- 0-dimensional manifolds (corners)
- 1-dimensional cobordisms between them
- 2-dimensional cobordisms between 1d cobordisms

The appropriate language for describing this kind of structures is the language of 2-categories. In this chapter, we give an overview of the theory of 2-categories; we refer the reader to [**Lei2004**] for details.

7.1. Strict 2-categories

Preliminary Definition 7.1.1. A 2-category is the following collection of data:

(1) A class $\mathcal{C}_0$, called objects of $\mathcal{C}$.
(2) For any $X, Y \in \mathcal{C}_0$, a set $\mathrm{Hom}_{\mathcal{C}}(X, Y)$, called 1-morphisms of $\mathcal{C}$.
(3) For any pair of 1-morphisms $f, g \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$, a set $\mathrm{Hom}^2_{\mathcal{C}}(f, g)$; elements of this set are called 2-morphisms of $\mathcal{C}$.
(4) Appropriate composition laws and identity morphisms for 1- and 2-morphisms.

Before giving the full definition, here are the motivating examples — whatever definition we give, it must cover these examples (possibly with minor technical modifications).

EXAMPLE 7.1.1. The 2-category **Cat** of all (small) categories. Objects of this category are categories, 1-morphisms are functors, and 2-morphisms are functorial morphisms.

EXAMPLE 7.1.2. The 2-category **Alg** of algebras and bimodules. Objects of this category are unital associative algebras (not necessarily finite-dimensional) over a fixed ground field $\mathbf{k}$, 1-morphisms $M\colon A \to B$ are B-A-bimodules, and 2-morphisms are morphisms of bimodules. Composition of 1-morphisms is given by tensor product over the algebra:

$$({}_C M_B, {}_B N_A) \mapsto {}_C M \otimes_B N_A$$

(we use subscripts such as ${}_B N_A$ to indicate that N is a left module over B and a right module over A.)

The identity 1-morphism from A to A is A itself considered as an A-bimodule.

EXAMPLE 7.1.3. Let X be a topological space. Define its *fundamental 2-groupoid* $\Pi_{\le 2}(X)$ to be the following 2-category:

- Objects: points $x \in X$
- 1-morphisms: paths $\gamma\colon I \to X$ between points
- 2-morphisms: equivalence classes of homotopies between paths (two homotopies are equivalent if there is a "homotopy between homotopies" between them)

Morphisms and 2-morphisms in a 2-category are frequently presented graphically. There are two approaches to it:

(1) Objects are represented by points, 1-morphisms by arrows, and 2-morphisms by 2-cells. This is the more common way.
(2) Objects are represented by regions in the plane, 1-morphisms by lines separating these regions ("domain walls"), and 2-morphisms by points where these lines meet (or by boxes). This approach is very familiar to physicists.

Following [**SP2009**, Appendix A.4], we will call diagrams of the first kind *pasting diagrams* and diagrams of the second kind *string diagrams*. In both cases, we will accept the convention that 1-morphisms go "from left to right", and 2-morphisms go "from top to bottom". The figure below (borrowed with modifications from [**SP2009**]) shows a comparison of these two types of graphical presentation.

To turn Definition 7.1.1 into a rigorous definition, we need to define compositions for 1-morphisms and 2-morphisms and state appropriate associativity and unit laws. There are a couple of subtle points; for example, for 2-morphisms we need not only composition of 2-morphisms shown below (vertical composition)
but also "horizontal composition", which we denote by $\beta \circ \alpha$ to distinguish it from the vertical composition:

We can save some effort if we start by saying that for given objects X, Y, all 1-morphisms between them, together with 2-morphisms between those, form a category. That takes care of (vertical) composition of 2-morphisms, identity 2-morphism, etc.

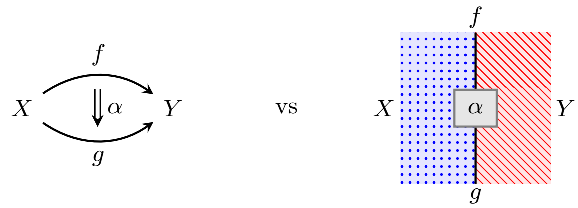


FIGURE 7.1.1. Pasting Diagrams vs String Diagrams

$$\begin{tikzcd} X \arrow[r, bend left=50, "f_1"] \arrow[r, bend right=50, "f_3"'] \arrow[r] \arrow[r, phantom, shift left=2, "\Downarrow\alpha"] \arrow[r, phantom, shift right=2, "\Downarrow\beta"] & Y \end{tikzcd} \rightsquigarrow \begin{tikzcd} X \arrow[r, bend left=50, "f_1"] \arrow[r, bend right=50, "f_3"'] \arrow[r, phantom, "\Downarrow\beta\alpha"] & Y \end{tikzcd}$$

$$\begin{tikzcd} X \arrow[r, bend left=50, "f_1"] \arrow[r, bend right=50, "g_1"'] \arrow[r, phantom, "\Downarrow\alpha"] & Y \arrow[r, bend left=50, "f_2"] \arrow[r, bend right=50, "g_2"'] \arrow[r, phantom, "\Downarrow\beta"] & Z \end{tikzcd} \rightsquigarrow \begin{tikzcd} X \arrow[r, bend left=50, "f_2 f_1"] \arrow[r, bend right=50, "g_2 g_1"'] \arrow[r, phantom, "\Downarrow\beta\circ\alpha"] & Z \end{tikzcd}$$

DEFINITION 7.1.1. A strict 2-category $\mathcal{C}$ is the following collection of data:

- A class of objects $\mathcal{C}_0$
- For any $X, Y \in \mathcal{C}_0$, a category $\mathrm{Hom}_{\mathcal{C}}(X, Y)$. Objects of this category are called 1-morphisms from X to Y; morphisms of $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ are called 2-morphisms of $\mathcal{C}$
- A (horizontal) composition functor

$$H\colon \mathrm{Hom}(X, Y) \times \mathrm{Hom}(Y, Z) \to \mathrm{Hom}(X, Z)$$

- For every $X \in \mathcal{C}$, an identity 1-morphism $1_X \in \mathrm{Hom}(X, X)$

such that the horizontal composition is associative and 1_X is a unit for this composition.

Instead of providing a 1-morphism $1_X \in \mathrm{Hom}(X, X)$, we can say "we have a functor $\mathbf{1} \to \mathrm{Hom}(X, X)$", where $\mathbf{1}$ is the category with a single object and a unique morphism (namely, the identity morphism).

Note that since composition is a functor, it also gives a horizontal composition of 2-morphisms; thus, this is automatically taken care of — no need to add horizontal composition of 2-morphisms by hand.

The exact meaning of "horizontal composition is associative" is as follows. For any $X, Y, Z, W \in \mathcal{C}$, one can construct two functors

$$\mathrm{Hom}(X, Y) \times \mathrm{Hom}(Y, Z) \times \mathrm{Hom}(Z, W) \to \mathrm{Hom}(X, W)$$

namely, $H \circ (H \times \mathrm{id})$ and $H \circ (\mathrm{id} \times H)$. Associativity means that these two functors are equal. Similarly one can write down the exact meaning of "1_X is a unit for horizontal composition".

Exercise 7.1.1. Consider the pasting diagram below. Then there are two obvious ways to compose the four 2-morphisms in this diagram into a single 2-morphism from $f_2 f_1$ to $g_2 g_1$: either using horizontal composition first or using vertical composition first. Prove that the resulting 2-morphisms are equal.

$$\begin{tikzcd}
X \arrow[r, bend left=50, "f_1"] \arrow[r] \arrow[r, bend right=50, "g_1"'] \arrow[r, phantom, shift left=2ex, "\Downarrow"] \arrow[r, phantom, shift right=2ex, "\Downarrow"] & Y \arrow[r, bend left=50, "f_2"] \arrow[r] \arrow[r, bend right=50, "g_2"'] \arrow[r, phantom, shift left=2ex, "\Downarrow"] \arrow[r, phantom, shift right=2ex, "\Downarrow"] & Z \arrow[r, rightsquigarrow] & X \arrow[r, bend left=40, "f_2 f_1"] \arrow[r, bend right=40, "g_2 g_1"'] \arrow[r, phantom, "\Downarrow"] & Z
\end{tikzcd}$$

This is known as the "exchange condition"; it is a special case of the so-called *pasting theorem*; broadly speaking, this theorem claims that given a composable pasting diagram, the result of composition does not depend on the order in which we are doing the compositions. Formulating it precisely is somewhat technical; see [**Pow1990**].

Lemma 7.1.2. *Let X be an object in a strict 2-category $\mathcal{C}$. Consider the monoid of 2-morphisms* $\mathrm{Hom}(1_X, 1_X)$. *Then horizontal and vertical compositions coincide, and the resulting composition is commutative.*

This is known as the *Eckmann–Hilton argument*; see [**Lei2004**, Lemma 1.2.4]. It can be viewed as a categorical analog of the well-known topological fact: for $k > 1$, homotopy groups $\pi_k(X)$ are commutative.

Exercise 7.1.2. Let A be an associative algebra, considered as an object in the 2-category **Alg**. Prove that then $\mathrm{Hom}_{\mathbf{Alg}}(1_A, 1_A) = z(A)$ is the center of A.

7.2. Weak 2-categories

There is one problem with the notion of strict 2-category: this definition is too strict. For example, in the proposed 2-category of algebras and bimodules, the identity 1-morphism $1_A \in \mathrm{Hom}(A, A)$ is A considered as an A-bimodule. But if M is an A-B-bimodule, $A \otimes_A M$ is not equal to M: it is only isomorphic. Thus, we need to relax the associativity condition for composition of 1-morphisms, similarly to what we did with the associativity condition for monoidal categories.

Definition 7.2.1. A *weak 2-category* (also called a bicategory) $\mathcal{C}$ is the following collection of data:

- A class of objects $\mathcal{C}_0$
- For any $X, Y \in \mathcal{C}_0$, a category $\mathrm{Hom}_{\mathcal{C}}(X, Y)$. Objects of this category are called 1-morphisms from X to Y; morphisms of $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ are called 2-morphisms of $\mathcal{C}$
- A (horizontal) composition functor
$$H\colon \mathrm{Hom}(X, Y) \times \mathrm{Hom}(Y, Z) \to \mathrm{Hom}(X, Z)$$
- For every $X \in \mathcal{C}_0$, an identity 1-morphism $1_X \in \mathrm{Hom}(X, X)$
- For any quadruple of objects $X, Y, Z, W \in \mathcal{C}$, an isomorphism α of functors $H \circ (H \times \mathrm{id})$ and $H \circ (\mathrm{id} \times H)$, both considered as functors
$$\mathrm{Hom}(X, Y) \times \mathrm{Hom}(Y, Z) \times \mathrm{Hom}(Z, W) \to \mathrm{Hom}(X, W).$$
- Left and right unit isomorphisms

which satisfy the pentagon and triangle axioms.

The pentagon and triangle axioms are similar to those for a monoidal category; details can be found, e.g., in [**Lei2004**, Section 1.5].

EXAMPLE 7.2.1.

(1) The 2-category **Alg** of algebras and bimodules, introduced in Example 7.1.2, is indeed a weak 2-category.
(2) The 2-category **Cat** of (small) categories, introduced in Example 7.1.1, is indeed a weak 2-category.
(3) As a variation of the above, we can define a weak 2-category $\mathbf{Ab_k}$ of $\mathbf{k}$-linear abelian categories, together with $\mathbf{k}$-linear functors between them. We can also impose some additional restrictions, such as local finiteness (all Hom spaces are finite-dimensional, and every object has finite length) or semisimplicity.

EXAMPLE 7.2.2. Let $\mathcal{C}$ be a monoidal category. Define the 2-category $\mathcal{BC}$ as the 2-category with one object $*$ and morphisms given by $\mathrm{Hom}_{\mathcal{BC}}(*,*) = \mathcal{C}$, with horizontal composition given by tensor product. It is easy to see that this is indeed a 2-category; conversely, every 2-category with only one object is obtained in this way. This is sometimes called *delooping* of $\mathcal{C}$.

REMARK 7.2.1. The notation $\mathcal{BC}$ is motivated by the following. Let M be a monoid (for example, a group). Define the category $\mathcal{B}M$ as the category with a single object $*$ and $\mathrm{Hom}(*,*) = M$. Then it is immediate from the definition that for a group G, the so-defined category is equivalent to the fundamental groupoid of the classifying space BG: $\mathcal{B}G \simeq \Pi_1(BG)$; therefore, $\mathcal{B}$ is the categorical analog of the classifying space construction.

The word "delooping" is justified by the following observation: for a path-connected topological space X, let ΩX be the space of based loops in X. Then we have a homotopy equivalence $\Omega BG \simeq G$; thus, in some sense B is the inverse operation to Ω, hence the name "delooping".

One can also define the notion of functor between (weak) 2-categories, which we skip, referring the reader to [**Lei2004**, Definition 1.5.8]. (Note: what we will be using is called "weak lax functors" in [**Lei2004**].) This allows one to define the notion of equivalence of weak 2-categories.

THEOREM 7.2.1. *Any weak 2-category is equivalent to a strict 2-category.*

This theorem is similar to the coherence theorem Theorem 2.3.3; we refer the reader to [**Lei2004**, Theorem 1.5.15] for the proof.

REMARK 7.2.2. Do not let this theorem give you the wrong idea. For $n = 3$, the notion of a weak 3-category (which is quite hard to define, see [**GPS1995**]) is not equivalent to that of a strict 3-category (which is relatively easy to define). We will return to this when discussing higher categories in Chapter 14.

CONVENTION 7.2.1. From now on, the name "2-category" will mean a weak 2-category unless explicitly stated otherwise.

Note that it is easy to go from a 2-category to a 1-category.

DEFINITION 7.2.2. Let $\mathcal{C}$ be a weak 2-category. Define the category $\tau_{\leq 1}\mathcal{C}$ as follows:

- Objects of $\tau_{\leq 1}\mathcal{C}$ are the same as objects of $\mathcal{C}$.
- Morphisms of $\tau_{\leq 1}\mathcal{C}$ are isomorphism classes of 1-morphisms in $\mathcal{C}$

We will call $\tau_{\leq 1}\mathcal{C}$ the 1-truncation of $\mathcal{C}$; it is also frequently called the *homotopy category* of $\mathcal{C}$. Alternative notation for $\tau_{\leq 1}\mathcal{C}$ is $h\mathcal{C}$.

7.3. Symmetric monoidal 2-categories

Our next goal is defining the notion of a monoidal weak 2-category. We will outline the construction but skip all the details, referring the reader to [**SP2009**, Appendix C].

Throughout this section, $\mathcal{C}$ is a weak 2-category.

As with a usual monoidal category, the starting point is a tensor product functor $\otimes\colon \mathcal{C}\times\mathcal{C}\to\mathcal{C}$. However, we need to define what it means for $\otimes$ to be associative. To put it into perspective, recall the notion of associativity at previous levels of complexity:

- when defining a monoid, we had a map of sets $M\times M\to M$ and we required $a(bc)=(ab)c$
- when defining a monoidal category, we had a functor $\otimes\colon\mathcal{C}\times\mathcal{C}\to\mathcal{C}$; the requirement $A\otimes(B\otimes C)=(A\otimes B)\otimes C$ does not make sense (objects in a category are never equal), so instead we introduced the associativity isomorphism $\alpha\colon (A\otimes B)\otimes C\to A\otimes(B\otimes C)$ and imposed an extra consistency relation, the pentagon axiom (which comes from trying to identify all 5 possible ways of bracketing the tensor product of 4 objects)
- now, in a 2-category, we need to have the tensor product functor and the associativity isomorphism α. But now each side of the pentagon axiom is a composition of 1-morphisms. By the same logic, it is meaningless to require that this diagram is commutative — 1-morphisms in a 2-category are rarely equal, just isomorphic. Thus, we need to provide a 2-isomorphism π between the two sides of the pentagon diagram; we will call it "pentagrammator".

$$
\begin{tikzcd}
 & (A\otimes(B\otimes C))\otimes D \arrow[rr] & & A\otimes((B\otimes C)\otimes D) \arrow[dr] & \\
((A\otimes B)\otimes C)\otimes D \arrow[ur] \arrow[drr] & & \Downarrow\pi & & A\otimes(B\otimes(C\otimes D)) \\
 & & (A\otimes B)\otimes(C\otimes D) \arrow[urr] & &
\end{tikzcd}
\tag{7.3.1}
$$

The so-defined pentagrammator must itself satisfy some compatibility conditions. Not surprisingly, it comes from a cell of dimension 3 in Stasheff's associahedron K_5 (see Remark 2.3.1). Recall that this polyhedron has 14 vertices, corresponding to 14 bracketings of the tensor product of 5 objects; edges of K_5 correspond to application of associativity isomorphisms, and 2-cells correspond to pentagon axiom and functoriality relation. As discussed before, the MacLane coherence axiom for monoidal categoires implies that the 2-skeleton of K_5 is simply-connected. In fact, it is homeomorphic to a 2-sphere: K_5 is topologically a 3-ball, and the 2-skeleton is the boundary of that ball.

We now add a relation corresponding to the single 3-cell in K_5; this relation has the form

(composition of 2-morphisms in the 2-cells in the upper hemisphere)

= (composition of 2-morphisms in the 2-cells in the lower hemisphere).

This gives us a consistency relation for the pentagrammator 2-morphism, called the *associahedron* equation. To give you some idea of what it looks like, Figure 7.3.1 shows the pasting diagram for one side of the associahedron equation, which corresponds to the upper hemisphere in the 14-vertex 3-cell. The lower side is similar. (This diagram is copied from [**SP2009**, Appendix C]; he uses a instead of α for the associativity 1-morphism.)

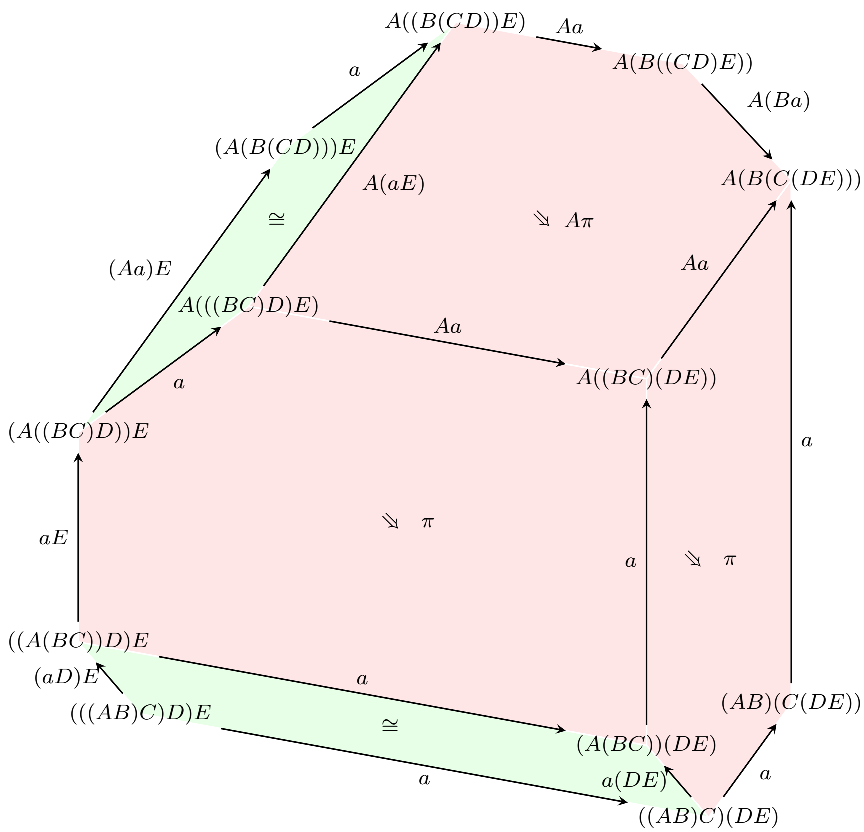


FIGURE 7.3.1. Pasting diagram for the left-hand side of the associahedron axiom. Green 2-cells correspond to commutative diagrams given by functoriality.

In a similar way, instead of requiring the commutativity of diagrams involving the left and right unit isomorphisms, we now need to introduce 2-morphisms ("unitors") filling these diagrams, and then impose additional relations on them coming from 3-cells.

We will not be providing any details of this, referring the reader to [**SP2009**]. Instead, we will show some examples.

EXAMPLE 7.3.1. Let **Alg** be the weak 2-category of algebras and bimodules as defined in Example 7.1.2. Then it has a natural structure of a symmetric monoidal 2-category, with $\otimes$ being the usual tensor product of algebras.

REMARK 7.3.1. Recall that a monoidal category $\mathcal{C}$ is essentially the same as a weak 2-category $\mathcal{BC}$ with one object. Similarly, a monoidal 2-category is the same as a weak 3-category with one object; the associahedron equation is a special case of a consistency equation for associativity of composition of 1-morphisms in a 3-category.

7.4. Locally finite categories and Deligne product

In this section, we give an example of a symmetric monoidal weak 2-category. Namely, recall that a $\mathbf{k}$-linear abelian category $\mathcal{A}$ is called *locally finite* if

- for any $X, Y \in \mathcal{A}$, the space $\mathrm{Hom}_{\mathcal{A}}(X, Y)$ is finite-dimensional, and
- every object $X \in \mathcal{A}$ has finite length

(see [**EGNO2015**, Section 1.8]). Denote by **Lin** the 2-category of locally finite $\mathbf{k}$-linear abelian categories, with 1-morphisms given by *right exact* $\mathbf{k}$-linear functors and 2-morphisms given by functorial morphisms (some motivations for restricting ourselves to right exact functors will be discussed in Chapter 16).

Our next goal is defining an analog of tensor product in **Lin**. Let $\mathcal{C}, \mathcal{D}, \mathcal{E}$ be $\mathbf{k}$-linear categories. Denote by $\mathbf{Bil}(\mathcal{C} \times \mathcal{D}, \mathcal{E})$ the category of $\mathbf{k}$-bilinear functors $F \colon \mathcal{C} \times \mathcal{D} \to \mathcal{E}$ which are right exact in each of the arguments.

DEFINITION 7.4.1. The Deligne product of $\mathbf{k}$-linear categories $\mathcal{C}$, $\mathcal{D}$ is a $\mathbf{k}$-linear category $\mathcal{C} \boxtimes \mathcal{D}$ together with a bilinear functor $\mathcal{C} \times \mathcal{D} \to \mathcal{C} \boxtimes \mathcal{D}$ such that for any $\mathcal{E}$, we have an equivalence

$$\mathbf{Bil}(\mathcal{C} \times \mathcal{D}, \mathcal{E}) \simeq \mathrm{Fun}(\mathcal{C} \boxtimes \mathcal{D}, \mathcal{E}).$$

where Fun is the category of $\mathbf{k}$-linear right exact functors.

This definition mimics the definition of tensor product of vector spaces as a universal object. As usual with such definitions, if such a category $\mathcal{C} \boxtimes \mathcal{D}$ exists, it is unique: any two such categories are canonically equivalent. However, existence is not obvious and in fact may fail without some assumptions on $\mathcal{C}$, $\mathcal{D}$.

THEOREM 7.4.1. *If $\mathcal{C}$, $\mathcal{D}$ are locally finite categories, then $\mathcal{C} \boxtimes \mathcal{D}$ exists and is a locally finite $\mathbf{k}$-linear category.*

We refer the reader to [**EGNO2015**, Section 1.11] for the proof of this theorem as well as results below.

LEMMA 7.4.2.
1. *For a finite-dimensional algebra A, denote by $A{-}\mathbf{mod}$ the category of finite-dimensional left A-modules. Then*
$$(A{-}\mathbf{mod}) \boxtimes (B{-}\mathbf{mod}) \simeq (A \otimes B){-}\mathbf{mod}.$$
2. *For any objects $X_1, Y_1 \in \mathcal{C}$, $X_2, Y_2 \in \mathcal{D}$, we have a canonical isomorphism*
$$\mathrm{Hom}_{\mathcal{C}\boxtimes\mathcal{D}}(X_1 \boxtimes X_2, Y_1 \boxtimes Y_2) \simeq \mathrm{Hom}_{\mathcal{C}}(X_1, Y_1) \otimes \mathrm{Hom}_{\mathcal{D}}(X_2, Y_2).$$
3. *If $\mathcal{C}$, $\mathcal{D}$ are semisimple $\mathbf{k}$-linear categories with simple objects X_i, Y_j respectively, then $\mathcal{C} \boxtimes \mathcal{D}$ is also semisimple, with simple objects $X_i \boxtimes Y_j$.*

THEOREM 7.4.3. *The 2-category* **Lin** *is a symmetric monoidal 2-category, with tensor product functor given by Deligne's product* $\boxtimes$ *and unit given by the category* $\mathbf{Vec_f}$.

We skip the proof, which is mostly straightforward; we just note that to show that $\mathbf{Vec_f}$ is the unit for this monoidal structure, we need to construct, for any $\mathcal{C} \in \mathbf{Lin}$, an equivalence of categories $\mathbf{Vec_f} \boxtimes \mathcal{C} \simeq \mathcal{C}$.

Another way of stating this is by saying that any locally finite $\mathbf{k}$-linear category has a natural structure of a module category over $\mathbf{Vec_f}$ (we will discuss the general notion of a module category later, in Section 16.4). Namely, for a finite-dimensional vector space V and an object $X \in \mathcal{C}$ in a locally finite category $\mathcal{C}$, we define the object $V \otimes X$ by the condition that for any object $Y \in \mathcal{C}$, we have a canonical isomorphism

$$\mathrm{Hom}(Y, V \otimes X) = V \otimes \mathrm{Hom}(Y, X)$$

(more formally, $V \otimes X$ is the object (co)representing the functor $F(Y) = V \otimes \mathrm{Hom}(Y, X)$).

CHAPTER 8

Duality in 2-categories

Recall that for an object X in a monoidal category, we have defined the notion of left and right duals. We will now extend this notion to objects and 1-morphisms in a weak 2-category $\mathcal{C}$. Doing it for objects requires defining a monoidal structure on $\mathcal{C}$; duality for 1-morphisms (also known as adjunction) can be defined in any 2-category.

Throughout this section $\mathcal{C}$ is a weak 2-category.

8.1. Adjunction for 1-morphisms

Definition 8.1.1. Let $\mathcal{C}$ be a weak 2-category. An *adjunction* in $\mathcal{C}$ is the following data:

- A pair of 1-morphisms $f \in \mathrm{Hom}(X, Y)$, $g \in \mathrm{Hom}(Y, X)$
- A 2-morphism $\eta\colon 1_X \to g \circ f$ (unit of adjunction)
- A 2-morphism $\varepsilon\colon f \circ g \to 1_Y$ (counit of adjunction)

satisfying the adjunction equations:

$$
\begin{aligned}
&(f \to f \circ 1_X \xrightarrow{1_f \circ \eta} f \circ g \circ f \xrightarrow{\varepsilon \circ 1} f \circ 1_Y \to f) = 1_f \\
&(g \to 1_X \circ g \xrightarrow{\eta \circ 1_g} g \circ f \circ g \xrightarrow{1 \circ \varepsilon} 1_X \circ g \to g) = 1_g
\end{aligned}
\tag{8.1.1}
$$

In this situation, we say that f is the *left adjoint* of g and g is the *right adjoint* of f; we will write

$$X \underset{g}{\overset{f}{\rightleftarrows}} Y.$$

Alternative notation for adjunction is $f \dashv g$; in both notations, the unit and the counit 2-morphisms are implicit.

To understand the adjunction equations, we can use pasting diagrams to describe the 2-morphisms in the left-hand side of these equations:

$$
\begin{tikzcd}
X \arrow[r, "f"] \arrow[rr, bend left=50, "1_X"] \arrow[rrr, bend left=70, "f"] \arrow[rrr, bend right=70, "f"'] \arrow[rr, phantom, shift left=4ex, "\Downarrow\eta"] & Y \arrow[r, "g"] \arrow[rr, bend right=50, "1_Y"'] \arrow[rr, phantom, shift right=4ex, "\Downarrow\varepsilon"] & X \arrow[r, "f"] & Y
\end{tikzcd}
\qquad
\begin{tikzcd}
Y \arrow[r, "g"] \arrow[rr, bend right=50, "1_Y"'] \arrow[rrr, bend left=70, "g"] \arrow[rrr, bend right=70, "g"'] \arrow[rr, phantom, shift right=4ex, "\Downarrow\varepsilon"] & X \arrow[r, "f"] \arrow[rr, bend left=50, "1_X"] \arrow[rr, phantom, shift left=4ex, "\Downarrow\eta"] & Y \arrow[r, "g"] & X
\end{tikzcd}
$$

Alternatively — and more intuitively — we can use string diagrams. Namely, we will illustrate 2-morphisms η and ε by string diagrams as in Figure 8.1.1. Then the adjunction equations become the equations of Figure 8.1.2, justifying our choice of graphical depiction.

FIGURE 8.1.1. String diagrams for adjunction unit and counit.

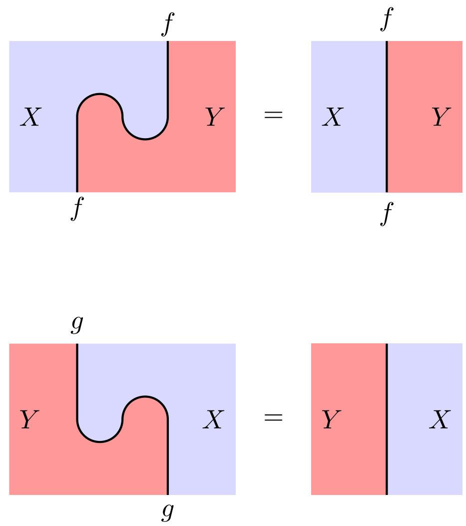


FIGURE 8.1.2. Adjunction equations via string diagrams

The definition of adjunction given above generalizes two well-known examples: duality in a monoidal category and adjunction of functors.

EXAMPLE 8.1.1. Let $\mathcal{C}$ be a monoidal category, and let $B\mathcal{C}$ be the corresponding 2-category with one object as defined in Example 7.2.2. Then the definition of adjunction for 1-morphisms in $B\mathcal{C}$ coincides with the notion of dual pairs of objects in $\mathcal{C}$: $A \dashv B$ as 1-morphisms in $B\mathcal{C}$ if and only if B is the right dual of A in $\mathcal{C}$.

THEOREM 8.1.1. *Let* **Cat** *be the 2-category of small categories. Let* $f\colon \mathcal{C} \to \mathcal{D}$, $g\colon \mathcal{D} \to \mathcal{C}$ *be 1-morphisms in* **Cat** (*i.e., functors between the corresponding categories*). *Then defining adjunction data* η, ε *as in Definition 8.1.1 is equivalent to defining a functorial isomorphism*

$$\mathrm{Hom}_{\mathcal{D}}(f(-), -) \simeq \mathrm{Hom}_{\mathcal{C}}(-, g(-)). \tag{8.1.2}$$

PROOF. (Sketch of proof)

Suppose we are given a functorial isomorphism (8.1.2). Then for any object $c \in \mathcal{C}$, we have isomorphism $\mathrm{Hom}_{\mathcal{D}}(f(c), f(c)) \simeq \mathrm{Hom}_{\mathcal{C}}(c, g(f(c)))$. In particular, applying it to the identity morphism $1_{f(c)}$, we get a canonical element $\eta \in \mathrm{Hom}(c, g(f(c)))$, or equivalently, a functorial morphism $\mathrm{id}_{\mathcal{C}} \to g \circ f$, which can be used as the unit of adjunction as in Definition 8.1.1. Similarly we get a functorial morphism $\varepsilon\colon fg(d) \to d$, which is the counit of adjunction. The adjunction relations shown in Figure 8.1.2 follow from the functoriality of (8.1.2).

We leave the construction in the opposite direction as an exercise to the reader. □

Adjoint functors between categories frequently appear as "restriction–induction" pairs; we give several examples below.

EXAMPLE 8.1.2. Let G be a finite group and $H \subset G$ a subgroup. Consider the restriction functor $\mathrm{Res}\colon \mathrm{Rep}\, G \to \mathrm{Rep}\, H$. Then it has a left adjoint called the induction functor $\mathrm{Ind}\colon \mathrm{Rep}\, H \to \mathrm{Rep}\, G$. The adjunction relation

$$\mathrm{Hom}_G(\mathrm{Ind}\, V, W) \simeq \mathrm{Hom}_H(V, \mathrm{Res}\, W), \qquad V \in \mathrm{Rep}\, H, W \in \mathrm{Rep}\, G$$

is known as *Frobenius reciprocity*; from our point of view, it is the definition of the induction functor.

EXAMPLE 8.1.3. Let $\mathcal{C}$ be the category of abelian groups. It has an obvious forgetful functor $F\colon \mathcal{C} \to \mathbf{Set}$. This functor has a left adjoint $I\colon \mathbf{Set} \to \mathcal{C}$; namely, $I(S)$ is the abelian group freely generated by elements of S. Again, we take it as the definition of "freely generated"; the nontrivial fact is that such a left adjoint exists.

Returning to adjunction in an arbitrary 2-category $\mathcal{C}$, here are some useful results about adjunctions.

LEMMA 8.1.2. *Let $f\colon X \to Y$ be a 1-morphism in a weak 2-category $\mathcal{C}$. Then the remaining adjunction data, g, η, ε (as in Definition 8.1.1), if it exists, is unique up to unique isomorphism: if $(f, g, \eta, \varepsilon)$ and $(f, \tilde{g}, \tilde{\eta}, \tilde{\varepsilon})$ are two different sets of adjunction data with the same f, then there is a unique 2-morphism $\alpha\colon g \to \tilde{g}$ which makes the appropriate diagrams commutative.*

We leave the proof as an exercise (see also [**ML1998**, Section IV.1]).

A variation of this is the following result.

LEMMA 8.1.3. *The unit of the adjunction is uniquely determined by the remaining adjunction data: if $(f, g, \eta, \varepsilon)$ and $(f, g, \eta', \varepsilon)$ are two sets of adjunction data, then $\eta = \eta'$.*

Similarly, the counit ε of adjunction is uniquely determined by (f, g, η).

Moreover, if $(f, g, \eta, \varepsilon)$ satisfies one of the adjunction equations (8.1.1), and $(f, g, \eta', \varepsilon)$ satisfies the other one, then $\eta = \eta'$ (and thus satisfies both equations).

Lemma 8.1.2 and Lemma 8.1.3 are analogs of Theorem 3.3.1 and Exercise 3.3.1 for monoidal categories, and the proofs are similar (see [**Lur2009a**, Lemma 2.3.8]).

The notion of adjunction is closely related to the notion of equivalence. Recall that a 1-morphism $f\colon X \to Y$ in a 2-category is called an *equivalence* if there exists $g\colon Y \to X$ such that $fg \simeq 1_Y$, $gf \simeq 1_X$. (This is a weaker version of an invertible morphism — requiring that $fg = 1_Y$ would be too strict, since 1-morphisms in a 2-category are usually only isomorphic, not equal.)

LEMMA 8.1.4. *If $f\colon X \to Y$, $g\colon Y \to X$ are 1-morphisms in a weak 2-category such that $fg \simeq 1_Y$, $gf \simeq 1_X$, then g is both a left and a right adjoint of f.*

Proof. Choose arbitrary isomorphisms $\alpha\colon fg \xrightarrow{\sim} 1_Y$, $\beta\colon gf \xrightarrow{\sim} 1_X$. Then the composition

$$\theta = (f \xrightarrow{\alpha^{-1}\circ 1_f} fgf \xrightarrow{1_f \circ \beta} f)$$

is an invertible 2-morphism $f \to f$. Thus, setting $\eta = (\theta^{-1} \circ 1_g)\alpha^{-1}$, $\varepsilon = \beta$ we see that the so-defined η, ε satisfy one of the adjunction equations. In a similar way we can find η', ε which satisfy the other adjunction equation. Now the result follows from Lemma 8.1.3. □

In other words, every equivalence can be upgraded to an adjoint equivalence.

8.2. Dual objects in 2-categories

Let us now talk about dual objects in $\mathcal{C}$. It only makes sense if $\mathcal{C}$ has a monoidal structure; thus, in this section we assume that $\mathcal{C}$ is a monoidal weak 2-category.

Recall that in a (usual) monoidal category, we defined a dual pair as the data $(A, B, \mathrm{ev}, \mathrm{coev})$, where $\mathrm{ev}\colon A \otimes B \to \mathbf{1}$, $\mathrm{coev}\colon \mathbf{1} \to B \otimes A$ are morphisms satisfying the zigzag (or rigidity) relations $z_A = 1_A$, $z_B = 1_B$, where

(8.2.1)
$$\begin{aligned} z_A &= (A \xrightarrow{1_A \otimes \mathrm{coev}} A \otimes B \otimes A \xrightarrow{\mathrm{ev} \otimes 1_A} A) \\ z_B &= (B \xrightarrow{\mathrm{coev} \otimes 1_B} B \otimes A \otimes B \xrightarrow{1_B \otimes \mathrm{ev}} B) \end{aligned}$$

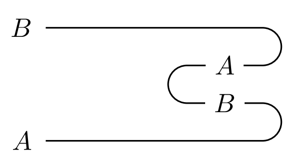
(see Definition 3.3.1).

We now want to upgrade this definition, defining the notion of a dual pair in a monoidal weak 2-category. In this case, it makes no sense to require $z_A = 1_A$, $z_B = 1_B$; instead, we need isomorphisms $\alpha\colon z_A \to 1_A$, $\beta\colon z_B \to 1_B$. Moreover, it is natural to expect that these isomorphisms themselves should satisfy some compatibility relations. Following [Pst2022], we will impose the following relations.

Let zz_+ be the "double zigzag" 1-morphism

$$zz_+ = (A \otimes B \xrightarrow{1_A \otimes \mathrm{coev} \otimes 1_B} A \otimes (B \otimes A) \otimes B \xrightarrow{\mathrm{ev} \otimes \mathrm{ev}} \mathbf{1})$$

(8.2.2)



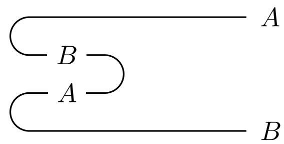
(we have turned the picture 90 degrees counterclockwise, to match our convention that in 2-categories, 1-morphisms go "from left to right").

Then we have two different 2-morphisms $zz_+ \to \mathrm{ev}$: $\alpha \otimes 1$, obtained by using α on the lower part of zigzag, and $1 \otimes \beta$, obtained by using β on the top part of zigzag (we leave it to the reader to write them algebraically).

It is natural to require that these 2-morphisms are equal:

(8.2.3)
$$\begin{tikzcd} A \otimes B \arrow[r, bend left, "zz_+"] \arrow[r, bend right, "\mathrm{ev}"'] \arrow[r, phantom, "\Downarrow \alpha \otimes 1"] & \mathbf{1} \end{tikzcd} \quad = \quad \begin{tikzcd} A \otimes B \arrow[r, bend left, "zz_+"] \arrow[r, bend right, "\mathrm{ev}"'] \arrow[r, phantom, "\Downarrow 1 \otimes \beta"] & \mathbf{1} \end{tikzcd}$$

Similarly, we can define a 1-morphism $zz_-\colon \mathbf{1} \to B \otimes A$ by the picture below



As with zz_+, there are two different 2-morphisms $zz_- \to \mathrm{coev}$, and we require that they are equal:

$$\begin{tikzcd} \mathbf{1} \arrow[rr, bend left, "zz_-"] \arrow[rr, bend right, "\mathrm{coev}"'] \arrow[rr, phantom, "\Downarrow 1\otimes\alpha"] & & B\otimes A \end{tikzcd} = \begin{tikzcd} \mathbf{1} \arrow[rr, bend left, "zz_-"] \arrow[rr, bend right, "\mathrm{coev}"'] \arrow[rr, phantom, "\Downarrow \beta\otimes 1"] & & B\otimes A \end{tikzcd} \tag{8.2.4}$$

We will call relations (8.2.3), (8.2.4) *swallowtail relations*. The name comes from their geometric interpretation in terms of 2-cobordisms, which we will discuss in the next section.

Thus, we are led to the following definition (we use the terminology of [**Pst2022**]).

Definition 8.2.1. A *coherent dual pair* in a weak monoidal 2-category $\mathcal{C}$ is the following collection of data:

- Objects $A, B \in \mathcal{C}$.
- 1-morphisms $\mathrm{ev}\colon A \otimes B \to \mathbf{1}$, $\mathrm{coev}\colon \mathbf{1} \to B \otimes A$.
- Invertible 2-morphisms $\alpha\colon z_A \to 1_A$, $\beta\colon z_B \to 1_B$, where z_A, z_B are defined by (8.2.1).

such that α, β satisfy the swallowtail relations (8.2.3), (8.2.4).

In this case, we say that B is a right dual of A, and A is a left dual of B.

Lemma 8.2.1. *Let $(A, B, \mathrm{ev}, \mathrm{coev}, \alpha, \beta)$ be as in the definition of coherent dual pair, but without the swallowtail relations. Then one can replace β by another 2-isomorphism $\tilde{\beta}$ so that $(A, B, \mathrm{ev}, \mathrm{coev}, \alpha, \tilde{\beta})$ is a coherent dual pair.*

In other words, if you have any isomorphisms $z_A \to 1_A$, $z_B \to 1_B$, then you can modify one of them to satisfy the swallowtail relations.

The proof of this lemma is similar to Lemma 8.1.4, where we also argued that given a pair of isomorphisms, we can modify one of them to satisfy an additional consistency requirement (in that case, the adjunction property). We refer the reader to [**Pst2022**, Theorem 2.7] for details of the proof.

Recall that for a weak 2-category $\mathcal{C}$, we defined a (usual) category $h\mathcal{C}$ by making 1-morphisms that were isomorphic in $\mathcal{C}$ equal in $h\mathcal{C}$, see Definition 7.2.2. Then Lemma 8.2.1 implies that any dual pair $(A, B, \mathrm{ev}, \mathrm{coev})$ in $h\mathcal{C}$ can be "upgraded" to a coherent dual pair in $\mathcal{C}$. In particular, this implies the following.

Corollary 8.2.2. *An object $A \in \mathcal{C}$ has a right (respectively, left) dual in $\mathcal{C}$ if and only if it has the right (respectively, left) dual in the homotopy category $h\mathcal{C}$.*

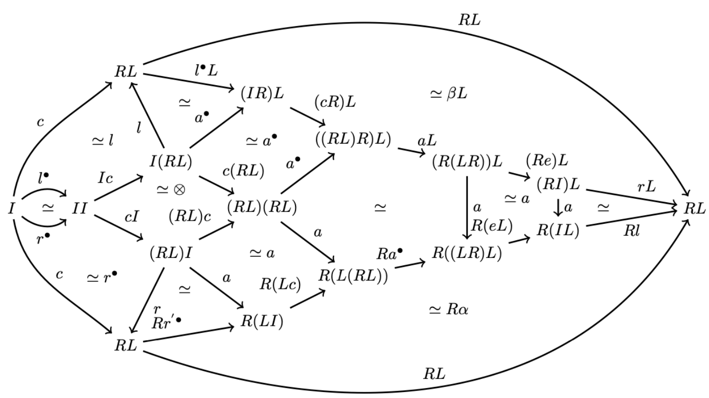
We can also ask whether, for a given A, all coherent dual pairs including it are equivalent. To do that, we define the notion of an isomorphism of dual pairs, which consists of a pair of isomorphisms of objects and 2-isomorphisms between evaluation and coevaluation; details can be found in [**Pst2022**, Section 2.2].

Theorem 8.2.3. *Let A be an object in $\mathcal{C}$, and let $(A, B, \dots)$, $(A, B', \dots)$ be two coherent dual pairs including A. Then there exists an isomorphism between them which is identity on A.*

(This is a simplified version; a more accurate statement, which also answers the question of whether such an isomorphism is unique, is given in [**Pst2022**, Theorem 2.14] and says that the 2-groupoid of coherent dual pairs is equivalent to the 2-groupoid of objects in $\mathcal{C}$ which have a right dual.)

In summary, we see that any dual pair in $h\mathcal{C}$ can be "upgraded" to a coherent dual pair in $\mathcal{C}$, and any two such upgrades are isomorphic. Thus, producing such a coherent dual pair is essentially equivalent to producing an object $A \in \mathcal{C}$ which has a right dual in $h\mathcal{C}$.

Remark 8.2.1. In all our discussions in this section, we ignored associativity and unit isomorphisms in the monoidal 2-category $\mathcal{C}$. Technically speaking, this is not correct, and we must include them. Unfortunately, if we do that, all diagrams become hopelessly complicated — for example, here is the diagram of one of the swallowtail relations in all glorious detail, taken from [**Pst2022**]:



Finally, we give the following definitions.

Definition 8.2.2. A symmetric monoidal weak 2-category $\mathcal{C}$ is called a *category with duals* if

(1) Every object $A \in \mathcal{C}$ is rigid, i.e., has a dual in $\mathcal{C}$.
(2) Every 1-morphism has both left and right adjoints.

Definition 8.2.3. An object A in a symmetric monoidal weak 2-category $\mathcal{C}$ is called *fully dualizable* if it lies in some subcategory $\mathcal{D} \subset \mathcal{C}$ which is a category with duals.

It is not difficult to show (see [**Lur2009a**, Proposition 4.2.3]) that an object A is fully dualizable if and only if A has a dual A^* and the corresponding evaluation and coevaluation 1-morphisms have left and right adjoints. Moreover, the existence of adjoints for ev implies the existence of adjoints for coev and vice versa.

8.3. Serre functor

Let $\mathcal{C}$ be a symmetric monoidal 2-category, and let A be a fully dualizable object in $\mathcal{C}$. Denote by $B = A^*$ the dual of A and by $\mathrm{ev}\colon A \otimes B \to \mathbf{1}$, $\mathrm{coev}\colon \mathbf{1} \to B \otimes A$ the corresponding evaluation and coevaluation morphisms.

By the definition of a fully dualizable object, the morphism ev must have right and left adjoints, which we will denote by ev^R, ev^L respectively. Both of them are morphisms $\mathbf{1} \to A \otimes B$; thus, we have adjunctions

$$A \otimes B \underset{\mathrm{ev}^R}{\overset{\mathrm{ev}}{\rightleftarrows}} \mathbf{1},$$

$$\mathbf{1} \underset{\mathrm{ev}}{\overset{\mathrm{ev}^L}{\rightleftarrows}} A \otimes B.$$

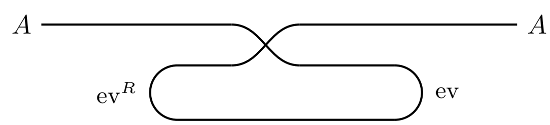
It is natural to ask what is the relation between these adjoints and the coevaluation morphism $\mathrm{coev}\colon \mathbf{1} \to B \otimes A$. Note that here A and B go in the opposite order, but since $\mathcal{C}$ is a symmetric monoidal category, we can use the commutativity isomorphism to exchange them, defining $\widetilde{\mathrm{coev}} = c_{B,A}\,\mathrm{coev}\colon \mathbf{1} \to A \otimes B$. It turns out that while in general, $\widetilde{\mathrm{coev}}$ is different from ev^L, ev^R, they can be related using the so-called *Serre automorphism*.

Definition 8.3.1. Let A be a fully dualizable object in $\mathcal{C}$. The Serre morphism S_A is the 1-morphism $A \to A$ defined by

$$S_A = \Big(A \xrightarrow{1_A \otimes \mathrm{ev}^R} A \otimes A \otimes B \xrightarrow{c_{A,A} \otimes 1_B} A \otimes A \otimes B \xrightarrow{1_A \otimes \mathrm{ev}} A\Big). \tag{8.3.1}$$

Graphically, this composition is represented by the picture (8.3.2) below:

(8.3.2)



where the input/output of S_A are the two horizontal strands labeled A, and the loop formed by ev^R and ev represents the auxiliary $A \otimes B$ pair created and then annihilated; the crossing in the middle is the symmetry isomorphism $c_{A,A}$.

Remark 8.3.1. The name "Serre automorphism" is motivated by the relation of this to Serre duality in the category of coherent sheaves on projective varieties; see [**Lur2009a**, Remark 4.2.4].

Theorem 8.3.1. *Let A be a fully dualizable object in $\mathcal{C}$ and let $S_A\colon A \to A$ be the Serre morphism defined by Definition 8.3.1. Then S_A is invertible, and we have an isomorphism of 1-morphisms*

$$\mathrm{ev}^L \simeq (S \otimes 1_B)\widetilde{\mathrm{coev}}$$
$$\mathrm{ev}^R \simeq (S^{-1} \otimes 1_B)\widetilde{\mathrm{coev}}.$$

We refer the reader to [**Pst2022**, Section 3.1] for the proof.

In particular, this theorem implies that if the Serre automorphism is trivial (i.e., one has a 2-isomorphism $S_A \simeq 1_A$), then $\mathrm{ev}, \widetilde{\mathrm{coev}}$ are two-sided adjoints of each other. More precisely, a choice of an isomorphism $S_A \simeq 1_A$ defines the unit and counit of adjunctions $\mathrm{ev} \dashv \widetilde{\mathrm{coev}}$, $\widetilde{\mathrm{coev}} \dashv \mathrm{ev}$.

8.4. Duality in Lin

As the simplest example of duality in a 2-category, let us consider the 2-category **Lin** of locally finite **k**-linear categories, introduced in Section 7.4. Recall that objects of that category are **k**-linear abelian categories such that all Hom spaces are finite-dimensional and every object has finite length. This 2-category has a natural symmetric monoidal structure, given by Deligne's product $\boxtimes$.

Let us now analyze which objects of **Lin** are rigid, i.e., for which categories $\mathcal{C}$ there exist a category $\mathcal{D}$ and functors

$$\eta\colon \mathbf{Vec_f} \to \mathcal{D} \boxtimes \mathcal{C}$$
$$\epsilon\colon \mathcal{C} \boxtimes \mathcal{D} \to \mathbf{Vec_f}$$

satisfying the rigidity axioms. Clearly, defining the functor η is equivalent to choosing an object

$$e = \eta(\mathbf{k}) = \bigoplus_{i=1}^{k} Y_i \boxtimes X_i \in \mathcal{D} \boxtimes \mathcal{C}$$

together with functorial isomorphisms

$$\begin{aligned} X &\simeq \bigoplus \varepsilon(X, Y_i) \otimes X_i \\ Y &\simeq \bigoplus \varepsilon(X_i, Y) \otimes Y_i \end{aligned} \tag{8.4.1}$$

for every $X \in \mathcal{C}$, $Y \in \mathcal{D}$.

Theorem 8.4.1. *Let $\mathbf{k}$ be an algebraically closed field of characteristic zero. Then a category $\mathcal{C} \in \mathbf{Lin}$ is rigid as an object of $\mathbf{Lin}$ if and only if it is semisimple and has finitely many isomorphism classes of simple objects. In this case, the dual category $\mathcal{D}$ is equivalent to the category of $\mathbf{k}$-linear functors $\mathcal{C} \to \mathbf{Vec_f}$; under this equivalence, the counit $\varepsilon\colon \mathcal{C} \boxtimes \mathcal{D} \to \mathbf{Vec_f}$ is given by*

$$\varepsilon(X, F) = F(X) \in \mathbf{Vec_f}, \qquad X \in \mathcal{C}, F \in \mathrm{Fun}(\mathcal{C}, \mathbf{Vec_f})$$

and the object $e = \bigoplus Y_i \boxtimes X_i$ is given by

$$e = \bigoplus X^i \boxtimes X_i$$

where the sum is over isomorphism classes of simple objects $X_i \in \mathcal{C}$, and for such a simple object, we define

$$X^i = \mathrm{Hom}(X_i, -) \in \mathrm{Fun}(\mathcal{C}, \mathbf{Vec_f}).$$

Proof. Our proof follows [**BDSPV2015**, Section A.6], where the proof is attributed to André Henriques. (Note that they work in greater generality: they do not require the category to be abelian, instead imposing a weaker condition of Cauchy completeness.)

First, finiteness easily follows from the observation that by (8.4.1), any $X \in \mathcal{C}$ is a subobject in a finite direct sum of copies of $X_1, \dots, X_k$. Thus, the only simple objects that can appear in the Jordan–Hölder series for X are those simple objects that appear in the Jordan–Hölder series for one of the X_i; which implies that there are only finitely many of them.

To prove semisimplicity, let $0 \to A_1 \to A_2 \to A_3 \to 0$ be a short exact sequence in $\mathcal{C}$. Using (8.4.1), we can rewrite it as a short exact sequence

$$0 \to \bigoplus \varepsilon(Y_i, A_1) \otimes X_i \to \bigoplus \varepsilon(Y_i, A_2) \otimes X_i \to \bigoplus \varepsilon(Y_i, A_3) \otimes X_i \to 0.$$

Since a direct sum of complexes is exact if and only if each summand is exact, this implies that each complex

$$K_i = \Big(0 \to \varepsilon(Y_i, A_1) \otimes X_i \to \varepsilon(Y_i, A_2) \otimes X_i \to \varepsilon(Y_i, A_3) \otimes X_i \to 0\Big)$$

is exact. This in turn is equivalent to the requirement that the complex of vector spaces

$$0 \to \varepsilon(Y_i, A_1) \to \varepsilon(Y_i, A_2) \to \varepsilon(Y_i, A_3) \to 0$$

is exact. But each short exact sequence of vector spaces splits; thus, each short exact sequence K_i splits, which implies that the original short exact sequence $0 \to A_1 \to A_2 \to A_3 \to 0$ also splits. Thus, $\mathcal{C}$ is semisimple.

We leave the proof of the remaining parts of the theorem as an exercise to the reader. □

After establishing duality for objects of **Lin**, we can turn to establishing duality for 1-morphisms. For simplicity, we give the answer for a slightly smaller 2-category of *finite* **k**-linear categories.

Definition 8.4.1. A locally finite **k**-linear category $\mathcal{A}$ is called *finite* if it satisfies the following additional conditions

- The set of isomorphism classes of simple objects in $\mathcal{A}$ is finite.
- There are enough projective objects.

A more detailed discussion of finite categories can be found, e.g., in [**EGNO2015**, Section 1.8].

Theorem 8.4.2. *Assume that* **k** *is an algebraically closed field of characteristic zero. Let* $\mathcal{C}$, $\mathcal{D} \in \mathbf{Lin}$ *be finite* **k***-linear categories. Then a functor* $F\colon \mathcal{C} \to \mathcal{D}$ *has a left adjoint if and only if it is left exact.*

A proof of this theorem can be found in [**DSPS2019**, Proposition 1.7] (strangely enough, even though the result was part of folklore for years, it was not written down prior to this paper).

Corollary 8.4.3. *Let* $\mathcal{C}, \mathcal{D}$ *be finite semisimple categories. Then any functor* $F\colon \mathcal{C} \to \mathcal{D}$ *admits both left and right adjoints.*

Exercise 8.4.1. Under the assumptions of Corollary 8.4.3, let X_i, $i = 1, \dots, n$ be simple objects of $\mathcal{C}$ and $Y_j, j = 1, \dots, m$ be simple objects of $\mathcal{D}$.

(1) Show that the category of functors $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is naturally equivalent to the category whose objects are $m \times n$ matrices of finite-dimensional vector spaces. (Hint: it suffices to consider functors $\mathbf{Vec_f} \to \mathbf{Vec_f}$).
(2) Show that the (two-sided) adjoint of such a matrix of vector spaces is the matrix obtained by transposing the original matrix and replacing each vector space by the dual.

Combining Theorem 8.4.1 with Corollary 8.4.3, we get the following result.

Theorem 8.4.4. *Assume that* **k** *is an algebraically closed field of characteristic zero. Then fully dualizable objects in* **Lin** *are finite semisimple* **k***-linear categories.*

CHAPTER 9

Separable Frobenius algebras

In this chapter, we study duality in the 2-category **Alg** of algebras and bimodules, which was introduced in Example 7.1.2. It turns out that the maximal subcategory with duals in **Alg** is the category of separable algebras (which in characteristic zero is the same as finite-dimensional semisimple algebras); if we also want to trivialize the Serre functor (in other words, if we want to have adjunction between evaluation and coevaluation 1-morphisms), we arrive at the notion of a separable symmetric Frobenius algebra. These results will be heavily used later, when we classify extended 2-dimensional TQFTs.

To simplify the exposition, throughout this chapter we assume that $\mathbf{k}$ is a field of characteristic zero.

9.1. Duality in Alg

Our goal is analyzing duality in the 2-category **Alg**, looking at duality both for objects and for 1-morphisms. Recall that objects of this category are associative algebras, and 1-morphisms $A \to B$ are B-A-bimodules.

Let us begin by looking at duality for objects. It turns out that each $A \in \mathbf{Alg}$ has a dual. Namely, for an algebra A, denote by A^{op} the *opposite algebra*, which coincides with A as a vector space but has the opposite multiplication. Then any A-bimodule can also be considered as a left module over the algebra $A \otimes A^{\mathrm{op}}$, or as a right module over $A \otimes A^{\mathrm{op}}$. In particular, A itself is a left module over $A \otimes A^{\mathrm{op}}$, and also a right module over $A \otimes A^{\mathrm{op}}$.

THEOREM 9.1.1. *Let $A \in \mathbf{Alg}$. Define the evaluation and coevaluation morphisms $A^{\mathrm{op}} \otimes A \to \mathbf{k}$, $\mathbf{k} \to A \otimes A^{\mathrm{op}}$ to be A, considered either as a left or a right $A \otimes A^{\mathrm{op}}$ module:*

$$\begin{aligned} \mathrm{ev} &= A_{A\otimes A^{\mathrm{op}}} \colon A \otimes A^{\mathrm{op}} \to \mathbf{k} \\ \mathrm{coev} &= {}_{A\otimes A^{\mathrm{op}}}A \colon \mathbf{k} \to A \otimes A^{\mathrm{op}}. \end{aligned} \tag{9.1.1}$$

Then $(A, A^{\mathrm{op}}, \mathrm{ev}, \mathrm{coev}, \alpha, \beta)$ is a coherent dual pair in $\mathbf{Alg}$, with isomorphisms $\alpha, \beta \colon A \otimes_A A \xrightarrow{\sim} A$ given by multiplication.

In other words, any algebra A has a dual, namely A^{op}. Note that it has nothing to do with the dual vector space, and it works for finite-dimensional as well as infinite-dimensional algebras.

Let us now look at duality for 1-morphisms, i.e., bimodules. Recall that a module P over an algebra A is called *projective* if it is a direct summand of a free module.

THEOREM 9.1.2. *Let ${}_BM_A$ be a B-A-bimodule, considered as a 1-morphism $A \to B$ in $\mathbf{Alg}$. Then the following are equivalent:*

(1) *M has a right adjoint.*

(2) *There exists an A-B-bimodule N, an element $e = \sum_{i=1}^{k} n_i \otimes m_i \in N \otimes M$, and a morphism of B-bimodules $(\ ,\ ) \colon M \otimes_A N \to B$ such that e is central: for any $x \in A$,*

$$\sum (x n_i) \otimes m_i = \sum n_i \otimes (m_i x)$$

and for any $m \in M$, $n \in N$ we have

$$\begin{aligned} m &= \sum (m, n_i) m_i, \\ n &= \sum n_i (m_i, n). \end{aligned} \tag{9.1.2}$$

(3) *M is a finitely generated projective B-module.*

In this case, its right adjoint is isomorphic to the module $N = M^\vee = \mathrm{Hom}_B(M, B)$ (note that it has a canonical A-B-bimodule structure).

Proof. Equivalence (1) $\iff$ (2) is obvious from the definition; the form $(\ ,\ )$ plays the role of evaluation morphism, and $e = \eta(1)$, where $\eta \colon A \to N \otimes M$ is the coevaluation.

To prove (2) $\implies$ (3), assume that we are given such an $e = \sum_{i=1}^{k} n_i \otimes m_i$. Consider the free B-module F of rank k: $F = \bigoplus B x_i$. Define morphisms of B-modules $s \colon M \to F$, $p \colon F \to M$ by

$$\begin{aligned} s(m) &= \sum (m, n_i) x_i \\ p(x_i) &= m_i. \end{aligned}$$

Then it follows from (2) that $p \circ s = \mathrm{id}_M$, which implies that M is a direct summand in F, so M is projective of finite rank. Thus, (2) $\implies$ (3).

Proof of (3) $\implies$ (2) is similar and is left to the reader.

Finally, to show that in this case, the right adjoint N is isomorphic to $M^\vee$, we construct morphisms $N \to M^\vee$, $M^\vee \to N$ as follows:

$$\begin{aligned} N \to M^\vee &\colon n \mapsto (-, n) \\ M^\vee \to N &\colon f \mapsto \sum n_i f(m_i) \end{aligned}$$

We leave it to the reader to verify that the so-defined maps are indeed morphisms of right A-modules, and that they are mutually inverse. □

Similarly, M has a left adjoint iff it is finitely generated and projective as a right A-module.

9.2. Separable algebras

Theorem 9.1.2 shows that in order for the coevaluation morphism $\mathbf{k} \to A^{\mathrm{op}} \otimes A$ (which is given by A considered as a left $A^{\mathrm{op}} \otimes A$ module) to have a right adjoint in **Alg**, it is necessary that A is a projective $A^{\mathrm{op}} \otimes A$ module. This naturally leads us to the notion of a *separable* algebra. We give a brief review of this notion here, referring the reader to [**Pie1982**, Chapter 10] for details.

Lemma 9.2.1. *Let A be an associative algebra over a field $\mathbf{k}$. Then the following are equivalent:*

(1) *A is projective as a left module over $A^e = A \otimes A^{\mathrm{op}}$.*

(2) *The multiplication map* $m\colon A \otimes A \to A$, *considered as a morphism of* A-*bimodules* ${}_AA \otimes A_A \to {}_AA_A$, *has a one-sided inverse: there exists a morphism of* A-*bimodules* $\Delta\colon A \to A \otimes A$ *such that* $m \circ \Delta = \mathrm{id}_A$.

(3) *There exists an element* $E = \sum x_i \otimes y_i \in A \otimes A$ *which is central:* $\sum ax_i \otimes y_i = \sum x_i \otimes y_i a$ *and satisfies* $m(E) = \sum x_i y_i = 1$.

Proof. (1) $\implies$ (2) is a standard fact of basic algebra: if P is a projective module, then any surjective morphism $X \to P \to 0$ has a one-sided inverse.

Similarly, (2) $\implies$ (1) is immediate from the definition: if such a morphism Δ exists, then A is a direct summand in the free A^e-module $A \otimes A \simeq A^e$.

Finally, (3) is just a rewriting of (2). □

Definition 9.2.1. An algebra satisfying one of the equivalent conditions of Lemma 9.2.1 is called *separable*.

An element $E \in A \otimes A$ as in part (3) of the lemma above is called a *separability idempotent*. This terminology is justified by the following exercise.

Exercise 9.2.1. Let A be a separable algebra with separability idempotent E. Then E, considered as an element in the algebra $A \otimes A^{\mathrm{op}}$, satisfies $E^2 = E$.

Note that the separability idempotent is not unique, nor does it have to be symmetric, as is illustrated by the following example.

Example 9.2.1. Let $A = \mathrm{Mat}_n(\mathbf{k})$ be the matrix algebra. Choose an index k and define $e_k = \sum_i E_{ik} \otimes E_{ki}$, where E_{ij} are matrix units. Then each of e_k is a separability idempotent, so A is separable.

We can also take as the separability idempotent the element

$$E = \tfrac{1}{n} \sum E_{ij} \otimes E_{ji}$$

which has the added benefit of being symmetric.

Remark 9.2.1. It can be shown that a *symmetric* separability idempotent, if it exists, is unique, see [**Agu2000**, Theorem 3.1].

One of the main results of this theory is the following.

Theorem 9.2.2. *If* $\mathbf{k}$ *has characteristic zero, an algebra* A *over* $\mathbf{k}$ *is separable if and only if it is finite-dimensional and semisimple. In this case, for any field extension* $\mathbf{k} \subset F$, *the algebra* $A_F = A \otimes_{\mathbf{k}} F$ *is also semisimple over* F.

We skip the proof of this result, referring the reader to [**Pie1982**, Section 10.7].

Remark 9.2.2. In fact, Theorem 9.2.2 can be generalized: it works for any perfect field, i.e., a field such that any finite extension of it is separable in the sense used in Galois theory. This includes not only fields of characteristic zero, but also finite fields.

Since any module over a semisimple algebra is projective, the results of the previous section immediately imply the following theorem.

Theorem 9.2.3. *Let* $\mathbf{ssAlg} \subset \mathbf{Alg}$ *be the category whose objects are separable algebras, and 1-morphisms are finite-dimensional bimodules. Then* $\mathbf{ssAlg}$ *is a category with duals as defined in Definition 8.2.2.*

Exercise 9.2.2. Show that for a separable algebra A, the Serre automorphism as defined in Definition 8.3.1 is given by A^* (dual vector space), considered as an A-bimodule.

9.3. Separable symmetric Frobenius algebras

The results of the previous section show that fully dualizable objects in **Alg** are separable algebras. However, for future use, let us also consider this question: what are fully dualizable algebras for which the evaluation ev_A and coevaluation coev_A are adjoints of each other?

One way to answer that is to use the notion of Serre functor, introduced in Section 8.3. By the results of that section, if ev has left and right duals ev^L, ev^R, then they coincide with the coevaluation coev twisted by the Serre automorphism or its inverse. Thus, to construct adjunctions $\mathrm{ev} \dashv \mathrm{coev}$, $\mathrm{coev} \dashv \mathrm{ev}$ we need to construct an isomorphism $S_A \simeq 1_A$. Since by Exercise 9.2.2 the Serre automorphism for an algebra A is given by A^* considered as an A-bimodule, we see that trivialization of the Serre automorphism is the same as a choice of an isomorphism of A-bimodules $A \simeq A^*$.

It turns out that the answer is related to Frobenius algebras. Recall that a symmetric Frobenius algebra is a finite-dimensional algebra A together with a linear map $\varepsilon\colon A \to \mathbf{k}$ such that the pairing

$$A \otimes A \to \mathbf{k}\colon a \otimes b \mapsto (a, b) = \varepsilon(ab)$$

is a symmetric non-degenerate bilinear form, see Definition 5.1.1, Definition 5.1.2.

Exercise 9.3.1. Show that for a finite-dimensional algebra A, constructing an isomorphism of A-bimodules $A \simeq A^*$ is equivalent to defining on A a structure of a symmetric Frobenius algebra (compare with Exercise 5.2.1).

However, it is useful to also give a more direct construction, relating adjunctions $\mathrm{ev} \dashv \mathrm{coev}$, $\mathrm{coev} \dashv \mathrm{ev}$ defined by (9.1.1) with the Frobenius algebra structure.

Theorem 9.3.1. *Let A be an algebra considered as an object in* **Alg**. *Then defining adjunction data for $\mathrm{ev}_A \dashv \mathrm{coev}_A$ is equivalent to defining on A a structure of a symmetric Frobenius algebra.*

In particular, this theorem shows that such an adjunction is only possible if A is finite-dimensional.

Proof. By definition, adjunction data consists of the unit and the counit of adjunction, i.e., 2-morphisms $\varepsilon\colon \mathrm{ev}_A \circ \mathrm{coev}_A \to 1_{\mathbf{k}}$, $\eta\colon 1_{A\otimes A^{\mathrm{op}}} \to \mathrm{coev}_A \circ \mathrm{ev}_A$, where $\mathrm{ev}_A\colon A \otimes A^{\mathrm{op}} \to \mathbf{k}$, $\mathrm{coev}_A\colon \mathbf{k} \to A \otimes A^{\mathrm{op}}$ are evaluation and coevaluation 1-morphisms.

Recalling that ev_A is given by A considered as a right A^e module, and coev_A is given by A considered as a left A^e module, we see that composition of 1-morphisms $\mathrm{ev}_A \circ \mathrm{coev}_A$, considered as a $\mathbf{k}$-bimodule, in other words a vector space, is given by

$$\mathrm{ev}_A \circ \mathrm{coev}_A \simeq A \otimes_{A^e} A \simeq A/[A, A]$$

where $[A, A]$ is the subspace in A generated by $xy - yx$, $x, y \in A$. Note that $A/[A, A]$ is just a vector space; it has no natural algebra structure (unlike the situation with Lie algebras, where $L/[L, L]$ is a Lie algebra), since $[A, A]$ is not an ideal in A.

Thus, we see that the counit of adjunction $\varepsilon\colon \mathrm{ev}_A \circ \mathrm{coev}_A \to 1_{\mathbf{k}}$ is a linear map

$$\varepsilon\colon A/[A, A] \to \mathbf{k}$$

or equivalently, a map $\varepsilon\colon A \to \mathbf{k}$ such that $\varepsilon(xy) = \varepsilon(yx)$.

Similarly, the unit of adjunction should be a morphism of bimodules ${}_{A^e}A^e_{A^e} \to {}_{A^e}A \otimes A_{A^e}$. Unpacking this definition, we see that η is a map of modules over 4 copies of A:

$$\eta\colon {}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} \to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2} \tag{9.3.1}$$

where indices are used to indicate copies of A. It is immediate that such a morphism is uniquely determined by $e = \eta(1 \otimes 1)$, which must be *bicentral*: if $e = \sum x_i \otimes x^i$, then

$$\begin{aligned} \sum x x_i \otimes x^i &= \sum x_i \otimes x^i x, \\ \sum x_i y \otimes x^i &= \sum x_i \otimes y x^i. \end{aligned} \tag{9.3.2}$$

We leave it to the reader to check that the adjunction conditions on η, ε are equivalent to the conditions that for any $x \in A$, we have

$$x = \sum (x, x_i) x^i = \sum (x, x^i) x_i,$$

where $(a, b) = \varepsilon(ab)$ (compare with the proof of Theorem 5.1.1). This means that $(\ ,\ )$ is a non-degenerate pairing and x_i, x^i are dual bases with respect to this pairing, so A is a symmetric Frobenius algebra. Conversely, if A is a symmetric Frobenius algebra, we can define $e = \sum x_i \otimes x^i$ where x_i, x^i are dual bases with respect to the Frobenius pairing. By Theorem 5.1.3, the so-defined e is central; since it is symmetric, it is also bicentral and thus defines a morphism η as in (9.3.1).

□

REMARK 9.3.1. If you have difficulty keeping track of the many copies of A and in which order they appear, you are not alone. Fortunately, it will become clearer once we relate it to geometry: see Section 10.5.

So far we have discussed adjunction ev $\dashv$ coev. What about the other adjunction, coev $\dashv$ ev?

First, by Theorem 9.1.2, in order for coev $= {}_{A^e}A$ to have a right adjoint it is necessary that A is projective over A^e, that is, that A is separable. It turns out that this is also sufficient — if we assume we already have adjunction ev $\dashv$ coev.

THEOREM 9.3.2. *Let A be an algebra together with adjunction data $\mathrm{ev}_A \dashv \mathrm{coev}_A$, $\mathrm{coev}_A \dashv \mathrm{ev}_A$. Then A is separable and has a canonical structure of a symmetric Frobenius algebra, with the counit $\varepsilon\colon A/[A, A] \to \mathbf{k}$ given by the counit of adjunction* ev $\dashv$ coev.

Conversely, let A be a separable symmetric Frobenius algebra. Then we can define adjunction data $\mathrm{ev}_A \dashv \mathrm{coev}_A$, $\mathrm{coev}_A \dashv \mathrm{ev}_A$.

To prove this theorem, we will need two preliminary results.

THEOREM 9.3.3. *For an algebra A, adjunction data for adjunction $\mathrm{coev}_A \dashv \mathrm{ev}_A$ is equivalent to an element $C \in A/[A, A]$ and a bicentral element*

$$\tilde{e} = \sum \tilde{x}_i \otimes \tilde{x}^i \in A \otimes A$$

such that $\sum \tilde{x}_i C \tilde{x}^i = 1$.

Note that bicentrality of $\tilde{e}$ implies that $\sum \tilde{x}_i C \tilde{x}^i$ only depends on the class of C in $A/[A, A]$.

The proof of this theorem is similar to the proof of Theorem 9.3.1; the element C corresponds to the unit $\mathbf{k} \to \mathrm{ev}_A \circ \mathrm{coev}_A$, and $\tilde{e}$ corresponds to the counit $\mathrm{coev}_A \circ \mathrm{ev}_A \to 1_{A^e}$. We leave the details to the reader.

Before proving the next theorem, we need some notation. Let A be a symmetric Frobenius algebra and let x_i, x^i be dual bases with respect to the Frobenius pairing. Consider the map $f\colon A \to A$ given by $x \mapsto \sum x_i x x^i$. Then bicentrality of e implies that for any x, $f(x) \in z(A)$, where $z(A)$ is the center of A. Similarly, it also shows that $f(xy) = f(yx)$, so we can consider f as a map

$$\begin{aligned} A/[A,A] &\to z(A) \\ [x] &\mapsto \sum x_i x x^i. \end{aligned} \tag{9.3.3}$$

THEOREM 9.3.4. *Let A be a symmetric Frobenius algebra over a field $\mathbf{k}$ of characteristic zero. Let x_i, x^i be dual bases in A with respect to the Frobenius pairing. Then the following are equivalent:*

(1) *A is separable.*
(2) *A is semisimple.*
(3) *There exists $[C] \in A/[A,A]$ such that $\sum x_i C x^i = 1$.*
(4) *The Euler element (see (5.2.1))*

$$w = \sum x_i x^i \tag{9.3.4}$$

is invertible.

PROOF. Equivalence of (1) and (2) is Theorem 9.2.2 (note that a Frobenius algebra is automatically finite-dimensional).

(3) $\Longrightarrow$ (1) is immediate: we can take $\sum x_i C \otimes x^i$ as separability idempotent.

(4) $\Longrightarrow$ (3) is also clear: if $w = \sum x_i x^i$ is invertible, then its inverse w^{-1} must be central, and $\sum x_i w^{-1} x^i = 1$.

It remains to prove (1) $\Longrightarrow$ (4).

Assume that A is separable; then A is semisimple and moreover, if $\bar{\mathbf{k}}$ is the algebraic closure of $\mathbf{k}$, then $A_{\bar{\mathbf{k}}} = A \otimes_{\mathbf{k}} \bar{\mathbf{k}}$ is semisimple. By Lemma 5.2.1, $A_{\bar{\mathbf{k}}} \simeq \bigoplus \mathrm{Mat}_{n_i}(\bar{\mathbf{k}})$, and $\varepsilon(a) = \sum \lambda_i \operatorname{tr}(a_i)$. An easy explicit computation then shows that the element e is given by

$$e = \sum \lambda_i^{-1} E^i_{kl} \otimes E^i_{lk}$$

where E^i_{kl} are matrix units in $\mathrm{Mat}_{n_i}(\bar{\mathbf{k}})$, and thus, $w = \sum \lambda_i^{-1} n_i 1_i$, where 1_i is the identity matrix in $\mathrm{Mat}_{n_i}(\bar{\mathbf{k}})$. Therefore, w is invertible in $A_{\bar{\mathbf{k}}}$ and thus in A. □

REMARK 9.3.2. An alternative proof which does not rely on the classification of semisimple algebras is given in [**LP2007**, Theorem 2.14].

Now we can return to Theorem 9.3.2.

PROOF OF THEOREM 9.3.2. One direction is easy. Assume that we have adjunctions $\mathrm{ev}_A \dashv \mathrm{coev}_A$, $\mathrm{coev}_A \dashv \mathrm{ev}_A$. By Theorem 9.3.1, the first of these adjunctions defines on A a structure of a symmetric Frobenius algebra. By Theorem 9.3.3, the second adjunction gives an element $\sum \tilde{x}_i \otimes C\tilde{x}^i$ which is a separability idempotent (see Lemma 9.2.1); thus, A is separable.

Conversely, let A be a separable symmetric Frobenius algebra. By Theorem 9.3.1, this gives rise to adjunction $\mathrm{ev}_A \dashv \mathrm{coev}_A$, so we only need to construct the second adjunction $\mathrm{coev}_A \dashv \mathrm{ev}_A$. To do it, let $C = [w^{-1}] \in A/[A,A]$, where w

is the Euler element (see Theorem 9.3.4) and $\tilde{e} = e = \sum x_i \otimes x^i$; by Theorem 9.3.3 they define an adjunction $\mathrm{coev}_A \dashv \mathrm{ev}_A$. □

REMARK 9.3.3. The careful reader can note that the construction of bi-adjunction $\mathrm{ev} \dashv \mathrm{coev}$, $\mathrm{coev} \dashv \mathrm{ev}$ from a symmetric Frobenius algebra has an extra feature: the elements e, $\tilde{e}$, which are parts of these adjunctions, actually coincide. It is natural to ask whether it is always so: does having two adjunctions as above automatically imply that $\tilde{e} = e$?

The answer is "no": in general, $\tilde{e} \neq e$; however, if both adjunctions exist, then we can modify data for one of them so that the equality $\tilde{e} = e$ is satisfied. See the discussion of "cusp flip relation" at the end of Section 10.4.

For future use, we also give here some further properties of separable symmetric Frobenius algebras.

EXERCISE 9.3.2. Let A be a finite-dimensional semisimple algebra. Define the counit $\varepsilon\colon A \to \mathbf{k}$ by

$$\varepsilon(a) = \mathrm{tr}(L_a)$$

where $L_a\colon A \to A$ is the operator of left multiplication by a. Prove that this defines on A a structure of a symmetric Frobenius algebra, and for this structure, the Euler element (9.3.4) is $w = 1$.

(Hint: it suffices to consider the case $A = \mathrm{Mat}_n(\mathbf{k})$. Alternatively, to prove that $(\ ,\ )$ is non-degenerate, you can prove that the kernel of this bilinear form is a 2-sided ideal in A which consists of nilpotent elements, and thus for a semisimple algebra must be trivial.)

COROLLARY 9.3.5. *Let A be a separable symmetric Frobenius algebra, and let $w = \sum x_i x^i$ be the Euler element defined by (9.3.4). Then w^{-1} is central, and the map*

$$A/[A,A] \to z(A)\colon [a] \mapsto \sum x_i a x^i \tag{9.3.5}$$

is an isomorphism; its inverse is given by

$$z(A) \to A/[A,A]\colon z \mapsto w^{-1}z. \tag{9.3.6}$$

In particular, $\sum x_i w^{-1} x^i = 1$ and the element

$$E = \sum w^{-1} x_i \otimes x^i = \sum x_i \otimes x^i w^{-1}$$

is a symmetric separability idempotent.

Another reformulation of this corollary is the statement that the map

$$\begin{aligned} p\colon A &\to A \\ a &\mapsto w^{-1} \sum x_i a x^i \end{aligned} \tag{9.3.7}$$

is a projector onto the center of A, and $\ker(p) = [A,A]$. Using the graphical representation of operations in a Frobenius algebra as in Chapter 5, this projector can be represented by Figure 9.3.1.

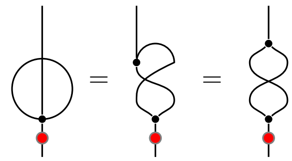


FIGURE 9.3.1. Projector onto $z(A)$, see (9.3.7). Red dot stands for operator of multiplication by w^{-1}.

9.4. Invariants and coinvariants

Corollary 9.3.5, which shows that for a separable symmetric Frobenius algebra we have a natural isomorphism $A/[A,A] \simeq z(A)$, is a special case of a more general result which we state here for future use.

Namely, let M be a bimodule over an associative algebra A (no further restrictions yet). Define the spaces of invariants and coinvariants by

$$\begin{aligned} M^A &= \mathrm{Hom}(A, M) = \{m \in M \mid am = ma \text{ for any } a \in A\} \\ M_A &= A \otimes_{A^e} M = M/\langle am - ma\rangle \end{aligned} \tag{9.4.1}$$

where Hom is computed in the category of A-bimodules, or equivalently, modules over $A^e = A \otimes A^{\mathrm{op}}$.

Standard results of algebra show that the functor of invariants is left exact, and the functor of coinvariants is right exact. The corresponding derived functors are called *Hochschild (co)homology* of A with coefficients in M (see, e.g., [**Pie1982**, Section 11.1]):

$$\begin{aligned} HH^n(A, M) &= \mathrm{Ext}^n_{A^e}(A, M), \\ HH_n(A, M) &= \mathrm{Tor}^{A^e}_n(A, M), \end{aligned}$$

so that $M^A = HH^0(A, M)$, $M_A = HH_0(A, M)$. For example, for $M = A$, we get

$$\begin{aligned} HH^0(A) &= A^{(A)} = z(A), \\ HH_0(A) &= A_{(A)} = A/[A, A]. \end{aligned}$$

For a general associative algebra, there is no relation between the functors of invariants and coinvariants. However, if A is a symmetric separable Frobenius algebra, things are simplified. Namely, consider the map $p\colon M \to M$ given by

$$m \mapsto \sum w^{-1} x_i m x^i = \sum x_i m x^i w^{-1}.$$

As in (9.3.3), it is easy to see that it descends to a map $M_A \to M^A$.

LEMMA 9.4.1. *If M is a bimodule over a separable symmetric Frobenius algebra A, then the map*

$$m \mapsto \sum x_i m x^i$$

is an isomorphism $M_A \xrightarrow{\sim} M^A$.

For $M = A$, this is exactly the statement of Corollary 9.3.5. We leave it to the reader to modify the proof of Corollary 9.3.5 to this more general case.

Note that since the functor of invariants is left exact and the functor of coinvariants is right exact, this shows that for a separable symmetric Frobenius algebra, both functors are exact so that higher Hochschild (co)homology vanishes.

As an application, given a right A-module M_A and a left A-module ${}_AN$, there are two versions of tensor product over A:

$$\begin{aligned} M \otimes_A N &= \operatorname{Coinv}(M \otimes N) = M \otimes N / \langle ma \otimes n - m \otimes an \rangle, \\ M \otimes^A N &= \operatorname{Inv}(M \otimes N) \\ &= \Big\{ x = \sum m_i \otimes n_i \in M \otimes N \mid \sum m_i a \otimes n_i = \sum m_i \otimes a n_i \Big\}. \end{aligned} \tag{9.4.2}$$

The first of these spaces (which is by far the more common) is a quotient of $M \otimes N$; the second is a subspace. As an immediate corollary of Lemma 9.4.1, we see that for a separable symmetric Frobenius algebra, we have a canonical isomorphism $M \otimes_A N \simeq M \otimes^A N$.

CHAPTER 10

Extended 2d TQFT

In this chapter we define the notion of extended 2d TQFT and classify such TQFTs. Our exposition is based on Schommer-Pries' thesis [**SP2009**].

Informally, an extended 2d TQFT is a theory in which we allow:

- 0-dimensional manifolds
- 1-dimensional cobordisms between 0d manifolds
- 2-dimensional cobordisms between them

and to each of those, assign corresponding algebraic data (to be specified later), subject to appropriate gluing laws.

As mentioned before, the natural language for this is the language of (weak) 2-categories: the 0-, 1-, and 2-dimensional manifolds and bordisms form a 2-category, which we will denote by $\mathbf{Bord}_2$; this category has a natural symmetric monoidal structure, given by disjoint union. Thus, we give the following preliminary definition.

Preliminary Definition 10.0.1. An extended 2d TQFT is a functor of symmetric monoidal 2-categories

$$Z\colon \mathbf{Bord}_2 \to \mathcal{C}$$

where $\mathcal{C}$ is a target symmetric monoidal 2-category.

Note that unlike the unextended case, where there is a standard choice of the target category $\mathcal{C}$ (namely, the category $\mathbf{Vec}$), in the extended case the choice of $\mathcal{C}$ is less clear. We will discuss it later.

To make this definition precise, we need to actually define the 2-category $\mathbf{Bord}_2$.

10.1. Defining $\mathbf{Bord}_2$

We begin with defining manifolds with corners. Note that there are two approaches to defining manifolds: one can either use coordinate charts and atlases, or sheaves of smooth functions. We will use the second approach.

For a topological space X, denote by $\mathcal{C}_X$ the sheaf of continuous functions on X.

Definition 10.1.1. An n-dimensional manifold with corners is a topological space M with a subsheaf $\mathcal{F} \subset \mathcal{C}_M$ such that for each point $x \in M$ there exists a homeomorphism of a neighborhood of x in M and a neighborhood of 0 in

$$\mathbb{R}^p_+ \times \mathbb{R}^{n-p}$$

which identifies the sheaf $\mathcal{F}$ with the sheaf of smooth functions on $\mathbb{R}^p_+ \times \mathbb{R}^{n-p}$ (we call a function on $\mathbb{R}^p_+ \times \mathbb{R}^{n-p}$ smooth if it is a restriction of a smooth function on $\mathbb{R}^n$).

It is easy to show that the number p is well-defined: for different p, the sets $\mathbb{R}_+^p \times \mathbb{R}^{n-p}$ are not diffeomorphic. We will call this number the index of x; the set

$$\partial^{(p)}(M) = \{x \mid \mathrm{index}(x) = p\}$$

is the codimension-p stratum of the boundary. In particular, we will call connected components of $\partial^{(1)}M$ *faces*.

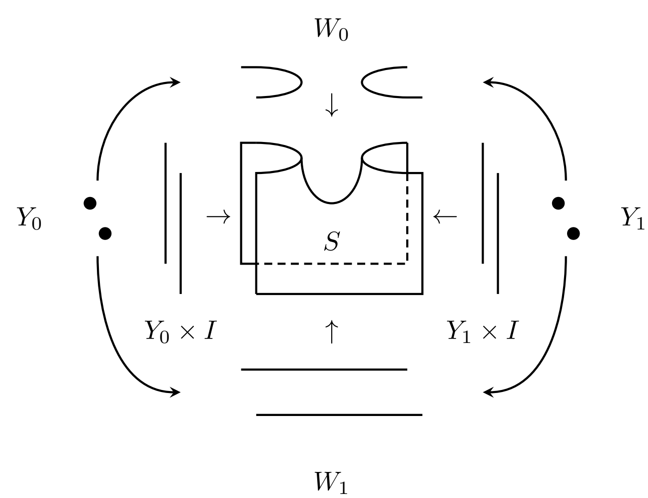
For technical reasons, we will impose an additional requirement that each point $x \in M$ of index p lies in the closure of exactly p **different** faces; thus, we do not allow a face to "intersect itself". Following [**SP2009**], we call such a manifold "manifold with faces".

Using this, we can give a definition of a 2-cobordism.

Definition 10.1.2. Let Y_0, Y_1 be closed $(n-2)$-dimensional manifolds, and let W_0, W_1 be two $(n-1)$-dimensional cobordisms between Y_0 and Y_1. A 2-cobordism from W_0 to W_1 is the following collection of data:

- An n-dimensional manifold S with corners of codimension up to 2.
- A decomposition of the set of faces of S into two disjoint sets, "horizontal faces" and "vertical faces", such that each $x \in \partial^{(2)}S$ is in the closure of one vertical and one horizontal face. We will denote by $\partial_h S$ the union of closures of horizontal faces, and similarly for vertical faces.
- An isomorphism $g\colon \overline{W_0} \sqcup W_1 \to \partial_h S$.
- An isomorphism $f\colon (Y_0 \times I) \sqcup (Y_1 \times I) \to \partial_v S$.

These isomorphisms must agree with the isomorphisms $\overline{Y_0} \sqcup Y_1 \xrightarrow{\sim} \partial W_i$ in the obvious way.



We define an isomorphism of two 2-cobordisms (S, g, f), $(\tilde{S}, \tilde{g}, \tilde{f})$ (for the same Y_0, Y_1, W_0, W_1) as a diffeomorphism $\varphi\colon S \to \tilde{S}$ such that $\varphi \circ g = \tilde{g}$, $\varphi \circ f = \tilde{f}$.

We can now give a definition of the 2-category $\mathbf{Bord}_2$.

Theorem 10.1.1. *There exists a weak 2-category* $\mathbf{Bord}_2$*, in which*

- *Objects are closed oriented* 0*-manifolds*
- *1-morphisms* $\mathrm{Hom}(X_0, X_1)$ *are oriented 1-dimensional cobordisms between* X_0, X_1

- *For* $W_0, W_1 \in \mathrm{Hom}(X_0, X_1)$*, the set of 2-morphisms* $\mathrm{Hom}(W_0, W_1)$ *is the set of isomorphism classes of 2-cobordisms* $W_0 \Rightarrow W_1$

To prove this theorem, one has to construct horizontal and vertical compositions, and prove strict associativity of vertical composition and weak associativity of horizontal composition. Vertical composition is relatively straightforward; it is done in the same way as the usual composition of cobordisms.

However, horizontal composition is much trickier. Recall that for two 1-cobordisms W_1, W_0, the smooth structure on the glued 1-manifold $W_0 \cup_{Y_1} W_1$ is not unique. It was not a problem for a 1-d TQFT, where 1-morphisms are isomorphism classes of cobordisms; however, to define extended 2d TQFT, we need to define 2-cobordism between 1-morphisms, and for that we need actual 1-cobordisms, not isomorphism classes.

There are two ways to deal with it. One can just randomly choose a smooth structure on any result of gluing $W_0 \cup_{Y_1} W_1$, using the axiom of choice; this makes composition not strictly associative but only associative up to isomorphism of 1-cobordisms (as is expected in a weak 2-category), but luckily creates no other issues. The other, less arbitrary, approach is to add more structure to 1-cobordisms — collars or a similar data; in [**SP2009**], he uses a structure he calls "halo". Both of these approaches give a weak 2-category, and in fact they give equivalent 2-categories, see [**SP2009**, Proposition 3.40].

We do not give details of either of these constructions, referring the reader to [**SP2009**]; this takes a significant part of his thesis.

Theorem 10.1.2. *The 2-category* $\mathbf{Bord}_2$ *defined in Theorem 10.1.1 has a natural structure of a symmetric monoidal 2-category, with tensor product given by disjoint union.*

This seems obvious, but remarkably, requires some machinery to do carefully. See [**Shu2010**], [**SP2009**, Section 3.1.4].

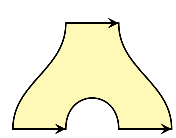
10.2. Definition of 2d extended TQFT

Now that we have defined the 2-category of bordisms, we can give a definition of an extended 2d TQFT.

Definition 10.2.1. Let $\mathcal{C}$ be a symmetric monoidal 2-category. A $\mathcal{C}$-valued extended 2d TQFT is a symmetric monoidal functor

$$Z\colon \mathbf{Bord}_2 \to \mathcal{C}. \tag{10.2.1}$$

Note that this is only one of several approaches to extended field theories. It has some limitations: for example, by definition, in a 2-category we only have 2-morphisms $\alpha\colon f \to g$ if f, g are 1-morphisms between the same objects. For instance, the 2-dimensional manifold with corners shown below cannot be considered a 2-cobordism:



There are alternative approaches; for $n = 2$, there is a notion of "open-closed" TQFT, studied in [**LP2007**], which allows 2-manifolds with corners similar to the one shown above. Yet one more approach is the notion of "TQFT with defects", as discussed, e.g., in [**DKR2011**], [**Car2016**]. We will talk about these approaches later.

10.3. Generators and relations of $\mathbf{Bord}_2$

THEOREM 10.3.1. *The 2-category* $\mathbf{Bord}_2$ *has the following generators*

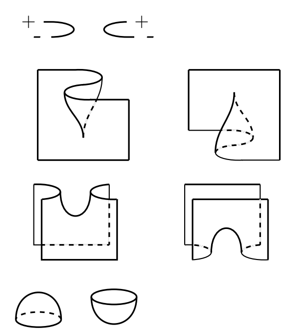
0d generators: $\overset{+}{\bullet}$ $\overset{-}{\bullet}$

1d generators:

2d generators: cusps

2d generators: saddles

2d generators: cap and cup



subject to the relations shown in Figure 10.3.1.

To make it easier to follow, another way to present the cusp flip relation is shown in Figure 10.3.2.

10.4. Classifying 2d extended TQFT

Having generators and relations for $\mathbf{Bord}_2$, we can now try to describe all extended 2d TQFTs with values in an arbitrary symmetric monoidal 2-category $\mathcal{C}$. Since there are a fair number of generators and relations, we will do it in steps.

Step 1: 0- and 1-d generators.

This is clear enough: these are equivalent to choosing a pair of objects $A = Z(\bullet^+)$, $B = Z(\bullet^-) \in \mathcal{C}$ and a pair of 1-morphisms

$$\begin{aligned} \mathrm{ev}\colon A \otimes B &\to \mathbf{1} \\ \mathrm{coev}\colon \mathbf{1} &\to B \otimes A \end{aligned}$$

corresponding to left and right elbows.

Note that since the target category $\mathcal{C}$ is a symmetric monoidal 2-category, we can switch the order of tensor product, e.g., considering ev as a morphism $B \otimes A \to \mathbf{1}$.

Step 2: cusps, cusp relations, and swallowtail relations

The cusp generators should give 2-morphisms

$$\begin{aligned} \alpha\colon z_A &\to 1_A \\ \alpha'\colon 1_A &\to z_A \\ \beta\colon z_B &\to 1_B \\ \beta'\colon 1_B &\to z_B \end{aligned}$$

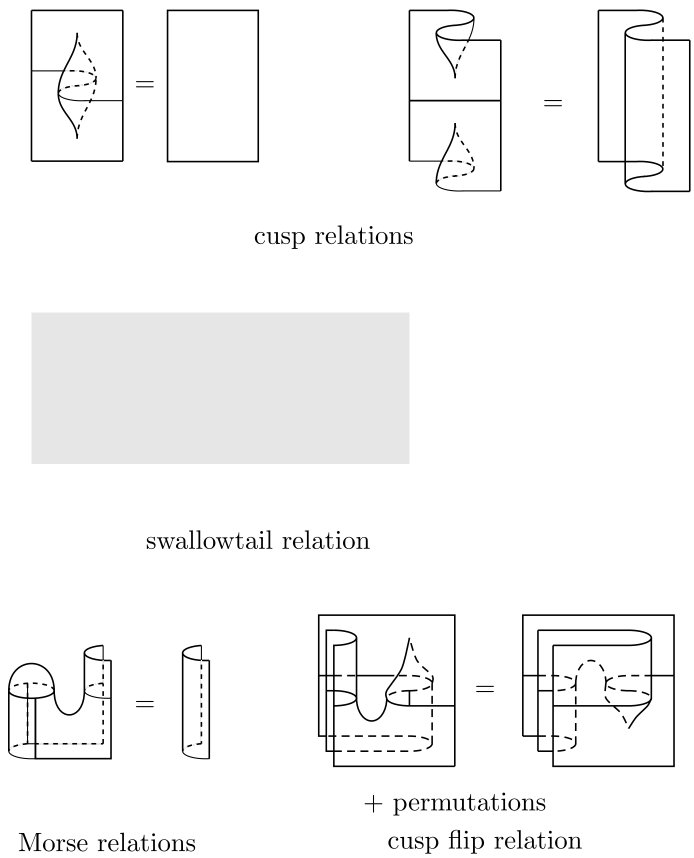


FIGURE 10.3.1. Relations between generators of $\mathbf{Bord}_2$.

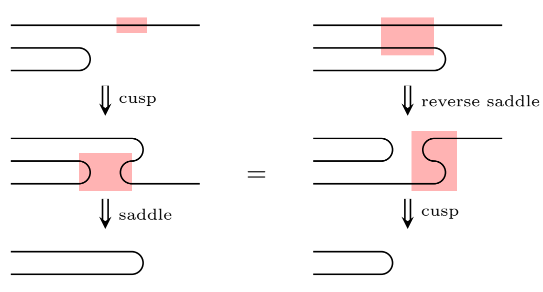


FIGURE 10.3.2. Alternative presentation of cusp flip relation. Each side is a sequence of horizontal slices of the 2-cobordisms, each arrow is an elementary 2-morphism. Red highlight shows the area where the next move will be applied.

where z_A, z_B are zigzag 1-morphisms:

$$
\begin{aligned}
z_A &= Z\left({}^{A} \quad {}_{A} \right) = \left(A \xrightarrow{1\otimes \mathrm{coev}} A\otimes B\otimes A \xrightarrow{\mathrm{ev}\otimes 1} A\right)\colon A\to A \\
z_B &= Z\left({}^{B} \quad {}_{B} \right) = \left(B \xrightarrow{\mathrm{coev}\otimes 1} B\otimes A\otimes B \xrightarrow{1\otimes \mathrm{ev}} B\right)\colon B\to B
\end{aligned}
\tag{10.4.1}
$$

(compare with Section 8.2). Cusp relations are equivalent to the statement that α' is the inverse of α, and β' is the inverse of β, so we do not need to specify α', β'; instead, we just need to require that α, β are isomorphisms.

Finally, the swallowtail relations are exactly relations (8.2.3), (8.2.4).

Thus, summarizing, we see that the data of Steps 1 and 2 is equivalent to a choice of a coherent dual pair in $\mathcal{C}$ (see Definition 8.2.1). As discussed in Section 8.2, such a pair is essentially uniquely determined by A; thus, Steps 1 and 2 are equivalent to choosing a rigid object $A \in \mathcal{C}$. We will denote by $B = A^*$ the dual object.

Note that the data above is already enough to define the value of Z on the circle, which we will denote by $\mathbf{dim}\, A$:

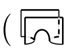
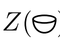
$$\mathbf{dim}\, A = Z(S^1) = \mathrm{ev} \circ \mathrm{coev} \colon \mathbf{1} \to \mathbf{1}$$

(in a 1d TQFT, it would be a number; in a 2d extended TQFT, it is a 1-morphism $\mathbf{1} \to \mathbf{1}$).

Now we are ready to discuss the remaining structures.

Step 3: saddles, cup and cap, and saddle relations

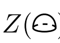
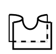
Let us begin with the cup and the reverse saddle. They give us 2-morphisms

(10.4.2)
$$\eta_+ = Z(\ \ ) \colon 1_{A\otimes B} \to \mathrm{coev} \circ \mathrm{ev},$$

$$\varepsilon_+ = Z(\ \ ) \colon \ \mathrm{ev} \circ \mathrm{coev} \to 1_{\mathbf{1}}.$$


Lemma 10.4.1. *The Morse relations involving cup and reverse saddle are equivalent to the requirement that η_+, ε_+ are the unit and the counit of adjunction* $\mathrm{coev} \dashv \mathrm{ev}$.

The proof is left to the reader as an exercise.

Similarly, the cap and the saddle give us 2-morphisms

(10.4.3)
$$\eta_- = Z(\ \ ) \colon 1_{\mathbf{1}} \to \mathrm{ev} \circ \mathrm{coev},$$

$$\varepsilon_- = Z(\ \ ) \colon \ \mathrm{coev} \circ \mathrm{ev} \to 1_{A\otimes B}.$$


and the Morse relations involving them are equivalent to the requirement that η_-, ε_- are the unit and the counit of adjunction $\mathrm{ev} \dashv \mathrm{coev}$.

Following [**Pst2022**], we will call a collection of data $(A, B, \mathrm{ev}, \mathrm{coev}, \alpha, \beta, \eta_\pm, \varepsilon_\pm)$, where $(A, B, \mathrm{ev}, \mathrm{coev}, \alpha, \beta)$ is a coherent dual pair, and $\eta_\pm, \varepsilon_\pm$ are unit and counit of adjunctions $\mathrm{ev} \dashv \mathrm{coev}$, $\mathrm{coev} \dashv \mathrm{ev}$, a *fully dual pair*.

Finally, the cusp flip relation gives a consistency relation between them (note that it involves a saddle and reverse saddle, i.e., ε_- and η_+). This can be written algebraically, which we skip, referring the reader to [**Pst2022**, Section 3.2] (Note that in that paper, he considers a more general situation: he requires that ev has both right and left adjoints, but does not require that both of them are given by ev. This is more suited to the study of the framed extended theory, as opposed to the oriented one we are considering.)

As in [**Pst2022**], we will call a fully dual pair *coherent* if it additionally satisfies the cusp flip relation.

This gives the following theorem.

Theorem 10.4.2. *An extended 2d TQFT $Z \colon \mathbf{Bord}_2 \to \mathcal{C}$ is equivalent to the data of a coherent fully dual pair $(A, B, \mathrm{ev}, \mathrm{coev}, \alpha, \beta, \eta_\pm, \varepsilon_\pm)$ in $\mathcal{C}$.*

However, this can be simplified. We have already discussed that the coherent dual pair $(A, B, \mathrm{ev}, \mathrm{coev}, \alpha, \beta)$ is essentially uniquely determined by A, which must be dualizable, i.e., have a dual in $h\mathcal{C}$. Similarly, one can show that if a fully dual pair exists, then one can make it coherent by modifying one of the counits and keeping the rest of the data intact (see [**Pst2022**, Theorem 3.16]). Thus, we get the following statement.

THEOREM 10.4.3. *An extended 2d TQFT* $Z\colon \mathbf{Bord_2} \to \mathcal{C}$ *is equivalent to the following data:*

- *A dualizable object* $A = Z(\bullet^+)$ *in* $\mathcal{C}$. *We denote by* A^* *the dual object and by* ev, coev *the corresponding evaluation and coevaluation morphisms* (*as discussed before, they are essentially unique*).
- *Adjunction data for* $\mathrm{ev} \dashv \mathrm{coev}$.

such that in addition, ev *is also the right adjoint of* coev*:* $\mathrm{coev} \dashv \mathrm{ev}$.

Note that the adjunction unit and counit for $\mathrm{coev} \dashv \mathrm{ev}$ are not part of the required data: we just need to know that they exist.

10.5. Extended TQFTs with values in Alg

As a special case of the general theory above, let us describe the 2d extended TQFTs with values in the 2-category **Alg** defined in Example 7.1.2. Recall that objects of that category are associative algebras over $\mathbf{k}$, and 1-morphisms are bimodules; monoidal structure is given by the usual tensor product of algebras.

For simplicity, in the remainder of this section we assume that the ground field $\mathbf{k}$ has characteristic zero.

Recall that by Theorem 10.4.2, to define an extended 2d TQFT $Z\colon \mathbf{Bord_2} \to \mathcal{C}$, we need the following data:

(1) A coherent dual pair $(A, B, \mathrm{ev}, \mathrm{coev}, \alpha, \beta)$, where $A = Z(\bullet^+)$
(2) Adjunction data for $\mathrm{ev} \dashv \mathrm{coev}$, $\mathrm{coev} \dashv \mathrm{ev}$, i.e., 2-morphisms

$$\varepsilon_+ = Z(\;)\colon \ \mathrm{ev} \circ \mathrm{coev} \to 1_{\mathbf{1}}$$
$$\eta_+ = Z(\;)\colon 1_{A\otimes B} \to \mathrm{coev} \circ \mathrm{ev}$$
$$\varepsilon_- = Z(\;)\colon \ \mathrm{coev} \circ \mathrm{ev} \to 1_{A\otimes B}$$
$$\eta_- = Z(\;)\colon 1_{\mathbf{1}} \to \mathrm{ev} \circ \mathrm{coev}$$

which must satisfy the adjunction relations (8.1.1) and an additional compatibility relation, the cusp flip relation. (In fact, any one of these 4 pieces of data uniquely determines the remaining three, if they exist).

Let us now apply it to the case when $\mathcal{C}$ is the 2-category **Alg**. In this case, by the results of Section 9.1, any algebra A has a dual, the opposite algebra $B = A^{\mathrm{op}}$, with evaluation and coevaluation given by

$$\mathrm{ev} = A_{A\otimes A^{\mathrm{op}}}\colon A \otimes A^{\mathrm{op}} \to \mathbf{k}$$
$$\mathrm{coev} = {}_{A\otimes A^{\mathrm{op}}}A\colon \mathbf{k} \to A \otimes A^{\mathrm{op}}.$$

This implies that for such a TQFT, we have

$$Z(S^1) = A \otimes_{A^e} A = A/[A, A]$$

Moreover, in Section 9.3 we discussed the data needed to define adjunctions $\mathrm{ev} \dashv \mathrm{coev}$, $\mathrm{coev} \dashv \mathrm{ev}$. In particular, it was shown that the data of adjunction

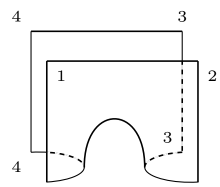
$\mathrm{ev}_A \dashv \mathrm{coev}_A$ is equivalent to defining on A a structure of a symmetric Frobenius algebra (thus, A must be finite-dimensional). Therefore, an extended 2d TQFT defines on A a structure of a symmetric Frobenius algebra, with counit

$$\varepsilon_+ = Z(\ )\colon A/[A,A] \to \mathbf{k}.$$

In this case, as discussed in the proof of Theorem 9.3.1, the operator $\eta_+ = Z(\ )$ is given by

$$\begin{aligned} \eta_+ = Z(\ )\colon {}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} &\to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2} \\ a \otimes b &\mapsto \sum ax_i \otimes bx^i \end{aligned}$$

where x_i, x^i are dual bases in A with respect to the Frobenius pairing $(a,b) = \varepsilon(ab)$. This makes it easier to see the role of all the copies of A in η_+: they correspond to four vertical segments in the cobordism shown below.



Similarly,

$$\begin{aligned} \varepsilon_- &= Z(\ )\colon {}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} \to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2} \\ \eta_- &= Z(\ )\colon \mathbf{k} \to A/[A,A] \end{aligned}$$

define a separability idempotent $E = \sum \tilde{x}_i C \tilde{x}^i \in A \otimes A$ where $\tilde{x}_i \otimes \tilde{x}^i = \varepsilon_-(1 \otimes 1)$, and $C = \eta_-(1)$, see Theorem 9.3.3.

We can now explain the role of the cusp flip relation.

Lemma 10.5.1. *The cusp flip relation is equivalent to the equality* $\varepsilon_- = \eta_+$ (*considered as a morphism of modules* ${}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} \to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2}$).

Thus, we arrive at the following theorem.

Theorem 10.5.2. *Let Z be an extended 2d TQFT with values in the 2-category* **Alg**, *and let $A = Z(\bullet^+)$. Then A is separable, $Z(S^1) \simeq A/[A,A]$, and A has a structure of a symmetric Frobenius algebra, with counit ε, such that*

$$\begin{aligned} &Z(\ ) = \varepsilon\colon A/[A,A] \to \mathbf{k} \\ &Z(\ )\colon {}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} \to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2} : a \otimes b \mapsto \sum ax_i \otimes bx^i \\ &Z(\ )\colon {}_{A_1}A_{A_2} \otimes {}_{A_3^{\mathrm{op}}}A_{A_4^{\mathrm{op}}} \to {}_{A_1}A_{A_3} \otimes {}_{A_4}A_{A_2} : a \otimes b \mapsto \sum ax_i \otimes bx^i \\ &Z(\ ) = [w^{-1}] \in A/[A,A] \end{aligned}$$

where x_i, x^i are dual bases in A with respect to the Frobenius pairing, and $w = \sum x_i x^i$ is the Euler element.

Conversely, every separable symmetric Frobenius algebra A defines an extended 2d TQFT such that the formulas above hold.

In other words, an extended 2d TQFT is equivalent to a separable symmetric Frobenius algebra A.

Note that "symmetric Frobenius algebra" is an additional structure on an algebra A, whereas "separable" is a property of the algebra A.

10.6. Comparison with non-extended theory

It is instructive to compare Theorem 10.5.2 with the classification of (non-extended) TQFTs. Recall that a non-extended 2d TQFT, by Theorem 4.3.3, is defined by a commutative Frobenius algebra. Since every extended TQFT, in particular, also gives a non-extended one, this implies that any extended TQFT defines a structure of a commutative Frobenius algebra on the vector space $V = Z(S^1)$.

Let us apply it to the extended TQFT defined by a separable symmetric Frobenius algebra A; we keep the notation of the previous section, using ε for the counit, etc.

Theorem 10.6.1. *Let Z be an extended TQFT defined by a separable symmetric Frobenius algebra A; let $V = Z(S^1) = A/[A, A]$. Then Z defines the following structure of a commutative Frobenius algebra on V:*

- *Unit: $[w^{-1}] \in V$ (here and below, for $a \in A$ we denote by $[a]$ its class in $V = A/[A, A]$).*
- *Multiplication: $[a] \cdot [b] = \sum [a x_i b x^i]$.*
- *Counit: $\varepsilon([a]) = \varepsilon(a)$, where ε is the counit in A.*

Alternatively, we can identify $Z(S^1) = z(A)$, using the isomorphism

$$\begin{aligned} A/[A, A] &\to z(A) \\ [a] &\mapsto \sum x_i a x^i. \end{aligned} \tag{10.6.1}$$

Then Theorem 10.6.1 can be rewritten as follows.

Theorem 10.6.2. *Under the identification $Z(S^1) \simeq z(A)$ given by (10.6.1), the unit and multiplication on $z(A)$ defined by the TQFT Z coincide with the unit and multiplication inherited from A, and the counit is given by*

$$\varepsilon_{z(A)}(c) = \varepsilon_A(w^{-1}c), \qquad c \in z(A) \subset A.$$

Note, however, that if A is separable (=semisimple), then $z(A)$ is also semisimple. Thus, we get the following corollary.

Corollary 10.6.3. *A (non-extended) 2d TQFT can be extended if and only if the algebra $Z(S^1)$ is semisimple.*

In particular, the TQFT defined by the commutative Frobenius algebra $\mathbb{C}[x]/(x^{n+1})$, described in Example 5.1.1, cannot be extended.

Example 10.6.1. Let G be a finite group, and let $A = \mathbf{k}[G]$ be its group algebra. Define the counit $\varepsilon \colon A \to \mathbf{k}$ by

$$\varepsilon(g) = \delta_{g,1} = \frac{1}{|G|} \operatorname{tr}_R(g), \qquad g \in G,$$

where R is the regular representation (note the difference in normalization with (6.2.4)).

This is a semisimple symmetric Frobenius algebra; thus, it defines an extended TQFT $Z_A \colon \mathbf{Bord}_2 \to \mathbf{Alg}$.

For this algebra, $e = \sum x_i \otimes x^i$ is given by

$$e = \sum_G g \otimes g^{-1}$$

so that $w = \sum x_i x^i = |G|$; thus, the isomorphism (9.3.5) $A/[A, A] \to z(A)$ becomes

$$x \mapsto \sum g x g^{-1}$$

and the projector onto $z(A)$ is given by $p(x) = \frac{1}{|G|} \sum g x g^{-1}$ which is, of course, a very familiar formula in representation theory.

In this case, we see that under identification $Z_A(S^1) \simeq z(\mathbf{k}[G])$, the counit becomes $\tilde{\varepsilon}(z) = \frac{1}{|G|^2} \operatorname{tr}_R(z)$ (note normalization!) which matches (6.2.4). Thus, the corresponding non-extended TQFT is the Dijkgraaf–Witten TQFT discussed in Chapter 6.

CHAPTER 11

PL manifolds, triangulations, and polygon decompositions

In this chapter, we start building an alternative approach to the construction of extended TQFTs. Instead of using Morse functions to decompose a cobordism into a product of elementary cobordisms, we will use triangulations and similar cell decompositions. This is naturally done in the category of piecewise-linear (PL) manifolds; thus, in this chapter we will work with PL manifolds. We refer the reader to [**RS1982**] for an overview of PL topology; another useful reference is Lurie's lecture notes [**Lur2009**].

11.1. Polyhedra and PL manifolds

Definition 11.1.1. A k-simplex $\sigma \subset \mathbb{R}^N$ is the convex hull of a set of $k+1$ points $x_0, x_1, \dots, x_k$ which are affinely independent, i.e., span an affine subspace of dimension k in $\mathbb{R}^N$.

A finite polyhedron is a subset $P \subset \mathbb{R}^N$ which can be presented as a finite union of simplices.

Note that for a given polyhedron, there are many ways to present it as a union of simplices. We will be mostly working with triangulations.

Definition 11.1.2. A triangulation of a polyhedron $P \subset \mathbb{R}^N$ is a finite collection $S = \{\sigma_i\}$ of simplices such that $P = \bigcup_S \sigma_i$ and

(1) Any face of a simplex $\sigma \in S$ is again a simplex in S.
(2) The intersection of any two simplices from S is a face of each of them.

(Note that by "face", we mean a face of any codimension, not necessarily codimension 1).

Theorem 11.1.1.

(1) *Any finite polyhedron $P \subset \mathbb{R}^N$ admits a triangulation.*
(2) *Any two triangulations of a finite polyhedron P admit a common subdivision: if T_1, T_2 are triangulations, then there exists a triangulation T_3 such that each simplex of T_1 is a union of simplices of T_3, and the same for T_2.*

We refer the reader to [**RS1982**, Theorem 2.11, Addendum 2.12] for the proof.

Definition 11.1.3. Let P be a polyhedron. A function $f\colon P \to \mathbb{R}^m$ is called *piecewise linear* (PL) if it is continuous and there exists a triangulation of P such that the restriction of f to each simplex of the triangulation is linear.

For polyhedra $P \subset \mathbb{R}^N$, $L \subset \mathbb{R}^m$, a map $f\colon P \to L$ is piecewise linear if it is piecewise linear as a map $P \to \mathbb{R}^m$.

It is easy to see that the composition of PL maps is again PL; thus, we get a category of PL polyhedra.

REMARK 11.1.1. Above, we defined polyhedra as subsets of $\mathbb{R}^N$. Choosing a triangulation gives a combinatorial description of P: all we need to know is how many simplices there are of each dimension and which faces are glued together. One can define an *abstract simplicial complex* K as such combinatorial data (we skip the precise definition). Conversely, given an abstract simplicial complex, we can construct a polyhedron $P = |K|$ by gluing together simplices along the face maps and embedding this union of simplices in $\mathbb{R}^N$ for sufficiently large N; moreover, such a polyhedron is unique up to PL isomorphism.

DEFINITION 11.1.4. Let M be a polyhedron. We will say that M is a *PL manifold* of dimension n if, for every point $x \in M$, there exists an open neighborhood $U \subset M$ containing x and a piecewise linear homeomorphism $f : U \to U'$ where U' is a neighborhood of 0 in $\mathbb{R}^n$.

REMARK 11.1.2. If M is a PL manifold, then its underlying topological space is automatically a topological manifold. The converse, however, is not true: it is possible to construct a polyhedron P such that the underlying topological space is a topological manifold, yet P is not a PL manifold.

Given a triangulation of a polyhedron P, it is possible to determine whether P is a PL manifold by looking at the so-called link of a point; we refer the reader to [**RS1982**] for details.

We can now also talk about triangulating smooth manifolds.

DEFINITION 11.1.5. Let P be a polyhedron and M a smooth manifold. We say that a map $f : P \to M$ is piecewise differentiable (PD) if there exists a triangulation of P such that the restriction of f to each simplex is smooth. We will say that f is a PD homeomorphism if f is piecewise differentiable, a homeomorphism, and the restriction of f to each simplex has injective differential at each point. In this case, we say that f is a Whitehead triangulation of M.

THEOREM 11.1.2. *Every smooth manifold admits a Whitehead triangulation $P \to M$. Moreover, in this case P is necessarily a PL manifold. Any two Whitehead triangulations give isomorphic PL manifolds.*

In other words, every smooth manifold has a unique PL structure.

Unfortunately, the converse is false. In general, not every PL manifold can be "smoothed" (i.e., obtained by triangulating a smooth manifold); moreover, if a smoothing exists, it might not be unique. The most famous examples of this are Milnor's exotic smooth structures on S^7: they are all isomorphic as PL manifolds but not as smooth manifolds. This problem, however, does not arise in low dimensions.

THEOREM 11.1.3. *In dimensions $n \leq 6$, every PL manifold admits a smooth structure, unique up to diffeomorphism.*

We refer the reader to [**Mil2011**] and [**Lur2009**, Lecture 23] for a review of the relations between PL manifolds and smooth manifolds.

It is easy to extend the definition above and define a PL manifold with boundary. Note that in the PL setting, there is no notion of a manifold with corners: a corner $\mathbb{R}_+^k \times \mathbb{R}^{n-k}$ is PL isomorphic to the halfspace $\mathbb{R}_+ \times \mathbb{R}^{n-1}$. We will use the standard notation B^n for the closed n-dimensional ball considered as an n-dimensional PL manifold with boundary, and $S^{n-1} = \partial B^n$ for the $(n-1)$-dimensional sphere. Note that in the PL category $B^n \simeq [0,1]^n$.

Some of the usual notions in the theory of smooth manifolds can be easily transferred to the PL setting; for example, it is not hard to introduce the notion of orientation and define oriented PL manifolds. Others are much harder; in particular, there is no easy way to define the notion of the tangent bundle if we want the fibers to be vector spaces. Instead, one uses an alternative language, that of microbundles (see [**Lur2009**, Lecture 10]). In fact, defining a vector bundle structure on the "tangent microbundle" of a PL manifold is essentially equivalent to defining a smooth structure ([**HM1974**]).

11.2. Pachner moves

Let M be a PL manifold. As mentioned before, it admits many different triangulations. It is natural to ask whether one can get from one triangulation to another by some operations ("moves"). The first result of this kind was established by Alexander in 1930, who proved that any two triangulations can be obtained from each other by a sequence of so-called "stellar moves" and their inverses. Unfortunately, there are infinitely many different stellar moves.

An improved version of this result was obtained by Pachner in the 1980s. For $n = 2$, Pachner's theorem can be stated as follows.

THEOREM 11.2.1. *Let M be a 2-dimensional PL manifold. Then any two triangulations of M can be obtained from each other by a finite sequence of the operations in Figure 11.2.1 and their inverses.*

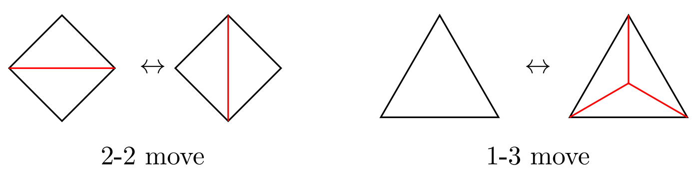
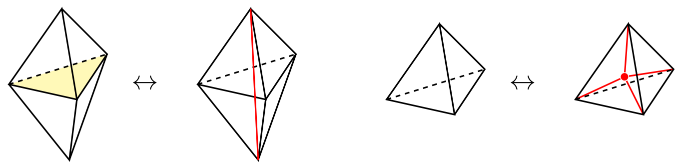


FIGURE 11.2.1. Pachner moves in dimension 2

These operations are called *bistellar moves*, or *Pachner moves*. They were introduced by Pachner in [**Pac1987**]; an exposition can be found in [**Lic1999**]. They can be easily constructed in any dimension; we show these moves in dimension 3 in Figure 11.2.2, and refer the reader to the papers cited above for the construction in arbitrary dimension.

This gives us another approach to constructing invariants of manifolds (and more generally, TQFTs): instead of using Morse functions to present a cobordism as a composition of elementary cobordisms, we can triangulate the manifold and construct the invariant by gluing together simplices. To check that this does not depend on the choice of triangulation, we need to check invariance under Pachner moves.



2-3 move 1-4 move

FIGURE 11.2.2. Pachner moves in dimension 3. In the second picture of the 2-3 move, the polyhedron is cut into 3 tetrahedra, all sharing the red edge; thus, any horizontal cross-section looks like the RHS of the 1-3 move in dimension 2. In the 1-4 move, the figure on the right is a tetrahedron cut into 4 tetrahedra, all sharing the center (red) vertex.

11.3. PLCW decompositions

However, for many purposes the notion of triangulation is too restrictive, and one would like to consider more general cell decompositions — for example, representing a torus as the result of gluing together opposite sides of a rectangle. In this section, we give exact definitions and results for such more general cell decompositions, following [**Kir2012**].

Recall that we denote by $B^k = [0,1]^k$ the k-ball with the canonical PL structure; we will also use the notation ∂B^k and $\operatorname{Int} B^k = B^k - \partial B^k$ for the boundary and interior of B^k respectively.

DEFINITION 11.3.1. A *generalized k-cell* in $\mathbb{R}^N$ is a subset $C \subset \mathbb{R}^N$ together with a decomposition $C = \mathring{C} \sqcup \dot{C}$ such that $\mathring{C} = \varphi(\operatorname{Int} B^k)$, $\dot{C} = \varphi(\partial B^k)$ (and thus $C = \varphi(B^k)$) for some PL map $\varphi\colon B^k \to \mathbb{R}^N$ which is injective on the interior of B^k.

In such a situation, the map φ is called a *characteristic map*.

It is easy to show that in fact φ is determined by C together with the decomposition $C = \mathring{C} \sqcup \dot{C}$, uniquely up to an automorphism of B^k.

DEFINITION 11.3.2. A generalized cell complex (g.c.c.) is a finite collection K of generalized cells in $\mathbb{R}^N$ such that

(1) For any distinct A, B in K, we have
$$\mathring{A} \cap \mathring{B} = \varnothing$$
(2) For any cell $C \in K$, $\dot{C}$ is a union of cells.

The support $|K| \subset \mathbb{R}^N$ of a generalized cell complex K is defined by
$$|K| = \bigcup_{C \in K} C.$$

A *generalized cell decomposition* of a compact polyhedron $P \subset \mathbb{R}^N$ is a generalized cell complex K such that $|K| = P$.

For a generalized cell complex K, we define the k-skeleton K^k of K as the generalized cell complex formed by all cells of dimension $\leq k$ in K.

This is still slightly too general; for example, this definition allows one to form a cell decomposition by taking a rectangle and folding one of the edges in half, gluing it to itself. To avoid that, we make the following (recursive) definition.

Definition 11.3.3. A generalized cell complex (respectively, a generalized cell decomposition) K will be called a *PLCW complex* (respectively, PLCW decomposition) if $\dim K = 0$, or $\dim K = n > 0$ and the following conditions hold:

(1) The $(n-1)$-skeleton K^{n-1} is a PLCW complex.
(2) For any n-cell $C \in K$, $C = \varphi(B^n)$, there exists a PLCW decomposition L of ∂B^n such that the restriction $\varphi|_{\partial B^n} : L \to K^{n-1}$ is a regular cellular map, i.e., for every cell D of L, $\varphi(D)$ is a generalized cell in K, and φ is injective on $\mathring{D}$.

For $n = 2$, this means the following:

- The 0-skeleton is a finite collection of points (vertices) in $\mathbb{R}^N$.
- The 1-skeleton K^1 is a finite union of 1-cells (i.e., PL images of an interval), with endpoints in vertices. 1-cells are only allowed to intersect at endpoints; we allow two endpoints of the same interval to coincide.
- 2-cells are k-gons, with $k \geq 1$, and the attachment map sends each side of the k-gon to one of the 1-cells, injectively on the interior of the side.

In particular, we allow gluing together two sides of the same polygon, but we do not allow, e.g., "folding" any side to glue it to itself.

Example 11.3.1. The usual construction of a torus by gluing together opposite sides of a rectangle gives a PLCW decomposition of the torus, consisting of one 0-cell, two 1-cells, and one 2-cell.

Three more examples of PLCW decompositions are shown in Figure 11.3.1.

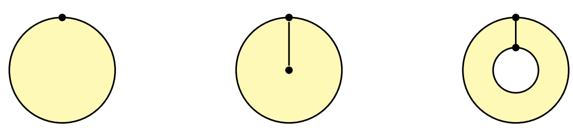


Figure 11.3.1. Examples of PLCW decompositions of a disk and an annulus.

For PLCW decompositions, we have an analog of Pachner moves. Informally, these operations consist of erasing a $(k-1)$-disk separating two different k-cells. More formally, they are defined as follows.

Let M be an n-dimensional PL manifold together with a PLCW decomposition K, and let $A = \varphi(B^k)$ be a k-cell in K. Let $L = \varphi^*(K)$ be the induced PLCW decomposition of the boundary $\partial B^k = S^{k-1}$. Let $H = \{x_k = 0\}$ be a hyperplane in $\mathbb{R}^k$; it cuts B^k into two half-balls, $B^k_\pm$, separated by the disk $B^k_0 \simeq B^{k-1}$. Let $E = S^{k-1} \cap H$ be the equator of S^{k-1}. Assume that E is a union of cells of L. Then we can construct a subdivision of K by replacing cell $A = \varphi(B^k)$ by two k-cells $A_\pm = \varphi(B^k_\pm)$ and a $(k-1)$-cell $A_0 = \varphi(B^k_0)$; in other words, we "cut" the k-ball B^k into two parts by a $(k-1)$-dimensional disk. We will call such subdivisions *elementary*.

Theorem 11.3.1. *Any two PLCW decompositions of a PL n-manifold M can be obtained from each other by a sequence of elementary subdivisions and their inverses.*

A proof of this fact can be found in [**Kir2012**].

For $n = 2$, there are two kinds of such subdivisions:

(1) Subdividing a 1-cell, i.e., inserting a new vertex dividing an edge into two.
(2) Subdividing a 2-cell, i.e., cutting a polygon into two by a diagonal.

Note that the word "diagonal" includes an arc connecting two adjacent vertices (but does not include a loop around a vertex). These two kinds of moves are shown in Figure 11.3.2.

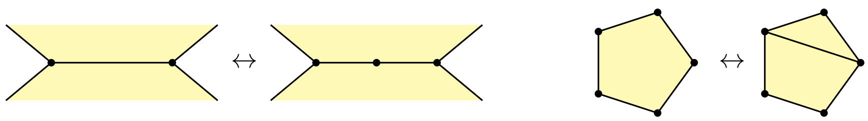


Figure 11.3.2. Elementary moves for $n = 2$

A sequence of such elementary moves relating two PLCW decompositions of a disk is shown in Figure 11.3.3.

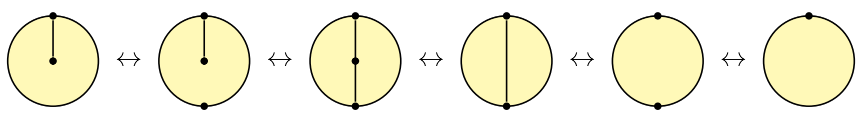


Figure 11.3.3. A sequence of elementary subdivisions of a disk

CHAPTER 12

Lattice construction of extended 2d TQFT

In this chapter, we give an alternative approach to constructing 2d TQFT, based on triangulating a manifold (or, more generally, choosing a PLCW decomposition, as described in the previous chapter). In this approach, an invariant of a closed 2-manifold is defined by a suitable sum over all ways of "coloring" cells of a given PLCW decomposition (we will explain it in more detail below). In physics, such a sum is usually referred to as "state sum"; it should be considered a discretized analog of the path integral.

In the special case of DW theory, this approach is known as "lattice gauge theory" and has been extensively studied by physicists. Generalization of this to an arbitrary separable symmetric Frobenius algebra was suggested in [**FHK1994**]. See also [**Oec2005**]. Generalization to arbitrary symmetric monoidal 2-categories is discussed in [**Dav2011**].

Throughout this chapter, we fix a separable symmetric Frobenius algebra A. As before, we assume that the ground field $\mathbf{k}$ has characteristic 0. We denote by $\varepsilon\colon A \to \mathbf{k}$ the counit. We also denote by $(a, b) = \varepsilon(ab)$ the corresponding bilinear pairing.

12.1. Definition of TQFT via state sum

As before, let x_i, x^i be dual bases in A with respect to the Frobenius pairing. We denote by g_{ij} the matrix of the pairing in the basis x_i: $g_{ij} = (x_i, x_j)$ and by g^{ij} the inverse matrix: $\sum_j g_{ij}g^{jk} = \delta_{ik}$. We will use g_{ij}, g^{ij} for raising and lowering indices of various tensors.

For any $k \geq 0$, define the map $A^k \to \mathbf{k}$ by

$$\langle a_1, \dots, a_k\rangle = \varepsilon(a_1 \cdot \dots \cdot a_k). \tag{12.1.1}$$

Lemma 12.1.1.

(1) *The map* (12.1.1) *is invariant under cyclic permutation of* a_i*:*

$$\langle a_1, \dots, a_k\rangle = \langle a_k, a_1, \dots, a_{k-1}\rangle.$$

(2) *The map* (12.1.1) *is multilinear over* $\mathbf{k}$*. Moreover, it is linear over* A*: for any* $i = 1, \dots, k$*,* $x \in A$*, we have*

$$\langle a_1, \dots, a_i x, a_{i+1}, \dots, a_k\rangle = \langle a_1, \dots, a_i, x a_{i+1}, \dots, a_k\rangle$$

(for $i = k$*, we take* $a_{i+1} = a_1$*).*

(3)

$$\sum_i \langle a_1, \dots, a_k, x_i\rangle\langle x^i, b_1, \dots, b_m\rangle = \langle a_1, \dots, a_k, b_1, \dots, b_m\rangle \tag{12.1.2}$$

The proof of this lemma is straightforward and is left as an exercise to the reader.

Let M be a closed oriented 2-dimensional manifold. Let us choose a PLCW decomposition K of M (see Section 11.3), with the set of 2-cells K_2, the set of 1-cells (edges) K_1, and the set of vertices K_0. We will denote by E the set of *oriented* edges, i.e., pairs consisting of an edge $r \in K_1$ and an orientation of r. Thus, every (unoriented) edge $r \in K_1$ gives rise to two oriented edges $r', r'' \in E$.

The invariant will be constructed by combining algebraic data defined by 2-cells, edges, and vertices of M, as defined below. The key ingredient of this construction is the tensor product of copies of A, one copy for each oriented edge r

$$A^E = \bigotimes_{r \in E} A_r \tag{12.1.3}$$

where we denote by A_r a copy of A corresponding to r.

Algebraic data determined by edges. Define, for each unoriented edge $r \in K_1$, an element

$$e_r = \sum x_i \otimes x^i = \sum g^{ij} x_i \otimes x_j \in A_{r'} \otimes A_{r''}$$

where r', r'' are two possible orientations of r. Since $g^{ij} = g^{ji}$ (which follows from the symmetry of the form $(\ ,\ )$), e_r does not depend on which of two possible orientations was denoted by r' and which by r''. Taking the product over all (unoriented) edges $r \in K_1(M)$, we get an element

$$e_M = \bigotimes_{r \in K_1} e_r \in A^E. \tag{12.1.4}$$

Algebraic data determined by vertices. Since A is a separable Frobenius algebra, we have an invertible Euler element $w = \sum x_i x^i \in z(A)$ (see Theorem 9.3.4). Choose now, for every vertex $v \in K_0$ of the PLCW decomposition, an oriented edge $r(v)$ incident to v (v can be either the head or the tail of $r(v)$). Let the operator

$$w_v^{-1} \colon A^E \to A^E \tag{12.1.5}$$

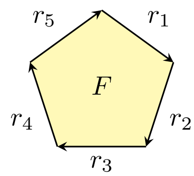
be the operator of left multiplication by w^{-1} in the component $A_{r(v)}$. Taking the product over all vertices, we get an operator

$$\prod_v (w_v^{-1}) \colon A^E \to A^E.$$

Note that this operator depends on the choice we made, namely the choice of oriented edge for each vertex.

Algebraic data determined by 2-cells. Every 2-cell $F \in K_2$ inherits a natural orientation from M. Thus, we can define its boundary ∂F. This is a collection of oriented edges, which has a natural counterclockwise cyclic order.

We will actually be more interested in the boundary with reversed orientation, writing $\overline{\partial F} = \{r_1, \dots, r_k\} \subset E$; if we draw F as a polygon in $\mathbb{R}^2$ with the standard orientation, then the edges r_i are oriented clockwise and taken in the natural clockwise order:

(12.1.6) 

Define the map

$$\begin{aligned}\langle\,\rangle_F\colon A^{\otimes\overline{\partial F}} &\to \mathbf{k}\\ a_1\otimes\cdots\otimes a_k &\mapsto \langle a_1,\dots,a_k\rangle.\end{aligned}$$

Lemma 12.1.1 implies that this map does not depend on which edge we denote by r_1 and thus is well-defined.

REMARK 12.1.1. The reason for choosing $\overline{\partial F}$ rather than ∂F is that we want to think of F as a cobordism from $\overline{\partial F}$ to the empty set.

Since M is a manifold, every unoriented edge appears twice as a boundary edge of a 2-cell, and every oriented edge appears once as part of ∂F. Thus, by taking the tensor product of $\langle\,\rangle_F$ over all 2-cells F, we get a linear map

$$\bigotimes_F \langle\,\rangle_F\colon A^E\to\mathbf{k}.$$

Let us now put it all together and define a number $Z_A(M,K)\in\mathbf{k}$ by

$$Z_A(M,K)=\Big(\bigotimes_F\langle\,\rangle_F\Big)\Big(\prod_v(w_v^{-1})\Big)e_M\in\mathbf{k} \tag{12.1.7}$$

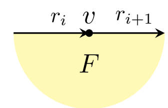
where e_M is defined by (12.1.4) and w_v is defined by (12.1.5).

THEOREM 12.1.2. *Let M be a closed oriented 2-manifold with PLCW decomposition K. Let $Z_A(M,K)$ be defined by* (12.1.7).

(1) *So defined $Z_A(M,K)$ does not depend on the choice of an edge $r(v)$ used to define w_v^{-1}.*
(2) *So defined $Z_A(M,K)$ does not depend on the choice of cell decomposition K.*

PROOF. Part (1) follows from two observations:

- $$\sum w^{-1}x_i\otimes x^i=\sum x_i\otimes x^i w^{-1}=\sum x_i\otimes w^{-1}x^i$$
  (the first identity uses the centrality of e, the second uses the fact that w^{-1} is central). This shows that replacing a choice of edge $r(v)$ by the same edge but with opposite orientation does not change the vector $w_v^{-1}e_M$ and thus, does not change $Z_A(M,K)$.
- If r_i, r_{i+1} are two edges of the boundary of some cell F both adjacent to vertex v, as shown below, then replacing $r(v)=r_i$ by $r(v)=r_{i+1}$ does not change the composition $\Big(\bigotimes_F\langle\,\rangle_F\Big)w_v^{-1}$ and thus, does not change $Z_A(M,K)$. Indeed, it follows from $\langle\dots,a_iw^{-1},a_{i+1},\dots\rangle=\langle\dots,a_i,w^{-1}a_{i+1},\dots\rangle$.



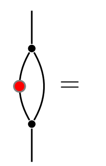
By using these two operations, we can replace any oriented edge adjacent to v by any other edge. This proves part (1) of the theorem.

To prove part (2), we need to verify that $Z_A(M, K)$ is invariant under elementary moves described in Figure 11.3.2.

Invariance under subdivision of a 2-cell follows from (12.1.2).

Invariance under subdivision of a 1-cell follows from

(12.1.8)

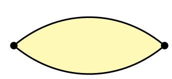


where the red dot stands for the operator of multiplication by w^{-1} (which comes from the operator w_v^{-1} for the newly added vertex). But this is immediate from the definition: indeed, comultiplication is given by $\Delta(a) = \sum ax_i \otimes x^i$, so the composition in the left-hand side is $a \mapsto \sum ax_i w^{-1} x^i = a$. □

Remark 12.1.2. Note that without the w_v^{-1} factor, this theorem would fail unless we assume $\sum x_i x^i = 1$.

Thus, we see that (12.1.7) defines an invariant of closed 2-manifolds. In the next section, we will generalize it to manifolds with boundary, thus defining an extended TQFT as a lattice theory.

Example 12.1.1. Consider the PLCW decomposition of S^2 obtained by gluing together opposite edges of the bigon shown below.



This cell decomposition has one 2-cell, one 1-cell, and two vertices. The state sum (12.1.7) in this case gives

$$Z_A(S^2) = \sum \varepsilon(w^{-2} x_i x^i) = \varepsilon(w^{-1}).$$

Example 12.1.2. Consider a PLCW decomposition of the torus T^2 obtained by gluing together opposite sides of the rectangle. This decomposition has a single 2-cell, two edges, and one vertex. The state sum (12.1.7) in this case gives

$$Z_A(T^2) = \sum \varepsilon(w^{-1} x_i x_j x^i x^j) = \sum (p(x_j), x^j) = \mathrm{tr}_A(p)$$

where $p\colon A \to A$ is given by $p(a) = \sum w^{-1} x_i a x^i$. By (9.3.7), the so-defined p is exactly the operator of projection onto the center of A; thus,

$$Z_A(T^2) = \mathrm{tr}_A(p) = \dim z(A).$$

12.2. Lattice theory as an extended TQFT

So far, we have defined $Z_A(M)$ for a closed 2-manifold M. However, we can easily turn it into a fully extended TQFT.

Namely, we define

$$Z_A(\bullet^+) = A, \qquad Z_A(\bullet^-) = A^{\mathrm{op}}$$

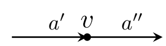
For an oriented 1-manifold N (possibly with boundary), we choose a cell decomposition of it (which for 1-dimensional manifolds just means choosing some points that divide N into intervals) and define

$$Z_A(N) = \Big(\bigotimes_{r\in E} A_r\Big)/\sum_v I_v \tag{12.2.1}$$

where for a vertex v, the subspace I_v is generated by the elements

$$(\dots \otimes a'x \otimes a'' \otimes \dots) - (\dots \otimes a' \otimes xa'' \otimes \dots), \qquad x \in A$$

where a', a'' are elements in the copies of A assigned to the two edges meeting at v as in the figure below.



It is trivial to check that the so-defined vector space does not depend on the choice of cell decomposition (up to a canonical isomorphism); namely,

$$\begin{aligned} Z_A(I) &= A \otimes_A A \otimes_A \dots \otimes_A A \simeq A \\ Z_A(S^1) &\simeq A/[A,A]. \end{aligned} \tag{12.2.2}$$

If I is an interval, then $Z_A(I)$ is an A-bimodule; thus, if we consider I as a cobordism $\bullet^+ \to \bullet^+$, then $Z_A(I)$ is a morphism $A \to A$ in **Alg**. We can also consider I as a cobordism $\bullet^+ \sqcup \bullet^- \Rightarrow \varnothing$, and $Z_A(I)$ as a right module over $A \otimes A^{\mathrm{op}}$, or consider I as a cobordism $\varnothing \Rightarrow \bullet^+ \sqcup \bullet^-$, and $Z_A(I)$ as a left module over $A \otimes A^{\mathrm{op}}$.

We can now define Z_A for a 2-cobordism. First, let M be a 2-manifold with boundary, considered as a cobordism $N \to \varnothing$, where $N = \overline{\partial M}$. Choose a cell decomposition K of M; let E_{bulk} be the set of oriented edges inside M (thus, every internal unoriented edge of K_1 appears twice in E_{bulk}) and E_N be the set of oriented edges on the boundary (each appearing with orientation inherited from $N = \overline{\partial M}$).

Let us modify (12.1.7) and define

$$Z_A(M,K) = \Big(\bigotimes_F \langle\,\rangle_F\Big)\Big(\prod_v (w_v^{-1})\Big)e_M \colon A^{\otimes E_N} \to \mathbf{k}. \tag{12.2.3}$$

(note that $e_M \in A^{\otimes E_{\mathrm{bulk}}}$, and $\bigotimes_F \langle\,\rangle_F$ is a map $A^{\otimes E_{\mathrm{bulk}} \sqcup E_N} \to \mathbf{k}$). Note that we only take the product $\prod_v w_v^{-1}$ over *internal* vertices of K and do not include the vertices on the boundary.

THEOREM 12.2.1.

(1) *The linear map $A^{\otimes E_N} \to \mathbf{k}$ defined by (12.2.3) descends to a map $Z_A(N) \to \mathbf{k}$.*
(2) *The so-defined operator is independent of the choice of a cell decomposition of M.*

(3) *For a closed oriented one-dimensional manifold N, let $M = N \times I$ considered as a cobordism $N \sqcup \overline{N} \to \varnothing$. Then the map*

$$Z_A(N \times I) \colon Z_A(N) \otimes Z_A(\overline{N}) \to \mathbf{k} \tag{12.2.4}$$

defined by (12.2.3)*, is a non-degenerate bilinear pairing.*

We leave the proof of this theorem to the reader; it is an easy modification of the proof of Theorem 12.1.2.

Since the pairing (12.2.4) is non-degenerate, it can be used to identify $Z_A(\overline{N}) \simeq (Z_A(N))^*$. In particular, it shows that if M is a cobordism $N_1 \overset{M}{\Longrightarrow} N_2$ — which can also be considered as a cobordism $N_1 \sqcup \overline{N_2} \Rightarrow \varnothing$ — then $Z_A(M) \colon Z_A(N_1) \otimes Z_A(\overline{N_2}) \to \mathbf{k}$ can also be considered as an operator $Z_A(N_1) \otimes Z_A(N_2)^* \to \mathbf{k}$, or equivalently, as an operator $Z_A(N_1) \to Z_A(N_2)$; more formally, for such a cobordism we define $Z_A(M) \colon Z_A(N_1) \to Z_A(N_2)$ by

$$\langle Z_A(M)x, y\rangle = Z_A(M)(x \otimes y), \qquad x \in Z_A(N_1), y \in Z_A(\overline{N_2}) \tag{12.2.5}$$

where $\langle\, , \rangle$ is the pairing (12.2.4).

Theorem 12.2.2. *Formulas* (12.2.3), (12.2.6) *define a (non-extended) 2-dimensional TQFT.*

Example 12.2.1. Let D be a disk considered as a cobordism $S^1 \to \varnothing$. Then $Z_A \colon A/[A, A] \to \mathbf{k}$ is given by the counit of the Frobenius algebra: $Z_A(x) = \varepsilon(x)$.

Example 12.2.2. Let $M = S^1 \times I$ be the cylinder considered as a cobordism $S^1 \sqcup S^1 \to \varnothing$. Choosing a decomposition of the cylinder obtained by gluing together two sides of the rectangle, we see that in this case we have

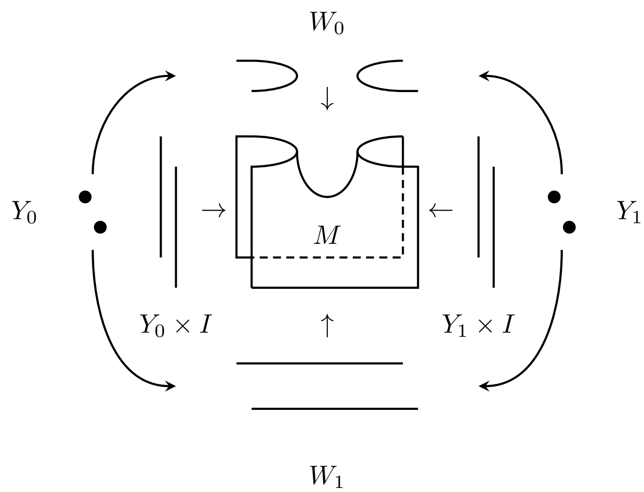
$$Z_A(M)(a \otimes b) = \sum \varepsilon(a x_i b x^i) = \sum \varepsilon(w^{-1} x_j a x^j x_i b x^i).$$

Exercise 12.2.1. Show that $Z_A($$) \colon V \otimes V \to V$, where $V = Z_A(S^1) = A/[A, A]$, is given by

$$Z_A([a] \otimes [b]) = \sum [a x_i b x^i].$$

Comparing it with Theorem 10.6.1, which gave an explicit description of the (non-extended) TQFT constructed using Morse decompositions of cobordisms, we see that these two descriptions match.

Finally, let us define an Z_A as an extended TQFT. We have already defined Z for 0-dimensional manifolds and 1d cobordisms; thus, to complete the story, we need to define $Z_A(M)$ for a 2-cobordism, i.e., a 2-manifold with corners (see Definition 10.1.2).



To do that, note that in this case, boundary ∂M consists of 3 pieces (possibly disconnected): W_0 (top), W_1 (bottom), and "vertical" boundary $Y \times I$, where $Y = Y_0 \sqcup Y_1$.

Choosing a cell decomposition K of M such that each corner is a vertex of that decomposition and using again the formula (12.2.3), we can define

$$\tilde{Z}_A(M, K)\colon A^{\otimes E_{\mathrm{top}}} \otimes A^{\otimes E_{\mathrm{bot}}} \otimes A^{\otimes E_v} \to \mathbf{k}$$

where E_{top}, E_{bot}, E_v are oriented edges at top, bottom, and vertical parts of the boundary respectively.

Now, for each vertical edge $e \in E_v$, take vector $1 \in A_e$. This defines a vector $1_v \in A^{\otimes E_v}$. Define

$$\begin{aligned} &Z_A(M, K)\colon A^{\otimes E_{\mathrm{top}}} \otimes A^{\otimes E_{\mathrm{bot}}} \to \mathbf{k} \\ &Z_A(M, K)(a_{\mathrm{top}} \otimes a_{\mathrm{bot}}) = \tilde{Z}_A(a_{\mathrm{top}} \otimes a_{\mathrm{bot}} \otimes 1_v). \end{aligned} \tag{12.2.6}$$

12.3. Finite group gauge theory as lattice theory

Let us consider a special example, which we have already discussed using a different approach in Example 10.6.1, namely when the algebra $A = \mathbf{k}[G]$ is the group algebra of a finite group G and Frobenius structure is given by

$$\varepsilon(g) = \delta_{g,1} = \frac{1}{|G|}\operatorname{tr}_R(g), \qquad g \in G,$$

where R is the regular representation.

In this case the element $e = \sum x_i \otimes x^i \in A \otimes A$ is given by

$$e = \sum_{g \in G} g \otimes g^{-1}$$

so that $w = |G|$.

In this case, the map $\langle\, , \,\rangle$ is given by

$$\langle g_1, \dots, g_k \rangle = \begin{cases} 1, & g_1 \dots g_k = 1 \\ 0, & \text{otherwise} \end{cases}$$

Thus, one easily sees that for a closed 2-manifold M the state-sum (12.1.7) becomes

$$Z_A(M,K) = \frac{1}{|G|^V} \sum_c 1$$

where V is the number of vertices of K and the sum is over all "colorings" of edges, i.e., ways of assigning a group element g_r to each oriented edge r so that for two opposite orientations of the same edge, $g_{r'} = g_{r''}^{-1}$ and such that the product of colors over the boundary of any 2-cell is equal to 1.

One easily sees that it is a discrete version of a connection in G-bundles: without the factor $\frac{1}{|G|^V}$, this counts G-bundles on M together with trivializations at every vertex. Thus, dividing by $|G|^V$ exactly gives the number of G-bundles on M counted as in Chapter 6; in other words, this recovers the Dijkgraaf–Witten TQFT.

CHAPTER 13

2d TQFT with defects

References: [**CDZR2025**]: review article; [**DKR2011**] one of first constructions; [**Car2016**]

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So far we only considered TQFTs defined on (unstratified) manifolds. Even the manifolds with corners, that we used to motivate the definition of extended TQFT, were meant to be glued back to form an unstratified manifold. Such (extended) TQFTs can be considered as describing physics on a homogenous spacetime. It is then natural to consider more general physical systems, where the different parts of the spacetime are described by different TQFTs. The codimension 1 surface separating the homogenous parts can be thought of as a defect that makes the spacetime non-homogenous. This motivates us to define the notion of TQFT with defects.

13.1. Data for TQFT with defects

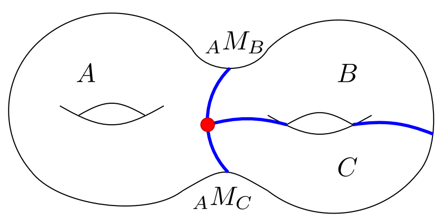


FIGURE 13.1.1. A stratified manfold decorated with algebras A,B and C on different regions. Labelling of defects (shown in blue) by bimodules requires a choice of orientation, which is suppressed here for brevity.

TQFT with defects can be defined as a functor of symmetric monoidal 2-categories:

$$Z^{\text{defect}}\colon \mathbf{Bord}_2(M, \text{Coloring}) \to \mathbf{Vec_k}.$$

Here $\mathbf{Bord}_2(M, \text{Coloring})$ is a 2-bordism decorated with certain extra data given as:

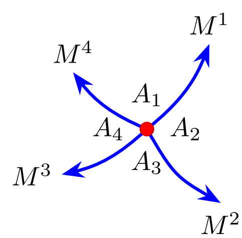
(1) Manfold M with a stratification

$M = M_2 \cup M_1 \cup M_0$

Here M_2 are the 2 dimensional open sets whose boundary constitutes M_1. Stratum M_1 is codimension 1 subspace with boundary M_0. Similarly, M_0 are codimension 2 subspace at the boundary of M_1. (For higher dimensional manifolds, the stratification can have higher codimension strata.)

For example, in the Figure 13.1.1, the stratum M_2 consists of three regions, labelled as A,B, and C. The stratum M_1 consists of blue lines. The red dot at the intersection of blue lines belongs to the stratum M_0.

(2) Coloring
- Connected components of M_2 are colored by semisimple symmetric Frobenius algebras. In Figure 13.1.1, A, B, and C, are three such algebras.
- Connected components of M_1 are colored by bimodules. These 1-dimensional components (called defects) come with a orientation. If the regions to the left and to the right of a defect are colored by algebras A and B respectively, the defect is labelled by a bimodule ${}_AM_B$. If we change the orientation of defect, it is labelled by ${}_BM^*_A$ bimodule. M^* is dual to M, as expected from orientation reversal.
- The vertices (also called junctions) are points where two or more defects meet. For each vertex $u \in M_0$, we need a multilinear map φ_u from cyclic tensor produce of incident bimodules to the field $\mathbf{k}$.



For example, if the junction looks like as given above, then it is assigned a map: $\varphi : \circlearrowleft_{A_1} (M^1 \otimes_{A_2} M^2 \otimes_{A_3} M^3 \otimes_{A_4} M^4) \to \mathbf{k}$. Here, $\circlearrowleft_A M$ is the cokernel of the map $A \otimes M \to M, a \otimes m \mapsto am - ma$. This property of φ ensures that the junction has no preferred starting edge. See [**Car2016**, Section 2.3] for motivation for this data.

13.2. State sum construction of TQFT with defects

In this section we sketch the construction of the functor Z^{defect} that maps the (decorated) bordism $M : U \to V$ to $\mathbf{Vec}_\mathbf{k}$ using a cell decomposition. We suppress the superscript defect for the remainder of the chapter for brevity. We follow the exposition of [**DKR2011**]. (Our $Z \equiv Z^{\text{defect}}$ is equivalent to T^{cw}).

We start by first associating a vector $Q_{p,e,or}$ to each oriented edge, within each polygon p. The contribution $Q_{p,e,or}$ of each edge is defined as:

(13.2.1)
$$Q_{p,e,\text{or}} = \begin{cases} A_a, & \text{if } e \text{ does not cross a defect line and lies within } M_2 \text{ labelled by algebra } a \text{ ,} \\ X_x, & \text{if } e \text{ crosses defect which is labelled } x \text{ with orientation into the polygon } p \text{ ,} \\ X^*_x, & \text{if } e \text{ crosses defect which is labelled } x \text{ with orientation out of the polygon } p \text{ .} \end{cases}$$

We define an intermediate vector space $Q(M)$ by tensoring the contribution of all edges that are not in incoming boundary of M.

$$Q(M) = \bigotimes_{(p,e,\mathrm{or}), e \notin \partial_{\mathrm{in}} M} Q_{p,e,\mathrm{or}} \tag{13.2.2}$$

Note that an internal edge e shared between two polygons p_1 and p_2 will thus contribute twice, as $Q_{p_1,e,\mathrm{or}} \otimes Q_{p_2,e,\bar{\mathrm{or}}}$.

Consider Figure 13.2.1 for example. Edge e_2, irrespective of its orientation, lies entirely in algebra b, so it is labelled A_b. Edge e_1, within polygon p_2, has defect x crossing it such that the defect is enterin p_2. So the triplet $(p_2, e_1, +)$ is labelled X_x. The same edge, within polygon p_1, has defect x going out of the polygon. So it is labelled X_x^* for the triplet $(p_1, e_1, +)$. We only consider those triplets where the orientation matches with that of the polygon. We would have had to consider both orientations of an edge (i.e. for example $(p_1, e_1, +)$ as well as $(p_1, e_1, -)$) only when the polygon closes on itself through that edge. See [**DKR2011**, Section 3.5] for an example.

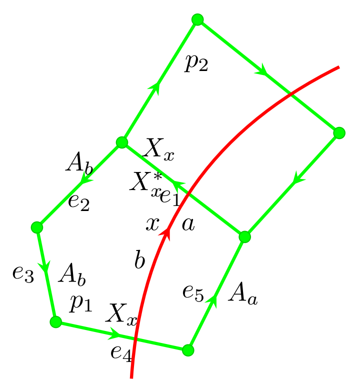


FIGURE 13.2.1. Example of vectors assigned to each edge within a 2-cell.

The functor Z is defined as:

$$Z(M) : Z(U) \xrightarrow{id_{Z(U)} \otimes P(M)} Z(U) \otimes Q(M) \otimes Z(V) \xrightarrow{E(M) \otimes id_{T(V)})} Z(V) \tag{13.2.3}$$

13.3. Independence of cell decomposition

Since the cell decomposition is not a part of the actual defect TQFT data, we need to show that the state sum construction is independent of our choice of cell decomposition.

Part 3

Cobordism hypothesis

Part 3 overview

In this part, we start discussion of extended TQFTs in any dimension, formulating the celebrated cobordims hypothesis.

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CHAPTER 14

Higher categories

In the previous chapters, we have defined extended 2d TQFTs using the language of weak 2-categories and symmetric monoidal 2-categories. In a similar way, it is clear that to define extended TQFTs in higher dimensions, we will need higher categories: if we have a good definition of an n-category, and we can construct the category of cobordisms with corners $\mathbf{Bord}_n$, we can define an extended n-dimensional TQFT as a symmetric monoidal functor $\mathbf{Bord}_n \to \mathcal{C}$, for some symmetric monoidal n-category $\mathcal{C}$.

In this chapter, we discuss various approaches to defining higher categories. We will restrict ourselves to a very brief overview, referring the reader to original papers for details.

14.1. Fundamental groupoid

Reversing the traditional order in mathematics, we begin our discussion of higher categories not with a definition but with an example: whatever definition we give later, it should cover that example. This example is that of a fundamental groupoid, described informally below.

Recall that for a topological space X, we denote by $\Pi(X)$ the fundamental groupoid of X; it is a category whose objects are points of X and morphisms are homotopy classes of paths in X.

We can extend it and define the fundamental infinity groupoid $\Pi_\infty(X)$ as follows:

- Objects of $\Pi_\infty(X)$ are points of X.
- 1-morphisms $x_0 \to x_1$ are paths in X, i.e., continuous maps $I \to X$ such that $\gamma(0) = x_0$, $\gamma(1) = x_1$ (here and below, $I = [0, 1]$).
- 2-morphisms $\gamma_0 \to \gamma_1$, where γ_i are paths from x_0 to x_1, are path homotopies, i.e., continuous maps $h\colon I^2 \to X$ such that $h(0, t) = \gamma_0(t)$, $h(1, t) = \gamma_1(t)$, and for all s, $h(s, 0) = x_0$, $h(s, 1) = x_1$.
- 3-morphisms are homotopies between homotopies, i.e., continuous maps $I^3 \to X$ with suitable restrictions to the boundary.
- ...

This defines an "infinity groupoid", which has morphisms of all orders. We can also define the fundamental n-groupoid $\Pi_{\le n}(X) = \tau_{\le n}(\Pi_\infty(X))$, which only has morphisms up to order n: namely, n-morphisms in $\Pi_{\le n}(X)$ are homotopy classes of n-morphisms in $\Pi_\infty(X)$ (two n-morphisms are homotopic if there exists an $(n+1)$-morphism between them in $\Pi_\infty(X)$).

Note that we have not yet discussed composition in $\Pi_\infty(X)$. It is easy to define composition in the same way as it is done in the usual fundamental groupoid $\Pi_{\le 1}(X)$; however, so defined composition is not associative in Π_∞. For example, for 1-morphisms $\gamma_1, \gamma_2, \gamma_3$, the compositions $\gamma_3(\gamma_2\gamma_1)$ and $(\gamma_3\gamma_2)\gamma_1$ are not equal as maps $I \to X$: in the first case, γ_1 is traveled over $t \in [0, 1/4]$, γ_2 over $t \in [1/4, 1/2]$, and γ_3 over $[1/2, 1]$. For the second composition, the intervals are $[0, 1/2]$, $[1/2, 3/4]$, and $[3/4, 1]$ respectively, so these compositions are not equal; however, they are homotopic to each other. Thus, any definition of an n-category should allow for compositions "associative up to homotopy".

Fundamental n-groupoids $\Pi_{\leq n}(X)$ were suggested by Grothendieck in [**Gro1983**]. He also proposed the following hypothesis.

Conjecture 14.1.1 (Homotopy hypothesis). *Homotopy n-types up to weak equivalence are classified by $\Pi_{\leq n}(X)$ up to equivalence.*

(Recall that a topological space X is called an *n-type* if its homotopy groups $\pi_k(X)$ are trivial for $k > n$.)

Of course, in order to discuss this hypothesis, one needs first to give a definition of n-categories and equivalences between them; thus, we should consider this hypothesis as a natural requirement which should be satisfied by any definition of an n-category.

One can also state an ∞ version of the homotopy hypothesis; namely, it says that topological spaces up to weak equivalence are classified by their ∞ fundamental groupoids up to equivalence.

14.2. Strict and weak n-categories

After giving the motivating example of n-groupoids, we can try to define the notion of an n-category.

The simplest approach is considering strict n-categories. These categories are not very hard to define. Namely, a strict n-category must have objects, and for any objects X, Y, we must have an $(n-1)$-category $\mathrm{Hom}(X, Y)$, together with a composition functor of $(n-1)$-categories $\mathrm{Hom}(X, Y) \times \mathrm{Hom}(Y, Z) \to \mathrm{Hom}(X, Z)$, which must be associative. In a similar way one defines identity morphisms. (Category theorists would say that an n-category is a category enriched over $(n-1)$-categories.)

However, this definition is not of much use. First, virtually all known examples are non-strict — compositions are not strictly associative but only "associative up to higher order isomorphisms", as we have already seen in the example of fundamental groupoids. Moreover, for $n \geq 3$, it turns out that this is more than just a technical problem: not every weak 3-category is equivalent to a strict 3-category (unlike the case of 2-categories). For example, it can be shown that $\Pi_{\leq 3}(S^2)$, the truncated fundamental groupoid of S^2, is not equivalent to a strict 3-groupoid (see [**Sim2012**, Theorem 2.7.2]).

Thus, we have no choice but to forget strict categories and start developing the theory of weak n-categories, where compositions are "associative up to higher order isomorphisms".

Unfortunately, giving an algebraic definition of a weak n-category is very hard to do. A definition of a weak 3-category exists (see [**GPS1995**]), but it is so long that I was never able to get to the end of it; for $n \geq 4$, the algebraic definitions are even more cumbersome. Some summary of algebraic approaches is collected in [**Lei2004**]. On the other hand, there are a number of rather easy to define examples of (weak) 3- and 4-categories, some of which we will give later.

14.3. Weak categories and infinity categories

An alternative approach to the notion of a weak n-category is to use the ideas of topology, or more precisely, homotopy theory. The key idea of this approach is that instead of defining a single composition operation which is weakly associative, we allow many composition operations, which must form a contractible space. Paths in this contractible space correspond to $(k+1)$-isomorphisms between the corresponding compositions, homotopies between paths correspond to $(k+2)$-isomorphisms, etc. Different compositions give different points in this contractible space; the fact that they are all isomorphic follows from the fact that this space is connected, and all compatibility relations between these isomorphisms follow from the fact that the space of compositions is contractible.

This idea can be used to define weak n-categories; it can also be used to define (∞, n) categories. Informally, an (∞, n) category is a category which has morphisms of all orders $k \geq 1$, but for $k > n$, all k-morphisms are invertible.

Of course, the above is not yet a definition; even for $n = 1$, developing this idea and turning it into a precise definition is non-trivial. In fact, there are many competing ways to define the notion of $(\infty, 1)$ category, all giving different formalizations of the above idea. Among them:

- Quasicategories, also known as weak Kan complexes
- Complete Segal spaces
- Segal categories

Fortunately, it turns out that in a certain sense all these definitions are equivalent. There are a number of papers discussing this, including [**Toë2004**], [**BSP2021**], and the book [**RV2022**]; for a short introduction, there is also Emily Riehl's article [**Rie2023**] in the Notices of the AMS. We will not be discussing this at all; we will restrict ourselves to giving just one definition, based on Segal categories. This construction takes some time; we will do it in the next several sections. Our exposition follows the paper [**SP2014**]; a detailed exposition of the theory of Segal categories as a model for (∞, n) categories can be found in [**Sim2012**].

14.4. Nerve of a category

Before giving a definition of a weak category, let us take another look at the definition of a usual category. Let $\mathcal{C}$ be a category; define a collection of sets X_n, $n \geq 0$ by

$$\begin{aligned} &X_0 = \operatorname{Obj}\mathcal{C} \quad \text{(objects of } \mathcal{C}) \\ &X_1 = \{(x_0, x_1, f) \mid x_i \in \mathcal{C}, f\colon x_0 \to x_1\} \quad \text{(morphisms in } \mathcal{C}) \\ &\quad \dots \\ &X_n = \{(x_0, \dots, x_n, f_1, \dots, f_n) \mid x_i \in \mathcal{C}, f_i\colon x_{i-1} \to x_i\} \\ &\qquad \text{(composable } n\text{-chains of morphisms in } \mathcal{C}) \end{aligned} \tag{14.4.1}$$

Then all compositions and units in $\mathcal{C}$ can be described in terms of maps between these sets. For example, we have the composition map

$$X_2 \to X_1$$
$$(x_0 \xrightarrow{f_1} x_1 \xrightarrow{f_2} x_2) \mapsto (x_0 \xrightarrow{f_2 f_1} x_2)$$

and a unit map

$$X_0 \to X_1$$
$$x \mapsto (x \xrightarrow{1_x} x)$$

More generally, we have *face maps*

$$\begin{aligned} d_i \colon X_n &\to X_{n-1}, \qquad 0 \le i \le n \\ (\dots x_{i-1} \xrightarrow{f_i} x_i \xrightarrow{f_{i+1}} x_{i+1} \to \dots) &\mapsto (\dots x_{i-1} \xrightarrow{f_{i+1} f_i} x_{i+1} \to \dots) \end{aligned} \tag{14.4.2}$$

(for $i = 0$, this map deletes both object x_0 and morphism f_1; for $i = n$, it deletes x_n and f_n).

We also have *degeneracy maps*

$$\begin{aligned} s_i \colon X_n &\to X_{n+1}, \qquad 0 \le i \le n \\ (\dots \xrightarrow{f_i} x_i \xrightarrow{f_{i+1}} \dots) &\mapsto (\dots \xrightarrow{f_i} x_i \xrightarrow{1_{x_i}} x_i \xrightarrow{f_{i+1}} \dots) \end{aligned} \tag{14.4.3}$$

One should think of d_i as "erasing" the object x_i in the chain, and s_i as doubling x_i and inserting the identity morphism. (The names "face map" and "degeneracy map" are motivated by the relation to homotopy theory; see Section 14.5.)

These maps satisfy a number of relations, which follow from (and in fact are equivalent to) the associativity and unit axioms for the category $\mathcal{C}$. One can write these relations explicitly; however, there is a more elegant way to do that, using the notion of *simplicial set*.

Definition 14.4.1. The *simplex category* $\mathbf{\Delta}$ is the category whose objects are totally ordered sets $[n] = \{0, 1, \dots, n\}$, $n \ge 0$, and morphisms are (non-strictly) order preserving functions.

In particular, for each $n \ge 0$, there are $n+1$ injections $d^i \colon [n-1] \to [n]$, called the coface maps, and $n+1$ surjections $s^i \colon [n+1] \to [n]$, called the codegeneracy maps, defined as follows:

$$d^i(k) = \begin{cases} k & k < i \\ k+1 & k \ge i \end{cases} \qquad\qquad s^i(k) = \begin{cases} k & k \le i \\ k-1 & k > i \end{cases}$$

The i-th coface map d^i misses the element i in the image, while s^i sends two elements to i. It is easy to show that these maps generate all morphisms in $\mathbf{\Delta}$: any order-preserving map is a composition of coface and codegeneracy maps.

Definition 14.4.2. A simplicial set is a functor $X \colon \mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Set}$.

It is common to use notation $X_n = X([n])$ and $d_i = X(d^i) \colon X_n \to X_{n-1}$, $s_i = X(s^i) \colon X_n \to X_{n+1}$.

(Again, the motivation for the name "simplicial set" will be given in Section 14.5.)

It is easy to see that simplicial sets form a category, with morphisms being morphisms of functors; in other words, given two simplicial sets $X_\bullet$, $Y_\bullet$, a morphism $f \colon X_\bullet \to Y_\bullet$ is a collection of maps $f_n \colon X_n \to Y_n$ which commute with all d_i, s_i. We denote by **sSet** the category of simplicial sets.

Theorem 14.4.1. *Let $\mathcal{C}$ be a category. Then formulas (14.4.1)–(14.4.3) define a simplicial set, called the* nerve *of $\mathcal{C}$.*

EXERCISE 14.4.1. Write explicitly the associativity axiom in $\mathcal{C}$ as an equality of two maps $X_3 \to X_1$, and write the corresponding relation between coface maps d^i in $\mathbf{\Delta}$.

One can ask if it is possible to describe which simplicial sets appear as nerves of categories. To answer that, we recall the notion of fiber product.

Assume that we have two sets A_1, A_2 and maps $p_i \colon A_i \to B$. The fiber product $A_1 \times_B A_2$ is defined by

$$A_1 \times_B A_2 = \{(a_1, a_2) \in A_1 \times A_2 \mid p_1(a_1) = p_2(a_2)\}$$

Clearly, we have canonical maps $A_1 \times_B A_2 \to A_i$ which make the diagram below commutative (this diagram is frequently called the *pullback square*)

$$\begin{tikzcd} A_1 \times_B A_2 \arrow[r] \arrow[d] & A_2 \arrow[d] \\ A_1 \arrow[r] & B \end{tikzcd} \tag{14.4.4}$$

A better definition — which also works in other situations, for example when A_i, B are topological spaces and all maps are continuous — defines $A_1 \times_B A_2$ as the universal object in the category of diagrams: for every diagram below, there is a unique map $P \to A_1 \times_B A_2$

$$\begin{tikzcd} P \arrow[r] \arrow[d] & A_2 \arrow[d] \\ A_1 \arrow[r] & B \end{tikzcd}$$

It is easy to see that for the nerve of a category $\mathcal{C}$, in other words, when X_n is given by (14.4.1), we have maps

$$p_i \colon X_n \to X_1$$
$$(\dots \to x_{i-1} \xrightarrow{f_i} x_i \to \dots) \mapsto (x_{i-1} \xrightarrow{f_i} x_i).$$

Thus, by the universality property of the fiber product, we have the *Segal map*

$$s_n \colon X_n \to X_1 \times_{X_0} X_1 \times_{X_0} X_1 \times \dots \times_{X_0} X_1. \tag{14.4.5}$$

Maps p_i (and thus, the Segal map s_n) can be defined for any simplicial set: they correspond to the map $[1] \to [n] \colon 0 \mapsto (i-1), 1 \mapsto i$ in $\mathbf{\Delta}$.

THEOREM 14.4.2. *A simplicial set X is a nerve of some category $\mathcal{C}$ if and only if the following* Segal condition *holds: for every $n \geq 1$, the Segal map* (14.4.5) *is a bijection*

$$X_n \simeq X_1 \times_{X_0} X_1 \times_{X_0} X_1 \times \dots \times_{X_0} X_1. \tag{14.4.6}$$

The proof of this theorem is straightforward: it is just another way of saying that any composable chain of morphisms consists of an n-tuple of morphisms where the target of the i-th morphism coincides with the source of the $(i+1)$-st one.

Thus, (usual) categories can be described as simplicial sets satisfying the Segal condition (14.4.6).

14.5. Simplicial sets as models of topological spaces

In the previous section, simplicial sets (i.e., functors $\mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Set}$) were introduced as a natural language to describe nerves of categories. However, historically simplicial sets appeared in topology. In this section, we briefly review that, referring the reader to the books [**May1992**], [**GJ1999**] for details.

Throughout this section, we will denote by **sSet** the category of simplicial sets.

EXAMPLE 14.5.1. Let X be a topological space. For any $n \geq 0$, define the set $\mathrm{Sing}(X)_n$ by

$$\mathrm{Sing}(X)_n = \{\text{continuous maps } \Delta^n \to X\}$$

where Δ^n is an n-dimensional simplex

$$\Delta^n = \Big\{(t_0, t_1, \ldots, t_n) \in \mathbb{R}^{n+1} \mid 0 \leq t_i \leq 1, \quad \sum t_i = 1\Big\}. \tag{14.5.1}$$

Vertices of Δ^n are in natural bijection with $[n]$, and any order-preserving map $f\colon [m] \to [n]$ defines an affine linear map $f\colon \Delta^m \to \Delta^n$. Precomposition with f defines on $\mathrm{Sing}_\bullet$ the structure of a simplicial set.

Thus, we have a functor

$$\mathrm{Sing}\colon \mathbf{Top} \to \mathbf{sSet}$$

usually called the *singular complex* of X. Moreover, many usual constructions for topological spaces, such as the notion of homotopy of maps, or the notion of direct product, can be easily generalized to simplicial sets. (In fact, it can be shown that for any topological space X, the singular complex $\mathrm{Sing}(X)$ has an additional property, existence of certain extensions; such simplicial sets are called *Kan complexes*.)

It turns out that we also have a *geometric realization functor* $|\quad|\colon \mathbf{sSet} \to \mathbf{Top}$; roughly, for a simplicial set K we take the disjoint union of (topological) simplices, one copy of Δ^n for each point $x \in K_n$, and then "glue" them together using the face maps. Details can be found, e.g., in [**May1992**, Chapter III]. In particular, $|K|$ is always a CW complex.

The functors $\mathrm{Sing}\colon \mathbf{Top} \to \mathbf{sSet}$ and $|\quad|\colon \mathbf{sSet} \to \mathbf{Top}$ are adjoints of each other: we have a functorial isomorphism

$$\mathrm{Hom}_{\mathbf{sSet}}(K, \mathrm{Sing}(X)) \simeq \mathrm{Hom}_{\mathbf{Top}}(|K|, X), \qquad X \in \mathbf{Top}, K \in \mathbf{sSet}$$

([**May1992**, §15]). The unit and counit of this adjunction, therefore, give us morphisms

$$\Psi\colon K \to \mathrm{Sing}(|K|),$$
$$\Phi\colon |\mathrm{Sing}(X)| \to X.$$

The following theorem is well-known (see e.g., [**May1992**, §16]; [**GJ1999**, Theorem I.11.4]).

THEOREM 14.5.1.

(1) *For any simplicial set K, $\Psi\colon K \to \mathrm{Sing}(|K|)$ is an embedding of simplicial sets, which is a weak homotopy equivalence. If K is a Kan complex, then Ψ is a homotopy equivalence.*
(2) *For any topological space X, $\Phi\colon |\mathrm{Sing}(X)| \to X$ is surjective and is a weak homotopy equivalence. If X is a CW complex, then Φ is a homotopy equivalence.*

Thus, one can think of simplicial sets as a combinatorial model for the study of homotopy of topological spaces. In many ways, **sSet** is easier to work with than the category **Top**; for example, **sSet** has a natural structure of a Quillen closed model category (which we are not going to discuss; interested readers should check, e.g., [**Sim2012**, Section 7] or [**GJ1999**, Section I.11]).

14.6. Enriched categories

In the previous sections, we gave a description of a (usual) category $\mathcal{C}$ as a simplicial set satisfying additional Segal condition. In a similar way, one can describe *enriched categories*, i.e., categories where Hom spaces are not just sets, but have additional structure — e.g., they are topological spaces. For example, if we denote by **Top** the category of topological spaces with continuous maps, a **Top**-enriched category is the following data:

(1) A class S of objects of $\mathcal{C}$
(2) For any pair of objects $x, y \in S$, a topological space $\mathrm{Hom}(x, y)$
(3) For any triple of objects, a continuous map $\mathrm{Hom}(x, y) \times \mathrm{Hom}(y, z) \to \mathrm{Hom}(x, z)$
(4) For any object $x \in S$, a point $1_x \in \mathrm{Hom}(x, x)$

satisfying the usual (strict) associativity and unit axioms. As usual, we restrict our attention to small categories, so we assume that objects form a set.

As before, we can describe such categories using simplicial sets — or more generally, simplicial objects in an appropriate category.

Definition 14.6.1. Let $\mathcal{M}$ be a category. A *simplicial object* in $\mathcal{M}$ is a functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathcal{M}$.

We denote the category of simplicial objects in $\mathcal{M}$ by $\mathbf{sSet}(\mathcal{M})$.

In particular, for $\mathcal{M} = \mathbf{Top}$, simplicial objects in **Top** are usually called *simplicial spaces*.

Recall that for a category $\mathcal{C}$, we have defined the simplicial set $X_\bullet = N\mathcal{C}$ (the nerve of $\mathcal{C}$) by (14.4.1). If $\mathcal{C}$ is a **Top**-enriched category, then $(N\mathcal{C})_n$ is not just a set but a topological space (since we have not defined any topology on the set of objects, we consider $X_0 = \mathrm{Obj}\,\mathcal{C}$ as a topological space with the discrete topology); thus, $N\mathcal{C}$ is a simplicial space.

The following theorem is an immediate generalization of Theorem 14.4.2.

Theorem 14.6.1. *A simplicial space $X_\bullet$ is the nerve of a* **Top**-*enriched category if and only if the following properties hold:*

- *X_0 is discrete*
- *For any $n \geq 1$, Segal map (14.4.5) is a homeomorphism.*

In a similar way, we can consider other enriched categories. Namely, if we have a category $\mathcal{M}$ which has finite products (and thus has a terminal object $\mathbf{1}$ corresponding to the product of the empty collection of objects in $\mathcal{M}$, see e.g., [**ML1998**, Section III.5]), an $\mathcal{M}$-enriched category $\mathcal{C}$ is the following data:

(1) A class S of objects of $\mathcal{C}$
(2) For any pair of objects $x, y \in S$, an object $\mathrm{Hom}(x, y) \in \mathcal{M}$
(3) For any triple of objects, an $\mathcal{M}$-morphism $\mathrm{Hom}(x, y) \times \mathrm{Hom}(y, z) \to \mathrm{Hom}(x, z)$
(4) For any object $x \in S$, an $\mathcal{M}$-morphism $\mathbf{1} \to \mathrm{Hom}(x, x)$

satisfying the usual (strict) associativity and unit axioms.

To describe such categories as simplicial objects, we need to deal with one minor issue. Namely, in the definition above, we assumed that objects of $\mathcal{C}$ form a class (or a set, for small categories). For **Top**-enriched categories, we considered this set as an object in **Top**, considering a set as a topological space with the discrete topology; however, for general $\mathcal{M}$ we cannot do it. Thus, we will need to modify our construction of the nerve of a category. One way to do that is using the following trick.

Let S be a set. Define the category of S-labelled simplices $\mathbf{\Delta}_S$ as the category whose objects are finite ordered sequences $(x_0, \dots, x_n)$, $x_i \in S$, and morphisms $(x_0, \dots, x_m) \to (y_0, \dots, y_n)$ are (non-strict) order-preserving maps $f\colon [m] \to [n]$ which satisfy $y_{f(i)} = x_i$. One should think of the category $\mathbf{\Delta}_S$ as finite ordered sets together with a labeling of elements of the ordered set by elements of S.

Definition 14.6.2. Let $\mathcal{M}$ be a category with finite products, and let $\mathcal{C}$ be a small $\mathcal{M}$-enriched category with set of objects S. Define, for any $n \geq 0$,

$$N\mathcal{C}\colon \mathbf{\Delta}_S^{\mathrm{op}} \to \mathcal{M}$$

by

$$N\mathcal{C}(x_0, \dots, x_n) = \mathrm{Hom}(x_0, x_1) \times \dots \times \mathrm{Hom}(x_{n-1}, x_n) \in \mathcal{M}$$

(in particular, for $n = 0$ it gives $N\mathcal{C}(x_0) = \mathbf{1}$, the terminal object in $\mathcal{M}$). Then $N\mathcal{C}$ is a functor $\mathbf{\Delta}_S^{\mathrm{op}} \to \mathcal{M}$, called the nerve of $\mathcal{C}$.

Theorem 14.6.2. *Let $\mathcal{M}$ be a category with finite products, and let S be a set. A functor $X\colon \mathbf{\Delta}_S^{\mathrm{op}} \to \mathcal{M}$ is a nerve of some $\mathcal{M}$-enriched category $\mathcal{C}$ with set of objects S if and only if it satisfies the following two properties:*

(1) $X(x_0) = \mathbf{1}$ *for any* x_0 (*considered as a coloring of simplex* $[0]$)
(2) *Segal condition: the morphism*

$$s_n\colon X(x_0, \dots, x_n) \to X(x_0, x_1) \times \dots \times X(x_{n-1}, x_n)$$

is an isomorphism in $\mathcal{M}$.

The proof is very similar to the proof of Theorem 14.4.2, and as before, is left to the reader.

14.7. Segal categories

We can now give a definition of one of the possible models of (∞, n) categories, namely, the Segal category. We begin with the definition of $(\infty, 1)$ category. It is based on the description of a category enriched over **Top** as a simplicial space as in Theorem 14.6.2, but with one important change.

Recall that we have defined a simplicial space as a simplicial object in the category **Top**, i.e., a functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathbf{Top}$. Recall also that a continuous map of topological spaces $f\colon X \to Y$ is called a *weak homotopy equivalence* if for every $n \geq 0$, it defines an isomorphism of homotopy groups $\pi_n(X) \simeq \pi_n(Y)$. In particular, if f is a homotopy equivalence, then it is automatically a weak homotopy equivalence. By a famous result of Whitehead, if X, Y are CW complexes, then the converse is also true; for general topological spaces, it is not so.

Definition 14.7.1. A *Segal category* $\mathcal{C}$ is a simplicial space which satisfies the following additional conditions:

(1) X_0 is discrete.
(2) The Segal map (14.4.5)

$$s_n : X_n \to X_1 \times_{X_0} X_1 \times_{X_0} X_1 \times \cdots \times_{X_0} X_1$$

is a weak homotopy equivalence.

We call X_0 the set of objects of $\mathcal{C}$ and for $x_0, x_1 \in X_0$, we call the space $X_1(x_0, x_1) = \{x_0\} \times_{X_0} X_1 \times_{X_0} \{x_1\}$ the space of morphisms $\mathrm{Hom}_{\mathcal{C}}(x_0, x_1)$.

We denote by **Seg** the category of Segal categories; it is a full subcategory in **sSet**(**Top**) whose objects are Segal categories.

Informally, if we think of $X_1 \times_{X_0} X_1 \times_{X_0} X_1 \times \cdots \times_{X_0} X_1$ as the space of composable chains of morphisms, then X_n is "composable chain of morphisms plus some additional data needed to define composition", and we require that the space of "additional data" be contractible.

Example 14.7.1. Let X be a topological space. Then the fundamental groupoid $\Pi_\infty(X)$, informally described in Section 14.1, can be rigorously defined as a Segal category. Namely, we let $X_0 =$ set of points of X (considered with discrete topology), and for $n \geq 1$, we define

$$X_n = \{\text{continuous functions } \Delta^n \to X\}$$

where Δ^n is the topological n-dimensional simplex (14.5.1); in other words, this is exactly the definition of singular complex functor Sing, but now we consider $X_n, n \geq 1$, as a topological space rather than a set.

In particular, we see that X_1 is the topological space of paths in X, and Segal map $s_n : X_n \to X_1 \times_{X_0} \cdots \times_{X_0} X_1$ sends a map $f : \Delta^n \to X$ to its restriction on the collection of edges $[i-1, i]$. Thus, one can think of X_n as composable collection γ_i of paths in X together with a map $\gamma : \Delta^n \to X$ whose restriction to edge $[i-1, i], i = 1 \dots n$, is equal to γ_i. Segal condition immediately follows from the fact that the union of these edges is a retract of the n-simplex.

In particular, given such a map γ, we can restrict it to the edge $[0, n]$, which gives us a path connecting x_0 with x_n in X. Any such path can be called a composition of γ_i; thus, we see that we have not one but many possible compositions, but the space of all possible compositions (for given paths γ_i) is contractible.

Segal category defined above is usually called the *Poincaré–Segal groupoid*; see [**Sim2012**, Sections 15.2, 15.3]. We will denote it $\Pi_S(X)$ to distinguish it from (informally defined) Π_∞.

14.8. Segal n-categories

The notion of Segal category, defined in the previous section, is one possible way of defining $(\infty, 1)$ categories. To define (∞, n) categories, we need to iterate this construction: informally, $(\infty, 2)$ category $\mathcal{C}$ should be given by a definition similar to that of a Segal category, but now 1-morphisms, instead of being topological spaces, must form Segal categories. However, to do that, we need to have a notion of weak equivalence between Segal categories (which we have not defined so far) and so on.

Following [**SP2014**], we give the following definition.

Definition 14.8.1. A *Segalic relative category* is a category $\mathcal{M}$ together with the following additional data:

- A subset $\mathcal{W}$ of morphisms in $\mathcal{M}$, called the set of *weak equivalences*.
- A functor $\tau_{\le 0}\colon \mathcal{M} \to \mathbf{Set}$.

This data has to satisfy the following conditions:

- $\mathcal{M}$ has finite products, and $\tau_{\le 0}$ preserves finite products.
- $\mathcal{W}$ includes all identity morphisms.
- $\mathcal{W}$ is closed under products: if $f\colon x \to y$ is in $\mathcal{W}$, then for any $z \in \mathcal{M}$, $f \times 1_z \in \mathcal{W}$.
- We have "2 out of 3" condition: given morphisms $f\colon x_0 \to x_1$, $g\colon x_1 \to x_2$ in $\mathcal{M}$, if any two out of three morphisms $f, g, g \circ f$ are in $\mathcal{W}$, then so is the third one.
- For any $f \in \mathcal{W}$, $\tau_{\le 0}(f)$ is a bijection.

Example 14.8.1. Let $\mathcal{M} = \mathbf{Top}$; define $\mathcal{W}$ to be the set of weak homotopy equivalences, and let $\tau_{\le 0} = \pi_0$ be the set of connected components.

Definition 14.8.2. Let $\mathcal{M}$ be a Segalic relative category. An $\mathcal{M}$-enriched Segal category $\mathcal{C}$ is a pair (S, X), where S is a set (called the set of objects of $\mathcal{C}$) and X is a functor $\mathbf{\Delta}_S^{\mathrm{op}} \to \mathcal{M}$ (compare with Theorem 14.6.2) which satisfies the following conditions:

(1) For any $x_0 \in S$, $X(x_0) = \mathbf{1}$ is the terminal object in $\mathcal{M}$.
(2) For any $n \ge 1$, the Segal map

$$s_n\colon X(x_0, \dots, x_n) \to X(x_0, x_1) \times \cdots \times X(x_{n-1}, x_n)$$

is a weak equivalence in $\mathcal{M}$ (i.e., is in $\mathcal{W}$).

We define functors between $\mathcal{M}$-enriched Segal categories in the same way as before; we denote by $\mathbf{Seg}(\mathcal{M})$ the category of all $\mathcal{M}$-enriched Segal categories.

Example 14.8.2. For $\mathcal{M} = \mathbf{Top}$, the definition of $\mathbf{Top}$-enriched Segal category is equivalent to that of a Segal category given in Definition 14.7.1: $\mathbf{Seg}(\mathbf{Top}) = \mathbf{Seg}$.

Example 14.8.3. Let $\mathcal{M} = \mathbf{Cat}$ be the category of small categories. Define the functor $\tau_{\le 0}\colon \mathbf{Cat} \to \mathbf{Set}$ by $\tau_{\le 0}(\mathcal{C})$ = set of isomorphism classes of objects in $\mathcal{C}$, and let $\mathcal{W}$ be the set of equivalences between categories. Then a $\mathbf{Cat}$-enriched Segal category is the same as a weak 2-category.

Details can be found in [**Tam1999**, Section 4] (see also [**LP2008**]).

For an $\mathcal{M}$-enriched Segal category $\mathcal{C} = (S, X)$, we define the (usual) category $\tau_{\le 1}\mathcal{C}$ which has the same set of objects as $\mathcal{C}$, and $\mathrm{Hom}_{\tau_{\le 1}\mathcal{C}}(x_0, x_1) = \tau_{\le 0} X(x_0, x_1)$. This category is also called the *homotopy category* of $\mathcal{C}$.

Theorem 14.8.1. *Let $\mathcal{M}$ be a Segalic relative category. Then $\mathbf{Seg}(\mathcal{M})$ is also a Segalic relative category, with $\tau_{\le 0}$, $\mathcal{W}$ defined as follows:*

- *For an $\mathcal{M}$-enriched category $\mathcal{C}$, we define*

$$\tau_{\le 0}(\mathcal{C}) = \text{ set of isomorphism classes of objects in } \tau_{\le 1}(\mathcal{C})$$

- *For $\mathcal{M}$-enriched categories $\mathcal{C}$, $\mathcal{D}$, a functor $F\colon \mathcal{C} \to \mathcal{D}$ is a weak equivalence if it satisfies the following conditions:*
  (1) *F induces equivalence of usual categories $\tau_{\le 1}\mathcal{C} \to \tau_{\le 1}\mathcal{D}$*
  (2) *F induces a weak equivalence of objects in $\mathcal{M}$*

$$\mathrm{Hom}_{\mathcal{C}}(x_0, x_1) \simeq \mathrm{Hom}_{\mathcal{D}}(F(x_0), F(x_1))$$

A proof of this theorem can be found in [**SP2014**, Theorem 5.13].

Thus, we can iterate the construction of Segal categories: starting with a Segalic relative category $\mathcal{M}$, we can successively define categories $\mathbf{Seg}(\mathcal{M})$, $\mathbf{Seg}^2(\mathcal{M}) = \mathbf{Seg}(\mathbf{Seg}(\mathcal{M}))$, etc.

The following definition was given in [**Tam1999**].

Definition 14.8.3. A Tamsamani weak n-category is an object of $\mathbf{Seg}^n(\mathbf{Set})$, where $\mathbf{Set}$ is considered as a Segalic relative category with $\tau_{\le 0} = \mathrm{id}$ and $\mathcal{W} =$bijections.

We denote by $\mathbf{Tam}^n = \mathbf{Seg}^n(\mathbf{Set})$ the category of all Tamsamani weak n-categories.

For example, a Tamsamani weak 0-category is a set, a Tamsamani weak 1-category is the usual category, a Tamsamani weak 2-category is the usual weak 2-category, or bicategory (see Example 14.8.3), etc.

Tamsamani also defines the notion of weak n-groupoid and defines the fundamental n-groupoid of a topological space. He then shows that in this theory of weak n-categories, the homotopy hypothesis holds: the functor $X \mapsto \Pi_{\le n}(X)$ gives rise to an equivalence of homotopy theories of topological n-types and n-groupoids (see [**Tam1999**, Theorem 8.0]). Thus, this theory is a reasonable candidate for the theory of weak n-categories.

Remark 14.8.1. The definition above is inductive: we define a Tamsamani n-category as a Segal category enriched over $(n-1)$-categories. Alternatively, one can also define Tamsamani weak n-categories as a suitable subcategory in the category of all functors $(\mathbf{\Delta}^{\times n})^{\mathrm{op}} \to \mathbf{Set}$, where $\mathbf{\Delta}^{\times n}$ is the n-fold product of simplex categories; see [**Pao2019**] for detailed discussion.

The same ideas can now be also used to define (∞, n) categories.

Definition 14.8.4. A Segal n-category is an object of $\mathbf{Seg}^n(\mathbf{Top})$, i.e., a Segal category enriched over $\mathbf{Seg}^{n-1}(\mathbf{Top})$.

We will take this as a definition of an (∞, n) category:

Definition 14.8.5. An (∞, n) category is the same as a Segal n-category.

In particular:

- An $(\infty, 0)$ category is the same as a topological space.
- An $(\infty, 1)$ category is the same as a Segal category as defined in Section 14.7.
- ...

Note that by the definition of $\mathbf{Seg}^n$, if $\mathcal{C}$ is an (∞, n) category, $n > 0$, then for any two objects $x, y \in \mathcal{C}$, $\mathrm{Hom}_{\mathcal{C}}(x, y)$ is an $(\infty, n-1)$ category. We can use it to define k-morphisms in an (∞, n) category: for $n = 0$ (i.e., when $\mathcal{C} = X$ is a topological space), k-morphisms are defined as in Section 14.1; for $n > 0$, k-morphisms in $\mathcal{C}$ are $(k-1)$-morphisms in the $(\infty, n-1)$ category $\mathrm{Hom}_{\mathcal{C}}(x, y)$ for some objects $x, y \in \mathcal{C}$.

We also note that the functor $\tau_{\le 0} = \pi_0 \colon \mathbf{Top} \to \mathbf{Set}$ gives rise to a truncation functor $\mathbf{Seg}^n(\mathbf{Top}) \to \mathbf{Seg}^n(\mathbf{Set})$; thus, any (∞, n) category can be truncated to a (Tamsamani) weak n-category. Moreover, combining it with truncation functors $\tau_{\le k} \colon \mathbf{Tam}^n \to \mathbf{Tam}^k$, $k \le n$, we can define a truncation functor $\tau_{\le k}$ from (∞, n) categories to weak k-categories.

14.9. Bordism category

Recall that we have defined the notion of manifolds with corners.

THEOREM 14.9.1. *There exists an* (∞, n) *category* $\mathbf{Bord}_n$ *whose objects are points, ..., n-morphisms are n-dimensional manifolds with corners.*

This construction was first suggested by Lurie in [**Lur2009a**]; an improved version, fixing some shortcomings of Lurie's, is given in [**CS2019**]. Note, however, that both references define (∞, n) categories in terms of n-fold complete Segal spaces, which is slightly different from the Segal n-category approach we take here. However, as mentioned before, all models of (∞, n) categories are equivalent, so one should be able to rewrite their construction in the language of Segal n-categories (however, to the best of my knowledge, it has not been explicitly done yet).

CHAPTER 15

Cobordism hypothesis

Main reference: [**Lur2009a**]; short review paper: [**Fre2013**].

15.1. Definition of extended TQFT and cobordism hypothesis

In this section, we finally give a definition of an extended TQFT in any dimension n.

Recall that in Section 14.9 we have defined the (∞, n) category $\mathbf{Bord}_n$ of n bordisms with corners; to be precise, we have defined two versions: framed bordism category $\mathbf{Bord^{fr}}_n$ and oriented bordism category $\mathbf{Bord^{or}}_n$. Moreover, each of these categories has a structure of symmetric monoidal (∞, n) category, with the monoidal structure given by the disjoint union.

We can now give the key definition of these lectures.

Definition 15.1.1. Let $\mathcal{C}$ be a symmetric monoidal (∞, n) category. A framed extended n-dimensional TQFT is a symmetric monoidal functor

$$Z\colon \mathbf{Bord^{fr}} \to \mathcal{C}.$$

In a similar way one defines the notion of an oriented TQFT.

We denote the set of all such functors by

$$\mathrm{Fun}^{\otimes}(\mathbf{Bord^{fr}}, \mathcal{C})$$

(respectively, $\mathrm{Fun}^{\otimes}(\mathbf{Bord^{fr}}, \mathcal{C})$). By FIXME, this is itself an $(\infty, 0)$ category, so we can talk, e.g., about isomorphism between two TQFTs.

A fundamental result in this area is the following result.

Theorem 15.1.1 (Cobordism hypothesis). *Let $\mathcal{C}$ be a symmetric monoidal (∞, n) category. Then we have an equivalence of categories*

$$\mathrm{Fun}^{\otimes}(\mathbf{Bord^{fr}}, \mathcal{C}) \simeq \widetilde{\mathcal{C}^{\mathrm{fd}}}$$

given by $Z \mapsto Z(\bullet)$, where $\mathcal{C}^{\mathrm{fd}}$ the full subcategory of $\mathcal{C}$ whose objects are fully dualizable objects in $\mathcal{C}$, and $\widetilde{\mathcal{C}^{\mathrm{fd}}}$ is the subcategory obtained from $\mathcal{C}^{\mathrm{fd}}$ by discarding all non-invertible morphisms (of all orders). By definition, $\widetilde{\mathcal{C}^{\mathrm{fd}}}$ is an $(\infty, 0)$ category.

Note that here it is crucial that we use the framed version of bordism category.

Before discussing the proof, let us note that it immediately implies the following.

Corollary 15.1.2. *An n-dimensional framed extended TQFT is determined uniquely up to isomorphism by the object $Z(\bullet) \in \mathcal{C}$ which must be fully dualizable.*

This is a very far-reaching generalization of Theorem 3.3.2 which proved a similar statement for 1-dimensional TQFTs.

The cobordism hypothesis has a very long history. It was first stated as a conjecture by Baez and Dolan [**BD1995**] in 1995. In 2008, Jacob Lurie published a famous paper [**Lur2009a**], where he restated this conjecture in the language of (∞, n) categories and outlined the proof. The crucial ingredient of the proof is result of Igusa on framed Morse functions [**Igu1987**]. Since then, many people worked on filling the details of the proof and extending it to other situations, such as [**AF**].

15.2. $SO(n)$ action and oriented cobordism hypothesis

The cobordism hypothesis gives a description of framed extended TQFTs. Now, let us discuss other versions of extended TQFT — most importantly, oriented TQFTs.

Since we have an obvious action of the group $\mathrm{GL}(n, \mathbb{R})$ on $\mathbb{R}^n$, it gives an action of $\mathrm{GL}(n, \mathbb{R})$ on the set of all framings of a given manifold and thus on the set (or rather ∞ category) $\mathrm{Fun}^{\otimes}(\mathbf{Bord^{fr}}, \mathcal{C})$ of all framed TQFTs. By cobordism hypothesis, $\mathrm{Fun}^{\otimes}(\mathbf{Bord^{fr}}, \mathcal{C}) \simeq \widetilde{\mathcal{C}^{\mathrm{fd}}}$; thus, we get an action of $\mathrm{GL}(n, \mathbb{R})$ on the $(\infty, 0)$ category $\widetilde{\mathcal{C}^{\mathrm{fd}}}$.

Let us now consider oriented extended TQFTs, i.e. monoidal functors $\mathbf{Bord^{or}} \to \mathcal{C}$. Since any framed manifold is automatically oriented, we have a functor $\mathbf{Bord^{fr}} \to \mathbf{Bord^{or}}$; thus, any oriented theory gives a framed theory. Moreover, since two framings related by the action of the group $\mathrm{GL}_+(n)$ give the same orientation, we see that an oriented TQFT gives rise to a $\mathrm{GL}_+(n)$-invariant framed TQFT. Therefore, it is natural to expect some relation between $\mathrm{Fun}^{\otimes}(\mathbf{Bord^{or}}, \mathcal{C})$ and (properly defined) invariant subcategory $(\widetilde{\mathcal{C}^{\mathrm{fd}}})^{\mathrm{GL}_+(n)}$.

To make these ideas precise, we should begin by defining what we mean by action of a group on a category and by "invariant subcategory" under this action. Unlike group actions on sets, this is far from obvious. To give the reader some taste of this, here is the simplest example.

Example 15.2.1. Let $\mathcal{C}$ be a category (usual 1-category) and let G be a discrete group. We define an action of G on $\mathcal{C}$ to be a collection of functors $T_g \colon \mathcal{C} \to \mathcal{C}$, $g \in G$, such that $T_1 = 1$, together with functorial isomorphisms

$$\alpha_{g_1 g_2} \colon T_{g_1} T_{g_2} \simeq T_{g_1 g_2}$$

which themselves should satisfy the compatibility condition: two ways of identifying

$$T_{g_1 g_2 g_3} \simeq T_{g_1} T_{g_2} T_{g_3}$$

should be equal.

If we have such an action, we define the category $\mathcal{C}^G$ as the category with the following objects and morphisms:

- Objects: an object $X \in \mathcal{C}$ together with a collection of isomorphisms $\varphi_g \colon T_G(X) \simeq X$, $g \in G$ such that $\varphi_1 = 1_X$ and $\varphi_{gh} = \varphi_g T_g(\varphi_h)$ (both sides are morphisms $T_{gh}(X) \simeq T_g T_h(X) \to X$).
- Morphisms:

$$\mathrm{Hom}_{\mathcal{C}^G}(X, X') = \{f \in \mathrm{Hom}_{\mathcal{C}}(X, X') \mid \varphi'_g f = \rho_g(f)\varphi_g\}$$

  (FIXME: verify)

This construction is sometimes called "equivariantization", see [**EGNO2015**, Section 2.7]. In particular, if we consider the category $\mathcal{C} = \mathbf{Vec_f}$ of finite-dimensional vector spaces with trivial action of a group G, then ${\mathbf{Vec_f}}^G = \operatorname{Rep} G$ is the category of finite-dimensional representations of G.

In a similar fashion, one can write explicitly a definition of an action of a group G on a 2-category and definition of $\mathcal{C}^G$ in this case; one can find all details in [**Hes2017**]. However, in the case of interest to us, there is a way around it. Namely, we are only interested in the action of a group on the $(\infty, 0)$ category $\tilde{\mathcal{C}}$. By FIXME, every $(\infty, 0)$ category is a fundamental groupoid of a topological space: $\mathcal{C} = \pi_{\leq\infty}(X)$. Thus, it is natural to define the action of G on $\pi_{\leq\infty}(X)$ as the action induced by an action of G on X.

Now, we need to define the notion of fixed points. The naive definition of fixed points of action of G on X is not homotopy invariant, so we cannot use it. Instead, we will use the notion of *homotopy fixed point*.

Let G be a topological group (for example, a Lie group). Recall that then one can define a contractible space EG with a free action of G; the quotient $BG = EG/G$ is called the classifying space of G. The name is justified by the fact that for any space X, isomorphism classes of G-bundles on X are in bijection with homotopy equivalence classes of maps $X \to BG$.

Definition 15.2.1. Let X be a topological space with a continuous action of a topological group G. The *homotopy fixed point set* is defined by

$$X^{hG} = \operatorname{Hom}_G(EG, X)$$

where Hom_G stands for the space of G-equivariant continuous maps.

This definition is rather natural: note that the usual set of G-invariant points can be defined as $X^G = \operatorname{Hom}_G(pt, X)$, where pt is the one-point set with trivial action of G. One can show that unlike X^G, the homotopy fixed point set X^{hG} is homotopy invariant.

Now we can formulate the oriented version of the cobordism hypothesis.

Theorem 15.2.1. *Let $\mathcal{C}$ be an (∞, n) category, and let $\widetilde{\mathcal{C}^{\mathrm{fd}}}$ be the groupoid of fully dualizable objects as in Theorem 15.1.1. Then one has a natural action of the group $\mathrm{O}(n)$ on $\widetilde{\mathcal{C}^{\mathrm{fd}}}$, and we have an equivalence*

$$\operatorname{Fun}^{\otimes}(\mathbf{Bord^{or}}, \mathcal{C}) \simeq (\widetilde{\mathcal{C}^{\mathrm{fd}}})^{h\mathrm{SO}(n)}.$$

This theorem can be derived from the (framed) cobordism hypothesis; details can be found in [**Lur2009a**].

Remark 15.2.1. One might wonder why we use $\mathrm{SO}(n)$ instead of $\mathrm{GL}_+(n)$. In fact, it does not matter: inclusion $\mathrm{SO}(n) \subset \mathrm{GL}_+(n)$ is homotopy equivalence, so the set of homotopy fixed points for both actions is the same. However, it is more convenient to work with compact groups, hence the use of $\mathrm{SO}(n)$.

15.3. Cobordism hypothesis for $n = 2$

Reference: [**Hes2017**]. $SO(2)$ action on arbitrary 2-category, Serre automorphism, and homotopy fixed points. Calabi-Yau object.

Main claim:

Fully dualizable objects in **Alg** are separable algebras

SO(2) homotopy fixed points are separable symmetric Frobenius algebras.
Example: $\mathcal{C} = \mathbf{Alg}$. Symmetric separable Frobenius algebra

Part 4

Tensor categories and Turaev–Viro theory

Part 4 overview

In this part, we discuss extended TQFTs in deminsion 3; our main example of such TQFTs is Turaev–Viro theory.

FIXME

CHAPTER 16

3-category of tensor categories

In the previous chapters, we have given two constructions of an extended 2-dimensional TQFT with values in the 2-category **Alg** of algebras and bimodules: one construction was based on the work of Schommer-Pries, describing the 2-category $\mathbf{Bord^{or}}_2$ by generators and relations, and the other was based on triangulations (or more general PL cell decompositions) of a 2-manifold. In both cases, such a theory is determined by $A = Z(\bullet)$ which must be a separable symmetric Frobenius algebra. This is a special case of the general cobordism hypothesis, according to which extended framed n-dimensional TQFTs with values in $\mathcal{C}$ are described by fully dualizable objects in $\mathcal{C}$, and oriented extended TQFTs are described by $\mathrm{SO}(n)$ homotopy fixed points in $\mathcal{C}^{\mathrm{fd}}$. In particular, for $n = 2$, $\mathcal{C} = \mathbf{Alg}$, fully dualizable objects are separable algebras, and SO(2) homotopy fixed points are separable symmetric Frobenius algebras.

In this chapter, we start exploring the 3d analog of these constructions. As we will show, in this case one possible replacement for the 2-category **Alg** is the 3-category **Tens** of finite tensor categories, and the fully dualizable objects in it are exactly semisimple tensor categories, i.e., fusion categories; SO(3) homotopy fixed points are (conjecturally) spherical fusion categories. The corresponding 3-dimensional oriented extended TQFT is known as Turaev–Viro theory.

In this chapter, we will define and study the 3-category **Tens**, following the works [**DSPS2019**], [**DSPS2020**]; the construction of Turaev–Viro theory will be given in a later chapter.

To simplify the exposition, throughout this chapter we assume that $\mathbf{k}$ is an algebraically closed field of characteristic zero. As usual, we will skip most proofs, referring the reader to original papers.

16.1. Review of 2-category Alg

For comparison, we recall the main steps in the construction of the 2-category **Alg**:

- We start with the category **Vec** of $\mathbf{k}$-vector spaces and define the tensor product functor $\otimes\colon \mathbf{Vec} \times \mathbf{Vec} \to \mathbf{Vec}$
- Next, we define an algebra A as an object $A \in \mathbf{Vec}$ together with a morphism $m\colon A \otimes A \to A$ and an element $1 \in A$ satisfying associativity and unit axioms. We also define the notion of morphisms between algebras.
- We define the notion of left and right A-module and A-B-bimodules, together with module homomorphisms.
- Finally, we define the notion of tensor product over an algebra: given right A-module M and left A-module N, we define $M \otimes_A N$

Let us now proceed to defining categorical analogs of all of the above.

16.2. Finite k-linear categories

Our first step is defining a 2-category which will replace the category **Vec**. For that, recall that we have defined the notion of a locally finite **k**-linear abelian category as a **k**-linear abelian category where all objects have finite length and all Hom spaces are finite-dimensional, see Section 7.4. Such categories form a 2-category **Lin**; this 2-category should be considered a natural categorical analog of the category of vector spaces.

Let us now consider a more restrictive set of requirements.

Definition 16.2.1. A **k**-linear category $\mathcal{A}$ is called *finite* if it satisfies the following properties:

(1) For any $A, B \in \mathcal{A}$, the vector space $\mathrm{Hom}_{\mathcal{A}}(A, B)$ is finite-dimensional.
(2) Any object $A \in \mathcal{A}$ has finite length.
(3) $\mathcal{A}$ has enough projectives: any $A \in \mathcal{A}$ is a quotient of a projective object $P \in \mathcal{A}$.
(4) There are only finitely many isomorphism classes of simple objects.

Note that we do not assume that $\mathcal{A}$ is semisimple.

Example 16.2.1. Let A be a finite-dimensional algebra over **k**. Then the category $A-\mathbf{mod}$ of finite-dimensional A-modules is a finite **k**-linear category. Moreover, it is easy to show that conversely, every finite **k**-linear category is equivalent to the category $A-\mathbf{mod}$ for some finite-dimensional algebra A (see [**EGNO2015**, Section 1.8]).

Unless stated otherwise, all functors between **k**-linear categories will be assumed to be additive, **k**-linear, and right exact.

Definition 16.2.2. We denote by $\mathbf{Lin_f}$ the 2-category of finite **k**-linear abelian categories together with additive and **k**-linear right exact functors between them and functorial morphisms.

The so-defined 2-category $\mathbf{Lin_f}$ is the categorical analog of the category of finite-dimensional vector spaces.

Remark 16.2.1. One might notice that we are not quite consistent: in the $n = 2$ case, we started with category of all vector spaces, not necessarily finite-dimensional ones, so our 2-category **Alg** also included infinite-dimensional algebras — which, however, were irrelevant for constructing TQFTs: any fully dualizable (i.e., separable) algebra is automatically finite-dimensional. In the $n = 3$ case, we start with finite linear categories from the beginning.

The reason is mostly technical; many of the constructions below work for any locally finite category (i.e., a category only satisfying the first 2 conditions of Definition 16.2.1). I expect that one can in fact define a version of the (not yet constructed) 3-category **Tens** starting with locally finite linear categories rather than finite; however, it does create some technical difficulties, and since we expect that the only categories we will need for extended 3d TQFTs will be finite anyway, we can save some trouble and only work with finite linear categories from the beginning.

16.3. Finite tensor categories

Our next step is defining a categorical analog of the notion of associative algebra. The obvious candidate is the notion of a monoidal category as discussed in Chapter 2. However, in order to make some future constructions possible, we will add some additional conditions.

DEFINITION 16.3.1. A *finite tensor category* is a finite **k**-linear category $\mathcal{C}$ together with a structure of a monoidal category such that

(1) The tensor product functor $\otimes: \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is **k**-bilinear.
(2) $\mathrm{Hom}(\mathbf{1}, \mathbf{1}) = \mathbf{k}$.
(3) $\mathcal{C}$ is rigid: every $X \in \mathcal{C}$ has left and right dual (see Definition 3.3.2).

(Note that we do not give a definition of (not necessarily finite) tensor category since the only tensor categories we will need are the finite ones.)

Condition (1) is a natural compatibility between the monoidal structure and structure of **k**-linear category; condition (2) is a harmless condition which is there just for convenience — almost everything below can be repeated, possibly with minor changes, without that assumption (in fact, [**DSPS2019**] do not include this condition). But the rigidity condition (3) is absolutely crucial: without it, most of the theory below would not work.

LEMMA 16.3.1. *Let $\mathcal{C}$ be a finite tensor category. Then*

(1) *Functor $\otimes: \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is bi-exact.*
(2) *Object $\mathbf{1} \in \mathcal{C}$ is simple.*

A proof of this lemma can be found in [**EGNO2015**, Proposition 4.2.1, Theorem 4.3.8].

As an immediate corollary, we see that tensor product can be considered as a functor $\mathcal{C} \boxtimes \mathcal{C} \to \mathcal{C}$, where $\boxtimes$ is Deligne's product of **k**-linear categories, see Section 7.4 (in the same way as for algebras, multiplication can be considered as either a bilinear map $A \times A \to A$ or as a linear map $A \otimes A \to A$).

LEMMA 16.3.2. *Let $\mathcal{C}$, $\mathcal{D}$ be finite tensor categories. Then category $\mathcal{C} \boxtimes \mathcal{D}$ has a natural structure of a finite tensor category, with tensor product functor given by*

$$(c_1 \boxtimes d_1) \otimes (c_2 \boxtimes d_2) = (c_1 \otimes c_2) \boxtimes (d_1 \otimes d_2).$$

This lemma is an analog of the statement that tensor product of algebras has a natural structure of an algebra. We skip the proof.

Below are some examples of finite tensor categories.

EXAMPLE 16.3.1. The category $\mathbf{Vec_f}$ of finite-dimensional vector spaces over **k** is a finite tensor category. This theory is the categorical analog of the trivial algebra $A = \mathbf{k}$.

EXAMPLE 16.3.2. Let G be a finite group. The category $\mathrm{Rep}\, G$ of finite-dimensional representations of G is a finite tensor category.

Note that if we replaced finite group by a compact group, the resulting category would not be finite.

EXAMPLE 16.3.3. Let H be a finite-dimensional Hopf algebra. Then category of finite-dimensional H-modules is a finite tensor category.

EXAMPLE 16.3.4. Let G be a finite group. Let $\mathbf{Vec}_G$ be the category of finite-dimensional G-graded vector spaces as defined in Example 2.3.2. Then $\mathbf{Vec}_G$ is a finite tensor category.

As for algebras, for finite tensor categories we have the notion of "opposite" multiplication: if $\mathcal{C}$ is a finite tensor category, we denote by $\mathcal{C}^{\mathrm{mp}}$ the same category but with opposite tensor product: $A \otimes^{\mathrm{mp}} B = B \otimes A$. (Notation mp is an abbreviation for "monoidally opposite"; it should not be confused with $\mathcal{C}^{\mathrm{op}}$, which is obtained from $\mathcal{C}$ by reversing all arrows.)

16.4. Module categories

Our next step is defining a categorical analog of a module over an algebra.

DEFINITION 16.4.1. Let $\mathcal{C}$ be a finite tensor category. A finite left *$\mathcal{C}$-module category* is a finite $\mathbf{k}$-linear category $\mathcal{M}$ together with a $\mathbf{k}$-bilinear action functor

$$\otimes \colon \mathcal{C} \times \mathcal{M} \to \mathcal{M}$$

which is right exact in $\mathcal{C}$ argument, and with functorial isomorphisms

$$\begin{aligned}&(c_1 \otimes c_2) \otimes m \simeq c_1 \otimes (c_2 \otimes m)\\ &\mathbf{1} \otimes m \simeq m\end{aligned}$$

which satisfy the obvious pentagon and unit axioms.

Note that in many references (e.g., [**DSPS2019**]), condition that $\otimes$ is right exact is not part of the definition. However, this condition is necessary for most of what follows, so if we do not include it as part of the definition, we would need to add it as an additional assumption in our theorems.

THEOREM 16.4.1. *Let $\mathcal{M}$ be a finite module category over a finite tensor category $\mathcal{C}$. Then the action functor $\mathcal{C} \otimes \mathcal{M} \to \mathcal{M}$ is exact in $\mathcal{C}$ and also exact in $\mathcal{M}$.*

We refer the reader to [**DSPS2019**, Corollary 2.26] for the proof.

In a similar way, one defines the notion of a *right* module category and a notion of a $\mathcal{C}$-$\mathcal{D}$ bimodule category; for the latter, we also need to add an isomorphism

$$(c \otimes m) \otimes d \simeq c \otimes (m \otimes d)$$

again satisfying obvious compatibility conditions.

EXAMPLE 16.4.1. Any finite tensor category $\mathcal{C}$ is automatically a bimodule category over itself.

EXAMPLE 16.4.2. Any finite linear category has a natural structure of a module category over $\mathbf{Vec_f}$.

EXAMPLE 16.4.3. Let G be a finite group and let $H \subset G$ be a subgroup. Then the category $\operatorname{Rep} H$ is a module category over $\operatorname{Rep} G$.

EXAMPLE 16.4.4. Let A be an algebra in $\mathcal{C}$, i.e., an object $A \in \mathcal{C}$ together with morphisms $m \colon A \otimes A \to A$, $i \colon \mathbf{1} \to A$ satisfying obvious associativity and unit axioms (see [**EGNO2015**, Section 7.8]). A right A-module is an object $M \in \mathcal{C}$ together with the action morphism $M \otimes A \to M$, again satisfying obvious axioms. We denote by $\mathbf{Mod}_{\mathcal{C}}(A)$ the category of right modules over A, with the obvious morphisms between them. Then the category $\mathbf{Mod}_{\mathcal{C}}(A)$ is a finite module category over $\mathcal{C}$. Moreover, it can be shown (see [**EGNO2015**, Corollary 7.10.5]) that every finite $\mathcal{C}$ module category can be obtained in this way. For $\mathcal{C} = \mathbf{Vec_f}$, this is exactly the statement given in Example 16.2.1.

This construction allows one to reduce the study of module categories over $\mathcal{C}$ (which in general are difficult to describe) to the study of modules over an algebra, which is a much more explicit question. This is one of the main tools used to prove results about module categories; proofs of most theorems below are based on this approach. However, since we skip most proofs, we will not go into more details about algebras in categories.

Definition 16.4.2. Let $\mathcal{M}$, $\mathcal{N}$ be finite left module categories over a finite tensor category $\mathcal{C}$. A module functor $F\colon \mathcal{M} \to \mathcal{N}$ is a $\mathbf{k}$-linear functor together with functorial isomorphisms

$$J\colon F(c \otimes m) \xrightarrow{\sim} c \otimes F(m)$$

satisfying the obvious pentagon and triangle relations. Unless explicitly stated otherwise, we will always assume that F is right exact; we will denote the set of all right exact module functors $\mathcal{M} \to \mathcal{N}$ by

$$\mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{N}) = \text{right exact module functors } \mathcal{M} \to \mathcal{N}.$$

Given two functors $F_1, F_2 \in \mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{N})$, we define $\mathrm{Hom}(F_1, F_2)$ to be the set of functorial morphisms $F_1 \to F_2$ which agree with isomorphisms J_1, J_2; we leave it to the reader to write the compatibility condition explicitly (or find it in [**EGNO2015**, Section 7.2]). This turns $\mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{N})$ into a category and gives rise to the 2-category $\mathcal{C}-\mathbf{mod}$ of (left) finite $\mathcal{C}$-module categories, together with $\mathcal{C}$-module functors and functorial morphisms.

In a similar way one defines module functors for right module categories and bimodule categories.

Theorem 16.4.2. *Let $\mathcal{M}$, $\mathcal{N}$ be finite module categories over a finite tensor category $\mathcal{C}$. Then the category $\mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{N})$ is a finite $\mathbf{k}$-linear category.*

If, in addition, $\mathcal{C}, \mathcal{M}_1, \mathcal{M}_2$ are semisimple, then $\mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{N})$ is also semisimple.

Proof of the first part can be found in [**EGNO2015**, Proposition 7.11.6]. (Note that in the book, they did not spell out exact requirements on the module categories; however, if you read the proof, you will see that it works for any finite module categories).

The second part is [**ENO2005**, Theorem 2.16].

Example 16.4.5. Let $\mathcal{M}$ be a $\mathcal{C}$-$\mathcal{D}$ bimodule category. Define

$$\mathcal{M}^\vee = \mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{C}).$$

This category is naturally a finite $\mathcal{D}$-$\mathcal{C}$ bimodule category (compare with Exercise 5.2.1).

In a similar way, we can define category

$$^\vee\mathcal{M} = \mathrm{Fun}_{\mathcal{D}^{\mathrm{mp}}}(\mathcal{M}, \mathcal{D})$$

which is again a $\mathcal{D}$-$\mathcal{C}$ bimodule category.

Lemma 16.4.3. *Let $\mathcal{C}$, $\mathcal{D}$ be finite tensor categories. The 2-category of finite $\mathcal{C}$-$\mathcal{D}$ bimodule categories is equivalent to the 2-category of finite $(\mathcal{C} \boxtimes \mathcal{D}^{\mathrm{mp}})$ module* *categories. In particular, 2-category of left $\mathcal{C}$-module categories is equivalent to the 2-category of right $\mathcal{C}^{\mathrm{mp}}$-module categories.*

16.5. Balanced tensor product

So far, we have defined a notion of a finite tensor category and a notion of module category over a finite tensor category; these notions are natural analogs of the notion of a commutative algebra and modules over algebras. Our next goal is defining an analog of tensor product over A.

Recall that given an algebra A, a right A-module M_A, and a left A-module ${}_AN$, there are several ways to define tensor product $M \otimes_A N$. The most straightforward is defining $M \otimes_A N$ as a suitable quotient of $M \otimes N$. However, a better approach is defining $M \otimes_A N$ via a universal property: namely, it is uniquely determined by the condition that for every vector space L, we have a functorial isomorphism

$$\mathrm{Hom}_{\mathbf{k}}(M \otimes_A N, L) = \mathrm{Bal}_A(M \times N, L)$$

where Bal_A is the space of A-balanced maps, i.e., bilinear maps $M \times N \to L$ satisfying condition

$$f(ma, n) = f(m, an)$$

for any $m \in M$, $n \in N$, $a \in A$.

As usual, universality immediately implies that such a vector space, if it exists, is unique up to a unique isomorphism. Existence is not obvious and is proved by an explicit construction.

Let us now define a categorical analog of this.

Definition 16.5.1. Let $\mathcal{C}$ be a finite tensor category. Let $\mathcal{M}_{\mathcal{C}}$, ${}_{\mathcal{C}}\mathcal{N}$ be right and left finite $\mathcal{C}$-module categories, and let $\mathcal{L}$ be a $\mathbf{k}$-linear category.

A $\mathcal{C}$-balanced functor $F\colon \mathcal{M} \times \mathcal{N} \to \mathcal{L}$ is a $\mathbf{k}$-bilinear functor $F\colon \mathcal{M} \times \mathcal{N} \to \mathcal{L}$, right exact in each variable, together with a functorial isomorphism

$$F(m \otimes c, n) \simeq F(m, c \otimes n), \qquad c \in \mathcal{C}$$

satisfying usual axioms (see [**DSPS2019**, Definition 3.1]).

One can also define a notion of morphisms of $\mathcal{C}$-balanced functors. We denote by $\mathrm{Bal}_{\mathcal{C}}(\mathcal{M} \times \mathcal{N}, \mathcal{L})$ the category of $\mathcal{C}$-balanced functors $\mathcal{M} \times \mathcal{N} \to \mathcal{L}$.

Definition 16.5.2. In the assumptions of Definition 16.5.1, a balanced tensor product is a $\mathbf{k}$-linear category $\mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N}$ together with a $\mathcal{C}$-balanced functor $\boxtimes_{\mathcal{C}}\colon \mathcal{M} \times \mathcal{N} \to \mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N}$ such that for all $\mathbf{k}$-linear categories $\mathcal{L}$, the functor $\boxtimes_{\mathcal{C}}$ induces an equivalence of categories

$$\mathrm{Bal}_{\mathcal{C}}(\mathcal{M} \times \mathcal{N}, \mathcal{L}) \simeq \mathrm{Fun}(\mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N}, \mathcal{L})$$

It is easy to show that if such $\mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N}$ exists, it is unique; more precisely, the balanced tensor product is unique up to equivalence, and this equivalence is in turn unique up to unique natural isomorphism. Said another way, the 2-category of linear categories representing the balanced tensor product is either contractible or empty. The hard part is proving existence of $\mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N}$.

Theorem 16.5.1. *Let $\mathcal{C}$ be a finite tensor category over $\mathbf{k}$, and let $\mathcal{M}$, $\mathcal{N}$ be right and left finite $\mathcal{C}$-module categories. Then the balanced tensor product $\mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N}$ exists and is a finite $\mathbf{k}$-linear category. Moreover, the functor $\boxtimes_{\mathcal{C}}\colon \mathcal{M} \times \mathcal{N} \to \mathcal{M} \boxtimes_{\mathcal{C}} \mathcal{N}$ is exact in each argument.*

A proof of this theorem can be found in [**DSPS2019**, Theorem 3.3].

EXAMPLE 16.5.1. Let $\mathcal{C} = \mathbf{Vec_f}$ be the category of finite-dimensional vector spaces. Then the balanced tensor product functor $\boxtimes_{\mathbf{Vec_f}}$ coincides with Deligne product defined in Section 7.4.

EXAMPLE 16.5.2. Let $\mathcal{M}$ be a finite left $\mathcal{C}$-module category. Then one has a natural equivalence $\mathcal{C} \boxtimes_{\mathcal{C}} \mathcal{M} \simeq \mathcal{M}$ (compare with the isomorphism $A \otimes_A M \simeq M$ for a module M over an algebra A).

LEMMA 16.5.2. *Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ be finite tensor categories, and let $\mathcal{M}$ be a finite $\mathcal{A}$-$\mathcal{B}$ bimodule category, and $\mathcal{N}$ a finite $\mathcal{B}$-$\mathcal{C}$ bimodule category. Then the balanced tensor product $\mathcal{M} \boxtimes_{\mathcal{B}} \mathcal{N}$ has a natural structure of $\mathcal{A}$-$\mathcal{C}$ bimodule category.*

This lemma can be found in [**DSPS2019**, Remark 3.7].

16.6. Category Tens

THEOREM 16.6.1. *There exists a $(3, 3)$ category* **Tens** *with the following objects and morphisms:*

- *Objects: $\mathbf{k}$-linear finite tensor categories*
- *1-morphisms $\mathcal{C} \to \mathcal{D}$: finite linear $\mathcal{D}$-$\mathcal{C}$ bimodule categories; horizontal composition of 1-morphisms is given by balanced Deligne product.*
- *2-morphisms: right exact bimodule functors*
- *3-morphisms: functorial morphisms*

Category **Tens** *has a structure of a symmetric monoidal 3-category, with respect to Deligne product $\boxtimes$. The unit for this monoidal structure is the tensor category* $\mathbf{Vec_f}$.

This theorem is stated in this form in [**DSPS2020**, Theorem 2.2.18].

REMARK 16.6.1. In my opinion, the current state of this and related statements is somewhat unsatisfactory. [**DSPS2020**, Theorem 2.2.18] does not give a proof; instead, they refer the reader to [**JFS2017**], where this statement seems to be stated in Example 8.10. That example does not state the theorem in any detail but briefly mentions that one can construct such a 3-category by thinking about tensor categories as algebras in a (suitably defined) 2-category of $\mathbf{k}$-linear categories, and provides a reference to yet another 100-page paper where the theory of E_n algebras in the setting of infinity categories is discussed. I have no doubt that the result is correct, and can be extracted from these papers, but I would much appreciate a more direct proof.

Note that [**JFS2017**] use a different model of higher categories than we do (namely, their approach is based on complete Segal spaces); however, since, as we discussed in Chapter 14, all models of (∞, n) categories are equivalent, this should not be a problem.

CHAPTER 17

Duality in 3-category Tens

In the previous chapter, we had outlined the construction of the 3-category **Tens** of finite tensor categories. Our goal now is to try and construct extended 3d TQFTs $\mathbf{Bord}_3 \to \mathbf{Tens}$. By cobordism hypothesis, framed extended TQFTs are classified by fully dualizable objects in **Tens**, and oriented TQFTs are classified by SO(3) homotopy fixed points of the groupoid $\mathbf{Tens}^{\mathrm{fd}}$. Thus, our next goal is to determine which objects, 1- and 2-morphisms in **Tens** have duals/adjoint. We had done a similar study for 2-category **Alg** in Chapter 9, where we established that fully dualizable objects are separable algebras, and SO(2) homotopy fixed points are symmetric Frobenius algebras; here we will present analog of these results, but one level higher in our category ladder.

As before, we will be mostly skipping the proofs, referring the reader to original papers. Our main reference for this chapter is [**DSPS2020**].

Throughout this chapter, we assume that the ground field $\mathbf{k}$ is algebraically closed and has characteristic 0. All $\mathbf{k}$-linear categories discussed in this chapter are assumed to be finite; all functors between them are assumed to be $\mathbf{k}$-linear and right exact.

17.1. Duality for objects and morphisms

First, we study the question of duality for objects of **Tens**, i.e., for finite tensor categories. Here the answer is simple and very similar to the one for 2-category **Alg**.

Let $\mathcal{C}$ be a finite tensor category. It is naturally a $\mathcal{C}$-bimodule category; equivalently, we can consider it as either left or right module category over the tensor category $\mathcal{C}^e = \mathcal{C} \boxtimes \mathcal{C}^{\mathrm{mp}}$.

THEOREM 17.1.1. *Every finite tensor category is rigid as an object in* **Tens***; the dual is given by category $\mathcal{C}^{\mathrm{mp}}$. The evaluation and coevaluation 1-morphisms are given by $\mathcal{C}$, considered as either left or right module over $\mathcal{C}^e$.*

A proof can be found in [**DSPS2020**, Proposition 3.1.2].

This theorem is a direct analog of Theorem 9.1.1.

Let us now study duality for 1-morphisms in **Tens**. Recall that given finite tensor categories $\mathcal{C}$, $\mathcal{D}$, the 2-category of 1-morphisms $\mathcal{D} \to \mathcal{C}$ is the category of finite $\mathcal{C}$-$\mathcal{D}$ bimodule categories.

Warning: in [**DSPS2020**], they use opposite convention!

THEOREM 17.1.2. *Let ${}_{\mathcal{C}}\mathcal{M}_{\mathcal{D}}$ be a finite bimodule category considered as 1-morphism between finite tensor categories $\mathcal{D} \to \mathcal{C}$. Then it has both left and right adjoints, given by categories $\mathcal{M}^\vee = \mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{C})$, ${}^\vee\mathcal{M} = \mathrm{Fun}_{\mathcal{D}^{\mathrm{mp}}}(\mathcal{M}, \mathcal{D})$.*

Proof is given in [**DSPS2020**, Proposition 3.2.1].

The final step in our discussion of dualizability in **Tens** is determining which 2-morphisms (i.e., module functors between bimodule categories) have left and right adjoints. This is where things become non-trivial. Indeed, it is a standard result of homological algebra that if a functor between abelian categories has both left and right adjoints, then it must be exact — and not every functor between bimodule categories is exact. Thus, it is not surprising that we need to impose additional conditions to have adjoints.

THEOREM 17.1.3. *Let $\mathcal{M}$, $\mathcal{N}$ be finite* semisimple *$\mathcal{C}$-$\mathcal{D}$ bimodule categories. Then any $F \in \mathrm{Fun}(\mathcal{M}, \mathcal{N})$ has both left and right adjoints, where* Fun *is the category of $\mathcal{C}$-$\mathcal{D}$ bimodule functors; moreover, left and right adjoints of F are isomorphic.*

As before, word "semisimple" just refers to the structure of abelian category; it is not related to bimodule structure. In the case $\mathcal{C} = \mathcal{D} = \mathbf{Vec_f}$, bimodule categories are just finite **k**-linear categories; in this case, the statement of this theorem is exactly the statement of Corollary 8.4.3. Proof of the general case of this theorem is given in [**DSPS2020**, Proposition 3.4.1].

17.2. Fully dualizable objects in Tens

We can now put it all together and describe the maximal subcategory with duals in **Tens**.

THEOREM 17.2.1. *Let $\mathbf{Tens}^{\mathrm{ss}} \subset \mathbf{Tens}$ be the category with the following objects and morphisms:*

- *Objects:* semisimple *finite tensor categories*
- *1-morphisms: semisimple finite bimodule categories*
- *2-morphisms: all module functors between bimodule categories*
- *3-morphisms: functorial morphisms*

Then $\mathbf{Tens}^{\mathrm{ss}}$ is a category with duals.

This is [**DSPS2020**, Proposition 3.4.3].

Note that the word semisimple just refers to the structure of **k**-linear category; it is not related to monoidal structure or structure of a bimodule category.

Indeed, if $\mathcal{C}$ is a semisimple finite tensor category, then it has a dual $\mathcal{C}^{\mathrm{mp}}$ which is obviously also semisimple, and the evaluation and coevaluation 1-morphisms are themselves semisimple module categories (since both are given by $\mathcal{C}$ itself) and thus are 1-morphisms in $\mathbf{Tens}^{\mathrm{ss}}$. Similarly, it is easy to show that for a semisimple bimodule category ${}_{\mathcal{C}}\mathcal{M}_{\mathcal{D}}$ considered as a 1-morphism in $\mathbf{Tens}^{\mathrm{ss}}$, the left and right adjoints, which are given by $\mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{C})$ and $\mathrm{Fun}_{\mathcal{D}}(\mathcal{M}, \mathcal{D})$, are again semisimple and thus are 1-morphisms in $\mathbf{Tens}^{\mathrm{ss}}$. Finally, by Theorem 17.1.3, every 2-morphism in $\mathbf{Tens}^{\mathrm{ss}}$ has left and right adjoints.

REMARK 17.2.1. In [**DSPS2020**] this theorem is formulated slightly differently: instead of requiring objects, i.e., tensor categories, to be semisimple, they require these categories to be *separable*, which better matches the definition for algebras. However, in characteristic zero case, these two notions coincide, and defining notion of separable tensor category is more complicated than doing it for algebras (we cannot state it directly; instead we state in terms of an algebra in the category). Therefore, we have chosen to simplify the exposition and avoid introducing the notion of separable tensor category.

Semisimple finite tensor categories have another name: they are called *fusion categories*.

DEFINITION 17.2.1. A fusion category is a a semisimple finite tensor category, i.e., a semisimple finite $\mathbf{k}$-linear category with a rigid monoidal structure such that $\mathrm{End}(\mathbf{1}) = \mathbf{k}$.

As an immediate corollary of Theorem 17.2.1, we get the following result.

COROLLARY 17.2.2. *Any fusion category is a fully dualizable object in* **Tens**.

In fact, one can prove that it converse is also true: a finite tensor category is fully dualizable as an object in **Tens** if and only if it is semisimple; see [**DSPS2020**, Proposition 3.4.10].

Thus, by Cobordism Hypothesis, framed extended 3d TQFTs with values in **Tens** are classified by fusion categories.

17.3. SO(3) fixed points and spherical fusion categories

Finally, let us try and classify oriented 3d extended TQFTs. According to general theory, such TQFTs are classified by homotopy fixed points for SO(3) action on $\mathbf{Tens}^{\mathrm{ss}}$. Unfortunately, explaining how SO(3) acts on $\mathbf{Tens}^{\mathrm{ss}}$ is rather non-trivial. We will only show one small piece of this data.

FIXME: this section is a very rough draft

Recall that $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. Arguing the same way as in FIXME, we see that this shows that we must have an action of $\mathbb{Z}_2$ on $\mathbf{Tens}^{\mathrm{ss}}$; the action of generator of $\mathbb{Z}_2$ is called the *Serre automorphism*. [**DSPS2020**] has a detailed description of this action in terms of framing.

It takes some time to unravel that definition, but the end result is that the Serre automorphism gives a twist: it sends a $\mathcal{C}$-$\mathcal{D}$ bimodule category $\mathcal{M}$ to the category ${}_{\langle rr \rangle}\mathcal{M}$, which is again a $\mathcal{C}$-$\mathcal{D}$ bimodule category. It coincides with $\mathcal{M}$ as abelian category, but now the right action of $\mathcal{C}$ is given by $c \times m \to c^{**} \otimes m$.

This suggests that a homotopy fixed point for SO(3) action should be a fusion category $\mathcal{C}$ together with functorial isomorphism $c \to c^{**}$ (and possibly some other structures).

REMARK 17.3.1. These ideas suggest that in any fusion category there should be a natural isomorphism $c \to c^{****}$. This is indeed true, but highly non-trivial; see [**EGNO2015**, Corollary 7.19.3]. It is closely related with classical result of Radford that in a finite-dimensional Hopf algebra, fourth power of antipode is an inner automorphism. We refer the reader to [**EGNO2015**, Section 7.19] for discussion of this.

This naturally leads us to the following definition.

DEFINITION 17.3.1. A *pivotal* category is a rigid monoidal category $\mathcal{C}$ together with a functorial isomorphism $\delta\colon X \to X^{**}$ such that $\delta_{X\otimes Y} = \delta_X \otimes \delta_Y$, $\delta_{\mathbf{1}} = 1$.

EXAMPLE 17.3.1. The category $\mathbf{Vec_f}$ of finite-dimensional vector spaces has a canonical pivotal structure. Same holds for the category $\mathrm{Rep}\, G$ of finite-dimensional representations of a group G.

EXERCISE 17.3.1. Show that δ is automatically compatible with duals:
$\delta_{X^*} = \big((\delta_X)^*\big)^{-1}$.

Note that pivotal structure is extra data: the same monoidal category can have more than one pivotal structure. It is currently unknown whether every fusion category admits at least one pivotal structure.

EXAMPLE 17.3.2. Let $\mathcal{C}$ be the category of finite-dimensional representations of Lie algebra $\mathfrak{sl}(2, \mathbb{C})$. Define the pivotal structure by

$$\delta = \delta_0 \circ z$$

where δ_0 is the standard vector space isomorphism $V \simeq V^{**}$ and $z = \exp(\pi i h)$, $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \in \mathfrak{sl}(2)$; thus, $z = +1$ on irreducible representations with even highest weight (or integer spin) and $z = -1$ on representations wiht odd highest weight (half-integer spin).

This example might seem artificial, but in fact it (or rather similar pivotal structure for representations of the corresponding quantum group) frequently appears in the literature in implicit form. For example, in the standard definition of the Kauffman bracket invariant of links (see [**Kau1987**]), the invariant of the unknot is equal to $-(A^2 + A^{-2})$, so even in the classical limit $A = 1$, we get answer -2. The reason for that is that in fact that invariant can be described in terms of knot invariants coming from the category of representations of $U_q\mathfrak{sl}(2)$ but with the pivotal structure defined as in this example.

The pivotal structure allows us to define a trace of an endomorphism. Recall that rigidity provides evaluation and coevaluation morphisms (see Section 3.3)

$$\mathrm{ev}_X : X^* \otimes X \to \mathbf{1}, \qquad \mathrm{coev}_X : \mathbf{1} \to X \otimes X^*,$$

and similarly $\mathrm{ev}_{X^*} : X^{**} \otimes X^* \to \mathbf{1}$, $\mathrm{coev}_{X^*} : \mathbf{1} \to X^* \otimes X^{**}$ for the dual object X^*.

DEFINITION 17.3.2. Let $\mathcal{C}$ be a pivotal category with pivotal structure $\delta_X : X \xrightarrow{\sim} X^{**}$. For $f \in \mathrm{End}(X)$, the *left trace* and *right trace* $\mathrm{tr}^L(f), \mathrm{tr}^R(f) \in \mathrm{End}(\mathbf{1})$ are defined by

$$\mathrm{tr}^L(f) = \mathrm{ev}_{X^*} \circ \big((\delta_X \circ f) \otimes \mathrm{id}_{X^*}\big) \circ \mathrm{coev}_X,$$
$$\mathrm{tr}^R(f) = \mathrm{ev}_X \circ \big(\mathrm{id}_{X^*} \otimes (f \circ \delta_X^{-1})\big) \circ \mathrm{coev}_{X^*}.$$

(see Figure 17.3.1). The *left* and *right dimensions* of X are $\dim^L(X) = \mathrm{tr}^L(\mathrm{id}_X)$ and $\dim^R(X) = \mathrm{tr}^R(\mathrm{id}_X)$.

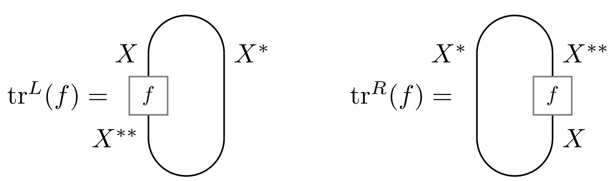


FIGURE 17.3.1. Left and right traces; pivotal isomorphism δ_X and its inverse are not shown.

Since for a finite tensor category $\mathrm{End}(\mathbf{1}) = \mathbf{k}$, the traces and dimensions are elements of $\mathbf{k}$.

LEMMA 17.3.1. *Left and right traces in a pivotal category $\mathcal{C}$ have the following properties (stated for tr^L; the same holds for tr^R):*

(1) *(Cyclicity) For any $f\colon X \to Y$ and $g\colon Y \to X$, $\mathrm{tr}^L(g \circ f) = \mathrm{tr}^L(f \circ g)$.*
(2) *(Multiplicativity) For $f \in \mathrm{End}(X)$, $g \in \mathrm{End}(Y)$, $\mathrm{tr}^L(f \otimes g) = \mathrm{tr}^L(f)\,\mathrm{tr}^L(g)$; in particular, $\dim^L(X \otimes Y) = \dim^L(X)\dim^L(Y)$.*
(3) *(Additivity) If $\mathcal{C}$ is an abelian category, and $0 \to X' \to X \to X'' \to 0$ is a short exact sequence and $f \in \mathrm{End}(X)$ preserves X', inducing $f' \in \mathrm{End}(X')$ and $f'' \in \mathrm{End}(X'')$, then $\mathrm{tr}^L(f) = \mathrm{tr}^L(f') + \mathrm{tr}^L(f'')$; in particular, $\dim^L(X) = \dim^L(X') + \dim^L(X'')$.*
(4) *(Duality) $\mathrm{tr}^L(f) = \mathrm{tr}^R(f^*)$, where $f^* \in \mathrm{End}(X^*)$ is the dual morphism (see Exercise 3.3.3); in particular, $\dim^L(X) = \dim^R(X^*)$.*

Proofs, together with a more detailed discussion of pivotal structures and dimensions, can be found in [**EGNO2015**, Section 4.7].

DEFINITION 17.3.3. A pivotal category $\mathcal{C}$ is called *spherical* if for any $X \in \mathcal{C}$, $f \in \mathrm{End}(X)$ we have

$$\mathrm{tr}^L(f) = \mathrm{tr}^R(f).$$

In this case, we will just use notation $\mathrm{tr}(f)$ for the trace of a morphism f, and $\dim(X) = \dim^L(X) = \dim^R(X)$ for dimension of an object.

Note that it is immediate from Lemma 17.3.1 that in a spherical category, $\mathrm{tr}(f) = \mathrm{tr}(f^*)$; in particular,

$$\dim(X) = \dim(X^*). \tag{17.3.1}$$

REMARK 17.3.2. All known examples of pivotal categories are spherical. It is an open problem if there exist non-spherical pivotal categories.

REMARK 17.3.3. Let $\mathcal{C}$ be a pivotal fusion category. It can be shown that this pivotal structure is spherical if and only if $\delta^2\colon V \to V^{****}$ coincides with the canonical isomorphism $V \xrightarrow{\sim} V^{****}$ mentioned in Remark 17.3.1, see [**DSPS2020**, Proposition 3.5.4].

CONJECTURE 17.3.2. *Homotopy fixed points of* $\mathrm{SO}(3)$ *action on* $\mathbf{Tens}^{\mathrm{ss}}$ *are spherical fusion categories.*

Reference: [**DSPS2020**, Conjecture 3.5.10].

CHAPTER 18

Spherical fusion categories

In this chapter we will study some algebraic properties of spherical fusion categories. We will use these results in the next chapter for constructing a lattice extended TQFT, the Turaev–Viro theory.

Throughout this chapter, $\mathcal{C}$ is a spherical fusion category over an algebraically closed field $\mathbf{k}$ of characteristic zero (see Definition 17.3.1, Definition 17.3.3).

We denote by $\mathcal{O}(\mathcal{C})$ the set of isomorphism classes of simple objects in $\mathcal{C}$; by definition of a fusion category, it is a finite set, and we will choose a representative $X_i \in \mathcal{C}$ for every $i \in \mathcal{O}(\mathcal{C})$. Abusing the language, we will write things such as $\bigoplus_{X\in\mathcal{O}(\mathcal{C})} \mathrm{End}(X)$ instead of $\bigoplus_{i\in\mathcal{O}(\mathcal{C})} \mathrm{End}(X_i)$. All our formulas will be independent of the choice of representatives.

It is known that any pivotal category is monoidally equivalent to a strictly pivotal category, i.e., a category in which $V^{**} = V$. Therefore in most formulas below we will suppress δ, in the same way as when working with monoidal categories, we frequently suppress the associativity isomorphisms.

18.1. Graphical calculus

In the previous chapters, we had already used graphical presentations of morphisms in a monoidal category. However, until now it was informal: it was only used as an illustration of our constructions and proofs. In this section, we will finally make some precise statements about such graphical presentations.

The idea of using such graphics is quite old; it seems that it was first suggested by Penrose, and then made popular by Joyal and Street [**JS1991**], [**JS1995**] and by Reshetikhin and Turaev [**RT1991**]. Our exposition follows the recent book [**TV2017**, Chapter 2], which provides the cleanest exposition of these results. Note, however, that in that book their convention is that graphs should be read "from the bottom up": incoming objects are shown at the bottom, and outgoing, at the top. We use the opposite convention, so we slightly modify their definitions.

Let $\mathcal{C}$ be a pivotal category. Following [**TV2017**], we define a $\mathcal{C}$-colored *Penrose diagram* to be the following data:

- a finite collection of non-intersecting "boxes", i.e., rectangles with vertical/horizontal sides in the strip $\mathbb{R} \times [0, 1]$
- a finite collection of non-intersecting smooth directed arcs (embedded intervals) and directed circles (images of S^1) in the strip $\mathbb{R}\times[0, 1]$. Endpoints of arcs are all distinct and can lie either on the horizontal sides of boxes or the top and bottom of the strip, i.e., lines $\mathbb{R} \times \{1\}$, $\mathbb{R} \times \{0\}$. At the endpoints, the lines must be transversal to the horizontal lines. Other than the endpoints, arcs and circles do not intersect the boxes or top and bottom lines.

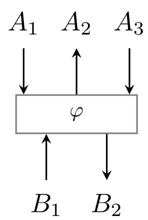
- coloring, i.e., assignment of an object of $\mathcal{C}$ to every arc or circle, and assignment of a morphism in $\mathcal{C}$ to every box. Namely, if $A_1, \dots, A_k$ are the colors of the arcs ending at the top side of a box, and $B_1, \dots, B_m$ are colors at the bottom of the box (both read left to right), then the box must be colored by a morphism

$$\varphi \in \mathrm{Hom}_{\mathcal{C}}(A_1^{\varepsilon_1} \otimes \cdots \otimes A_k^{\varepsilon_k}, B_1^{\varepsilon'_1} \otimes \cdots \otimes B_m^{\varepsilon'_m})$$

where

$$A_i^{\varepsilon_i} = \begin{cases} A_i, & \text{if the corresponding arc is directed downward} \\ A_i^* & \text{otherwise} \end{cases} \tag{18.1.1}$$

and similarly for B's, see Figure 18.1.1.



$$\varphi\colon A_1 \otimes A_2^* \otimes A_3 \to B_1^* \otimes B_2$$

Figure 18.1.1. Graphical notation for a morphism in a monoidal category.

An example of such a diagram is shown in Figure 18.1.2.

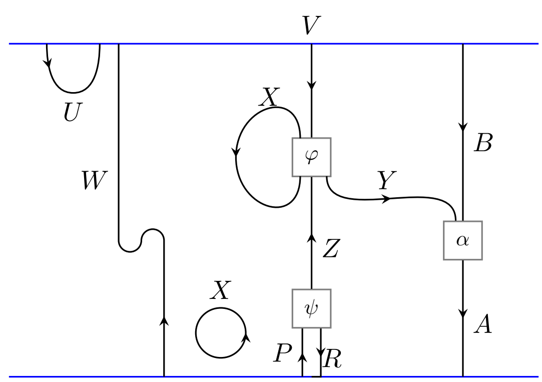


Figure 18.1.2. An example of $\mathcal{C}$-colored diagram

For such a diagram Γ, we can define objects $F_{\mathrm{top}}(\Gamma)$, $F_{\mathrm{bot}}(\Gamma) \in \mathcal{C}$ by taking the tensor product of colors of arcs with endpoints on the top horizontal line $\mathbb{R} \times \{1\}$ (respectively, $\mathbb{R} \times \{0\}$), with the same sign convention as in (18.1.1). Our goal is to define, for any such diagram Γ, a morphism $F(\Gamma)\colon F_{\mathrm{top}}(\Gamma) \to F_{\mathrm{bot}}(\Gamma)$ in $\mathcal{C}$.

To do that, note that for such diagrams (or, to be precise, for isotopy classes of diagrams), we can define two operations.

First, if Γ_1, Γ_2 are diagrams such that the top of Γ_2 matches the bottom of Γ_1, we can "stack" them, placing Γ_1 on top of Γ_2 and then rescaling vertically; we will denote this composition $\Gamma_2 \circ \Gamma_1$. As with gluing cobordisms, this requires smoothing the resulting joined arcs, which is well defined on the set of isotopy classes.

Second, we can define the tensor product $\Gamma_1 \otimes \Gamma_2$ by placing Γ_2 on the right of Γ_1 in the same strip.

FIGURE 18.1.3. Composition and tensor product of diagrams

THEOREM 18.1.1. *Let $\mathcal{C}$ be a pivotal category. Then there is a unique way to assign to each $\mathcal{C}$-colored diagram Γ a morphism $F(\Gamma)\colon F_{\mathrm{top}}(\Gamma) \to F_{\mathrm{bot}}(\Gamma)$ such that*

(1) *$F(\Gamma)$ only depends on isotopy class of Γ*
(2) *$F(\Gamma_2 \circ \Gamma_1) = F(\Gamma_2)F(\Gamma_1)$*
(3) *$F(\Gamma_2 \otimes \Gamma_1) = F(\Gamma_2) \otimes F(\Gamma_1)$*
(4) *For a diagram that consists of a single box colored by φ as in Figure 18.1.1, $F(\Gamma) = \varphi$*
(5) *For elementary diagrams (interval, cap, cup), the values of F are as shown in Figure 18.1.4.*

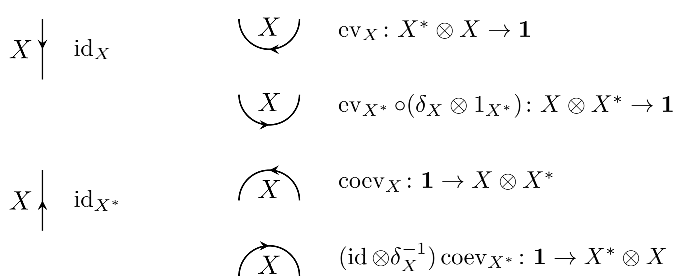


FIGURE 18.1.4. Morphisms corresponding to elementary diagrams.

Note that while in the definition of a Penrose diagram we require that boxes are oriented vertically, the isotopy is allowed to rotate the boxes (by multiples of 360 degrees).

This theorem is a reformulation of [**TV2017**, Theorem 2.6]; we refer the reader to that book for a proof. It can be restated as follows. Let $\mathrm{Diag}(\mathcal{C})$ be the category whose objects are finite sequences $\mathbf{A} = ((A_1, \varepsilon_1), \dots, (A_k, \varepsilon_k))$, where A_i are objects of $\mathcal{C}$ and $\varepsilon_i = \pm 1$, and morphisms $\mathbf{A} \to \mathbf{B}$ are isotopy classes of $\mathcal{C}$-colored diagrams with colors $A_1, \dots, A_k$ at the top and colors $B_1, \dots, B_m$ at the bottom and signs ε are determined by the orientation of the corresponding arcs as in (18.1.1). The stacking and tensor product operations defined above show that it is a monoidal category, with unit object given by empty collection of objects of $\mathcal{C}$. Then we can restate Theorem 18.1.1 as follows.

THEOREM 18.1.2. *The correspondence shown in Figure 18.1.4 can be uniquely extended to a monoidal functor $F\colon \mathrm{Diag}(\mathcal{C}) \to \mathcal{C}$.*

In particular, for a closed diagram Γ (i.e., with empty top and empty bottom) the above theorem defines $F(\Gamma) \in \mathrm{End}_{\mathcal{C}}(\mathbf{1})$. If $\mathcal{C}$ is a tensor category, so that $\mathrm{End}(\mathbf{1}) = \mathbf{k}$, this gives a numerical invariant of a diagram.

This theorem makes it possible to give graphical proofs of various equalities: in order to show that two morphisms, defined by $\mathcal{C}$-colored diagrams, are equal, it suffices to show that the diagrams themselves are isotopic. Such proofs are much easier to follow than algebraic arguments.

18.2. Dimensions in a spherical category

As before, let $\mathcal{C}$ be a spherical fusion category. Recall that by the Schur lemma (Lemma 2.2.1), for any simple X, we have $\mathrm{End}(X) = \mathbf{k}$.

For an object $X \in \mathcal{C}$ and a simple object X_i, we define the multiplicity $[X : X_i]$ by

$$[X : X_i] = \dim \mathrm{Hom}(X_i, X).$$

Semisimplicity immediately implies that then

$$X \simeq \bigoplus_{\mathcal{O}(\mathcal{C})} n_i X_i, \quad n_i = [X : X_i].$$

In fact, one has a stronger statement.

Lemma 18.2.1. *One has a functorial isomorphism*

$$X \simeq \bigoplus_{i \in \mathcal{O}(\mathcal{C})} \mathrm{Hom}(X_i, X) \otimes X_i.$$

It is obvious that for a simple object X, its dual X^* is also simple; this gives an involution $*$ on the set $\mathcal{O}(\mathcal{C})$ such that $X_i^* \simeq X_{i^*}$.

Exercise 18.2.1. Show that $[X_i \otimes X_j : \mathbf{1}] = \delta_{i,j^*}$.

Remark 18.2.1. It is tempting (and is frequently done in physics papers) to choose isomorphisms $X_i^* \simeq X_{i^*}$ and write $X_i^* = X_{i^*}$. However, this can lead to trouble, so we will not be doing this.

We denote by $\dim X \in \mathbf{k}$ dimension of an object $X \in \mathcal{C}$ (see Definition 17.3.3); in particular, we will use notation

$$d_i = \dim X_i$$

for dimensions of simple objects. Obviously, $\dim \mathbf{1} = 1$.

Lemma 18.2.2.

(1) $\dim(X \oplus Y) = \dim X + \dim Y$
(2) $\dim(X \otimes Y) = \dim X \dim Y$
(3) *For a simple $X = X_i$, its dimension d_i is non-zero.*

Remark 18.2.2. In general, dimensions are not integers; for example, in so-called Fibonacci category we have an object X satisfying $X \otimes X \simeq X \oplus \mathbf{1}$, which implies that $\dim X = (1 \pm \sqrt{5})/2$. It can be shown, however, that in a spherical fusion category dimensions are always algebraic integers (see [**EGNO2015**, Corollary 4.7.13]).

DEFINITION 18.2.1. The dimension of a spherical fusion category is defined by

$$\dim(\mathcal{C}) = \sum_{X \in \mathcal{O}(\mathcal{C})} d_X^2.$$

EXAMPLE 18.2.1. Let $\mathcal{C} = \mathrm{Rep}(G)$ be the category of finite-dimensional representations of a finite group G. Then $\dim(\mathrm{Rep}\, G) = |G|$.

REMARK 18.2.3. If $\mathcal{C}$ is pivotal but not necessarily spherical, we can still define its dimension by $\dim \mathcal{C} = \sum d_X d_{X^*}$. Moreover, one can show that $|X|^2 = d_X d_{X^*}$ is independent of a choice of the pivotal structure and in fact can be defined without using pivotal structure at all ([**EGNO2015**, Section 7.21]); thus, $\dim \mathcal{C}$ is well-defined for any fusion category. However, we will only be using it for spherical categories.

THEOREM 18.2.3. *Let $\mathcal{C}$ be a pivotal fusion category over a field $\mathbf{k}$ of characteristic zero. Then*

(1) $\dim \mathcal{C} \neq 0$.
(2) *If $\mathbf{k} \subset \mathbb{C}$, then for any simple X, d_X is real and non-zero, and $\dim \mathcal{C}$ is real and ≥ 1.*

This result is non-trivial, since a priori, all dimensions are just elements of $\mathbf{k}$. We refer the reader to [**EGNO2015**, Theorem 7.21.12] for the proof. Note that it is easily shown to be false in positive characteristic.

18.3. Grothendieck ring

DEFINITION 18.3.1. Let $\mathcal{C}$ be a fusion category. Its Grothendieck group $\mathbf{Gr}(\mathcal{C})$ is the abelian group freely generated by isomorphism classes of simple objects. We define, for an object $X \in \mathcal{C}$, its class $[X] \in \mathbf{Gr}(\mathcal{C})$ by

$$[X] = \sum_{i \in \mathcal{O}(\mathcal{C})} [X : X_i][X_i].$$

It is obvious from the definition that $[X \oplus Y] = [X] + [Y]$.

Since $\mathcal{C}$ is a fusion category, $\mathbf{Gr}(\mathcal{C})$ has an additional structure of a ring, with unit given by $1 = [\mathbf{1}]$ and multiplication given by

$$[X] \cdot [Y] = [X \otimes Y].$$

This ring comes with a distinguished basis $[X_i]$ such that the structure constants N_{ij}^k, defined by the formula below, are non-negative integers:

$$[X_i][X_j] = \sum_k N_{ij}^k [X_k]$$

(these constants are sometimes called the *Clebsch–Gordan coefficients*). One can also turn the ring $\mathbf{Gr}(\mathcal{C})$ into an algebra over $\mathbf{k}$ by defining

$$\mathbf{Gr}_{\mathbf{k}}(\mathcal{C}) = \mathbf{Gr}(\mathcal{C}) \otimes_{\mathbb{Z}} \mathbf{k},$$

in other words, replacing the abelian group with basis $[X_i]$ by a vector space over $\mathbf{k}$ with the same basis; in physics, this algebra is sometimes called the *Verlinde algebra* of $\mathcal{C}$.

EXERCISE 18.3.1. Show that for a spherical fusion category, $x \mapsto \dim(x)$ gives an algebra homomorphism

$$\mathbf{Gr}_{\mathbf{k}}(\mathcal{C}) \to \mathbf{k}.$$

EXERCISE 18.3.2. Let $\mathcal{C}$ be a spherical fusion category. Let $\mathbf{d} \in \mathbf{Gr_k}(\mathcal{C})$ be defined by

$$\mathbf{d} = \sum_{X \in \mathcal{O}(\mathcal{C})} d_X [X]. \tag{18.3.1}$$

Prove that then, for any $y \in \mathbf{Gr_k}(\mathcal{C})$, we have

$$y \cdot \mathbf{d} = \mathbf{d} \cdot y = \dim(y)\mathbf{d}.$$

One can think of $\mathbf{d}$ as an analog of the regular representation: for $\mathcal{C} = \operatorname{Rep} G$, $\mathbf{d}$ is the class of the regular representation of G.

18.4. Analog of Frobenius pairing

Our next goal is to show that for spherical fusion categories, we can repeat many of the same constructions we did for Frobenius algebras. We begin with the most fundamental one, the Frobenius pairing.

Recall that for a symmetric Frobenius algebra A, we have a counit $\varepsilon\colon A \to \mathbf{k}$ which gives rise to non-degenerate symmetric bilinear pairing $(a, b) = \operatorname{tr}(ab)$. More generally, for any $n \geq 1$ we defined the map

$$A^{\otimes n} \to \mathbf{k}$$
$$a_1 \otimes \cdots \otimes a_n \mapsto \varepsilon(a_1 \dots a_n)$$

which was invariant under cyclic permutation of arguments.

It is easy to define an analog of this for spherical categories.

DEFINITION 18.4.1. Let $\mathcal{C}$ be a spherical fusion category. We define the functor $\mathcal{C}^{\boxtimes n} \to \mathbf{Vec_f}$ by

$$\langle A_1 \boxtimes \cdots \boxtimes A_n \rangle = \operatorname{Hom}(\mathbf{1}, A_1 \otimes \cdots \otimes A_n).$$

We will also frequently write $\langle A_1, \dots, A_n \rangle$ instead of $\langle A_1 \boxtimes \cdots \boxtimes A_n \rangle$.

In particular, for $n = 2$ we get a functor $\mathcal{C} \boxtimes \mathcal{C} \to \mathbf{Vec_f}$. Passing to Grothendieck rings, it gives a bilinear map

$$A \otimes A \to \mathbf{k}, \quad A = \mathbf{Gr_k}(\mathcal{C}).$$

It is obvious from Exercise 18.2.1 that this bilinear pairing is non-degenerate; it is an analog of the Frobenius pairing $\varepsilon(ab)$.

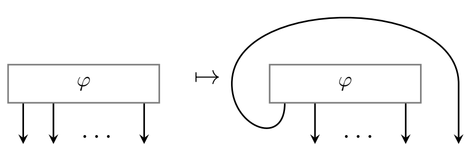
The following lemma, whose proof is immediate from Theorem 18.1.1, is an analog of the cyclic invariance of the trace for symmetric Frobenius algebras.

LEMMA 18.4.1. *Define the functorial isomorphism*

$$z\colon \langle A_1 \boxtimes \cdots \boxtimes A_n \rangle \xrightarrow{\sim} \langle A_2 \boxtimes \cdots \boxtimes A_n \boxtimes A_1 \rangle \tag{18.4.1}$$

by (18.4.2). *Then* $z^n = 1$.

(18.4.2)



Note that the definition of z requires an isomorphism $c \to c^{**}$, i.e., a pivotal structure.

As a corollary, we see that we have a well-defined functor $\langle\ \rangle\colon \mathcal{C}^{\boxtimes S} \to \mathbf{Vec_f}$ for any cyclically ordered set S.

For $\varphi \in \langle A_1 \boxtimes \cdots \boxtimes A_m, X\rangle$, $\psi \in \langle X^*, B_1, \ldots, B_n\rangle$ we can define their composition

$$\varphi \underset{X}{\circ} \psi = (1 \otimes \mathrm{ev}_X \otimes 1)\varphi \otimes \psi \in \langle A_1, \ldots, A_m, B_1, \ldots, B_n\rangle. \tag{18.4.3}$$

In particular, taking direct sum over all simple X, we get a morphism

$$\bigoplus_{X\in\mathcal{O}(\mathcal{C})} \langle A_1, \ldots, A_m, X\rangle \otimes \langle X^*, B_1, \ldots, B_n\rangle \simeq \langle A_1, \ldots, A_m, B_1, \ldots, B_n\rangle \tag{18.4.4}$$

Lemma 18.4.2. *In a spherical fusion category $\mathcal{C}$, the map* (18.4.4) *is an isomorphism.*

Indeed, it immediately follows from Lemma 18.2.1 that we have a canonical isomorphism

$$\bigoplus_{X\in\mathcal{O}(\mathcal{C})} \langle X^*, B_1, \ldots, B_n\rangle \otimes X \simeq B_1 \otimes \cdots \otimes B_n$$

which implies the statement of the lemma.

Note that for any objects $A, B \in \operatorname{Obj}\mathcal{C}$, we have a non-degenerate pairing $\operatorname{Hom}_{\mathcal{C}}(A, B) \otimes \operatorname{Hom}_{\mathcal{C}}(A^*, B^*) \to \mathbf{k}$ defined by

$$(\varphi, \varphi') = (\mathbf{1} \xrightarrow{\mathrm{coev}_A} A \otimes A^* \xrightarrow{\varphi\otimes\varphi'} B \otimes B^* \xrightarrow{\mathrm{ev}_B} \mathbf{1}). \tag{18.4.5}$$

In particular, this gives us a non-degenerate pairing

$$\langle A_1, \ldots, A_n\rangle \otimes \langle A_n^*, \ldots, A_1^*\rangle \to \mathbf{k} \tag{18.4.6}$$

and thus, functorial isomorphisms $\langle A_1, \ldots, A_n\rangle^* \simeq \langle A_n^*, \ldots, A_1^*\rangle$ compatible with the cyclic permutations (18.4.1).

Remark 18.4.1. Note that pairing (18.4.5) requires a pivotal structure. For a general fusion category, we have a partial replacement for that. Namely, let X be a simple object in a fusion category $\mathcal{C}$. Define the pairing

$$(\ ,\ )_X\colon\ \operatorname{Hom}(A, X) \otimes \operatorname{Hom}(A^*, X^*) \to \mathbf{k} \tag{18.4.7}$$

by

$$(\mathbf{1} \xrightarrow{\mathrm{coev}_A} A \otimes A^* \xrightarrow{\varphi\otimes\varphi'} X \otimes X^*) = (\varphi, \psi)_X \cdot \mathrm{coev}_X$$

(note that since X is simple, Schur's lemma implies that the space $\operatorname{Hom}_{\mathcal{C}}(\mathbf{1}, X\otimes X^*)$ is one-dimensional). In the pivotal case, this differs from the pairing (18.4.5) by a factor of $\dim X$: for $\varphi \in \operatorname{Hom}(A, X)$, $\psi \in \operatorname{Hom}(A^*, X^*)$, we have $(\varphi, \psi)_X = \frac{1}{\dim X}(\varphi, \psi)$.

18.5. Graphical calculus continued: graphs on surfaces

Lemma 18.4.1 shows that in a pivotal fusion category, the vector space $\langle A_1 \boxtimes \cdots \boxtimes A_n\rangle = \operatorname{Hom}(\mathbf{1}, A_1 \otimes \cdots \otimes A_n)$, up to a canonical isomorphism, only depends on the cyclic order of $A_1, \ldots, A_n$. Moreover, we have canonical isomorphisms

$$\operatorname{Hom}(A_1 \otimes \ldots A_k, B_1 \otimes \cdots \otimes B_m) \simeq \langle B_1, \ldots, B_m, A_k^*, \ldots, A_1^*\rangle.$$

This makes it possible to modify our graphical calculus discussed in Section 18.1, replacing all rectangular boxes by small disks, or even easier, just by vertices; orientation of the plane gives a counterclockwise cyclic order on the set of edges meeting at this vertex. Thus, we make the following definition, which is a minor modification of the notion of $\mathcal{C}$-colored diagram discussed in Section 18.1.

DEFINITION 18.5.1. Let M be an oriented 2-manifold with boundary (not necessarily compact), and let $\mathcal{C}$ be a spherical fusion category.

A $\mathcal{C}$-colored graph on M is the following collection of data:

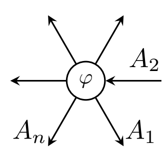
- a finite collection of points (vertices) in the interior of M
- a finite collection of non-intersecting smooth directed arcs (embedded intervals) and directed circles (images of S^1) in M. Endpoints of arcs are all distinct and can lie either on the boundary of M or at the vertices. At the endpoints on the boundary of M, the lines must be transversal to the boundary. Arcs meeting at a vertex must have distinct tangent vectors. Other than the endpoints, arcs and circles do not intersect the boundary of M or vertices.
- coloring, i.e., assignment of an object of $\mathcal{C}$ to every arc or circle, and assignment of a morphism in $\mathcal{C}$ to every vertex. Namely, if $A_1, \dots, A_k$ are the colors of the arcs meeting at a vertex v, listed in counterclockwise order, then the vertex must be colored by a morphism

  $$\varphi \in \langle A_1^{\varepsilon_1}, \dots, A_k^{\varepsilon_k} \rangle$$

  where

  $$A_i^{\varepsilon_i} = \begin{cases} A_i, & \text{if the corresponding arc is directed outward} \\ A_i^* & \text{if the corresponding arc is directed inward} \end{cases} \tag{18.5.1}$$

  see Figure 18.5.1.



$$\varphi \in \langle A_1 \boxtimes A_2^* \boxtimes \cdots \boxtimes A_n \rangle$$

FIGURE 18.5.1. Coloring the vertices of $\mathcal{C}$-colored graphs. We put a small disk at the vertex.

For presentation purposes, we will frequently draw small circles in place of vertices and put the corresponding color (i.e., a morphism of $\mathcal{C}$) inside the disk.

Such colored graphs, with minor modifications, have appeared in the physics literature under the name "string-nets", see [**LW2005**].

The following theorem is a modification of Theorem 18.1.1; we give it here in the form written down in [**Kir2011**].

THEOREM 18.5.1. *There is a unique way to assign to every $\mathcal{C}$-colored planar graph Γ in a disk $D \subset \mathbb{R}^2$ a vector*

$$\langle \Gamma \rangle_D \in \langle A_1^{\varepsilon_1} \boxtimes \cdots \boxtimes A_k^{\varepsilon_k} \rangle \tag{18.5.2}$$

where $A_1, \dots, A_k$ are colors of edges meeting the boundary of D (legs), taken in counterclockwise order, with the same sign conventions as before so that the following conditions are satisfied:

(1) $\langle \Gamma \rangle$ *only depends on the isotopy class of* Γ.

(2) *If* Γ *is a single vertex colored by* $\varphi \in \langle A_1^{\varepsilon_1} \boxtimes \cdots \boxtimes A_k^{\varepsilon_k} \rangle$*, then* $\langle \Gamma \rangle = \varphi$*.*
(3) *Local relations shown in Figure 18.5.2 hold.*

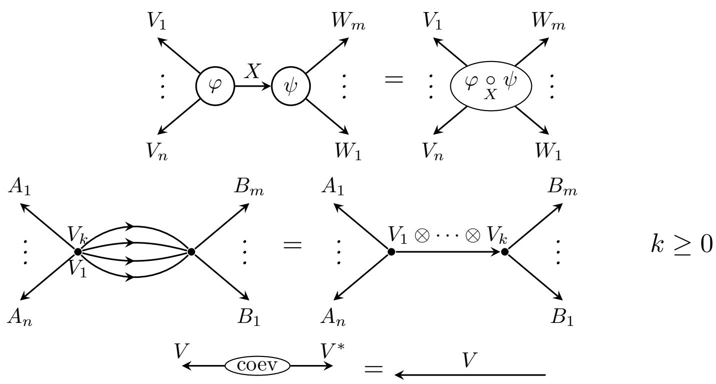


FIGURE 18.5.2. Local relations for colored graphs. Here $\varphi \circ \psi$ is defined by (18.4.3).

Local relations should be understood as follows: for any pair Γ, Γ' *of colored graphs which are identical outside a subdisk* $D' \subset D$*, and in this disk are homeomorphic to the graphs shown in Figure 18.5.2, we must have* $\langle \Gamma \rangle = \langle \Gamma' \rangle$*.*

We will call the vector $\langle \Gamma \rangle$ *the* evaluation *of* Γ*.*

In particular, for a closed $\mathcal{C}$-colored graph $\Gamma \subset \mathbb{R}^2$, $\langle \Gamma \rangle \in \mathbf{k}$ is a number.

The evaluation map $\Gamma \to \langle \Gamma \rangle$ defined in Theorem 18.5.1 has a number of useful properties which can be found, e.g., in [**Kir2011**]. We only give here one of them.

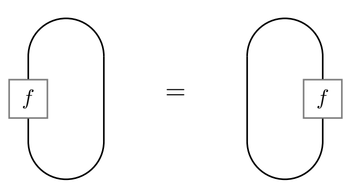
LEMMA 18.5.2. *Let* Γ *be a* $\mathcal{C}$*-colored graph in a disk* D*, and let* $D' \subset D$ *be a subdisk such that* $\partial D'$ *does not contain vertices of* Γ *and meets edges of* Γ *transversally. Then* $\langle \Gamma \rangle_D$ *will not change if we replace subgraph* $\Gamma \cap D'$ *by a single vertex colored by* $\langle \Gamma \cap D' \rangle_{D'}$*.*

Moreover, it turns out that we can also define $\langle \Gamma \rangle$ for a $\mathcal{C}$-colored graph on the sphere.

LEMMA 18.5.3. *Let* Γ *be a* $\mathcal{C}$*-colored graph on the sphere* S^2*. Choose a point* $p \in S^2$ *which does not lie on any of the edges of* Γ*; then we can identify* $S^2 \backslash \{p\} \simeq \mathbb{R}^2$ *and thus define* $\langle \Gamma \rangle_{S^2 \backslash \{p\}} \in \mathbf{k}$*.*

Then so-defined $\langle \Gamma \rangle_{S^2 \backslash \{p\}}$ *does not depend on the choice of point* p*.*

Indeed, it suffices to show that the number $\langle \Gamma \rangle_{S^2 \backslash \{p\}}$ does not change when we move point p through one of the edges of Γ. But this is exactly the equality of left and right traces in Definition 17.3.3:

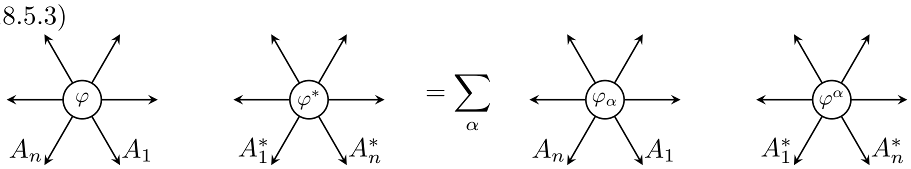


Lemma 18.5.3 was first proved in [**BW1996**]; in fact, this was the motivation for introducing the notion of a spherical category (which also explains the name).

To simplify the future constructions, we introduce two additional conventions for working with colored graphs.

CONVENTION 18.5.1. If a colored graph contains a pair of vertices, one with outgoing edges labeled $A_1, \dots, A_n$ and the other with edges labeled $A_n^*, \dots, A_1^*$, and the vertices are labeled by a pair of letters φ, φ^* (or ψ, ψ^*, or $\dots$), it will stand for summation over the dual bases:

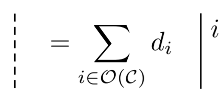
(18.5.3)



where $\varphi_\alpha \in \langle A_1, \dots, A_n \rangle$, $\varphi^\alpha \in \langle A_n^*, \dots, A_1^* \rangle$ are dual bases with respect to the pairing (18.4.6).

CONVENTION 18.5.2. A dashed line in a colored graph will stand for the sum of all colorings of an edge by simple objects i, each taken with coefficient $d_i = \dim X_i$:

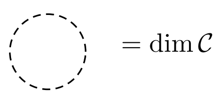
(18.5.4)



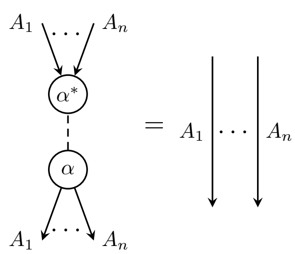
Using these conventions, we can formulate some properties of colored graphs.

LEMMA 18.5.4. *We have the following equalities* (*here and below, equality of two colored graphs should be understood as equality* $\langle \Gamma_1 \rangle = \langle \Gamma_2 \rangle$)*:*

(18.5.5)



(18.5.6)



18.6. Canonical algebra

For Frobenius algebras, non-degeneracy of the pairing $(\ ,\ )$ also gave us the coevaluation linear map $\mathrm{coev}\colon \mathbf{k} \to A \otimes A$, or equivalently, an element $e = \mathrm{coev}(1) = \sum x_i \otimes x^i \in A \otimes A$. Again, it has an obvious analog for spherical fusion categories.

LEMMA 18.6.1. *Let* $\mathcal{C}$ *be a spherical fusion category. Let*

$$E = \bigoplus_{i \in \mathcal{O}(\mathcal{C})} X_i \boxtimes X_i^* \in \mathcal{C} \boxtimes \mathcal{C}. \tag{18.6.1}$$

Then we have the following properties:

(1) *So defined E is independent of the choice of representatives of isomorphism classes of simple objects: different choices give canonically isomorphic objects E.*
(2) *It is symmetric: there is an isomorphism $c\colon E \simeq P(E)$, where $P\colon \mathcal{C}\boxtimes\mathcal{C} \to \mathcal{C}\boxtimes\mathcal{C}$ is given by $P(X\boxtimes Y) = Y\boxtimes X$, and $c^2 = 1$.*
(3) *For any $X \in \mathcal{C}$, we have a canonical functorial isomorphism*

$$X \simeq \bigoplus_i \langle X, X_i\rangle \otimes X_i^*$$

The proof is straightforward; details can be found, e.g., in [**BK2010**].

Remark 18.6.1. If we consider E as an object in $\mathcal{C}\boxtimes\mathcal{C}^{\mathrm{mp}}$, then one can show that it has a canonical structure of an algebra in that category; this algebra is sometimes called the *canonical algebra* of $\mathcal{C}$, see [**EGNO2015**, Example 7.9.14], where it is shown that $E \simeq \underline{\mathrm{Hom}}(\mathbf{1},\mathbf{1})$, where $\underline{\mathrm{Hom}}$ is the so-called inner hom.

We also have an analog of bicentrality (compare with Lemma 5.1.4).

Lemma 18.6.2. *Let $\mathcal{C}$ be a spherical fusion categor, and let $E \in \mathcal{C}\boxtimes\mathcal{C}$ be defined by (18.6.1). Then E is bicentral: for any $A \in \mathcal{C}$, we have canonical functorial isomorphisms*

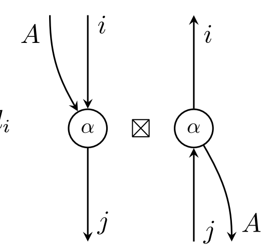
$$\gamma_A\colon \bigoplus (A\otimes X_i)\boxtimes X_i^* \simeq \bigoplus X_i \boxtimes (X_i^*\otimes A)$$
$$\gamma'_A\colon \bigoplus (X_i\otimes A)\boxtimes X_i^* \simeq \bigoplus X_i \boxtimes (A\otimes X_i^*)$$

which satisfy

$$\gamma_{A\otimes B} = (\gamma_A\otimes 1_B)\circ(1_A\otimes\gamma_B).$$

and similarly for γ'.

Proof. We define the isomorphism γ by the following picture:

(18.6.2) $$\gamma_A = \bigoplus_{i,j\in\mathcal{O}(\mathcal{C})} d_i$$ 

We refer the reader to [**BK2010**] for the proof that so defined γ satisfies $\gamma_{A\otimes B} = \gamma_A\circ\gamma_B$; another proof of centrality of E is given in [**EGNO2015**, Proposition 7.18.5].

Construction of isomorphism γ' is similar and is left to the reader. □

Note that both the symmetry of E and bicentrality use pivotal structure; without the pivotal structure, these result need some changes.

Exercise 18.6.1. Let $\mathcal{C}$ be a fusion category (not necessary pivotal), and let E be defined by (18.6.1).

(1) Show that $P(E) \simeq (D\otimes 1)E \simeq (1\otimes D^{-1})E$, where functor $D\colon \mathcal{C}\to\mathcal{C}$ is given by $D(X) = X^{**}$.

(2) Use pairing defined in Remark 18.4.1 to construct functorial isomorphisms

$$\gamma_A\colon \bigoplus (A\otimes X_i)\boxtimes X_i^* \simeq \bigoplus X_i \boxtimes (X_i^*\otimes A^{**})$$
$$\gamma'_A\colon \bigoplus (X_i\otimes A)\boxtimes X_i^* \simeq \bigoplus X_i \boxtimes (A\otimes X_i^*)$$

(note A^{**} in the first isomorphism!)

18.7. Drinfeld center

For an associative algebra A, we can define its center $z(A)$ (which is a commutative algebra) and $A/[A,A]$, which is just a vector space. Both constructions played an important role in the construction of extended 2d TQFTs; in particular, we have shown that for a separable symmetric Frobenius algebra one has a natural map $A\to z(A)\colon a\mapsto \sum x_i a x^i$, which descends to an isomorphism $A/[A,A]\simeq z(A)$.

In this section, we define similar notions for a spherical fusion category, starting with the categorical analog of the center, which in this case is called the *Drinfeld center*. Our exposition follows [**EGNO2015**, Section 7.13].

Definition 18.7.1. Let $\mathcal{C}$ be a monoidal category. Its *Drinfeld center* is the category $\mathcal{Z}(\mathcal{C})$ with the following objects and morphisms:

- Objects: pairs (Z,γ), where $Z\in\mathcal{C}$ and γ is a functorial isomorphism

$$\gamma_A\colon A\otimes Z\to Z\otimes A$$

satisfying the following conditions:

(1) Hexagon axiom: the following diagram is commutative:

$$\begin{tikzcd} A\otimes B\otimes Z \arrow[rr, "\gamma_{A\otimes B}"] \arrow[dr, "1_A\otimes\gamma_B"'] & & Z\otimes A\otimes B \\ & A\otimes Z\otimes B \arrow[ur, "\gamma_A\otimes 1_B"'] & \end{tikzcd} \tag{18.7.1}$$

(For better readability, we have dropped associativity isomorphisms from this diagram; compare with hexagon axiom in Definition 2.4.1.)

(2) Triangle axiom: $\gamma_{\mathbf{1}}\colon \mathbf{1}\otimes Z\to Z\otimes\mathbf{1}$ coincides with composition of unit isomorphisms $\mathbf{1}\otimes Z\simeq Z\simeq Z\otimes\mathbf{1}$.

γ is usually called the *half-braiding*.

- Morphisms:

$$\mathrm{Hom}_{\mathcal{Z}(\mathcal{C})}((Z,\gamma),(Z',\gamma')) = \{f\in \mathrm{Hom}_{\mathcal{C}}(Z,Z') \mid \gamma'(1\otimes f) = (f\otimes 1)\gamma\}.$$

Example 18.7.1. Let $\mathbf{Vec}_G$ be the category of G-graded vector spaces. Then an object of $\mathcal{Z}(\mathbf{Vec}_G)$ is a G-graded vector space together with a collection of isomorphisms $V_{gh}\simeq V_{hg}$ for any g. Equivalently, it is a G-graded vector space together with an action of G such that

$$gV_h\subset V_{ghg^{-1}}.$$

This can also be described as a category of representations of algebra $A=\mathbf{k}(G)\#\mathbf{k}[G]$, where $\mathbf{k}(G)$ is the algebra of functions on G with pointwise multiplication, $\mathbf{k}[G]$ is the group algebra, and $\#$ is the smash product (also called the cross product) of algebras with respect to natural action of $\mathbf{k}[G]$ on $\mathbf{k}(G)$.

More generally, it can be shown that for $\mathcal{C} = \operatorname{Rep} H$, where H is a finite-dimensional Hopf algebra, $\mathcal{Z}(\operatorname{Rep} H)$ is equivalent to the representation category of a new Hopf algebra $D(H)$, called the *Drinfeld double* of H. The algebra $D(H)$ contains both H and H^* as subalgebras, and as a vector space, $D(H) \simeq H \otimes H^*$; however, H and H^* do not commute in $D(H)$, so multiplication in $D(H)$ is non-trivial. We refer the reader to [**EGNO2015**, Section 7.14] for details.

LEMMA 18.7.1. *Let $\mathcal{C}$ be a monoidal category.*

(1) *$\mathcal{Z}(\mathcal{C})$ has a natural structure of a monoidal category; if $\mathcal{C}$ is rigid, so is $\mathcal{Z}(\mathcal{C})$.*
(2) *If $\mathcal{C}$ is a $\mathbf{k}$-linear category, then $\mathcal{Z}(\mathcal{C})$ is also a $\mathbf{k}$-linear category.*
(3) *If $\mathcal{C}$ is a pivotal (respectively, spherical) category, then $\mathcal{Z}(\mathcal{C})$ is also pivotal (respectively, spherical).*

The proof is mostly straightforward; details can be fond in [**EGNO2015**, Section 7.13].

LEMMA 18.7.2. *Let $\mathcal{C}$ be a finite tensor category. Then one has an equivalence of $\mathbf{k}$-linear categories*

$$\operatorname{Fun}_{\mathcal{C}^e}(\mathcal{C},\mathcal{C}) \simeq \mathcal{Z}(\mathcal{C})$$
$$F \mapsto F(\mathbf{1}),$$

where $\mathcal{C}^e = \mathcal{C} \boxtimes \mathcal{C}^{\mathrm{mp}}$.

The proof is again a straightforward comparison of the definitions; see details in [**EGNO2015**, Proposition 7.13.8].

THEOREM 18.7.3. *The Drinfeld center of a fusion category is itself a fusion category.*

PROOF. Indeed, by Lemma 18.7.1, $\mathcal{Z}(\mathcal{C})$ is a rigid monoidal $\mathbf{k}$-linear category, and the unit object of $\mathcal{Z}(\mathcal{C}) = \mathbf{1}$ is simple. Thus, it only remains to show that $\mathcal{Z}(\mathcal{C})$ is finite and semisimple. This immediately follows from Lemma 18.7.2 together with Theorem 16.4.2 □

THEOREM 18.7.4. *Let $\mathcal{C}$ be a spherical fusion category. Then $\mathcal{Z}(\mathcal{C})$ is also a spherical fusion category, and*

$$\dim \mathcal{Z}(\mathcal{C}) = (\dim \mathcal{C})^2.$$

See [**EGNO2015**, Proposition 9.3.4].

REMARK 18.7.1. This shows that contrary to naive intuition, the Drinfeld center of $\mathcal{C}$ is “larger” than $\mathcal{C}$ (at least, has larger dimension). In particular,

$$\mathbf{Gr}(\mathcal{Z}(\mathcal{C})) \neq z(\mathbf{Gr}(\mathcal{C})).$$

Note that in general, it is very hard to describe all simple objects in $\mathcal{Z}(\mathcal{C})$, even knowing all simple objects of $\mathcal{C}$.

Since every object of $\mathcal{Z}(\mathcal{C})$ is also an object of $\mathcal{C}$, it is easy to extend our graphical calculus, allowing $\mathcal{C}$-colored graphs where some arcs are colored by objects of $\mathcal{Z}(\mathcal{C})$. In these graphs, we will show objects of $\mathcal{Z}(\mathcal{C})$ by double green lines and the half-braiding isomorphism $\gamma_A\colon A \otimes Z \to Z \otimes A$ by a crossing as in Figure 18.7.1.

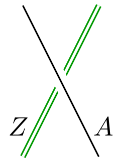
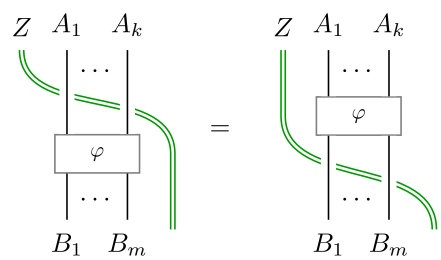
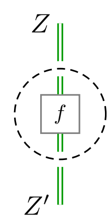


FIGURE 18.7.1. Graphical presentation of the half-braiding $\gamma_A\colon A\otimes Z\to Z\otimes A$, $A\in\mathcal{C}$, $Z\in\mathcal{Z}(\mathcal{C})$

EXERCISE 18.7.1. Show that for any morphism $\varphi\colon A_1\otimes\cdots\otimes A_k\to B_1\otimes\cdots\otimes B_m$ in $\mathcal{C}$, we have the following equality:



EXERCISE 18.7.2. Let $\mathcal{C}$ be a spherical fusion category and let (Z,γ), (Z',γ') be two objects of $\mathcal{Z}(\mathcal{C})$. Define a map $P\colon \operatorname{Hom}_{\mathcal{C}}(Z,Z')\to\operatorname{Hom}_{\mathcal{C}}(Z,Z')$ (note that at the moment we are not requiring any compatibility with half-braidings γ,γ') by the graph below:

(18.7.2) $$P(f)=\frac{1}{\dim\mathcal{C}}$$ 

Use Lemma 18.5.4 to prove that then P is a projector onto $\operatorname{Hom}_{\mathcal{Z}(\mathcal{C})}(Z,Z')\subset\operatorname{Hom}_{\mathcal{C}}(Z,Z')$.

Finally, note that for algebras, we had a natural embedding $z(A)\subset A$ and also the map $A\to z(A)\colon a\mapsto\sum x_i a x^i$. These two maps are not inverses of each other; their composition is multiplication by the Euler element $w=\sum x_i x^i$.

Similar constructions exist for categories. Namely, we have the obvious forgetful functor

$$\begin{aligned}F\colon \mathcal{Z}(\mathcal{C})&\to\mathcal{C}\\(Z,\gamma)&\mapsto Z.\end{aligned}\tag{18.7.3}$$

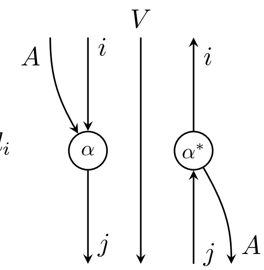
THEOREM 18.7.5. *Let $\mathcal{C}$ be a spherical fusion category. Then the forgetful functor F has a left adjoint functor*

$$I\colon\mathcal{C}\to\mathcal{Z}(\mathcal{C})$$

given by

$$I(V)=\bigoplus_{i\in\mathcal{O}(\mathcal{C})}X_i\otimes V\otimes X_i^*$$

with γ *given by:*

(18.7.4)
$$\gamma_A = \bigoplus_{i,j\in\mathcal{O}(\mathcal{C})} d_i$$


(*compare with* (18.6.2)).

We refer the reader to [**BK2010**, Theorem 2.3] for the proof.

Note that this implies that any object $Z \in \mathcal{Z}(\mathcal{C})$ is a direct summand of an object of the form $I(A)$, $A \in \mathcal{C}$: we can take $A = F(Z)$.

18.8. Center and universal trace

Recall from Section 9.4 that for a bimodule M over an associative algebra A, we defined the spaces of invariants and coinvariants by

$$\begin{aligned} M^A &= \mathrm{Hom}(A, M) = \{m \in M \mid am = ma \text{ for any } a \in A\} \\ M_A &= A \otimes_{A^e} M = M/\langle am - ma\rangle \end{aligned} \tag{18.8.1}$$

where Hom is computed in the category of A-bimodules, or equivalently, modules over $A^e = A \otimes A^{\mathrm{op}}$. In particular, for $M = A$, these definitions give

$$\begin{aligned} HH^0(A) &= A^{(A)} = z(A), \\ HH_0(A) &= A_{(A)} = A/[A, A]. \end{aligned}$$

It is natural to ask if we can define an analog of this for a bimodule category $\mathcal{M}$ over a finite tensor category $\mathcal{C}$. In the previous section, we have already given an answer in the special case $\mathcal{M} = \mathcal{C}$, defining the Drinfeld center $\mathcal{Z}(\mathcal{C})$, which is an analog of $z(A)$. In this section we consider the more general case. Our exposition follows [**FSS2017**]; other references for the material of this section include [**Gre2013**] (probably one of the earliest references on this topic) and [**BZFN2010**, Section 5] (in the context of ∞-categories).

DEFINITION 18.8.1. Let $\mathcal{M}$ be a finite bimodule category over a finite tensor category $\mathcal{C}$. Define the *categorical center* $\mathcal{Z}(\mathcal{M})$ as the category with the following objects and morphisms:

- Objects: pairs (M, γ), where $M \in \mathcal{M}$ and γ is a functorial isomorphism
$$\gamma_A \colon A \otimes M \to M \otimes A, \quad A \in \mathcal{C},$$
satisfying the hexagon and unit axioms similar to those in Definition 18.7.1.
- Morphisms:

$$\mathrm{Hom}_{\mathcal{Z}(\mathcal{M})}((M,\gamma),(M',\gamma')) = \{f \in \mathrm{Hom}_{\mathcal{M}}(M, M') \mid \gamma'(1 \otimes f) = (f \otimes 1)\gamma\}.$$

LEMMA 18.8.1. *Under the assumptions of Definition* 18.8.1, *one has an equivalence of categories*

$$\begin{aligned} \mathrm{Fun}_{\mathcal{C}^e}(\mathcal{C}, \mathcal{M}) &\simeq \mathcal{Z}(\mathcal{M}) \\ F &\mapsto F(\mathbf{1}) \end{aligned}$$

As an immediate corollary, we see that $\mathcal{Z}(\mathcal{M})$ is a finite $\mathbf{k}$-linear category; one can show (FIXME) that if $\mathcal{M}$ is semisimple, then so is $\mathcal{Z}(\mathcal{M})$.

Let us now try to define the analog of coinvariants M_A. For bimodules over an algebra, we had defined $M_A = M/\langle am - ma\rangle$. A better way to define M_A is by using the universal property: for any vector space L,

$$\mathrm{Hom}(M_A, L) = \mathrm{Bal}(M, L)$$

where Bal stands for the space of balanced maps, i.e., maps $f\colon M \to L$ satisfying the condition

$$f(am) = f(ma), \quad a \in A.$$

We can now define an analog of this for bimodule categories. First, we define an analog of balanced maps, essentially in the same way as we did in Section 16.5 when defining the balanced Deligne product of categories.

Definition 18.8.2. Let $\mathcal{M}$ be a bimodule category over a finite tensor category $\mathcal{C}$, and let $\mathcal{L}$ be a $\mathbf{k}$-linear category. A $\mathcal{C}$-balanced functor $\mathcal{M} \to \mathcal{L}$ is a pair (F, J), where $F\colon \mathcal{M} \to \mathcal{L}$ is a $\mathbf{k}$-linear right exact functor, and J is a functorial isomorphism

$$J\colon F(C \otimes M) \simeq F(M \otimes C), \quad M \in \mathcal{M}, C \in \mathcal{C}$$

satisfying the obvious compatibility conditions (compare with Definition 16.5.1).

One can also define a notion of morphisms of $\mathcal{C}$-balanced functors. We denote by $\mathrm{Bal}_{\mathcal{C}}(\mathcal{M}, \mathcal{L})$ the category of $\mathcal{C}$-balanced functors $\mathcal{M} \to \mathcal{L}$.

Definition 18.8.3. Under the assumptions of Definition 18.8.2, a categorical trace is a $\mathbf{k}$-linear category $\circlearrowleft\!\!\!\!\otimes\mathcal{M}$ together with a $\mathcal{C}$-balanced functor $F\colon \mathcal{M} \to \circlearrowleft\!\!\!\!\otimes\mathcal{M}$ such that for any $\mathbf{k}$-linear category $\mathcal{L}$, functor F induces an equivalence of categories

$$\mathrm{Bal}_{\mathcal{C}}(\mathcal{M}, \mathcal{L}) \simeq \mathrm{Fun}(\circlearrowleft\!\!\!\!\otimes\mathcal{M}, \mathcal{L}).$$

As with the definition of the balanced tensor product, it is easy to show that $\circlearrowleft\!\!\!\!\otimes\mathcal{M}$, if it exists, is unique up to a unique equivalence. However, existence is not obvious.

Example 18.8.1. Let $\mathcal{M}_{\mathcal{C}}$, ${}_{\mathcal{C}}\mathcal{N}$ be right and left finite module categories over $\mathcal{C}$. Then Deligne's product $\mathcal{M}\boxtimes\mathcal{N}$ is a $\mathcal{C}$-bimodule category, and $\circlearrowleft\!\!\!\!\otimes(\mathcal{M}\boxtimes\mathcal{N}) \simeq \mathcal{M}\boxtimes_{\mathcal{C}}\mathcal{N}$; thus, in this case the existence of the balanced product (Theorem 16.5.1) implies the existence of the categorical trace $\circlearrowleft\!\!\!\!\otimes(\mathcal{M}\boxtimes\mathcal{N})$.

Theorem 18.8.2. *Let $\mathcal{M}$ be a finite bimodule category over a finite tensor category $\mathcal{C}$. Then the categorical trace $\circlearrowleft\!\!\!\!\otimes\mathcal{M}$ exists; moreover, we have an equivalence of categories*

$$\mathcal{C} \boxtimes_{\mathcal{C}^e} \mathcal{M} \simeq \circlearrowleft\!\!\!\!\otimes\mathcal{M}$$

where, as before, $\mathcal{C}^e = \mathcal{C} \boxtimes \mathcal{C}^{\mathrm{mp}}$.

A proof of this theorem can be found in [**FSS2017**, Theorem 3.3].

Finally, recall that for a module over a separable symmetric Frobenius algebra, one has isomorphism $M_{(A)} \simeq M^{(A)}$ given by $m \mapsto \sum x_i m x^i$, see Section 9.4.1. The following theorem is a categorical analog of this statement.

Theorem 18.8.3. *Let $\mathcal{M}$ be a finite bimodule category over a pivotal fusion category $\mathcal{C}$. Let functor $I\colon \mathcal{M} \to \mathcal{Z}(\mathcal{M})$ be defined by*

$$I(M) = \bigoplus X_i \otimes M \otimes X_i^* \tag{18.8.2}$$

with the half-braiding on $I(M)$ defined in the same way as in Theorem 18.7.5.

Then I has a natural structure of a balanced functor, and it gives an equivalence of categories $\mathcal{Z}(\mathcal{M}) \simeq \circledast\mathcal{M}$.

Proof. FIXME

Balancing structure is given by functorial isomorphism

$$\gamma'_A \colon \bigoplus (X_i \otimes A) \boxtimes X_i^* \simeq \bigoplus X_i \boxtimes (A \otimes X_i^*)$$

defined in Lemma 18.6.2. □

CHAPTER 19

Turaev–Viro theory

In this chapter, we define an extended 3-dimensional TQFT called Turaev–Viro theory. It will be defined as a lattice theory: we will define $Z(M)$ by choosing a cell decomposition of M, writing $Z(M)$ as a state sum, and then proving that it is independent of the choice of cell decomposition. In many ways, it is very similar to the definition of 2-dimensional FHK theory in Chapter 12, but replacing a separable symmetric Frobenius algebra by a spherical fusion category $\mathcal{C}$.

This theory was introduced by Turaev and Viro [**TV1992**] in the special case when $\mathcal{C}$ is the category of representations of quantum group $U_q\mathfrak{su}(2)$ with q being a root of unity; in their work, they only considered the usual (unextended) theory. It was generalized in [**BW1996**] to arbitrary spherical fusion categories (again, as an unextended theory); in fact, this was the motivation for introducing the notion of a spherical fusion category. Extension to 3-2-1 theory was done in the series of papers [**BK2010**], [**Bal2010a**], [**Bal2010b**]. Extension down to a point is fairly obvious (we were just too lazy to write it down).

Throughout this chapter, $\mathcal{C}$ is a spherical fusion category over an algebraically closed field $\mathbf{k}$ of characteristic zero. We keep all notation from the previous chapter, in particular denoting by $\mathcal{O}(\mathcal{C})$ the set of isomorphism classes of simple objects in $\mathcal{C}$ and choosing a representative X_i for each $i \in \mathcal{O}(\mathcal{C})$. Our main reference is [**BK2010**].

19.1. Turaev–Viro theory in dimensions $0, 1$

Defining $Z(\bullet)$, $Z(interval)$, $Z(S^1)$.

19.2. Turaev–Viro theory in dimension 2

$Z(2cell)$. $Z(surface)$

19.3. Turaev–Viro theory: full definition

$Z(3cell)$

19.4. Example: Dijkgraaf–Witten in 3 dimensions

Part 5

Chern–Simons and Crane–Yetter theories

CHAPTER 20

3-2-1 TQFT and modular categories

References: for modular categories, [**EGNO2015**]; for modular functor, [**BK2001**] and works by Jurgen Andersen (FIXME). For correspondence between 321 theories and modular categories, [**BDSPV2015**].

20.1. 3-2-1 theories

Overview of 3-2-1 theory. Theorem: every 3-2-1 theory defines on $\mathcal{C}$ a structure of ribbon category.

20.2. Graphical calculus

20.3. Modular group action and modular categories

Verlinde algebra. Links. Theorem: a 3-2-1 theory defines on $\mathcal{C}$ structure of a modular category (assuming it is semisimple).

20.4. Reshetikhin–Turaev theory

Theorem: every modular category defines a (central extension of) 3-2-1 theory. Historical reference; surgery.

20.5. Chern–Simons theory

Category $\mathcal{C}(\mathfrak{g}, k)$. Result of Kazhdan–Lusztig–Finkelberg. Relation with path integral (Axelrod–Singer)

20.6. Link invariants

Invariants of links in S^3. Jones polynomial.

Index

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